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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D7

MLB

LAST_MODIFIED=Thu Jan 12 10:24:09 2012

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Schematic / PCB #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
051-9509	1	SCH, MLB, D7	SCH	CRITICAL	
820-3302	1	PCBF, MLB, D7	PCB	CRITICAL	

DRAWING

TITLE=D7

ABBREV=DRAWING

LAST_MODIFIED=Thu Jan 12 10:24:09 2012

REV

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DESCRIPTION OF REVISION

CK APPD

DATE

2012-01-12

DRAWING TITLE

SCH, D7, MLB

Apple Inc.

051-9509

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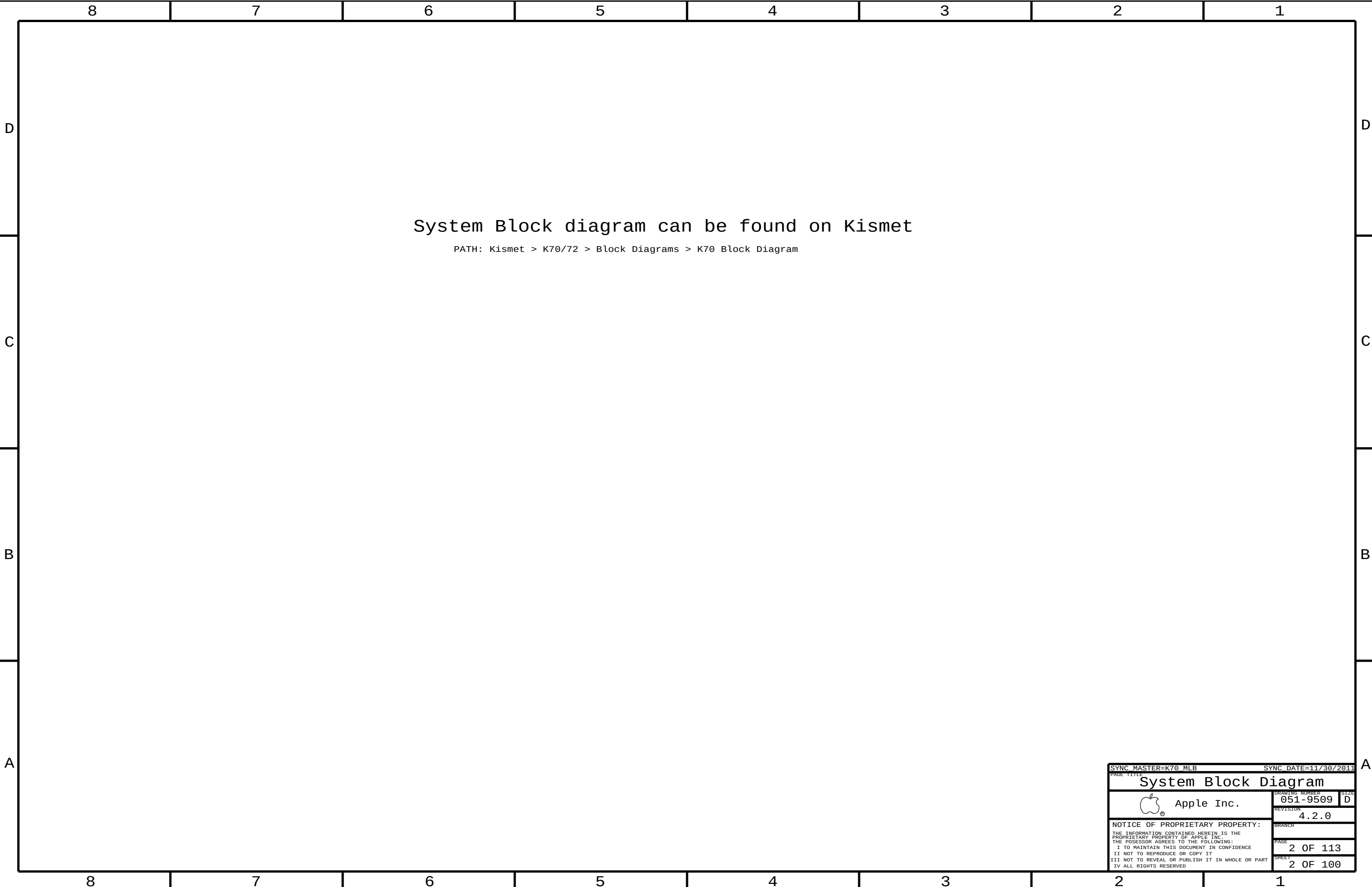
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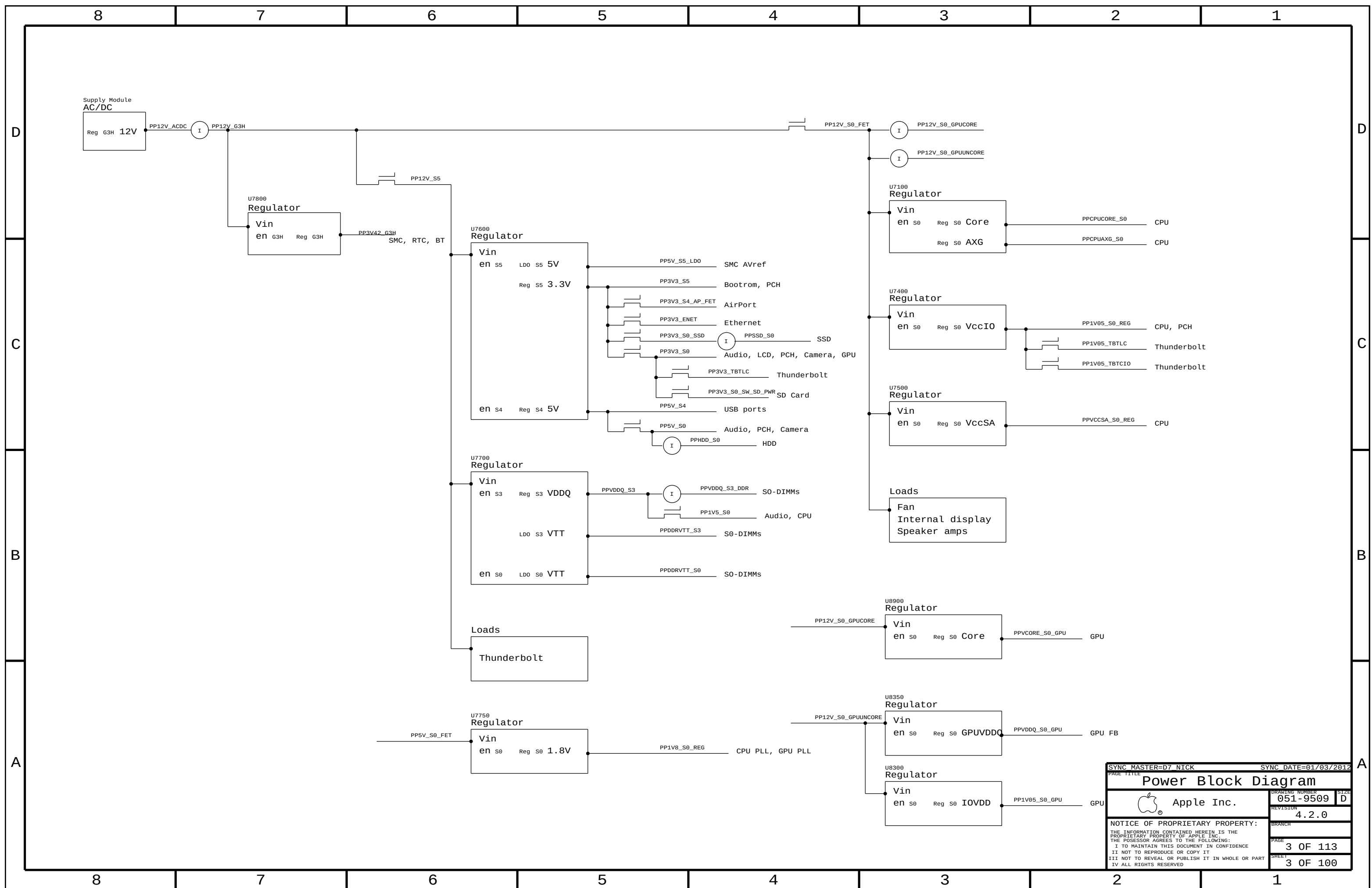
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System Block Diagram			
 Apple Inc.		DRAWING NUMBER	051-9509
		REVISION	4.2.0
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Main BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
085-4441	PCBA, MLB, DEV, D7	DEVELOPMENT, D7_DEVEL
639-3566	PCBA, MLB, D7, GSA, GOOD	D7_COMMON, CPU: GOOD, GPU: GSA, GS, FBA, SSD: N, EEEE: DF98
639-3668	PCBA, MLB, D7, GSB, GOOD	D7_COMMON, CPU: GOOD, GPU: GSB, GS, FBB, SSD: N, EEEE: F117
639-3567	PCBA, MLB, D7, GTX, BETTER	D7_COMMON, CPU: BETTER, GPU: 107GTX, FBA, FBB, SSD: Y, EEEE: DT42
639-3665	PCBA, MLB, D7, GTX, CTO	D7_COMMON, CPU: CTO, GPU: 107GTX, FBA, FBB, SSD: Y, EEEE: F116

Bar Code Labels / EEEE #'s

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_DF98	CRITICAL	EEEE:DF98
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_DT42	CRITICAL	EEEE:DT42
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F116	CRITICAL	EEEE:F116
825-7122	1	MLB LABEL, 48.0X4.8	EEEE_F117	CRITICAL	EEEE:F117

BOM Groups

BOM_GROUP	BOM_OPTIONS
D7_COMMON	COMMON,ALTERNATE,D7_COMMON1,D7_COMMON2,D7_PROGPARTS
D7_COMMON1	XDP,RSMRST:SMC,SPEAKERID,TBTHV:P12V
D7_COMMON2	SNS_CPUCORE:3PHASE,CPUCOREDRV:ISL6612,IG:N,GPU_ROM:YES,SNS_GPUS0:K70,SNS_VDDQS3_DDR:Y
D7_PROGPARTS	SMC:PROT01,B00TROM:PROG,T29ROM:PROG,CIVROM:PROG,CAMROM:PROG
D7_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,BKLT_PWM,DEVEL_SENSORS,DEVEL_AUDIO
DEVEL_SENSORS	SNS_VDDQS0_GPU:Y,SNS_VDDQS3:Y,TEMPSNSDEV
D7_PRODUCTION	SNS_VDDQS0_GPU:N,SNS_VDDQS3:N,VREFMRGN:N

Add 'K70_PRODUCTION' at RevA release

CPUs

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4240	1	IVB, QC13, QS, E0, 2. 8G, 65W, 4+1, 1. 10, 6M, LGA	CPU	CRITICAL	CPU:GOOD
337S4258	1	IVB, QC48, QS, E1, 2. 9G, 65W, 4+2, 1. 10, 6M, LGA	CPU	CRITICAL	CPU:BETTER
337S4246	1	IVB, SR0PN, PRQ, E1, 3. 16, 65W, 4+2, 1. 15, 8M, LGA	CPU	CRITICAL	CPU:CTO

Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
337S4234	1	IC, PCH, PPT-DT, Z77, QS, C1	U1800	CRITICAL	
338S1047	1	IC, TRT, CR-4C, ES1, 288 FCBGA, 12X12MM	U3600	CRITICAL	
337S4221	1	IC, GPU, NV, GK107-GS-2/1-QS-A	U8000	CRITICAL	GPU: GSA
337S4220	1	IC, GPU, NV, GK107-GS-2/1-QS-A	U8000	CRITICAL	GPU: GSB
337S4239	1	IC, GPU, NV, GK107-6TX-QS-A2	U8000	CRITICAL	GPU: 107GTX
343S0592	1	IC, BCM57766, CIV+, A0, 8X8	U3900	CRITICAL	
607-9432	1	K70, GDDR5, SAMSUNG	VRAM	CRITICAL	GPU: 107GTX

Programmable Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
341S3493	1	IC, CR, V24. 2, D7/D7I	U3690	CRITICAL	T29ROM:PROG
335S0865	1	IC, EEPROM, SERIAL, 256KB, MLP8	U3690	CRITICAL	T29ROM:BLANK
335S0897	1	IC, 64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK
341S3480	1	IC, PROGRAMD, EFI ROM, K70	U5110	CRITICAL	BOOTROM:PROG
335S0862	1	IC, SERIAL FLASH, 2MBIT, 2.7V, REV F	U3990	CRITICAL	CIVROM:BLANK
341S3487	1	IC, ENET 1MBITFLASH, CIV, PVT, 340	U3990	CRITICAL	CIVROM:PROG
338S1098	1	IC, SMC12-A3, BLANK, D7	U4900	CRITICAL	SMC:BLANK
341S3484	1	IC, SMC, PROGRAMD, PROTO1, D7	U4900	CRITICAL	SMC:PROTO1
341S3388	1	IC, SMC, PROGRAMD, EVT, D7	U4900	CRITICAL	SMC:EVT
341S3389	1	IC, SMC, PROGRAMD, DVT, D7	U4900	CRITICAL	SMC:DVT
341S3396	1	IC, SMC, PROGRAMD, PVT, D7	U4900	CRITICAL	SMC:PVT
341S3409	1	IC, SMC, PROGRAMD, PROD, D7	U4900	CRITICAL	SMC:PROD
341S3453	1	IC, CAMERA FLASH, K70/K72	U4202	CRITICAL	CAMROM:PROG
335S0852	1	IC, FLASH, SPI, 1MBIT, 3V3	U4202	CRITICAL	CAMROM:BLANK

Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
377S0107	377S0126		ALL	USB diodes
157S0055	157S0058		ALL	Enet magnetics
376S1081	376S0975		ALL	P/NCh dual FET
341S3486	341S3487		ALL	P/NCh dual FET

CPU Socket

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
511S0073	1	SOCKET, MOLEX, LGA1155, CPU	LF U1000	CRITICAL	

CPU Socket Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
511S0071	511S0073		ALL	TYCO SOCKET
511S0072	511S0073		ALL	FOXCONN SOCKET

VRAM BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
607-9432	K70, GDDR5, SAMSUNG	FB: BOTH_SAMSUNG
607-9435	K70, GDDR5, HYNIX	FB: BOTH_HYNIX
607-9433	K70, GDDR5, SAMSUNG_CH1	FB: CH1_SAMSUNG
607-9436	K70, GDDR5, HYNIX_CH1	FB: CH1_HYNIX
607-9434	K70, GDDR5, SAMSUNG_CH2	FB: CH2_SAMSUNG
607-9437	K70, GDDR5, HYNIX_CH2	FB: CH2_HYNIX

VRAM Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
333S0619	4	IC, SGRAM, GDDR5, 32MX32, 1.5GHZ, G-DIE, HF	U8400, U8450, U8500, U8550	CRITICAL	FB: BOTH_SAMSUNG
333S0620	4	IC, GDDR5, 32MX32, 1.5GHZ, VEGA 44M, G-DIE	U8400, U8450, U8500, U8550	CRITICAL	FB: BOTH_HYNIX
333S0631	2	IC, SGRAM, GDDR5, 64MX32, 4.2GBPS, D-DIE, HF	U8400, U8450	CRITICAL	FB: CH1_SAMSUNG
333S0630	2	IC, GDDR5, 2GB, M-DIE, 170B FBGA	U8400, U8450	CRITICAL	FB: CH1_HYNIX
333S0631	2	IC, SGRAM, GDDR5, 64MX32, 4.2GBPS, D-DIE, HF	U8500, U8550	CRITICAL	FB: CH2_SAMSUNG
333S0630	2	IC, GDDR5, 2GB, M-DIE, 170B FBGA	U8500, U8550	CRITICAL	FB: CH2_HYNIX

VRAM Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
607-9435	607-9432		VRAM	GDDR5_B0TH
607-9436	607-9433	GPU:GSA	VRAM	GDDR5_CH1
607-9437	607-9434	GPU:GSB	VRAM	GDDR5_CH2

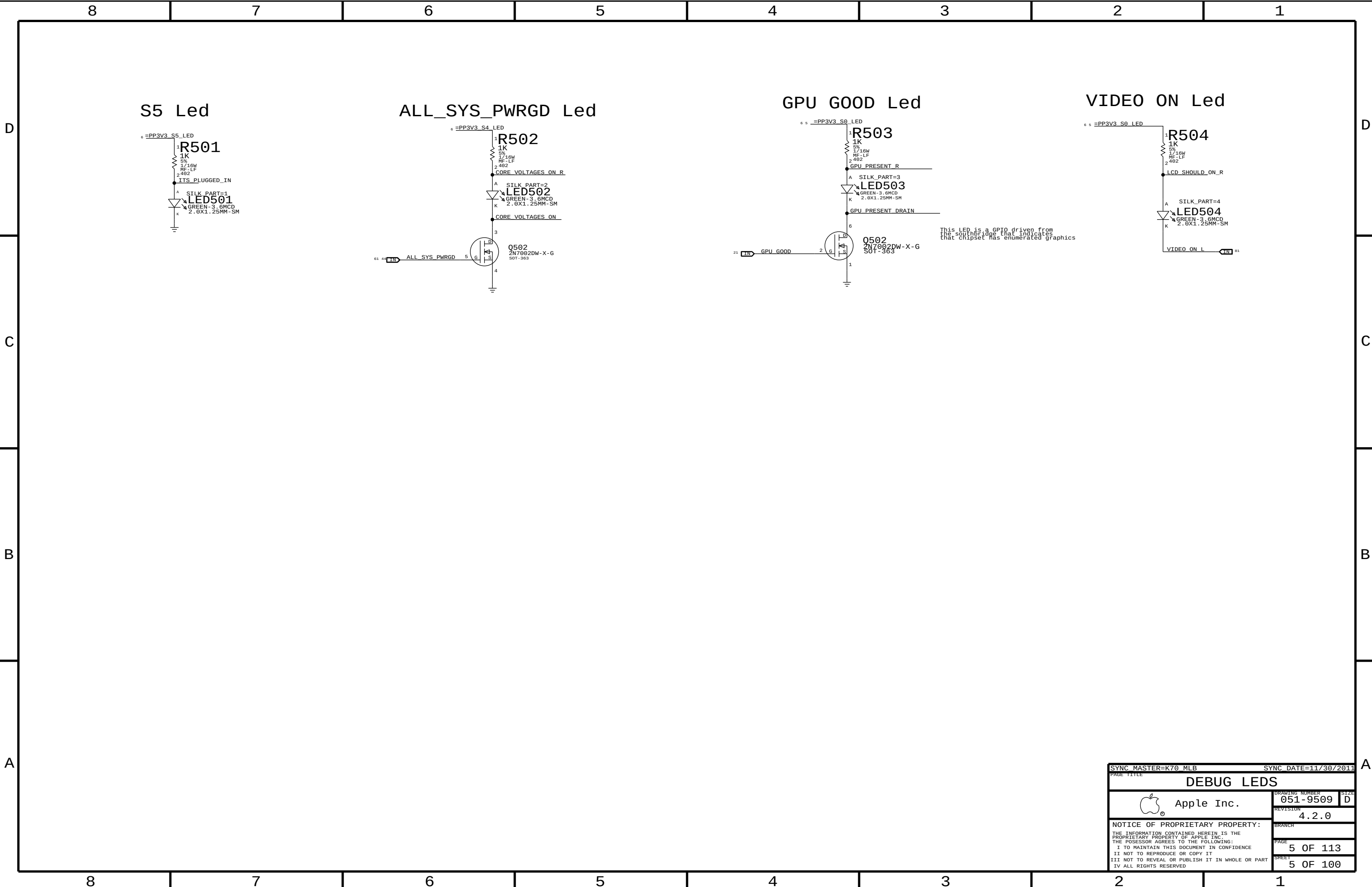
GPU Module Parts

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
607-9433	1	K70, GDORS, SAMSUNG_CH1	VRAM	CRITICAL	GPU: GSA
607-9434	1	K70, GDORS, SAMSUNG_CH2	VRAM	CRITICAL	GPU: GSB

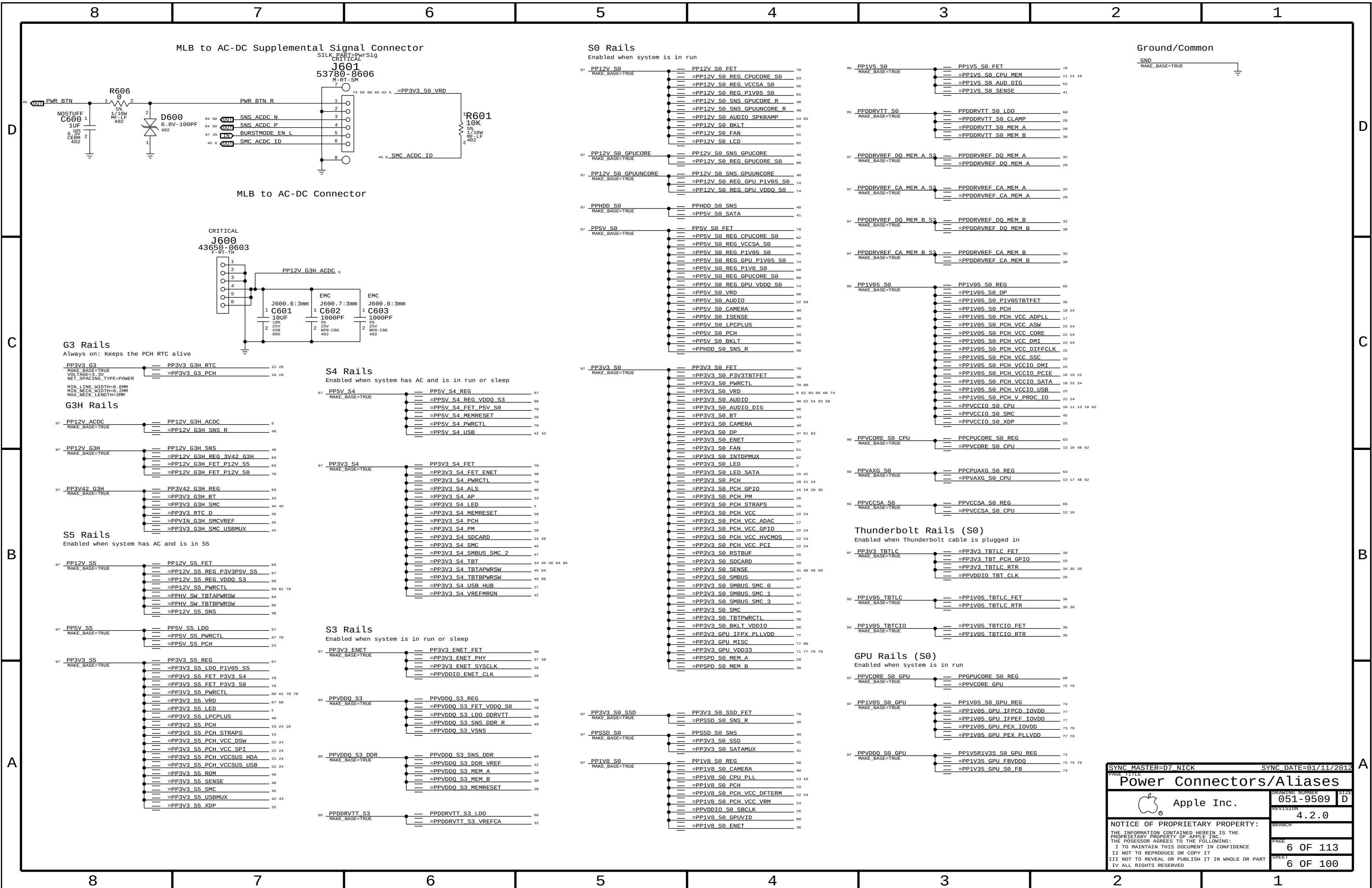
Programmable Parts (unused)

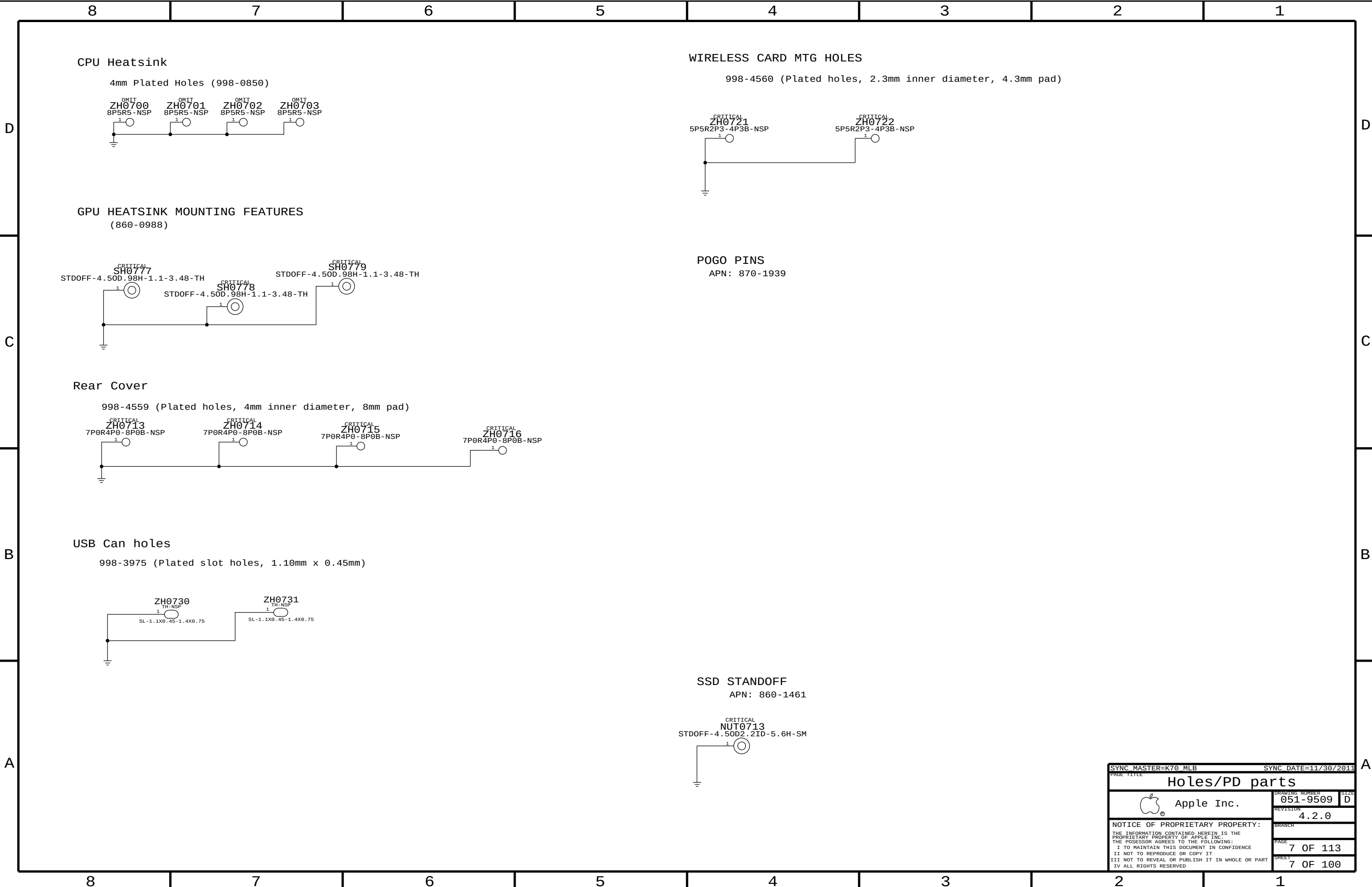
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S0724	1	IC,1 MBIT SERIAL FLASH	U8701	CRITICAL	GPUROM:BLANK

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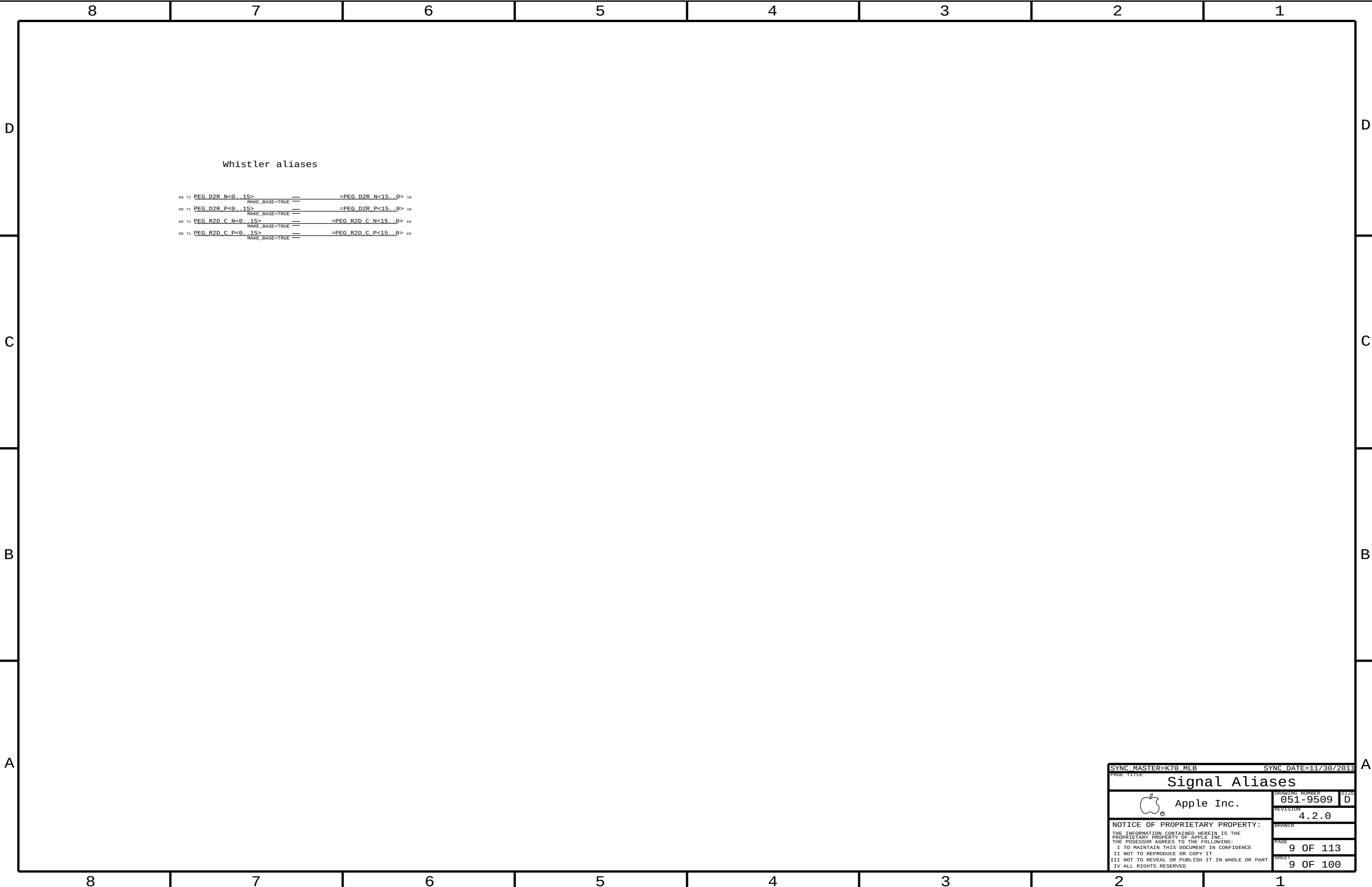
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SYNC MASTER=K70 MLB		SYNC DATE=11/30/2011	
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CPU Reserved		PCH Clocks		PCH Unused Display		PCH SATA		PCH Test Points	
10 TP_CPU_RSVD<16..1> == NC_CPU_RSVD<16..1> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE_CLK100M PE4N == NC_PCIE_CLK100M PE4N MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_RED == NC_CRT_IG_RED MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CN == NC_SATA_C_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP1 == NC_PCH_TP1 MAKE_BASE=TRUE NO_TEST=TRUE	
10 TP_CPU_RSVD<46..19> == NC_CPU_RSVD<46..19> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PCIE_CLK100M PE4P == NC_PCIE_CLK100M PE4P MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_GREEN == NC_CRT_IG_GREEN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_R2D_CP == NC_SATA_C_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP2 == NC_PCH_TP2 MAKE_BASE=TRUE NO_TEST=TRUE	
91 10 CPU_CFG<15..12> == TP_CPU_CFG<15..12> MAKE_BASE=TRUE		21 TP_PCIE_CLK100M PE6N == NC_PCIE_CLK100M PE6N MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_BLUE == NC_CRT_IG_BLUE MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RN == NC_SATA_C_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP3 == NC_PCH_TP3 MAKE_BASE=TRUE NO_TEST=TRUE	
		21 TP_PCIE_CLK100M PE6P == NC_PCIE_CLK100M PE6P MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_CRT_IG_HSYNC == NC_CRT_IG_HSYNC MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_C_D2RP == NC_SATA_C_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP4 == NC_PCH_TP4 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 TP_CRT_IG_VSYNC == NC_CRT_IG_VSYNC MAKE_BASE=TRUE NO_TEST=TRUE				21 TP_PCH_TP5 == NC_PCH_TP5 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 TP_CRT_IG_DDC_CLK == NC_CRT_IG_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CN == NC_SATA_D_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP6 == NC_PCH_TP6 MAKE_BASE=TRUE NO_TEST=TRUE	
				10 TP_CRT_IG_DDC_DATA == NC_CRT_IG_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_D_R2D_CP == NC_SATA_D_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP7 == NC_PCH_TP7 MAKE_BASE=TRUE NO_TEST=TRUE	
						10 TP_SATA_D_D2RN == NC_SATA_D_D2RN MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP8 == NC_PCH_TP8 MAKE_BASE=TRUE NO_TEST=TRUE	
						10 TP_SATA_D_D2RP == NC_SATA_D_D2RP MAKE_BASE=TRUE NO_TEST=TRUE		21 TP_PCH_TP9 == NC_PCH_TP9 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP10 == NC_PCH_TP10 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP11 == NC_PCH_TP11 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP12 == NC_PCH_TP12 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP13 == NC_PCH_TP13 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP14 == NC_PCH_TP14 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP15 == NC_PCH_TP15 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP16 == NC_PCH_TP16 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP17 == NC_PCH_TP17 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP18 == NC_PCH_TP18 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP19 == NC_PCH_TP19 MAKE_BASE=TRUE NO_TEST=TRUE	
								21 TP_PCH_TP20 == NC_PCH_TP20 MAKE_BASE=TRUE NO_TEST=TRUE	
CPU Memory									
12 TP_MEM_A_DQ_CB<7..0> == NC_MEM_A_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PE_TX_N<3..0> == NC_PE_TXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLN<3..0> == NC_DP_IG_B_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CN == NC_SATA_E_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE			
12 TP_MEM_A_DQS_N<8> == NC_MEM_A_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PE_TX_P<3..0> == NC_PE_TXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_MLP<3..0> == NC_DP_IG_B_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_R2D_CP == NC_SATA_E_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE			
12 TP_MEM_A_DQS_P<8> == NC_MEM_A_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PE_RX_N<3..0> == NC_PE_RXN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_HPD == NC_DP_IG_B_HPD MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RN == NC_SATA_E_D2RN MAKE_BASE=TRUE NO_TEST=TRUE			
12 TP_MEM_B_DQ_CB<7..0> == NC_MEM_B_DQ_CB<7..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_PE_RX_P<3..0> == NC_PE_RXP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_B_DDC_CLK == NC_DP_IG_B_DDC_CLK MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_E_D2RP == NC_SATA_E_D2RP MAKE_BASE=TRUE NO_TEST=TRUE			
12 TP_MEM_B_DQS_N<8> == NC_MEM_B_DQSN<8> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_B_DDC_DATA == NC_DP_IG_B_DDC_DATA MAKE_BASE=TRUE NO_TEST=TRUE					
12 TP_MEM_B_DQS_P<8> == NC_MEM_B_DQSP<8> MAKE_BASE=TRUE NO_TEST=TRUE									
88 12 MEM_A_CLK_N<2..3> == NC_MEM_A_CLKN<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_N == NC_DMI_MIDBUS_CLK100M_N MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLN<3..0> == NC_DP_IG_C_MLN<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_F_R2D_CN == NC_SATA_F_R2D_CN MAKE_BASE=TRUE NO_TEST=TRUE			
88 12 MEM_A_CLK_P<2..3> == NC_MEM_A_CLKP<2..3> MAKE_BASE=TRUE NO_TEST=TRUE		10 DMI_MIDBUS_CLK100M_P == NC_DMI_MIDBUS_CLK100M_P MAKE_BASE=TRUE NO_TEST=TRUE		10 DP_IG_C_MLP<3..0> == NC_DP_IG_C_MLP<3..0> MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_F_R2D_CP == NC_SATA_F_R2D_CP MAKE_BASE=TRUE NO_TEST=TRUE			
88 12 MEM_A_CS_L<2..3> == NC_MEM_A_CS_L<2..3> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_C_AUX_N == NC_DP_IG_C_AUXN MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_F_D2RN == NC_SATA_F_D2RN MAKE_BASE=TRUE NO_TEST=TRUE			
88 12 MEM_A_CKE<2..3> == NC_MEM_A_CKE<2..3> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_C_AUX_P == NC_DP_IG_C_AUXP MAKE_BASE=TRUE NO_TEST=TRUE		10 TP_SATA_F_D2RP == NC_SATA_F_D2RP MAKE_BASE=TRUE NO_TEST=TRUE			
88 12 MEM_B_CLK_N<2..3> == NC_MEM_B_CLKN<2..3> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_C_HPD == NC_DP_IG_C_HPD MAKE_BASE=TRUE NO_TEST=TRUE					
88 12 MEM_B_CLK_P<2..3> == NC_MEM_B_CLKP<2..3> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_C_CTRL_CLK == NC_DP_IG_C_CTRL_CLK MAKE_BASE=TRUE NO_TEST=TRUE					
88 12 MEM_B_CS_L<2..3> == NC_MEM_B_CS_L<2..3> MAKE_BASE=TRUE NO_TEST=TRUE				10 DP_IG_C_CTRL_DATA == NC_DP_IG_C_CTRL_DATA MAKE_BASE=TRUE NO_TEST=TRUE					
88 12 MEM_B_CKE<2..3> == NC_MEM_B_CKE<2..3> MAKE_BASE=TRUE NO_TEST=TRUE									
88 12 MEM_A_ODT<2..3> == NC_MEM_A_ODT<2..3> MAKE_BASE=TRUE NO_TEST=TRUE									
88 12 MEM_B_ODT<2..3> == NC_MEM_B_ODT<2..3> MAKE_BASE=TRUE NO_TEST=TRUE									



Whistler aliases

SYNC_MASTER=K70_MLB

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Signal Aliases

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4.2.0

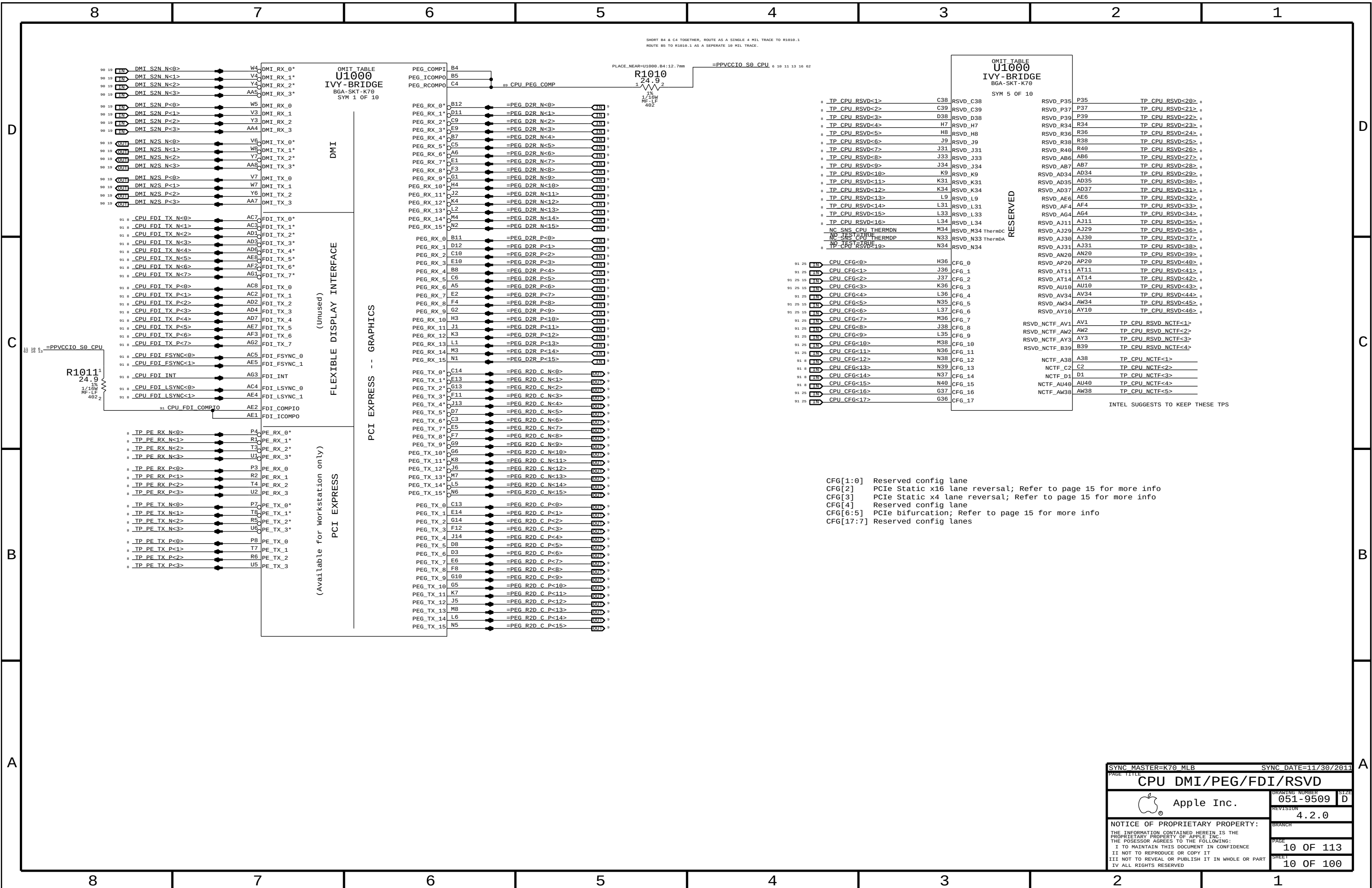
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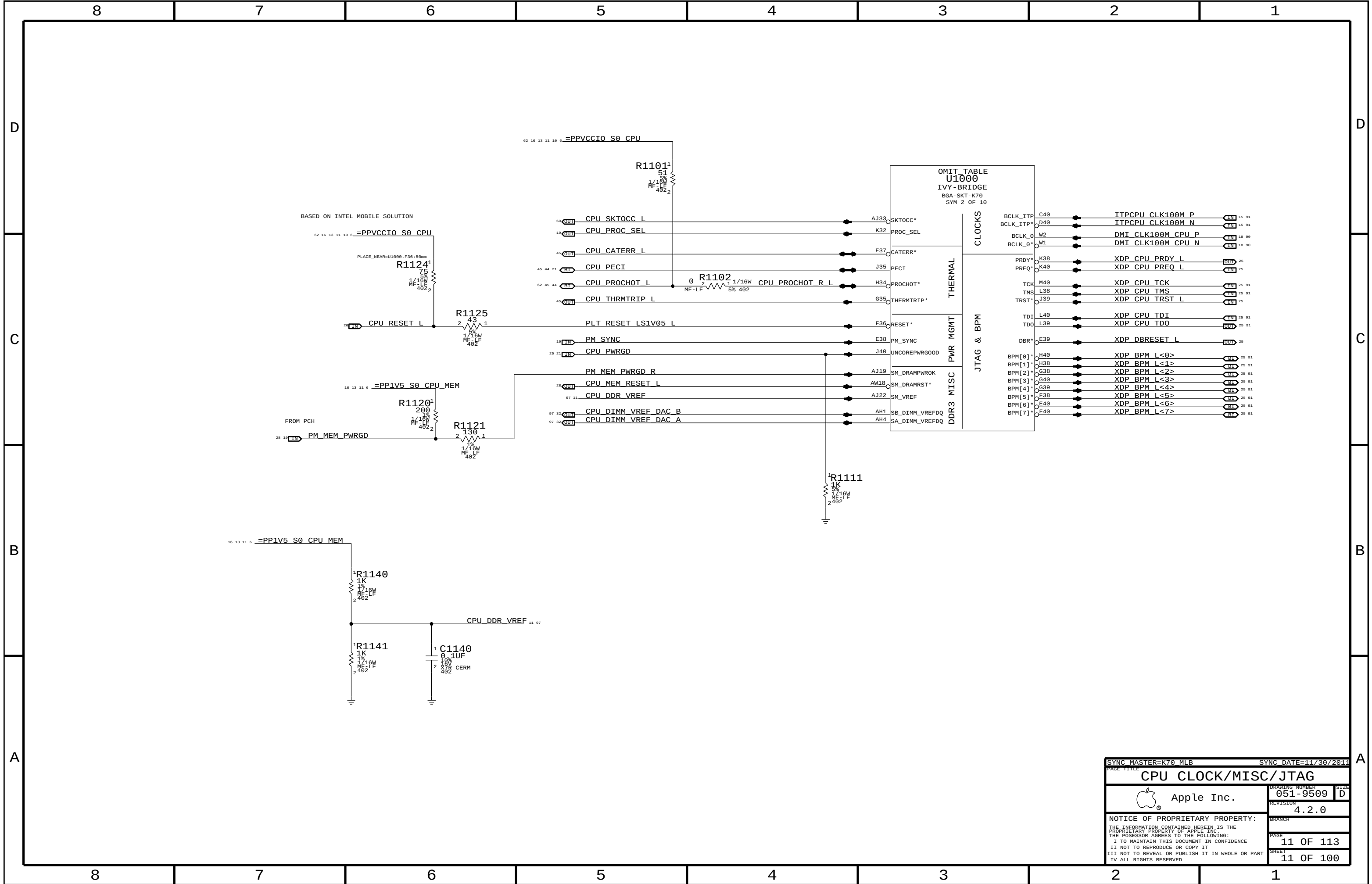
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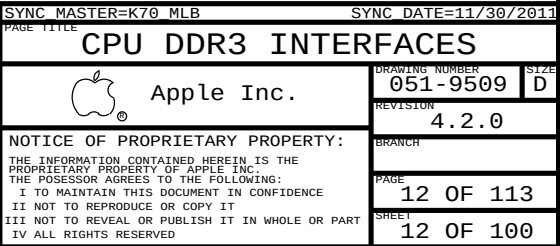
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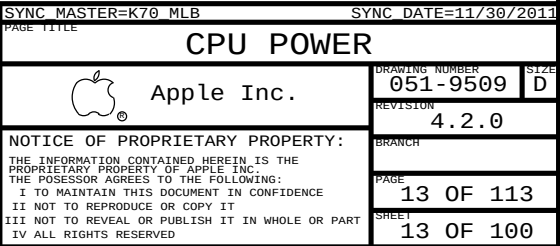
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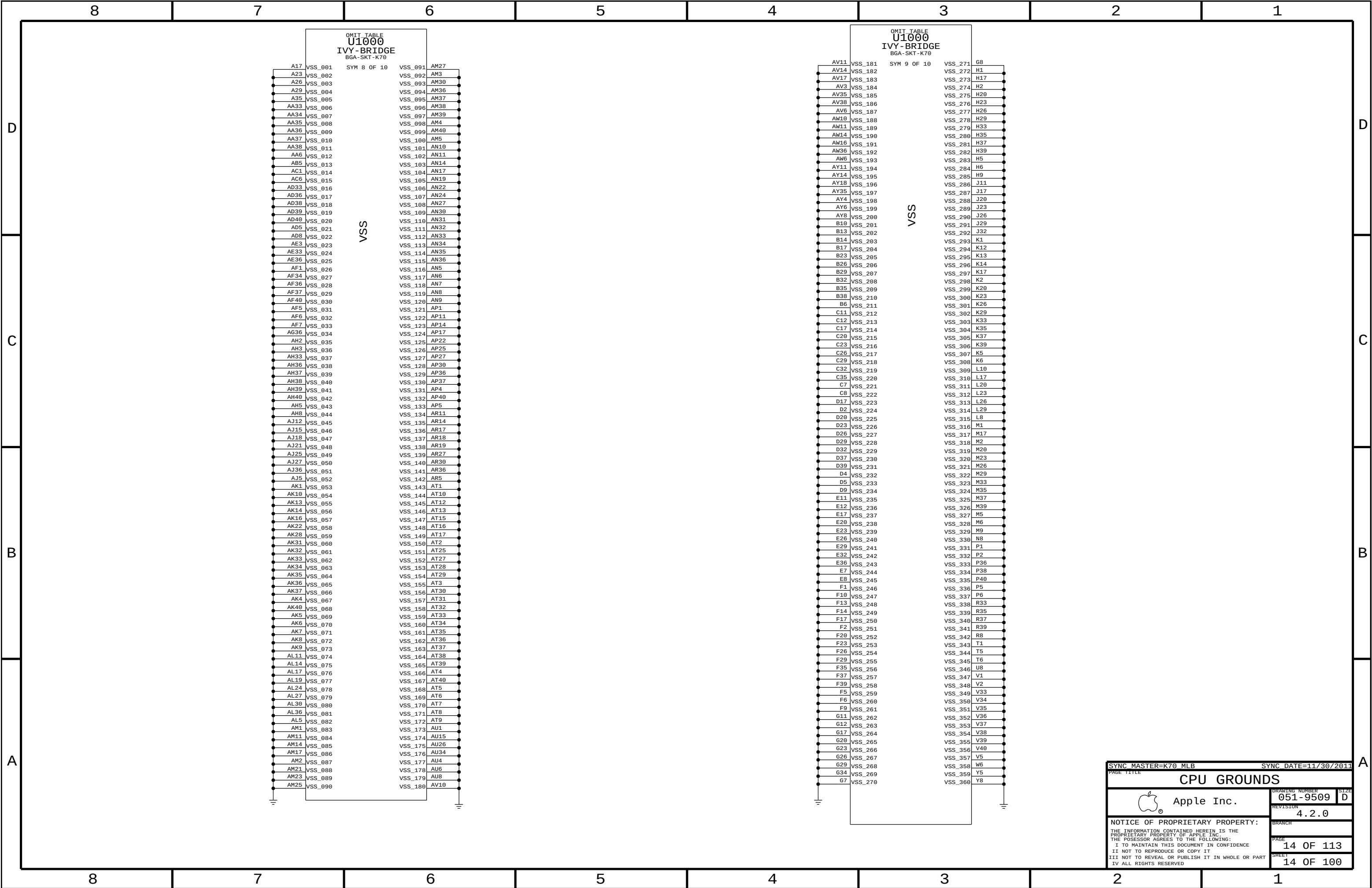
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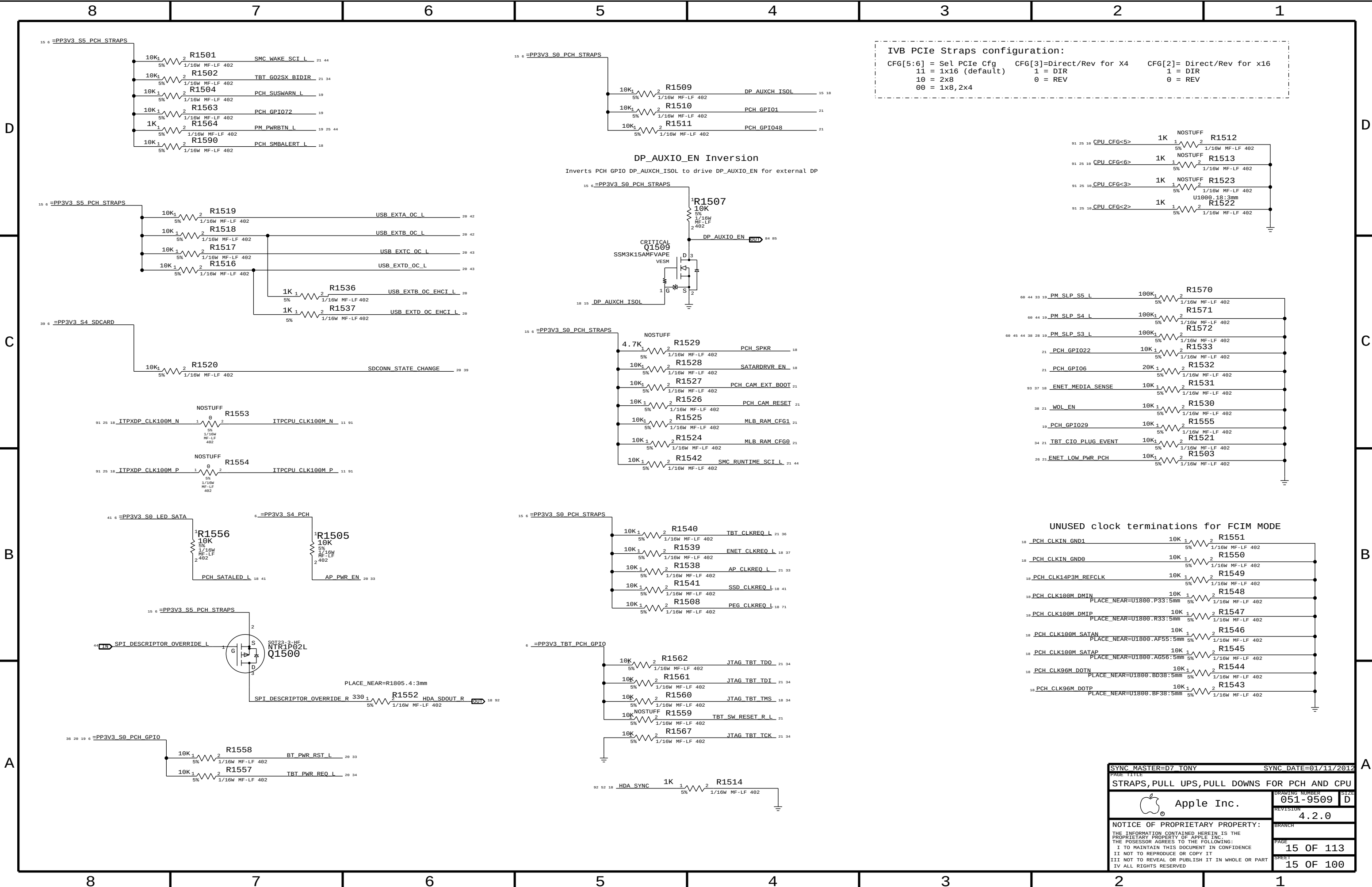
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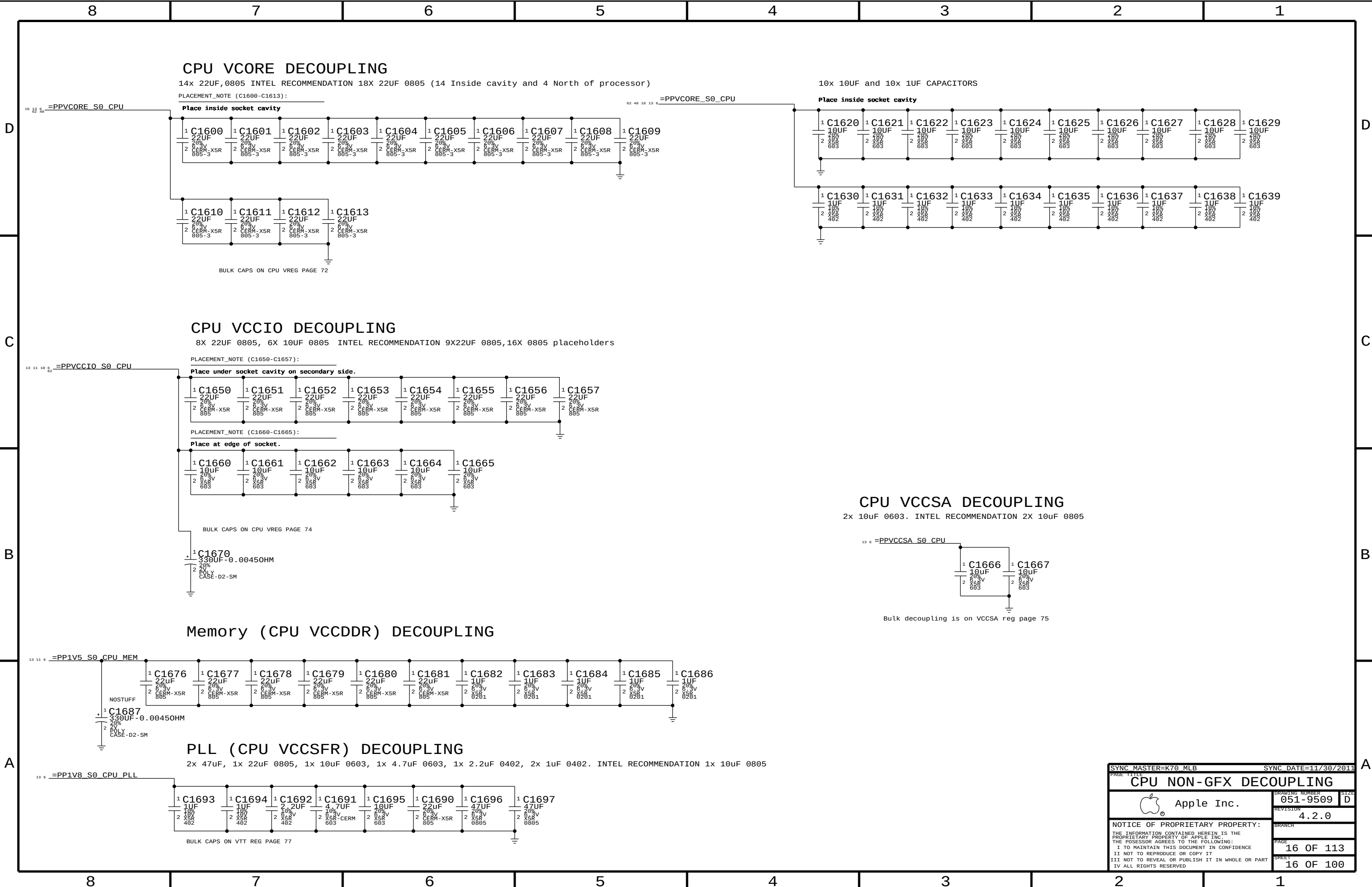
IVB PCIe Straps configuration:
CFG[5:6] = Sel PCIe Cfg CFG[3]=Direct/Rev for X4 CFG[2]= Direct/Rev for x16
11 = 1x16 (default) 1 = DIR 1 = DIR
10 = 2x8 0 = REV 0 = REV
00 = 1x8,2x4

DP_AUXIO_EN Inversion
Inverts PCH GPIO DP_AUXCH_ISOL to drive DP_AUXIO_EN for external DP

UNUSED clock terminations for FCIM MODE

18 PCH CLKIN_GND1	10K	1/16W MF-LF 402	R1551
18 PCH CLKIN_GND0	10K	1/16W MF-LF 402	R1550
18 PCH CLK14P3M_REFCLK	10K	1/16W MF-LF 402	R1549
18 PCH CLK100M_DMIN	10K	1/16W MF-LF 402	R1548
18 PCH CLK100M_DMIP	10K	1/16W MF-LF 402	R1547
18 PCH CLK100M_SATAN	10K	1/16W MF-LF 402	R1546
18 PCH CLK100M_SATAP	10K	1/16W MF-LF 402	R1545
18 PCH CLK96M_DOTN	10K	1/16W MF-LF 402	R1544
18 PCH CLK96M_DOTP	10K	1/16W MF-LF 402	R1543

SYNC MASTER=D7 TONY		SYNC DATE=01/11/2012	
PAGE TITLE			
STRAPS,PULL UPS,PULL DOWNS FOR PCH AND CPU			
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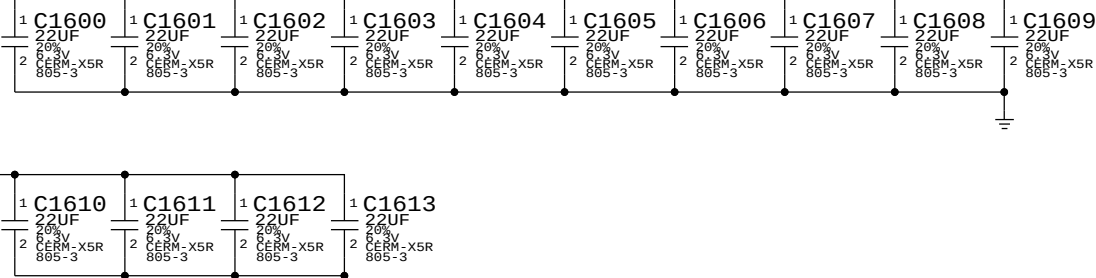


CPU Vcore DECOUPLING

14x 22uF,0805 INTEL RECOMMENDATION 18X 22uF 0805 (14 Inside cavity and 4 North of processor)

PLACEMENT_NOTE (C1600-C1613):

Place inside socket cavity



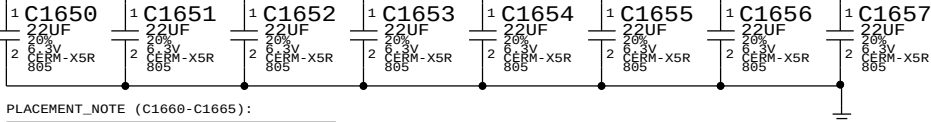
BULK CAPS ON CPU VREG PAGE 72

CPU VCCIO DECOUPLING

8X 22uF 0805, 6X 10uF 0805 INTEL RECOMMENDATION 9X22uF 0805,16X 0805 placeholders

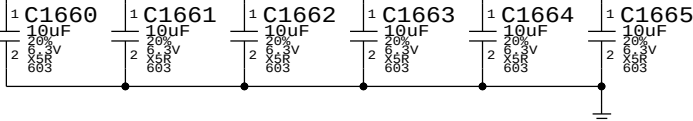
PLACEMENT_NOTE (C1650-C1657):

Place under socket cavity on secondary side.

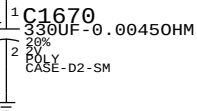


PLACEMENT_NOTE (C1660-C1665):

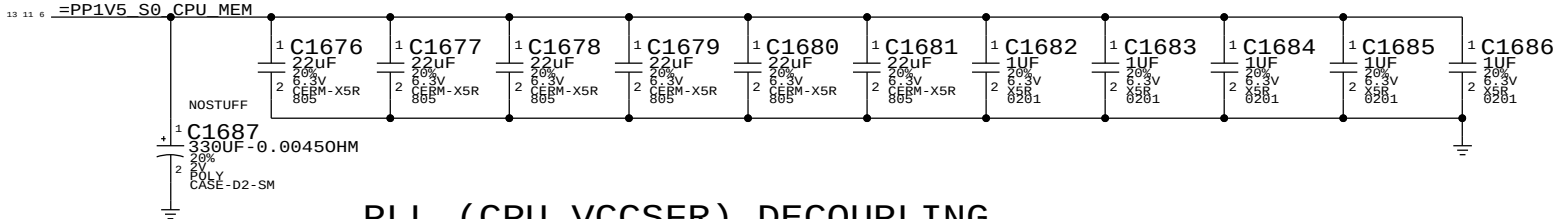
Place at edge of socket.



BULK CAPS ON CPU VREG PAGE 74

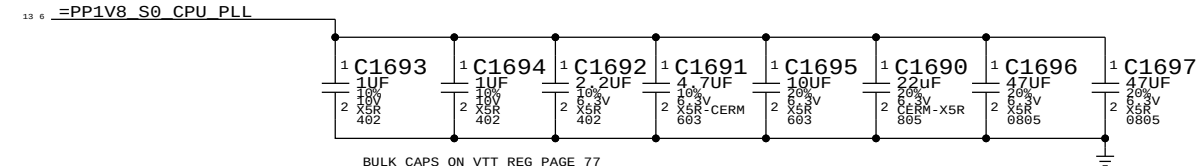


Memory (CPU VCCDDR) DECOUPLING



PLL (CPU VCCSFR) DECOUPLING

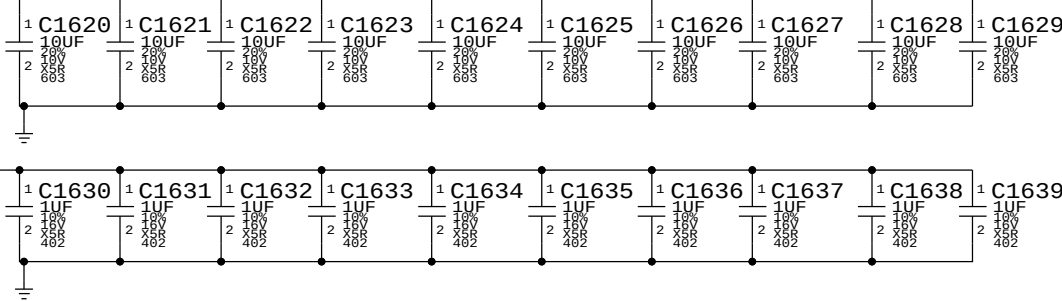
2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 1x 10uF 0805



BULK CAPS ON VTT REG PAGE 77

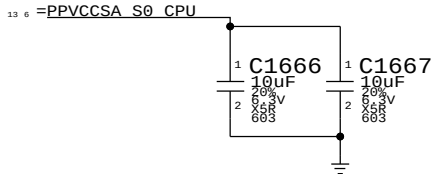
10x 10uF and 10x 1uF CAPACITORS

Place inside socket cavity



CPU VCCSA DECOUPLING

2x 10uF 0603. INTEL RECOMMENDATION 2X 10uF 0805



Bulk decoupling is on VCCSA reg page 75

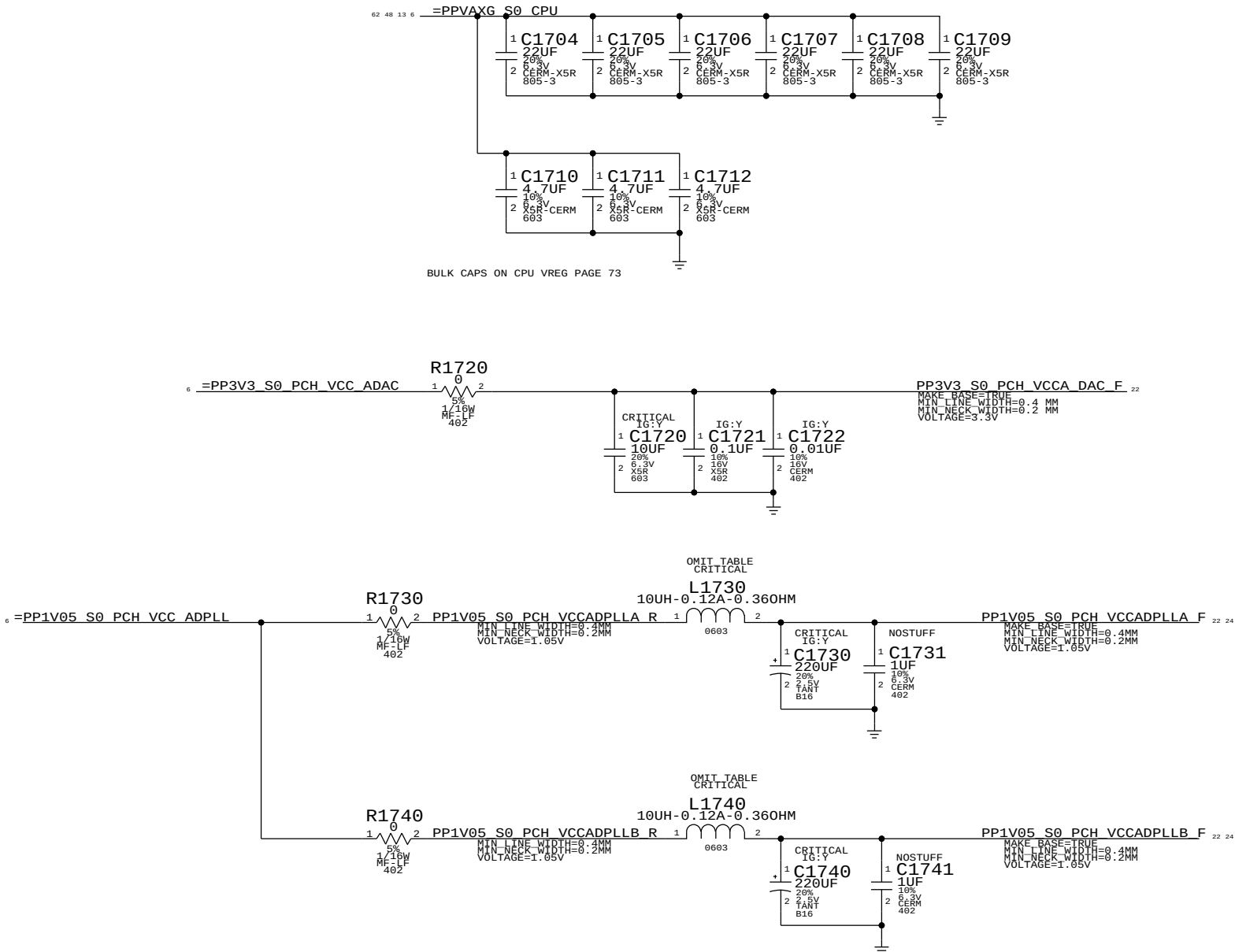
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		DRAWING NUMBER	051-9509
		REVISION	4.2.0
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VAXG DECOUPLING

INTEL RECOMMENDATION 4X22UF 0805,3X 4.7UF

PLACEMENT_NOTE (C1704-C1709):

Place inside socket cavity



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
152S1070	2	IND, WW, 10UH, 20%.120MA, 0.360HMS	L1730, L1740	CRITICAL	IG:Y
113S0022	2	RES, MF, 1/10W, 00HM, 5, 0603, SMD, LF	L1730, L1740		IG:W

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GFX DECOUPLING & PCH PWR ALIAS

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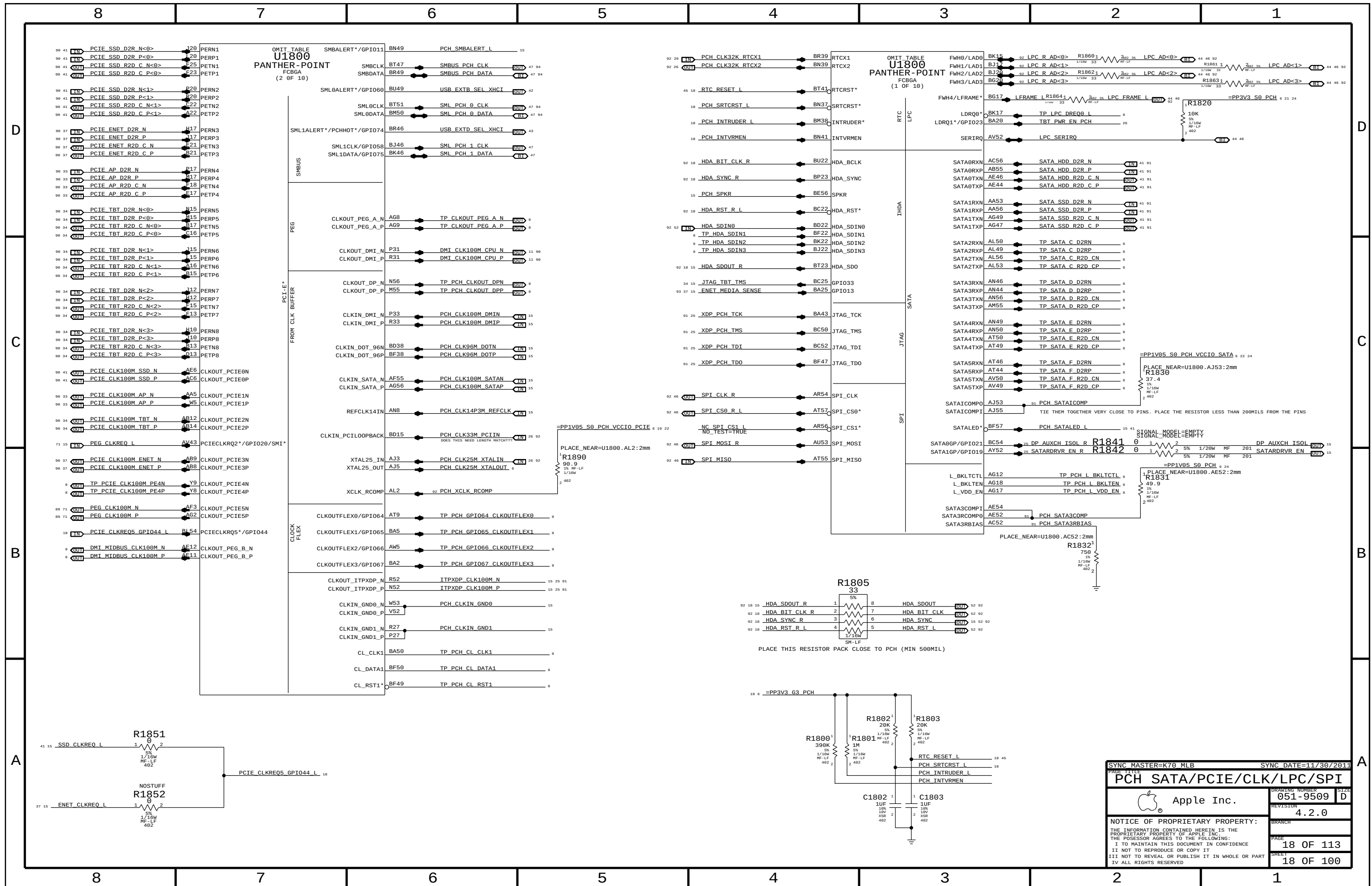
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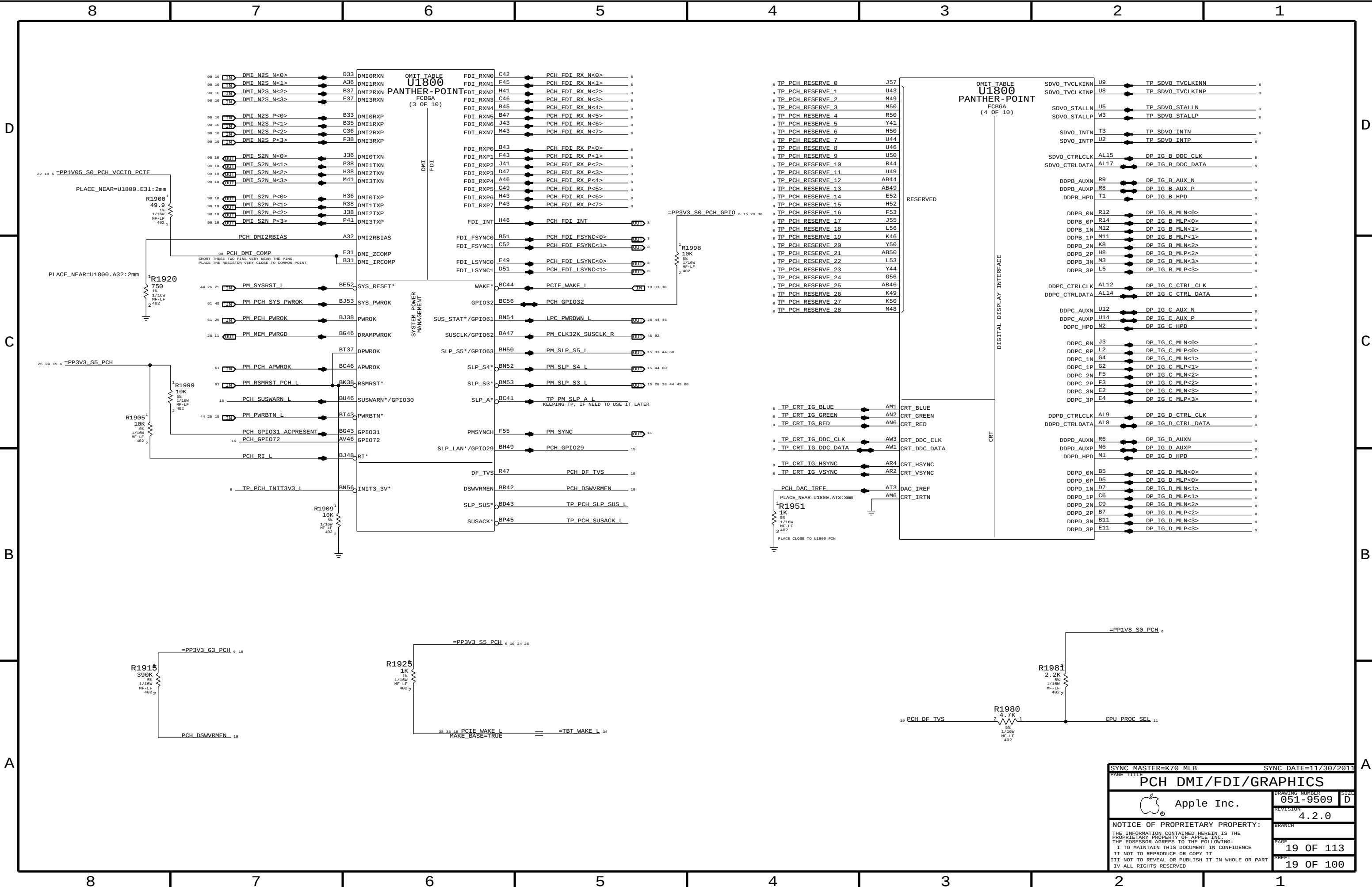
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PCH DMI/FDI/GRAPHICS

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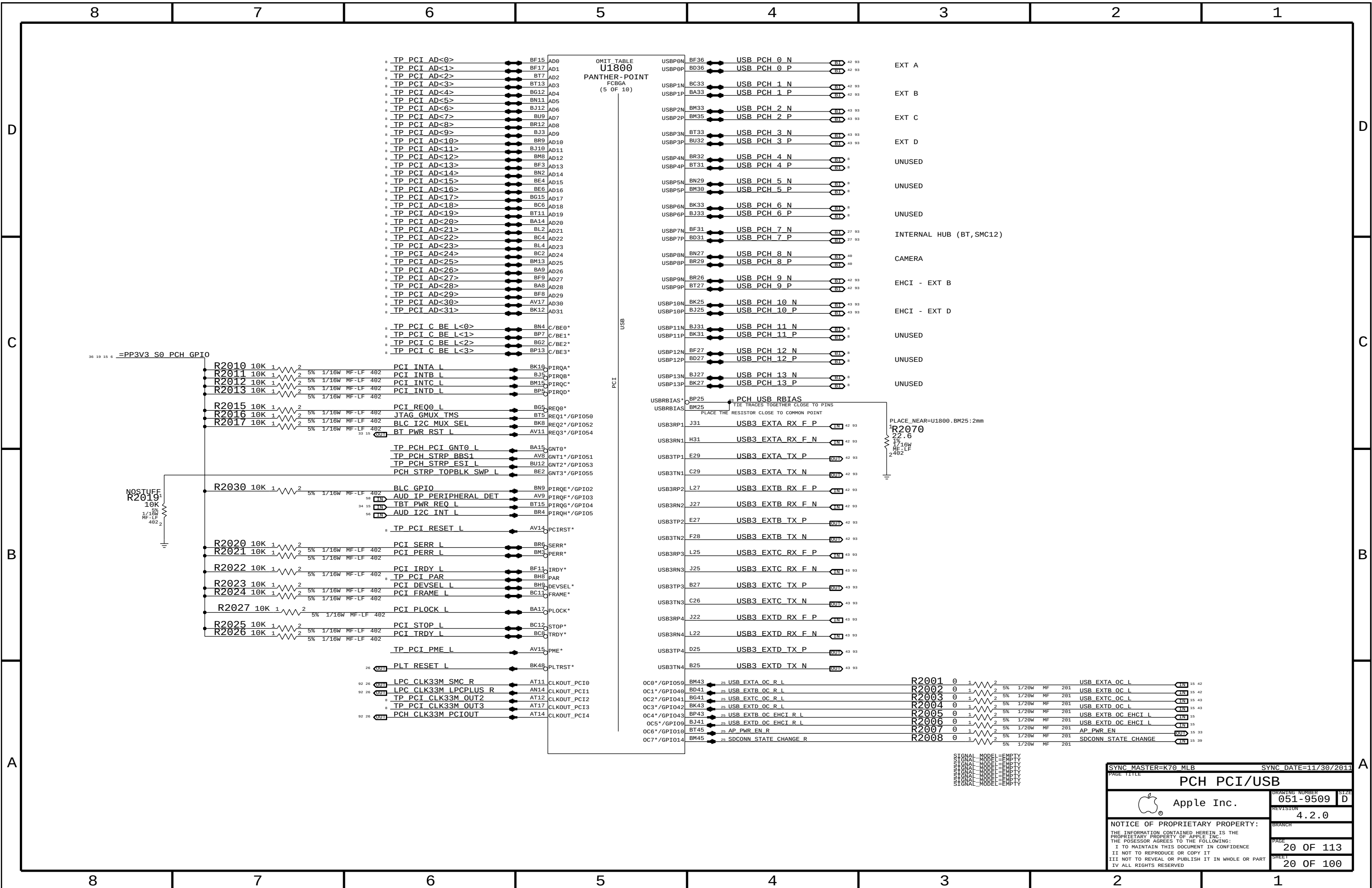
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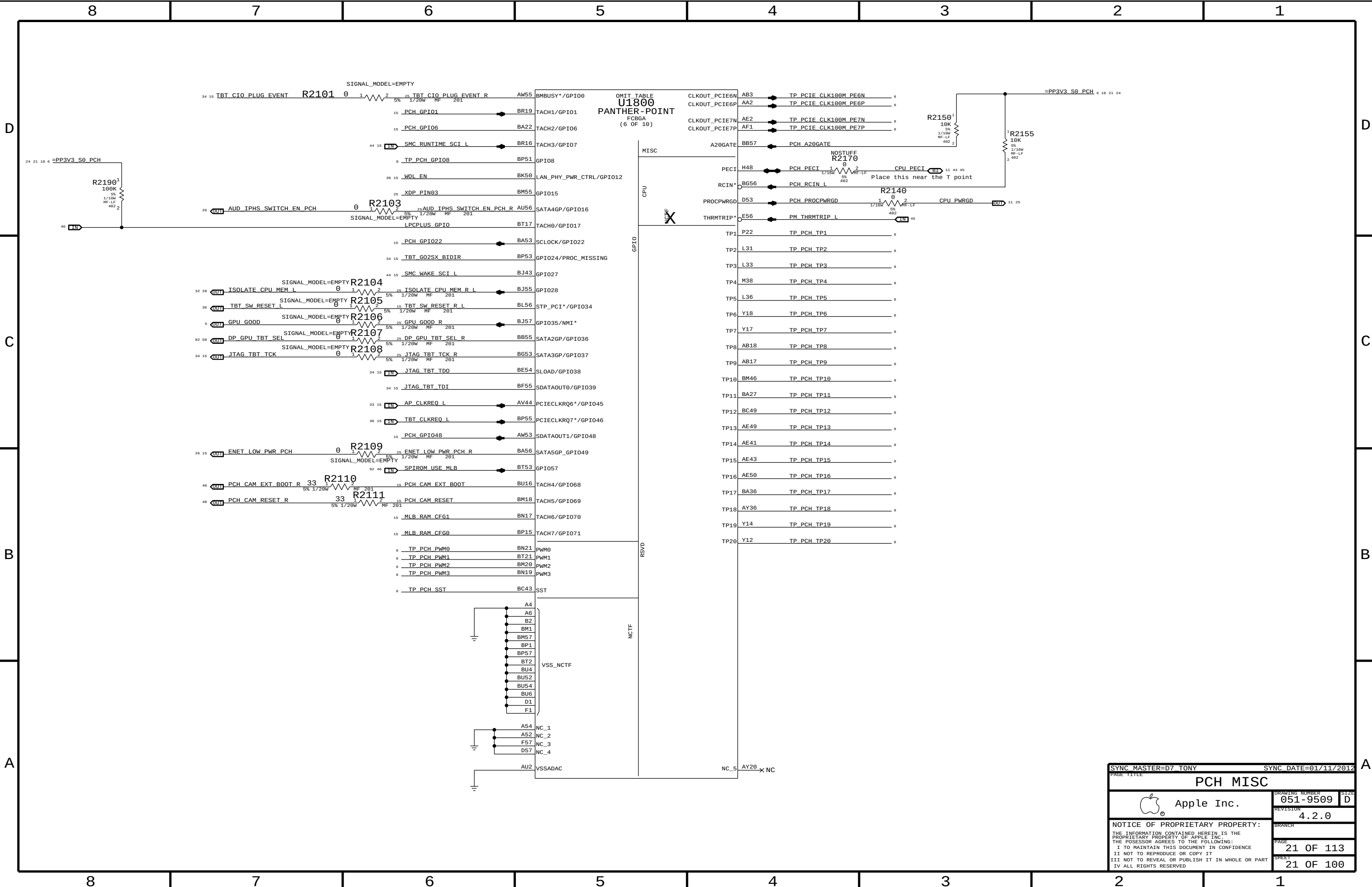
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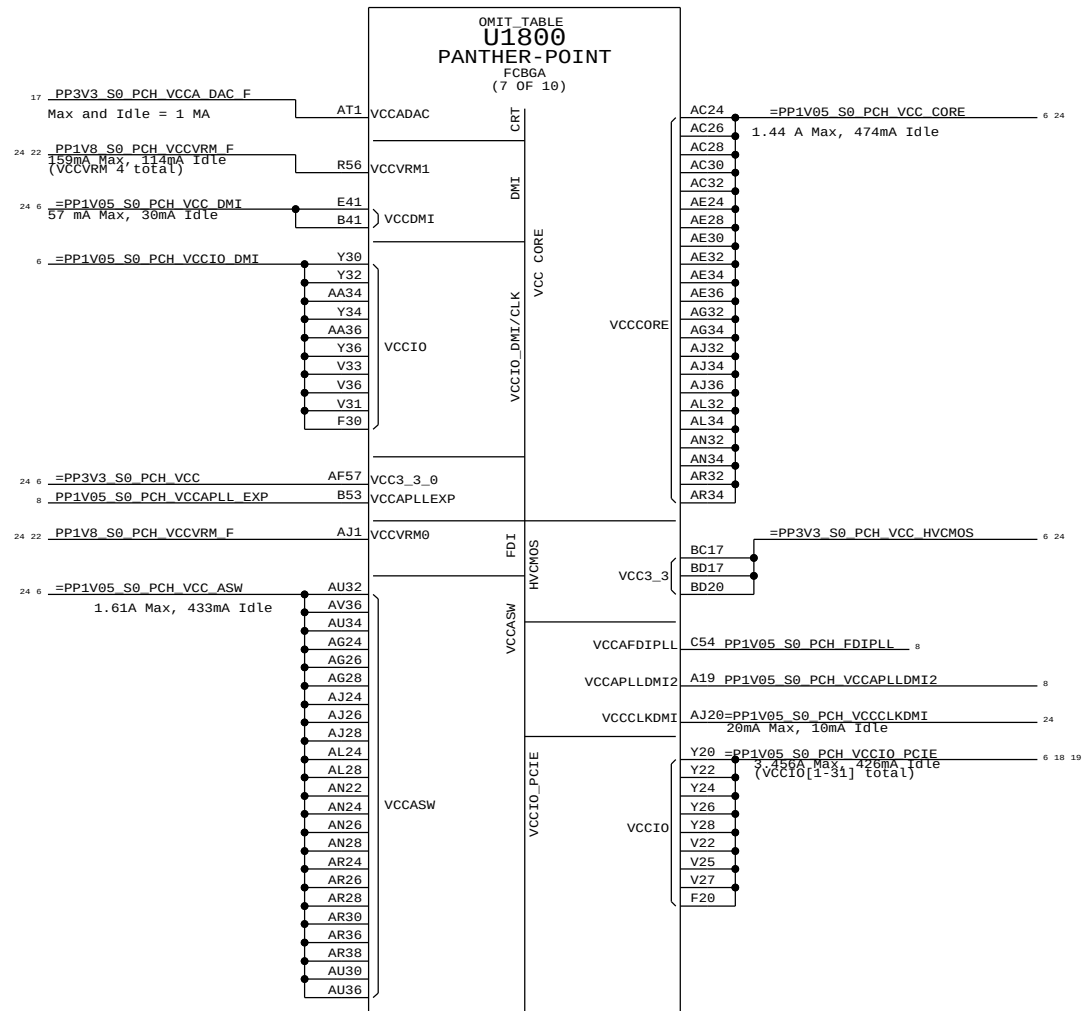
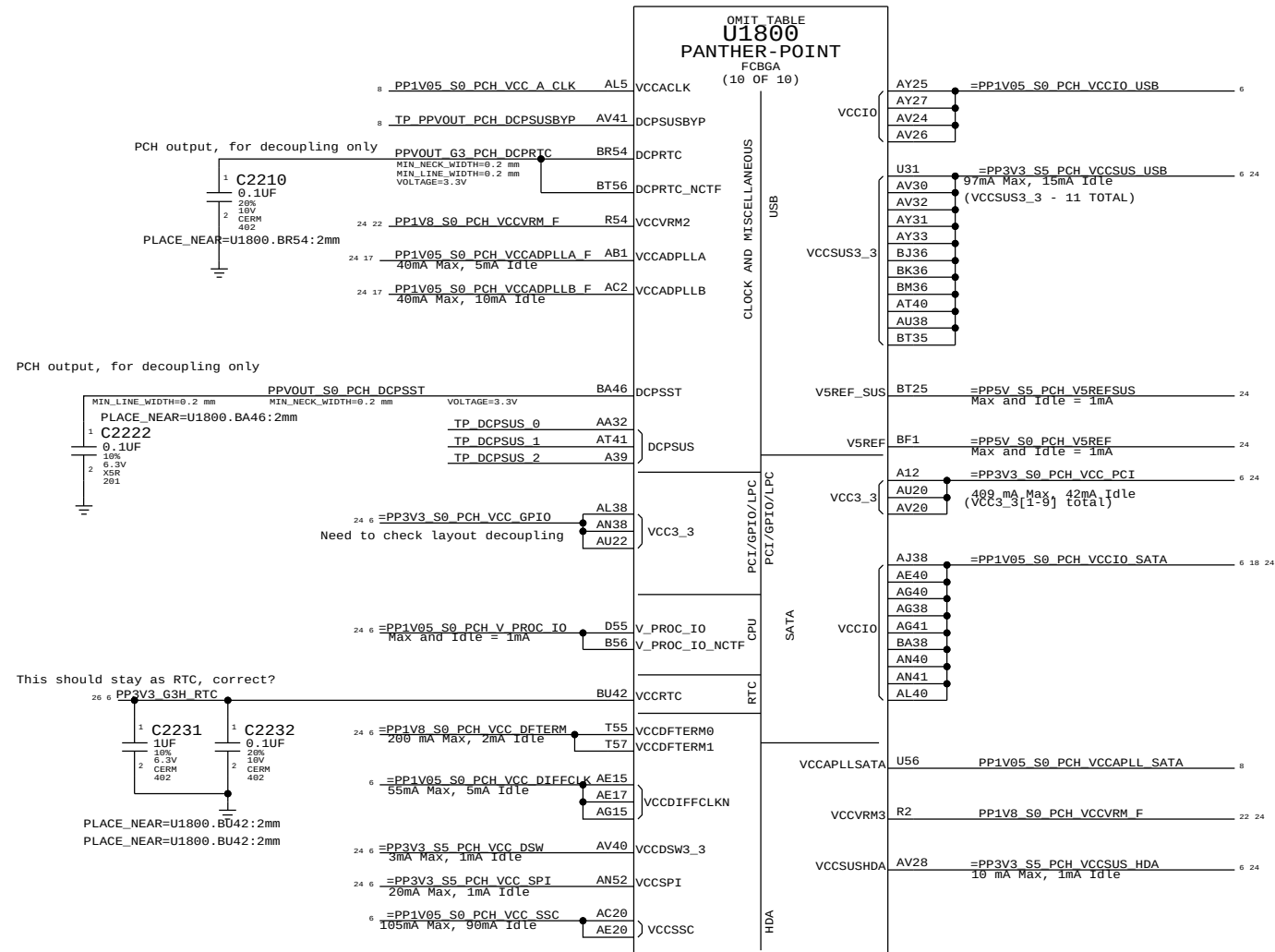
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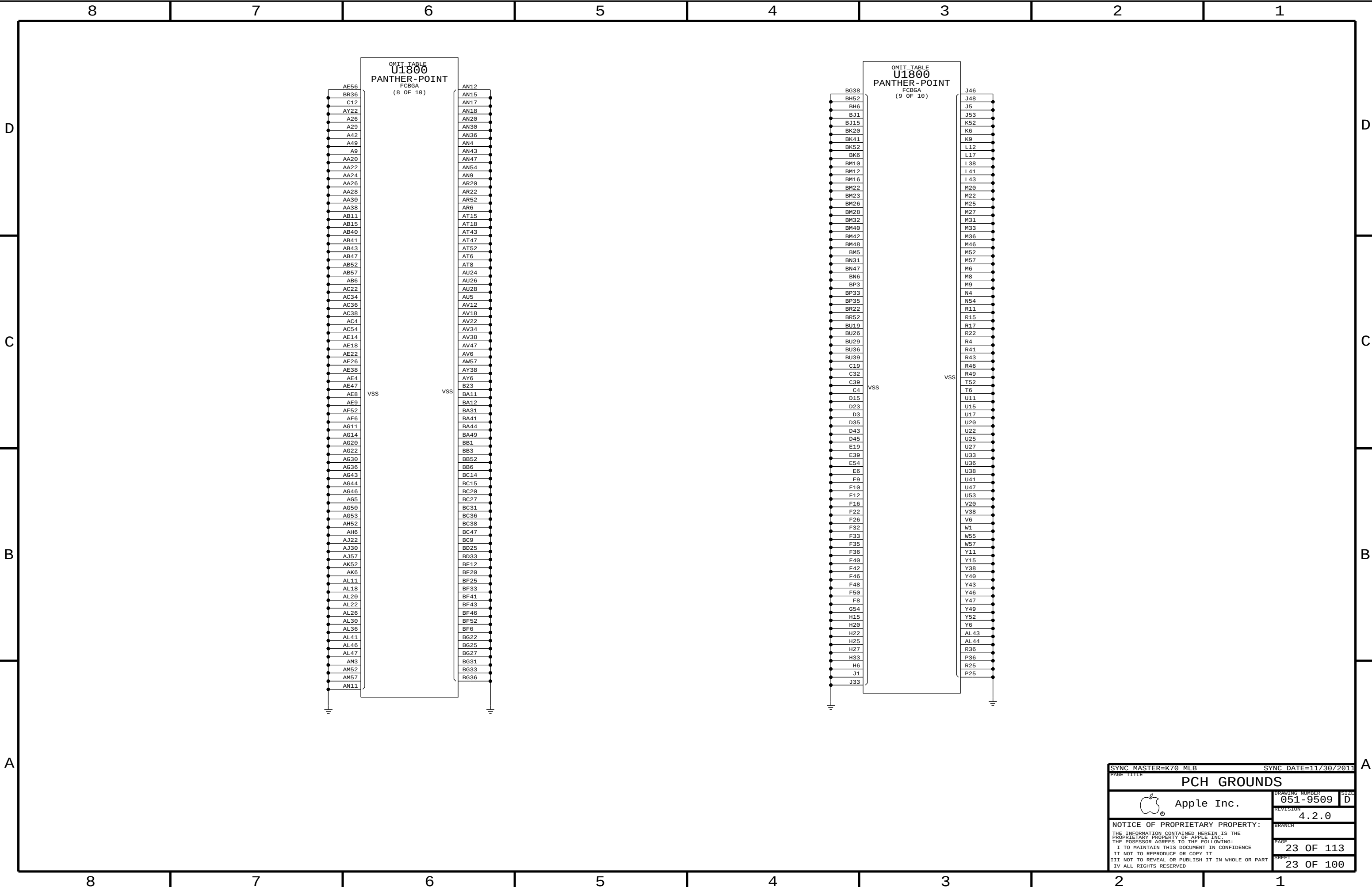
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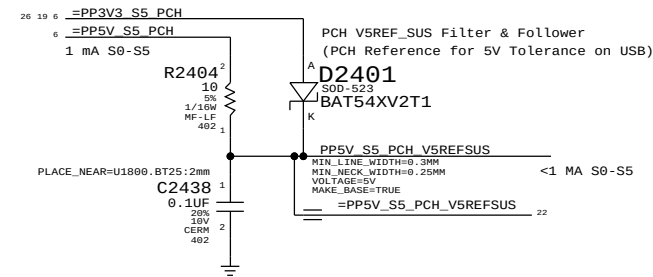
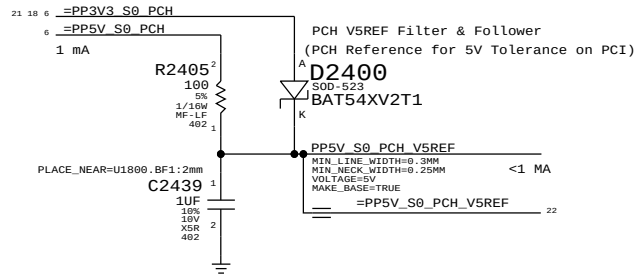




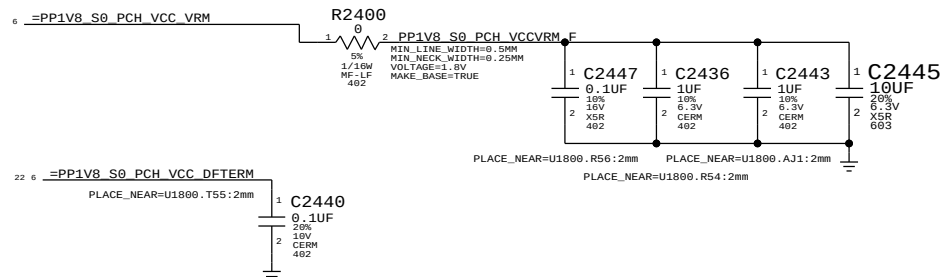




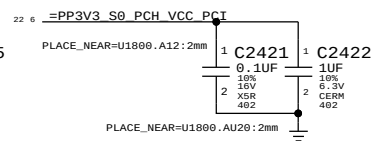
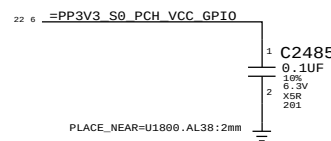
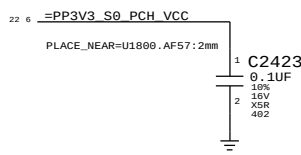
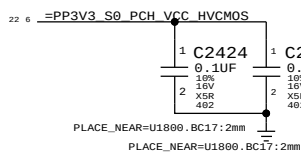
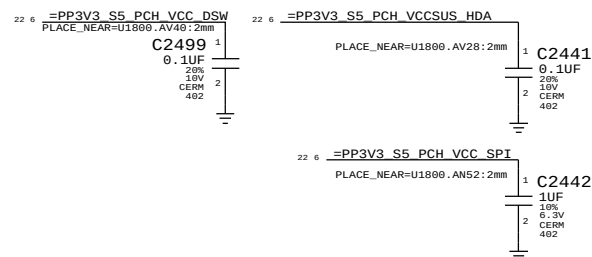
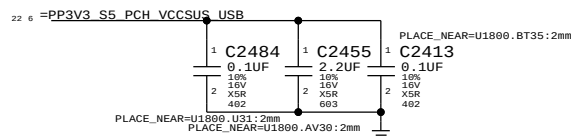
Power Sequencing



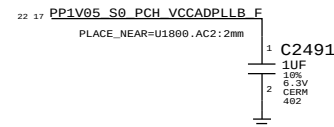
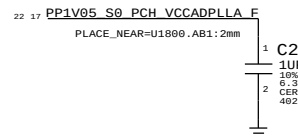
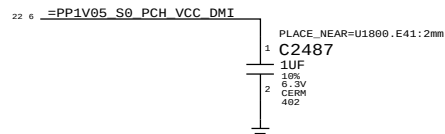
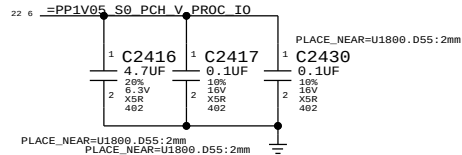
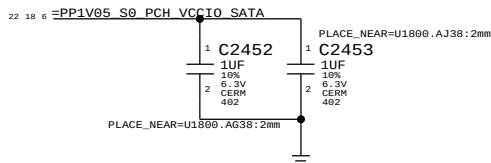
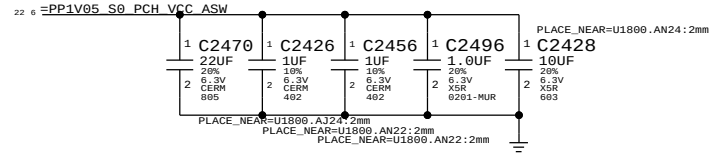
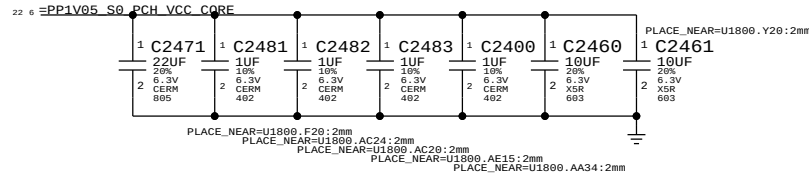
1V8 S0 Rails



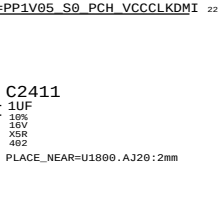
3V3 S5 Rails



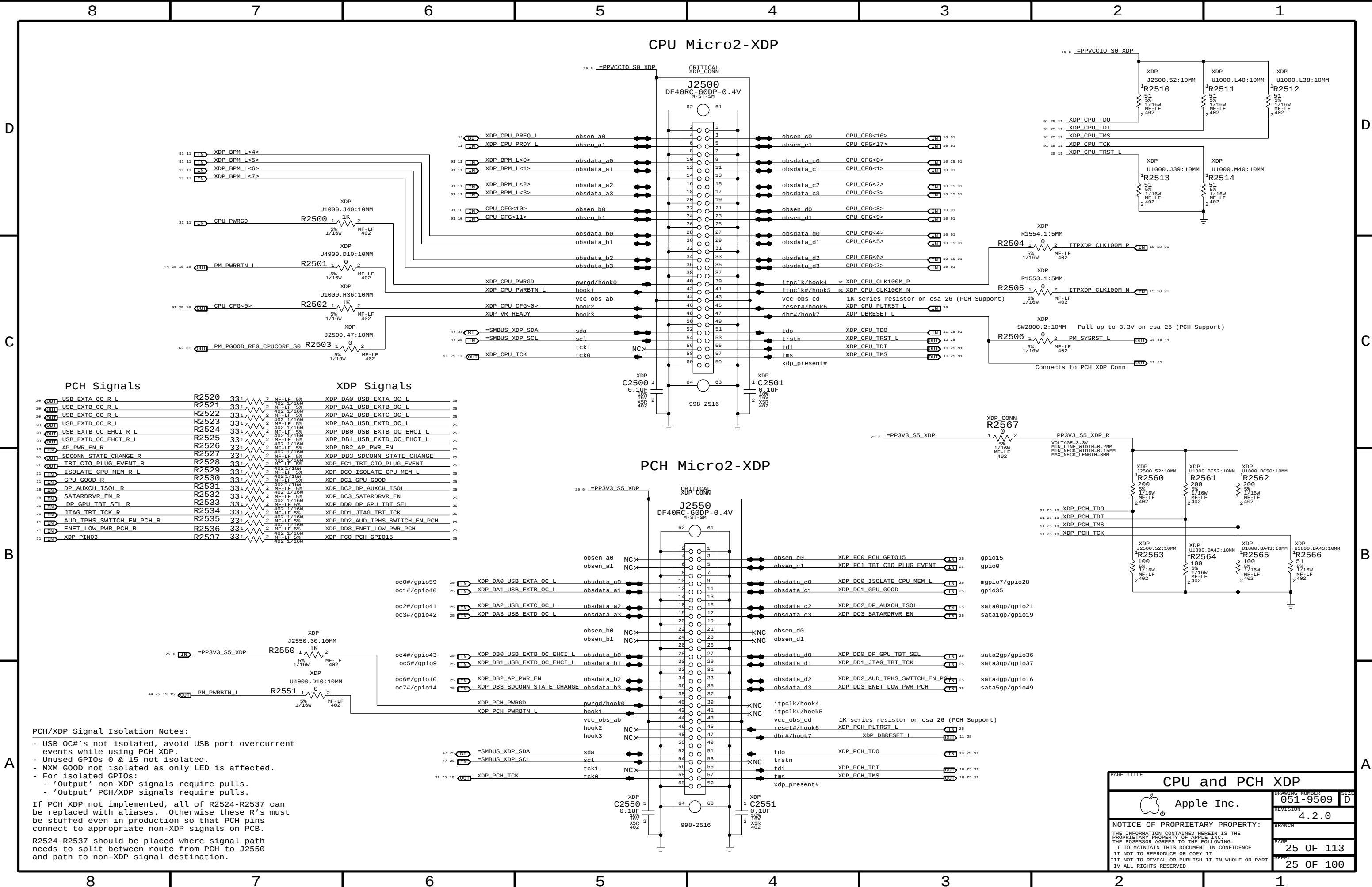
1V05 S0 Rails



3V3 S0 Rails



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PCH Signals

XDP Signals

OUT	USB_EXTB_OC_R_L	R2520	331	2	MF-LF	5%	XDP_DA0_USB_EXTB_OC_L	25
OUT	USB_EXTB_OC_R_L	R2521	331	2	MF-LF	5%	XDP_DA1_USB_EXTB_OC_L	25
OUT	USB_EXTB_OC_R_L	R2522	331	2	MF-LF	5%	XDP_DA2_USB_EXTB_OC_L	25
OUT	USB_EXTB_OC_R_L	R2523	331	2	MF-LF	5%	XDP_DA3_USB_EXTB_OC_L	25
OUT	USB_EXTB_OC_EHCI_R_L	R2524	331	2	MF-LF	5%	XDP_DB0_USB_EXTB_OC_EHCI_L	25
OUT	USB_EXTB_OC_EHCI_R_L	R2525	331	2	MF-LF	5%	XDP_DB1_USB_EXTB_OC_EHCI_L	25
IN	AP_PWR_EN_R	R2526	331	2	MF-LF	5%	XDP_DB2_AP_PWR_EN	25
OUT	SDCONN_STATE_CHANGE_R	R2527	331	2	MF-LF	5%	XDP_DB3_SDCONN_STATE_CHANGE	25
OUT	TBT_CIO_PLUG_EVENT_R	R2528	331	2	MF-LF	5%	XDP_FC1_TBT_CIO_PLUG_EVENT	25
IN	ISOLATE_CPU_MEM_R_L	R2529	331	2	MF-LF	5%	XDP_DC0_ISOLATE_CPU_MEM_L	25
IN	GPU_GOOD_R	R2530	331	2	MF-LF	5%	XDP_DC1_GPU_GOOD	25
IN	DP_AUXCH_ISOL_R	R2531	331	2	MF-LF	5%	XDP_DC2_DP_AUXCH_ISOL	25
IN	SATARDVR_EN_R	R2532	331	2	MF-LF	5%	XDP_DC3_SATARDVR_EN	25
IN	DP_GPU_TBT_SEL_R	R2533	331	2	MF-LF	5%	XDP_DD0_DP_GPU_TBT_SEL	25
IN	JTAG_TBT_TCK_R	R2534	331	2	MF-LF	5%	XDP_DD1_JTAG_TBT_TCK	25
IN	AUD_IPHS_SWITCH_EN_PCH_R	R2535	331	2	MF-LF	5%	XDP_DD2_AUD_IPHS_SWITCH_EN_PCH	25
IN	ENET_LOW_PWR_PCH_R	R2536	331	2	MF-LF	5%	XDP_DD3_ENET_LOW_PWR_PCH	25
IN	XDP_PIN03	R2537	331	2	MF-LF	5%	XDP_FC0_PCH_GPIO15	25

PCH/XDP Signal Isolation Notes:

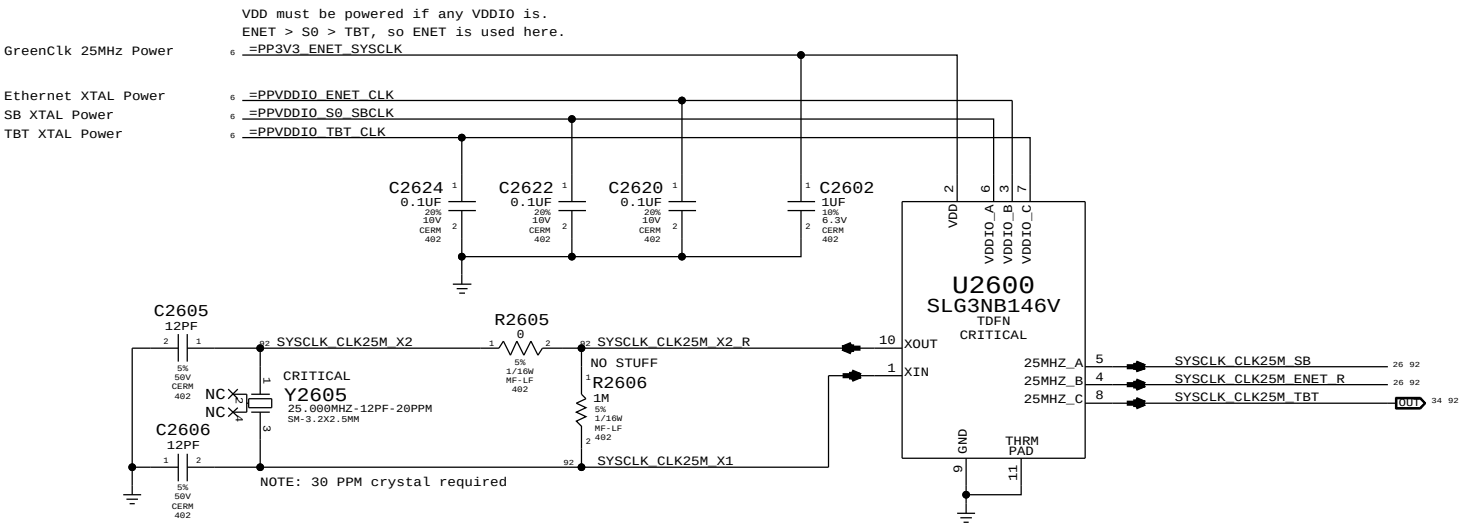
- USB OC#'s not isolated, avoid USB port overcurrent events while using PCH XDP.
- Unused GPIOs 0 & 15 not isolated.
- MXM_GOOD not isolated as only LED is affected.
- For isolated GPIOs:
 - 'Output' non-XDP signals require pulls.
 - 'Output' PCH/XDP signals require pulls.

If PCH XDP not implemented, all of R2524-R2537 can be replaced with aliases. Otherwise these R's must be stuffed even in production so that PCH pins connect to appropriate non-XDP signals on PCB.

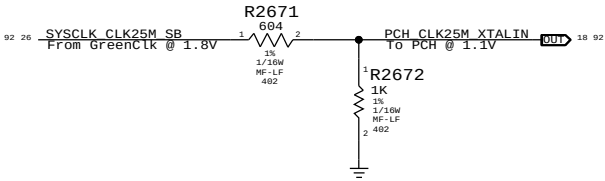
R2524-R2537 should be placed where signal path needs to split between route from PCH to J2550 and path to non-XDP signal destination.

PAGE TITLE		
CPU and PCH XDP		
Apple Inc.	DRAWING NUMBER	051-9509
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PAGE	25 OF 113	
SHEET	25 OF 100	

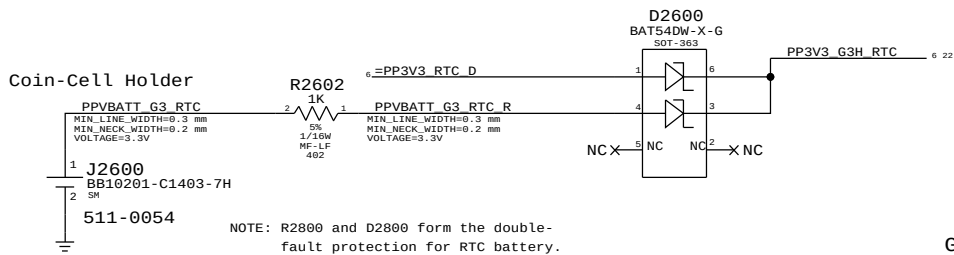
System 25MHz Clock Generator



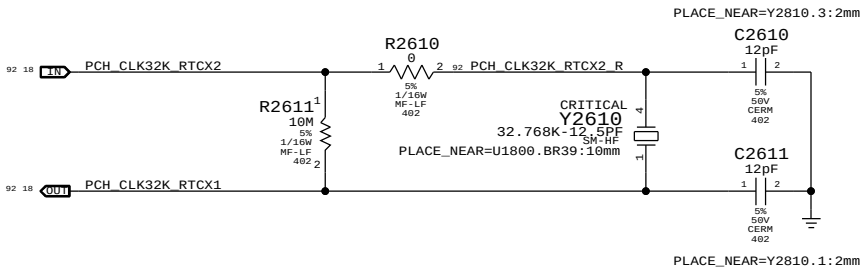
PCH 25MHZ CLOCK



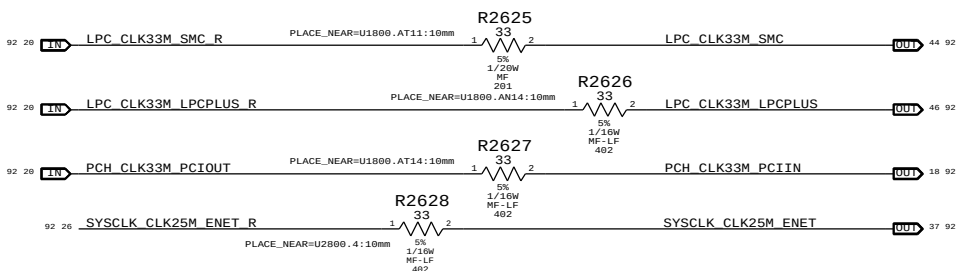
RTC Power Sources



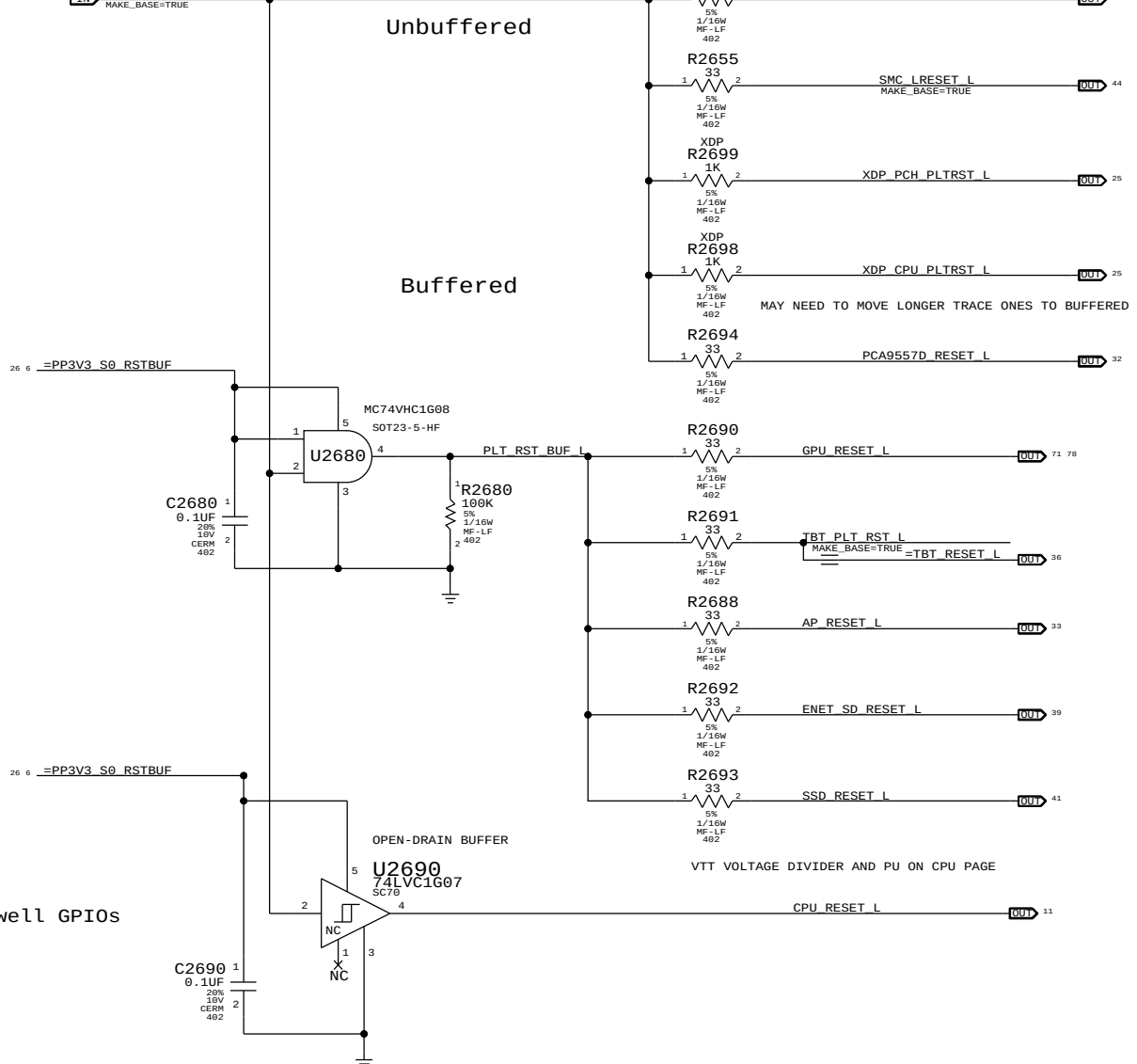
PCH RTC Crystal



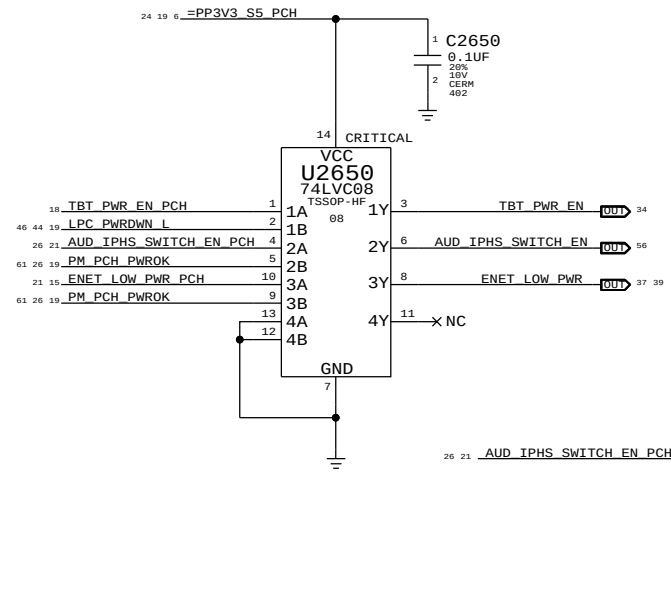
Clock series termination



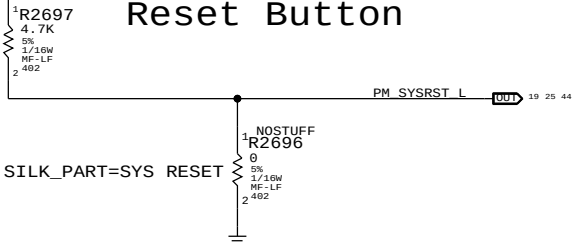
Platform Reset Connections



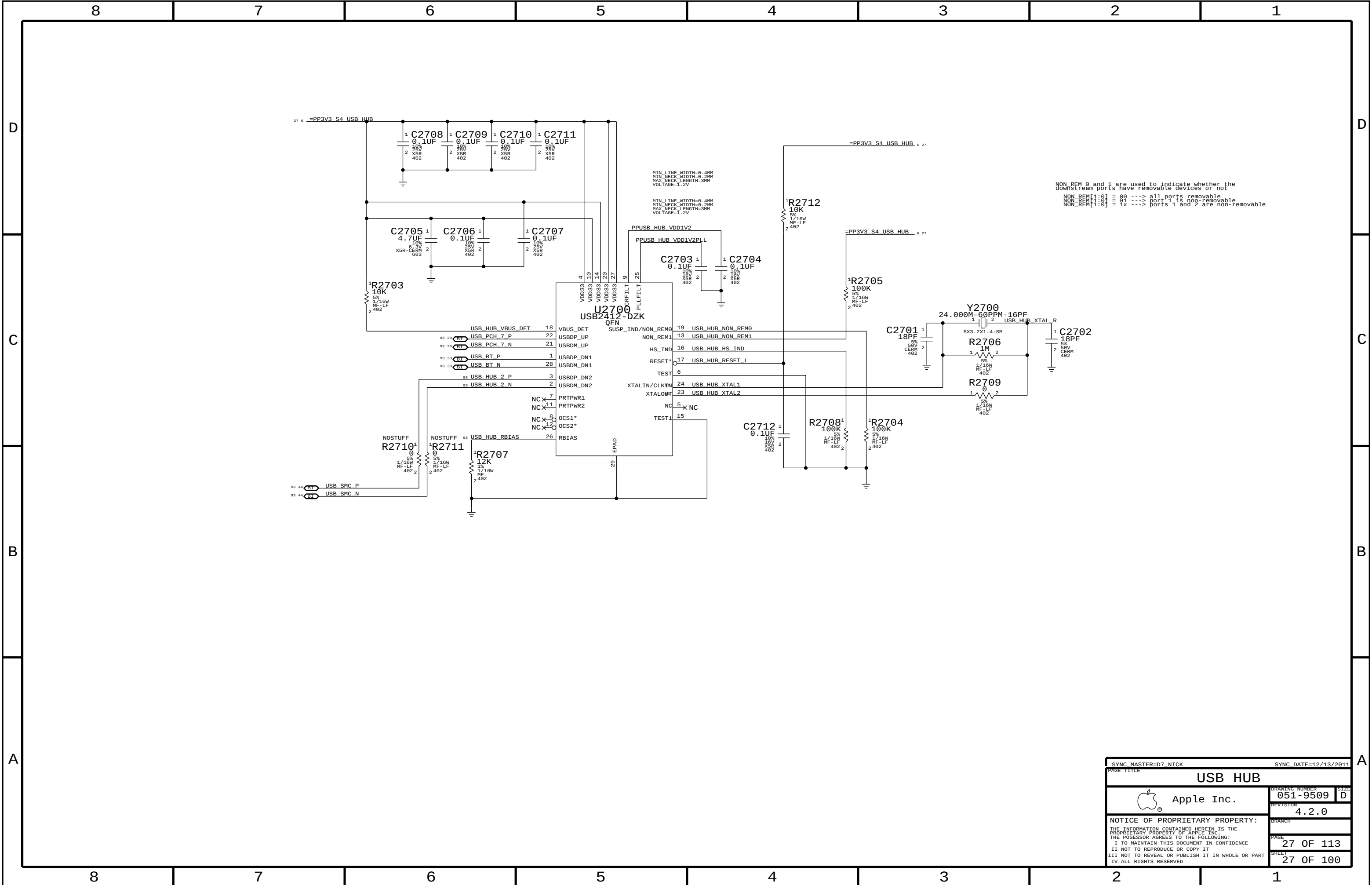
GPIO Isolation to prevent glitches on critical core well GPIOs



Reset Button



SYNC MASTER=K70 MLB		SYNC DATE=11/30/2011	
PAGE TITLE			
CHIPSET SUPPORT			
 Apple Inc.		DRAWING NUMBER	051-9509
		STATE	D
		REVISION	4.2.0
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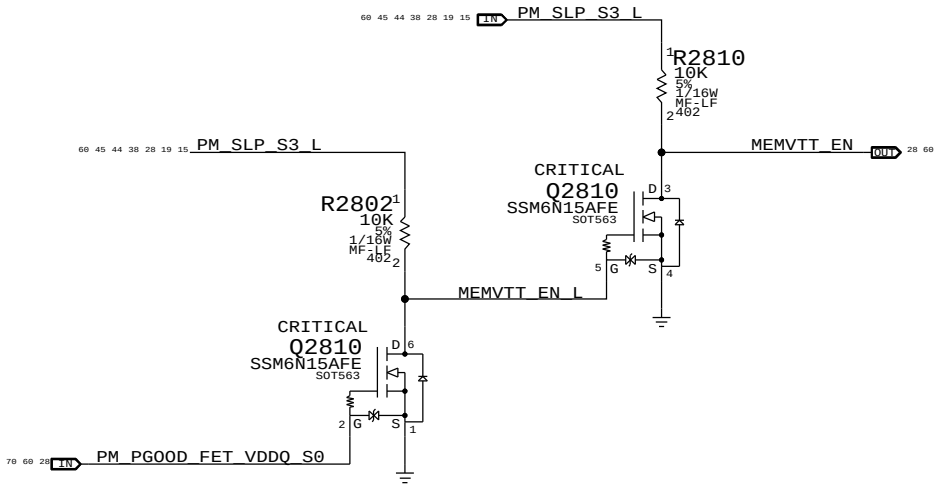
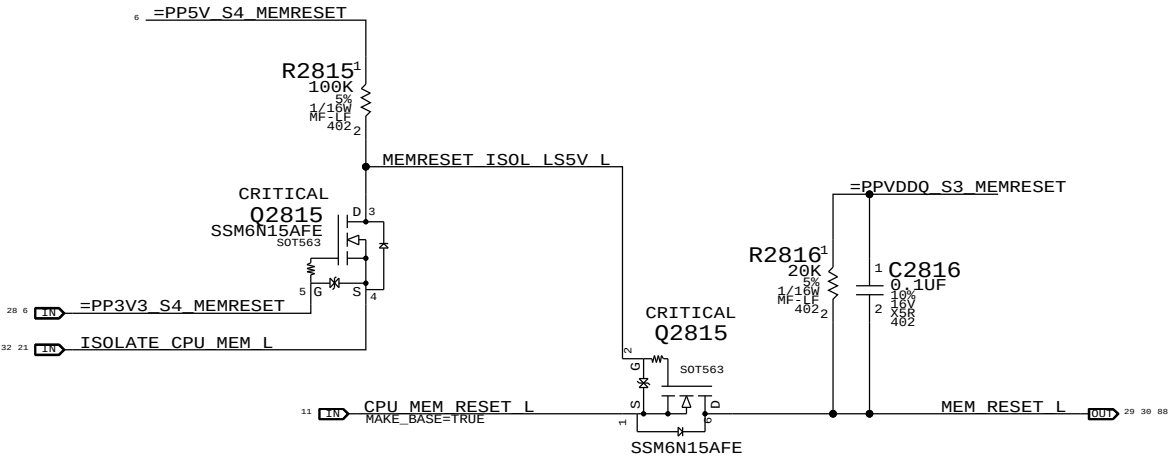
The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

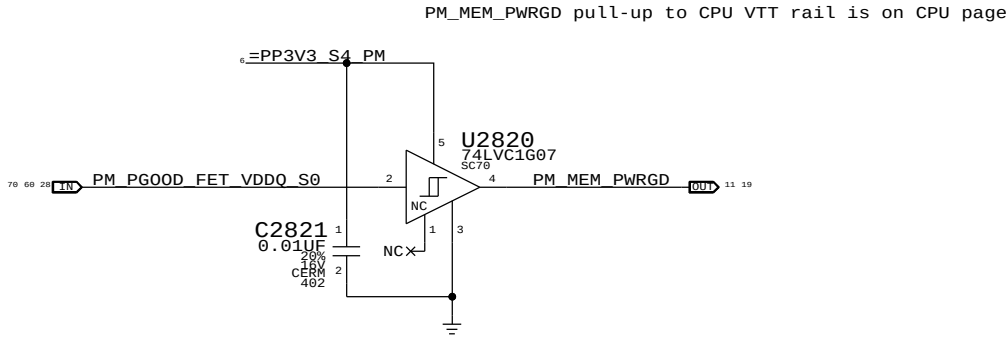
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

MEMVTT_EN = PM_PG00D_FET_VDDQ_S0 * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

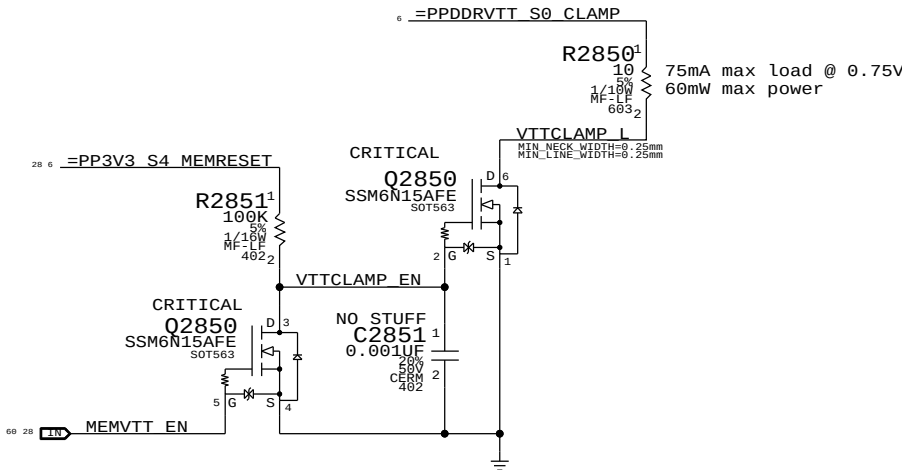


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

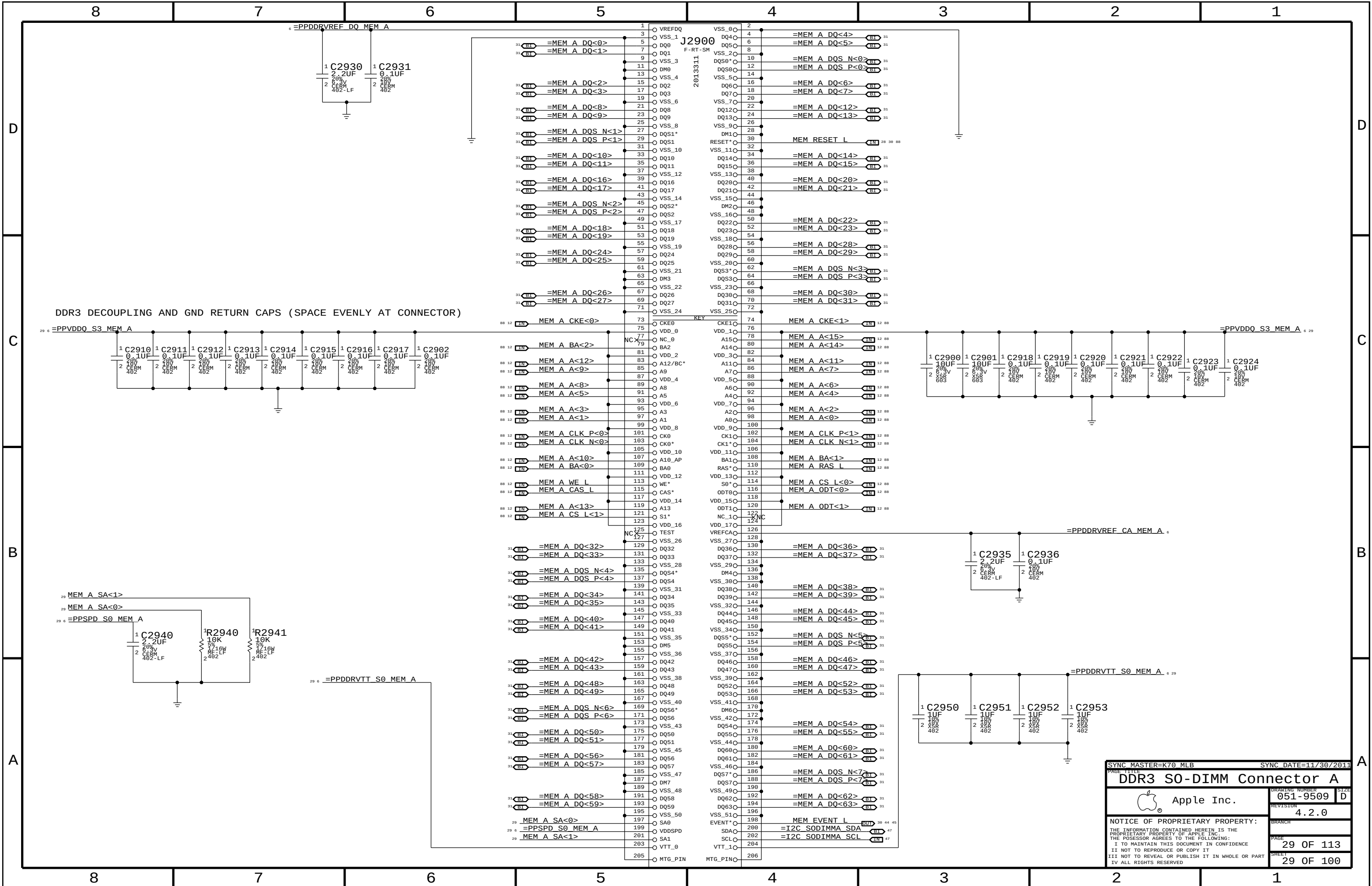


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	1	CPU_MEM_RESET_L	1
1	1	0	1	1	1	1
2	0	0	1	1	1	0
3	0	0	0	X	1	0
4	0	0	1	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	1	CPU_MEM_RESET_L	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

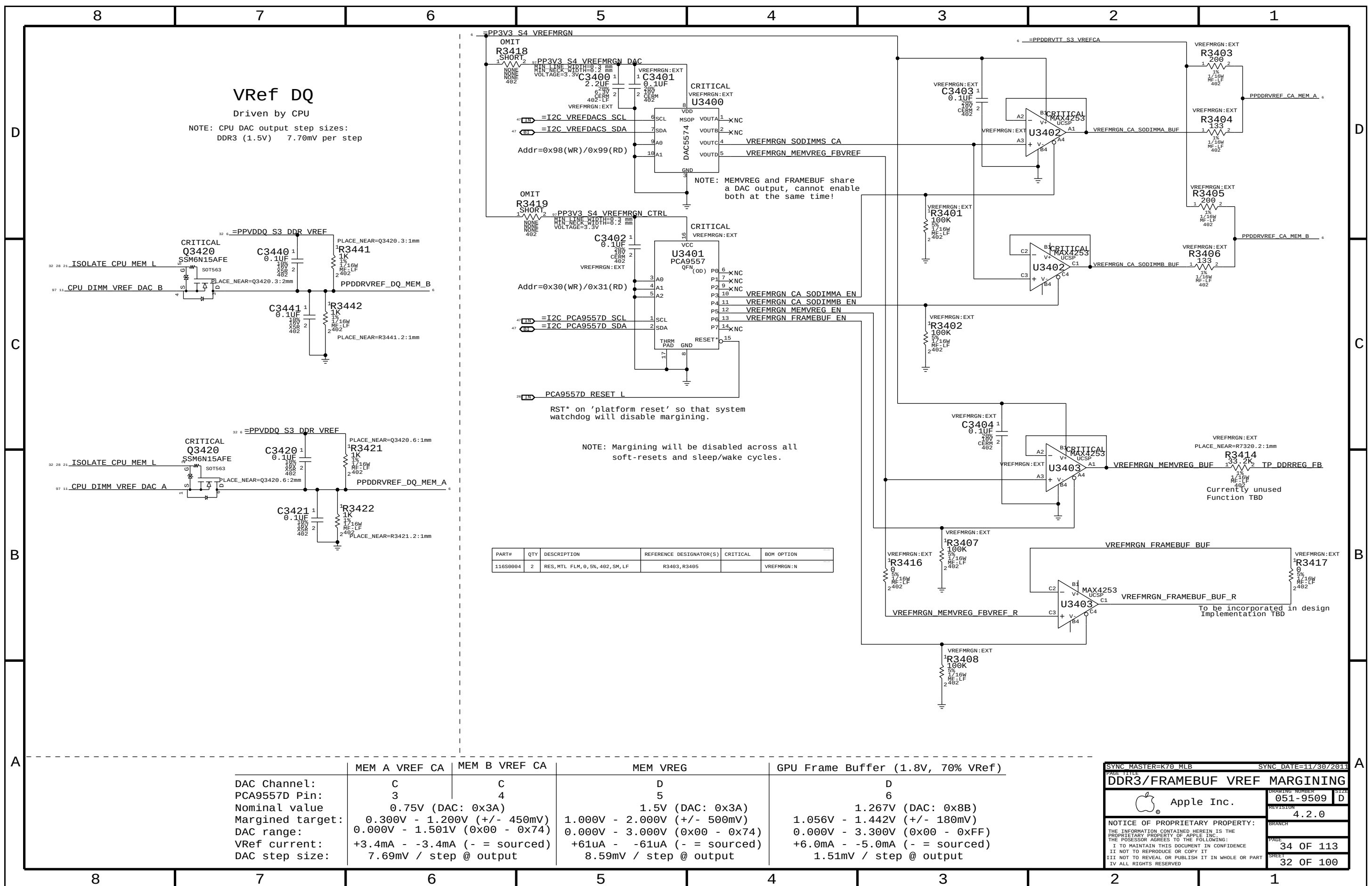
SYNC MASTER=K70 MLB		SYNC DATE=11/30/2011	
PAGE TITLE		CPU Memory S3 Support	
DRAWING NUMBER		051-9509	SIZE D
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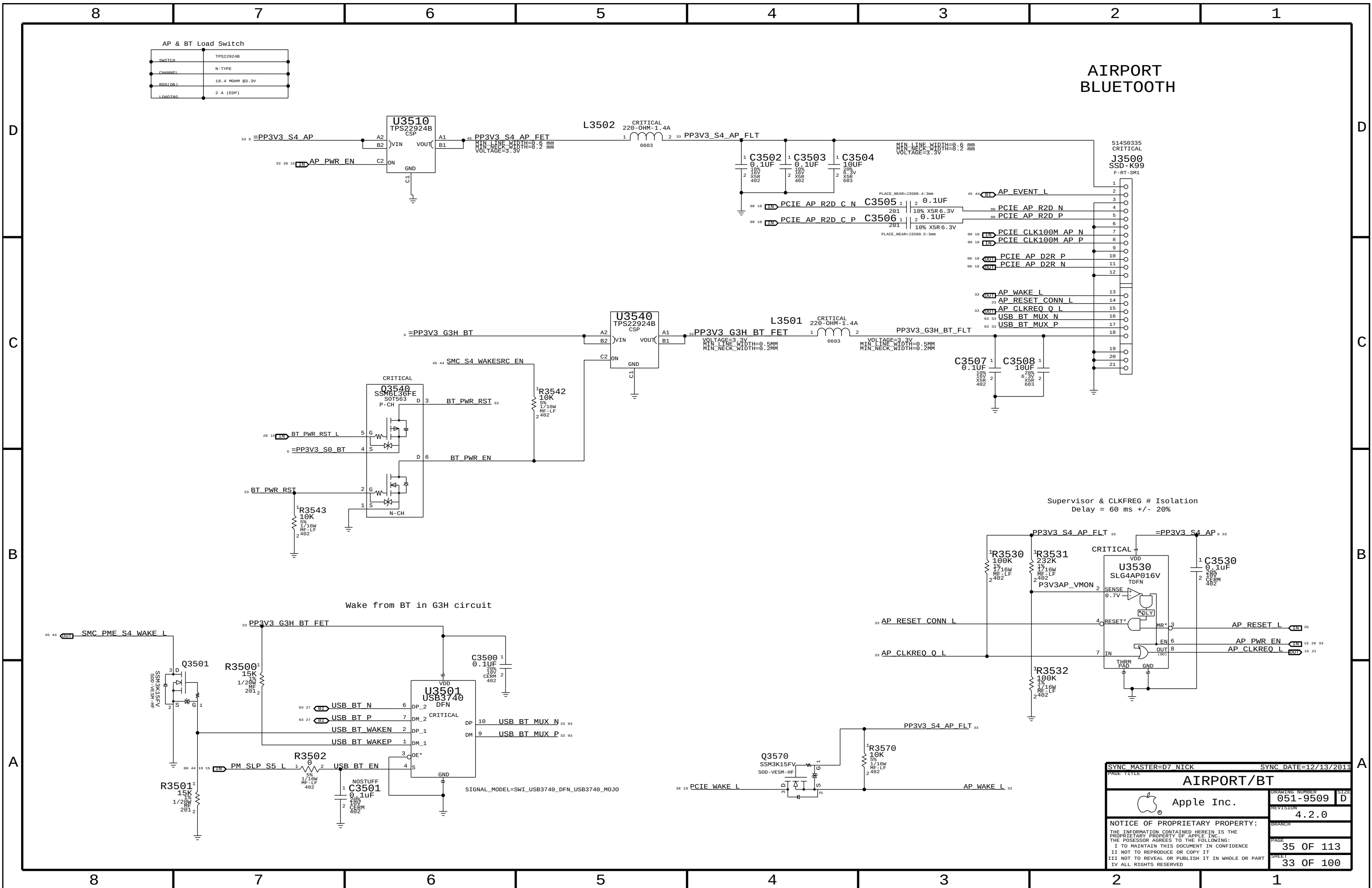


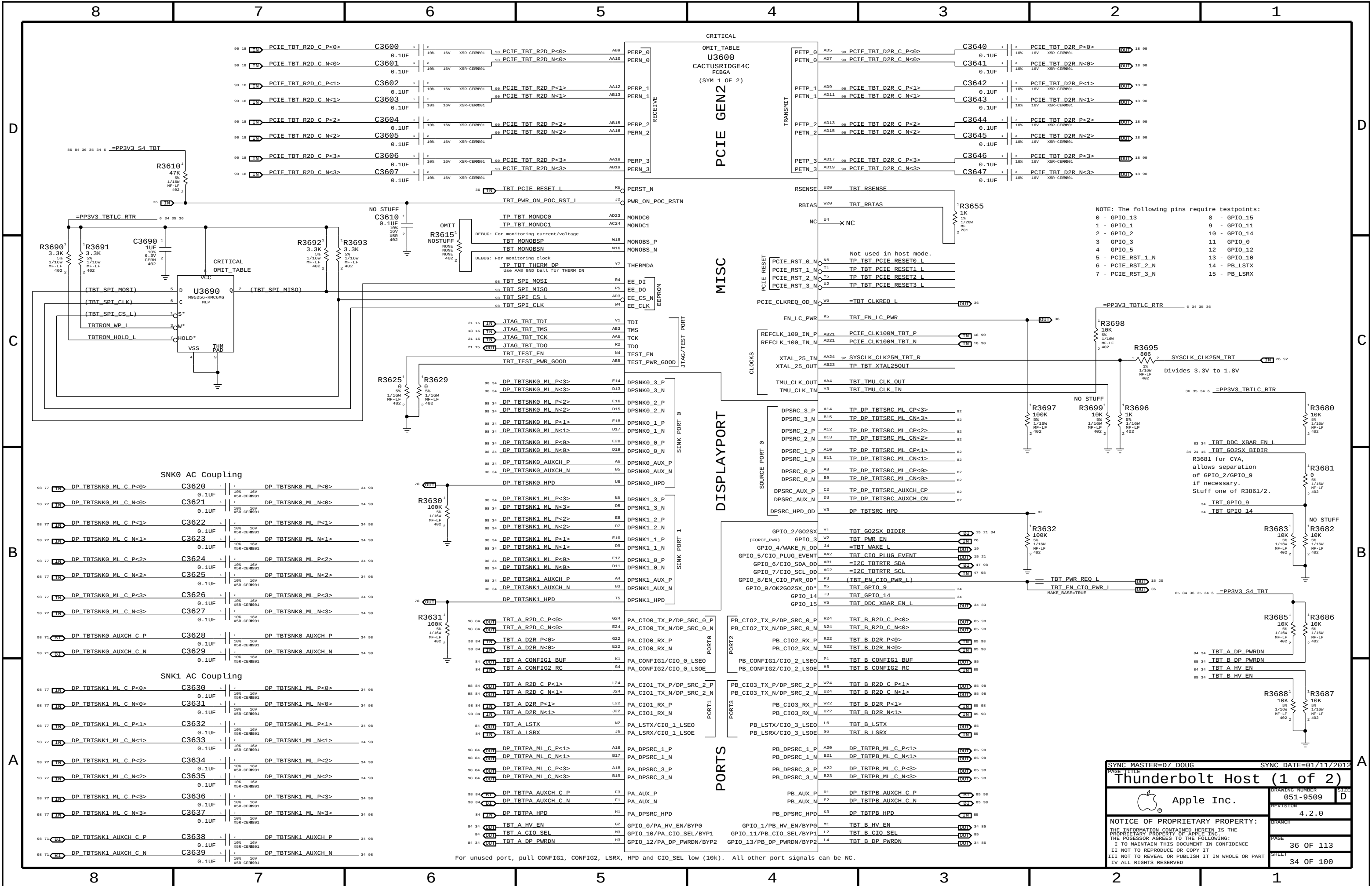
PAGE TITLE		SYNC DATE=11/30/2011	
DDR3 S0-DIMM Connector A		DRAWING NUMBER	
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BRANCH		PAGE	
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8		7		6		5		4		3		2		1	
D	88	12	MEM A DQS N<0>	==	=MEM A DQS N<0>	29	88	12	MEM B DQS N<0>	==	=MEM B DQS N<0>	30	D		
	88	12	MEM A DQS P<0>	==	=MEM A DQS P<0>	29	88	12	MEM B DQS P<0>	==	=MEM B DQS P<0>	30			
	88	12	MEM A DQ<7>	==	=MEM A DQ<7>	29	88	12	MEM B DQ<7>	==	=MEM B DQ<7>	30			
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	88	12	MEM A DQ<0>	==	=MEM A DQ<0>	29	88	12	MEM B DQ<0>	==	=MEM B DQ<0>	30			
C	88	12	MEM A DQS N<1>	==	=MEM A DQS N<1>	29	88	12	MEM B DQS N<1>	==	=MEM B DQS N<1>	30	C		
	88	12	MEM A DQS P<1>	==	=MEM A DQS P<1>	29	88	12	MEM B DQS P<1>	==	=MEM B DQS P<1>	30			
	88	12	MEM A DQ<15>	==	=MEM A DQ<15>	29	88	12	MEM B DQ<15>	==	=MEM B DQ<15>	30			
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SYNC MASTER=K70 MLB		SYNC DATE=11/30/2011	
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SYNC DATE=01/11/2012

Thunderbolt Host (1 of 2)

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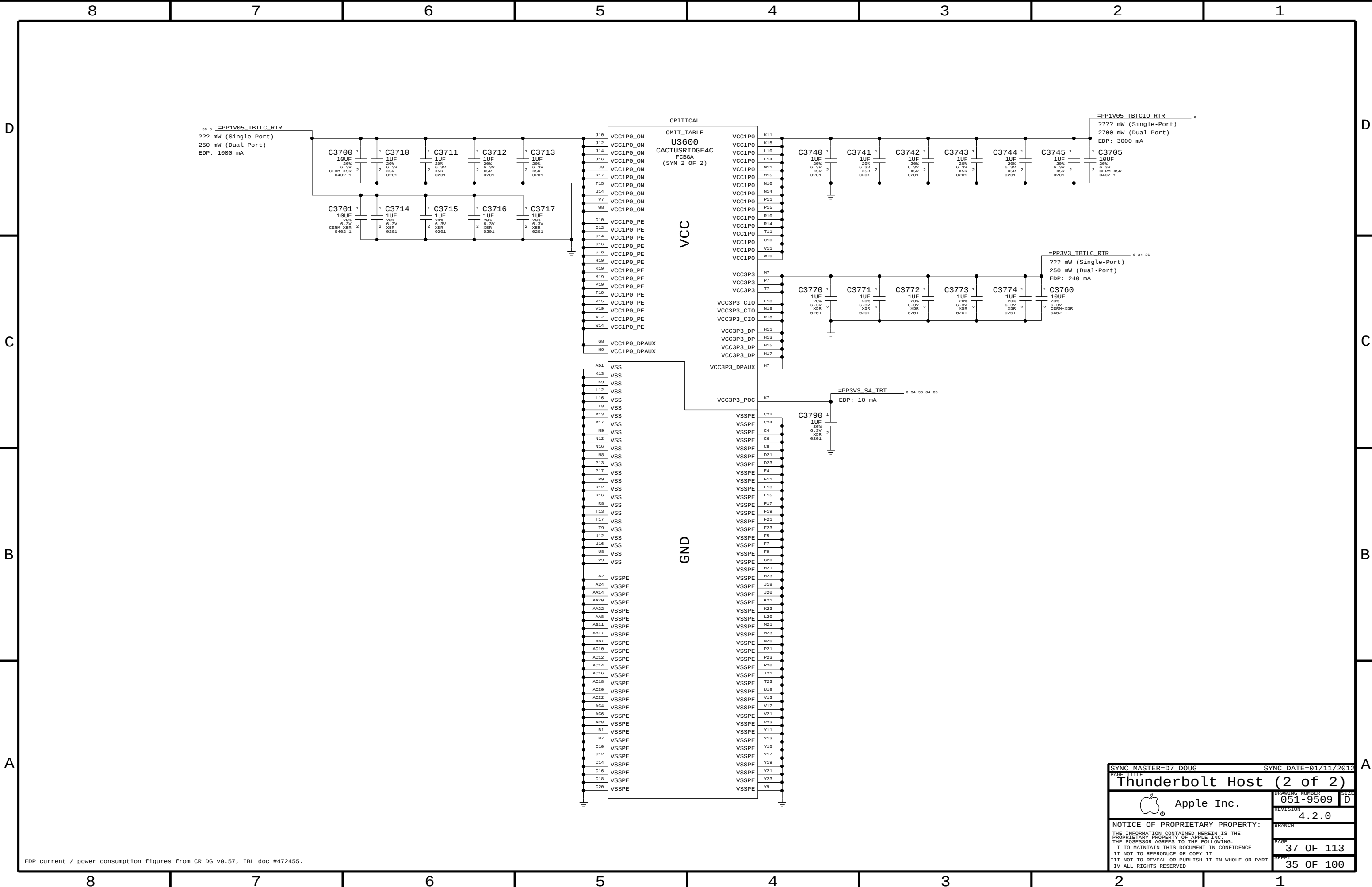
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For unused port, pull CONFIG1, CONFIG2, LSRX, HPD and CIO_SEL low (10k). All other port signals can be NC.



EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

SYNC MASTER=D7 DOUG		SYNC DATE=01/11/2012	
PAGE 11111			
Thunderbolt Host (2 of 2)		DRAWING NUMBER 051-9509	
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Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP15V_TBT_REG (15V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBTTPWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

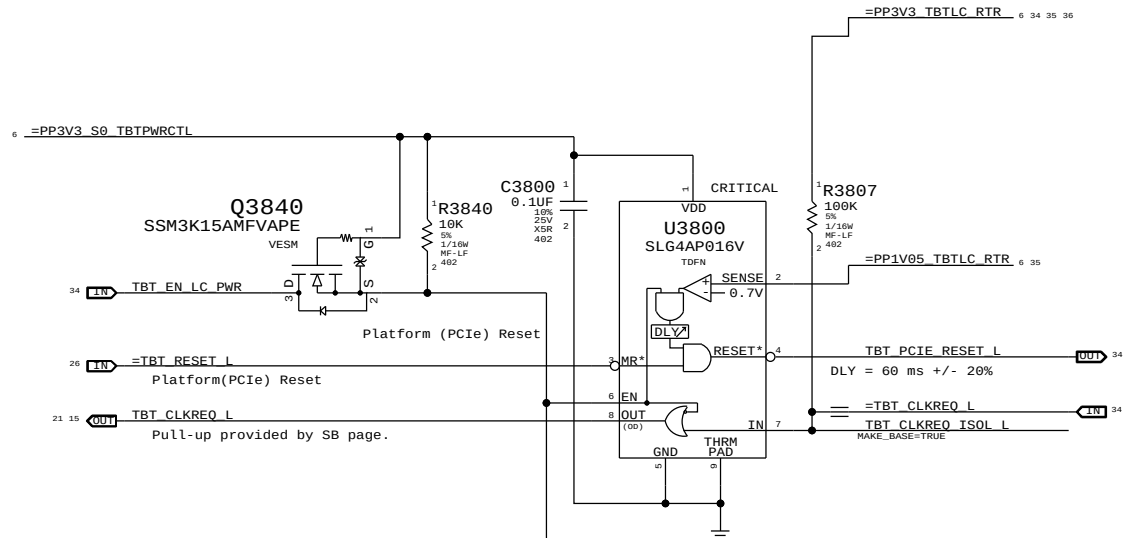
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

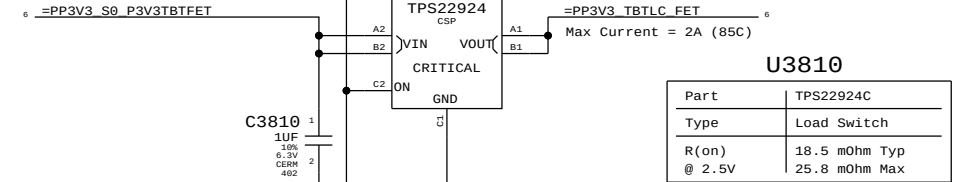
BOM options provided by this page:

TBTBST:Y - Stuffs 15V boost circuitry.

Supervisor & CLKREQ# Isolation

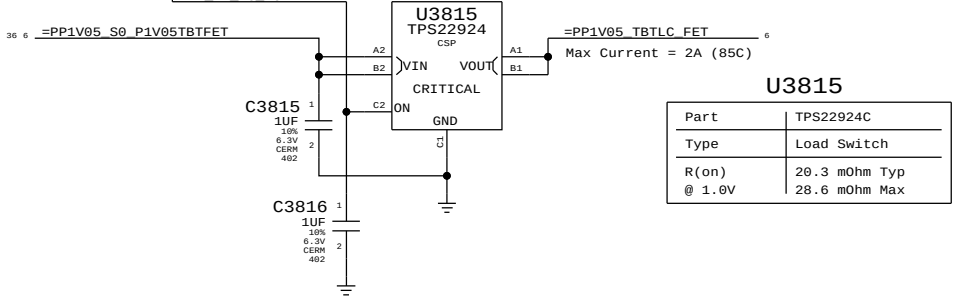


3.3V TBT "LC" Switch



Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

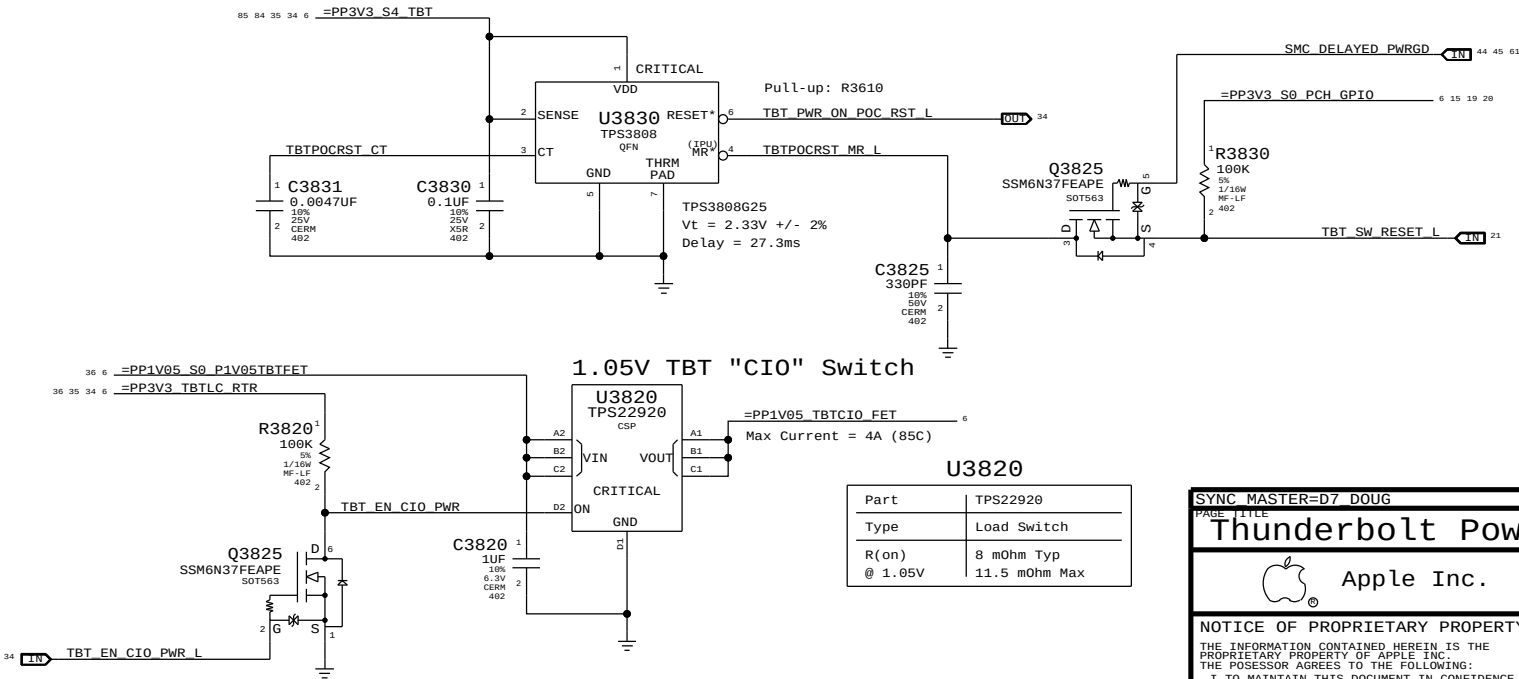
1.05V TBT "LC" Switch



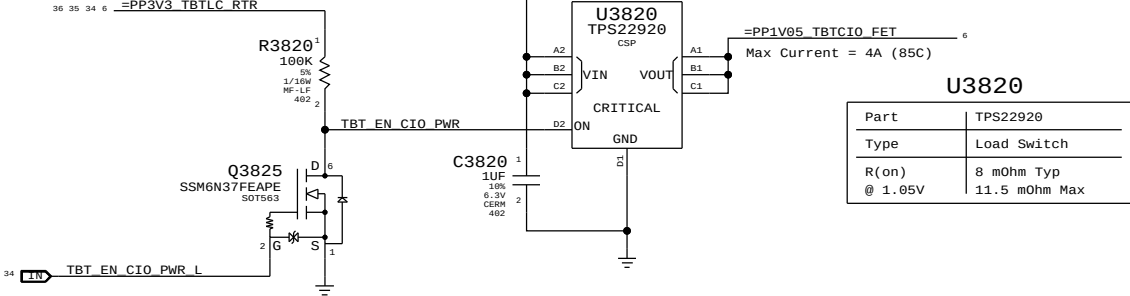
Part	TPS22924C
Type	Load Switch
R(on) @ 1.0V	20.3 mOhm Typ 28.6 mOhm Max

TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.



1.05V TBT "CIO" Switch



Part	TPS22920
Type	Load Switch
R(on) @ 1.05V	8 mOhm Typ 11.5 mOhm Max

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SYNC DATE=01/11/2012

Thunderbolt Power Support

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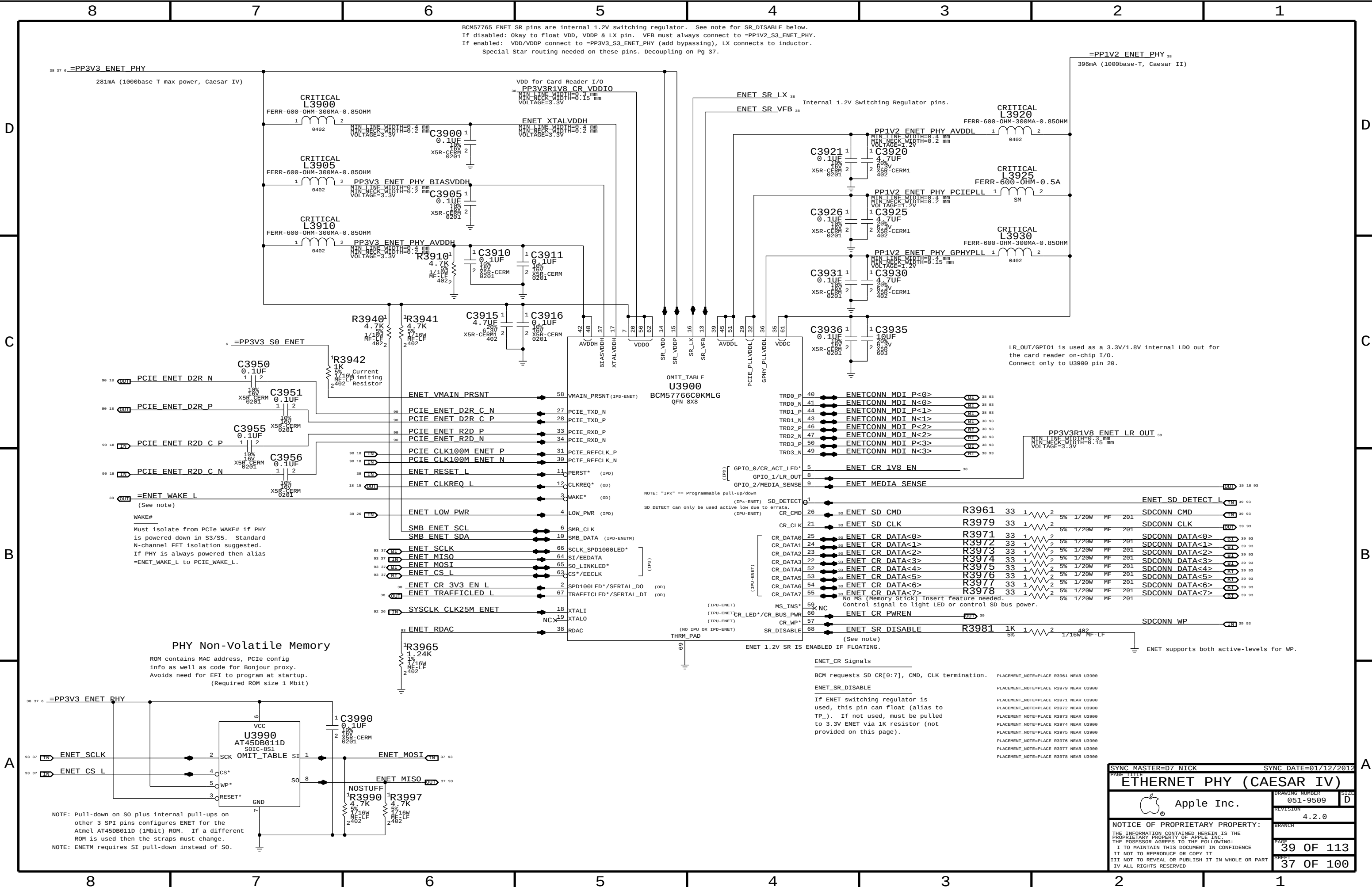
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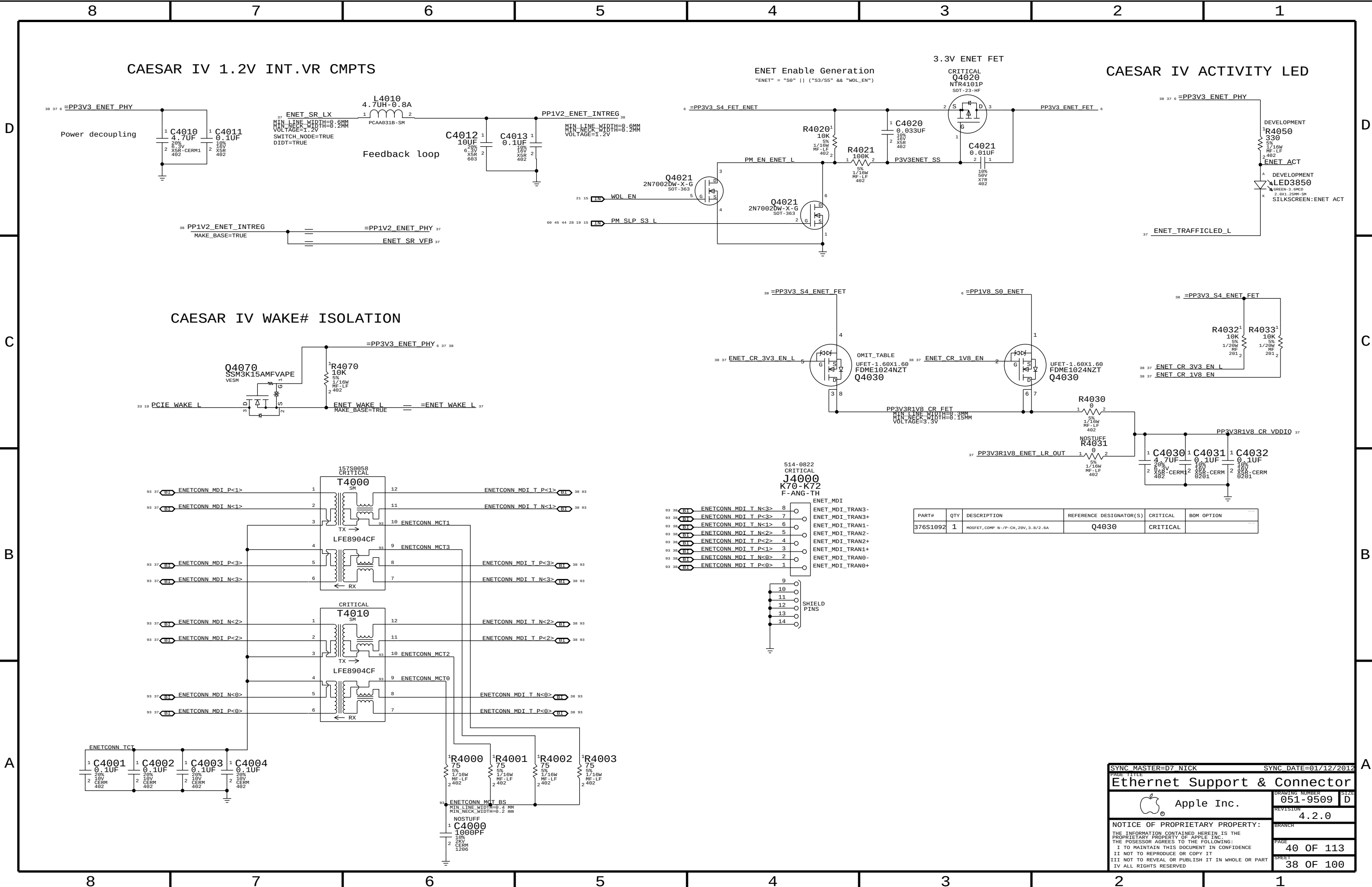
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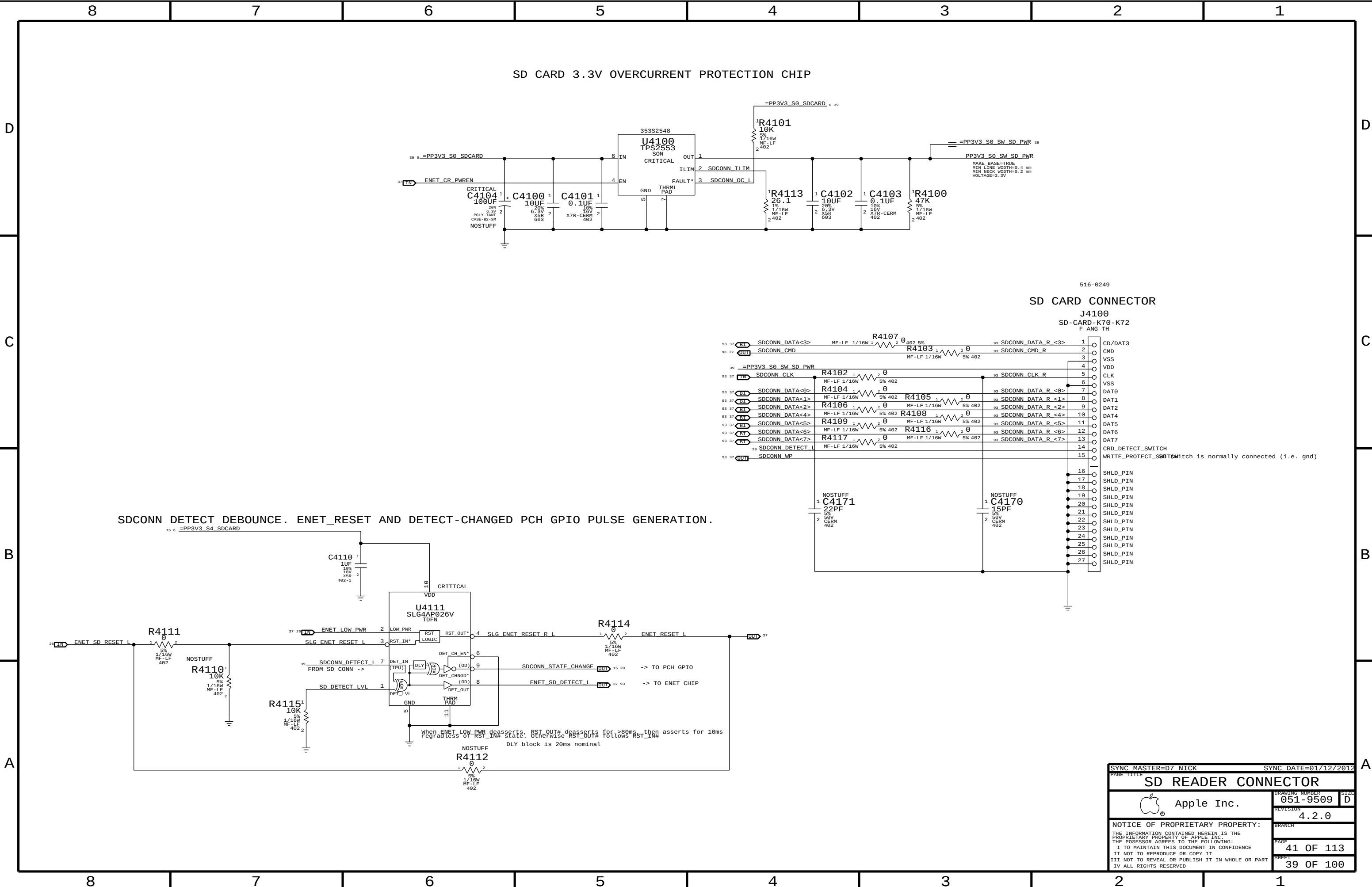
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SDCONN DETECT DEBOUNCE. ENET_RESET AND DETECT-CHANGED PCH GPIO PULSE GENERATION.

SD CARD CONNECTOR

J4100
SD-CARD-K70-K72
F-ANG-TH

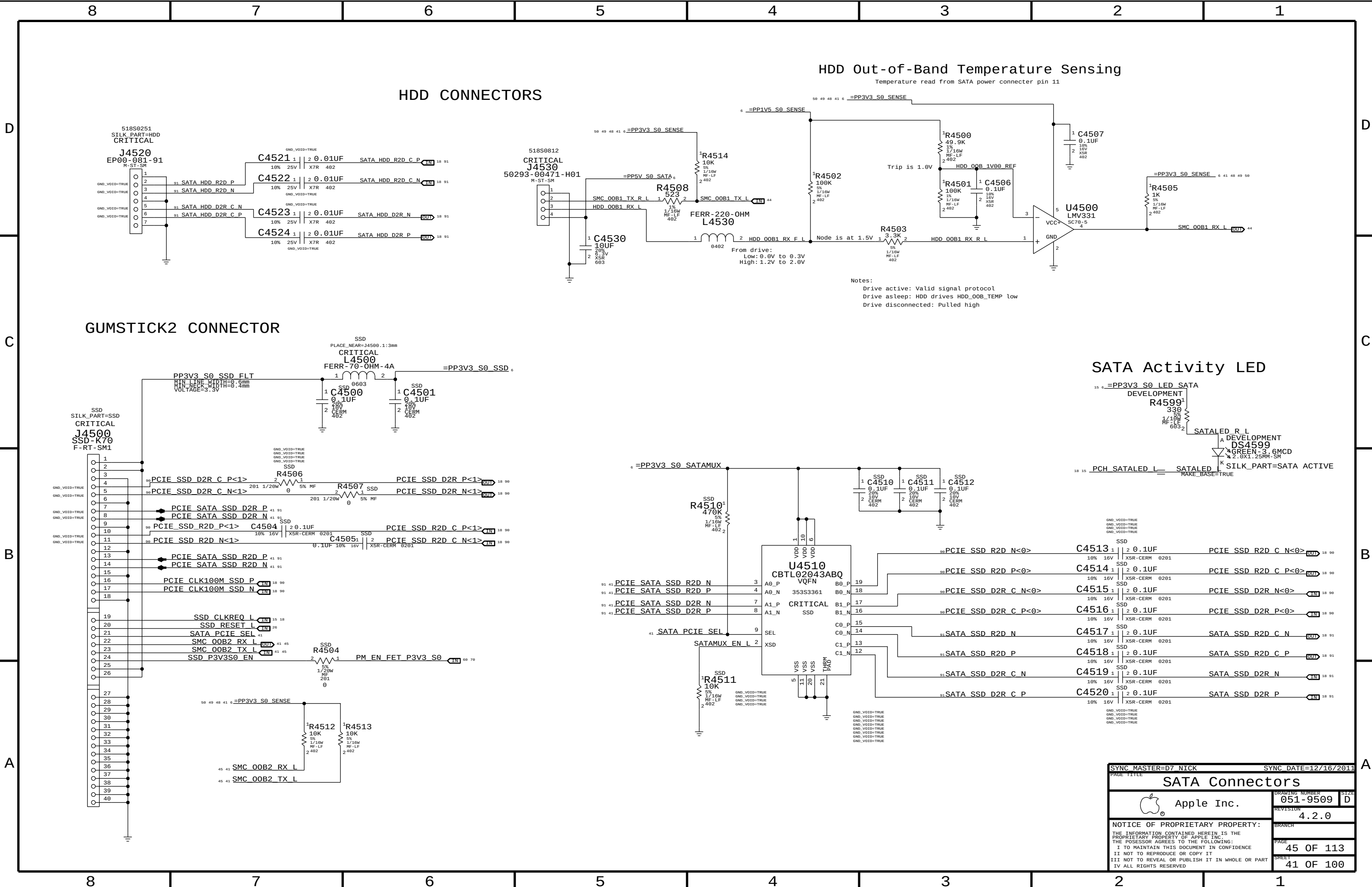
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SD READER CONNECTOR		DRAWING NUMBER	SIZE
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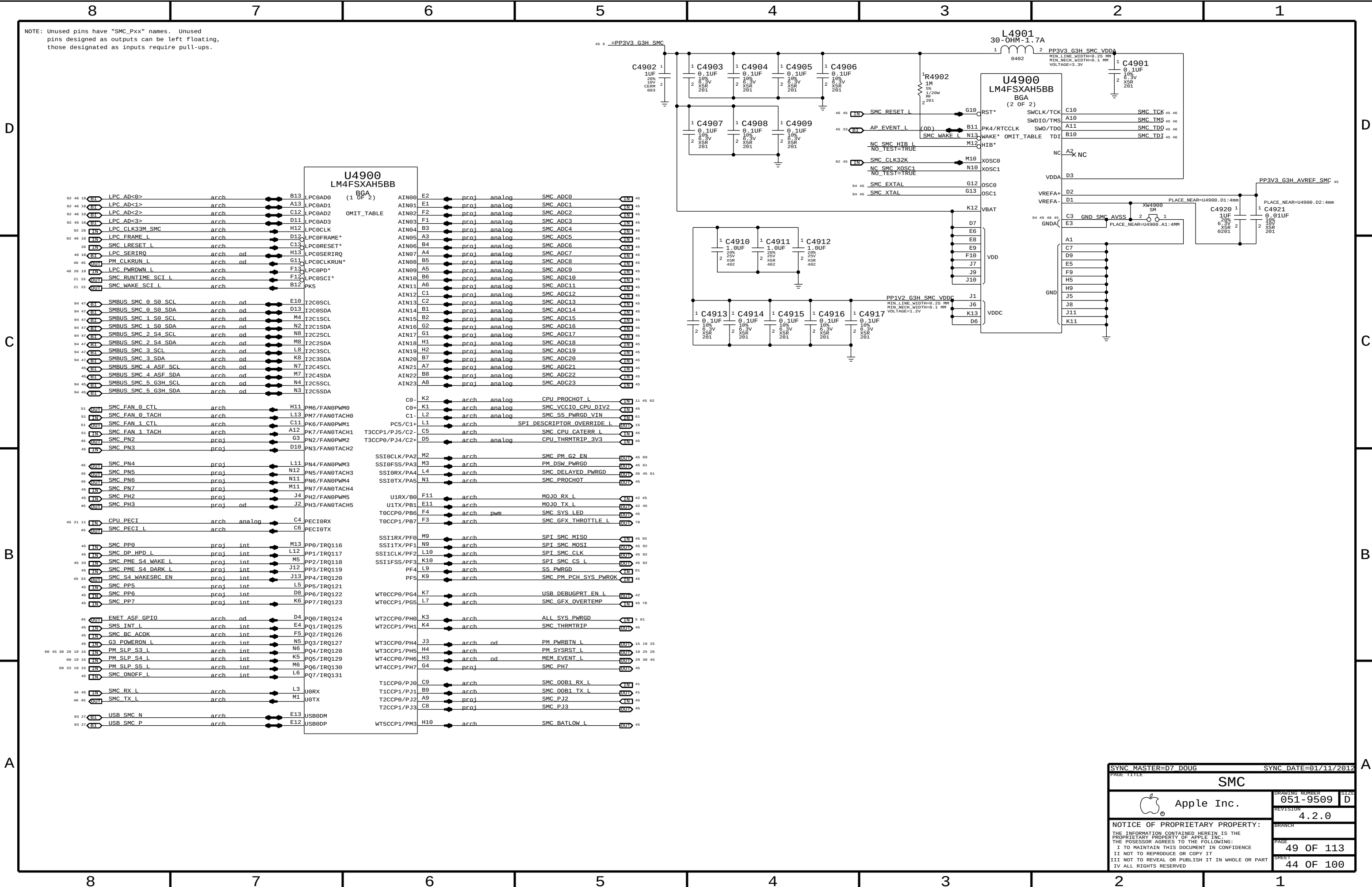
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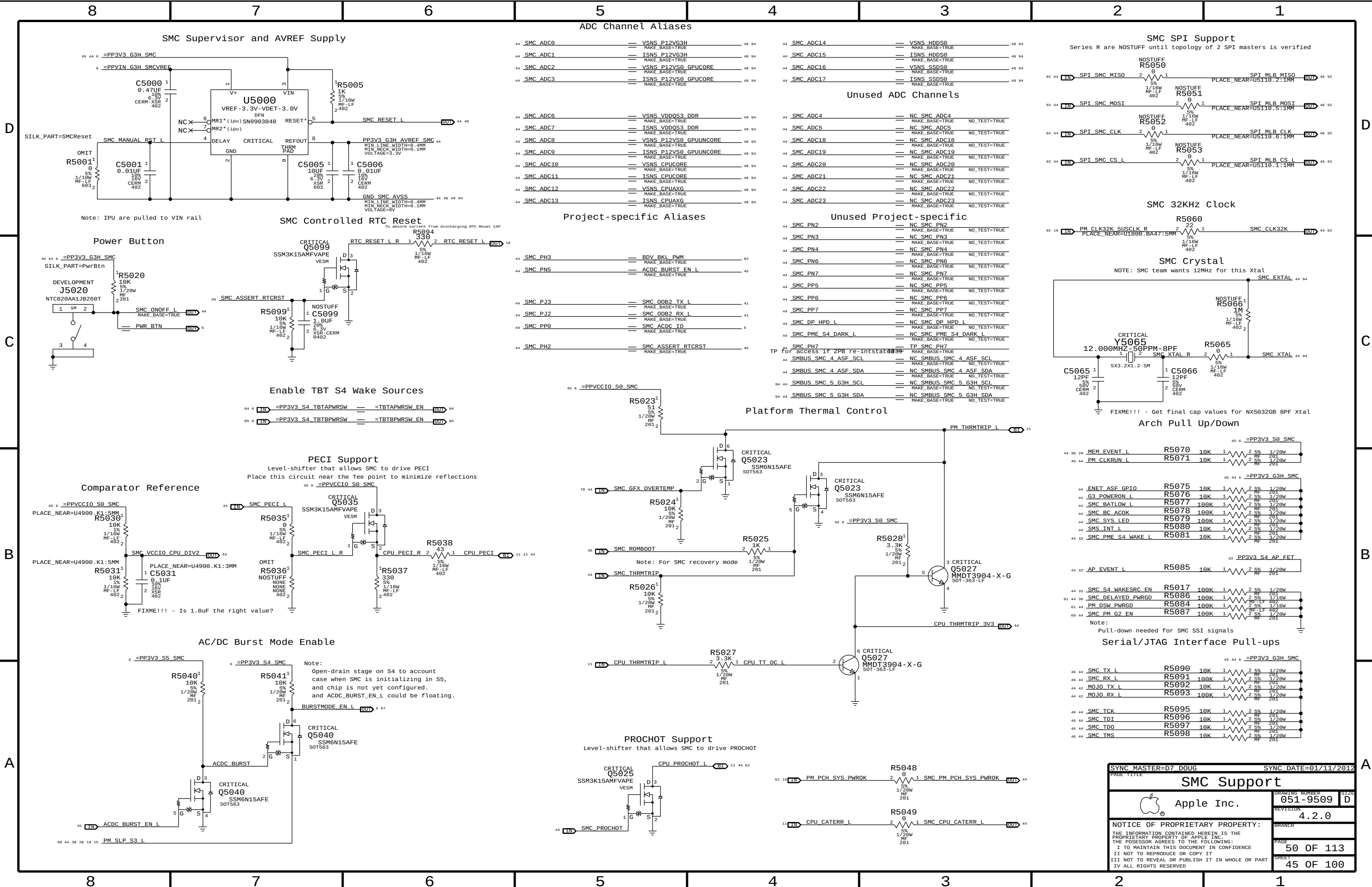


D









SMC Supervisor and AVREF Supply

ADC Channel Aliases

Unused ADC Channels

SMC SPI Support

SMC 32KHz Clock

SMC Crystal

SMC Controlled RTC Reset

Project-specific Aliases

Unused Project-specific

Enable TBT S4 Wake Sources

PECI Support

Comparator Reference

AC/DC Burst Mode Enable

PROCHOT Support

Platform Thermal Control

Arch Pull Up/Down

Serial/JTAG Interface Pull-ups

SYNC MASTER=D7 DOUG

SYNC DATE=01/11/2012

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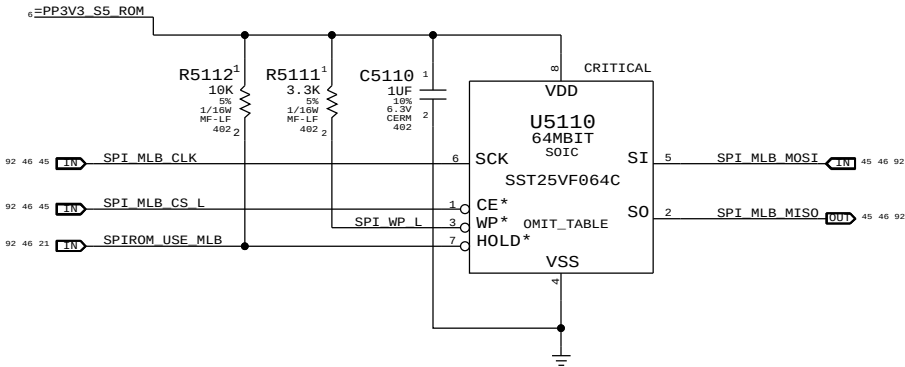
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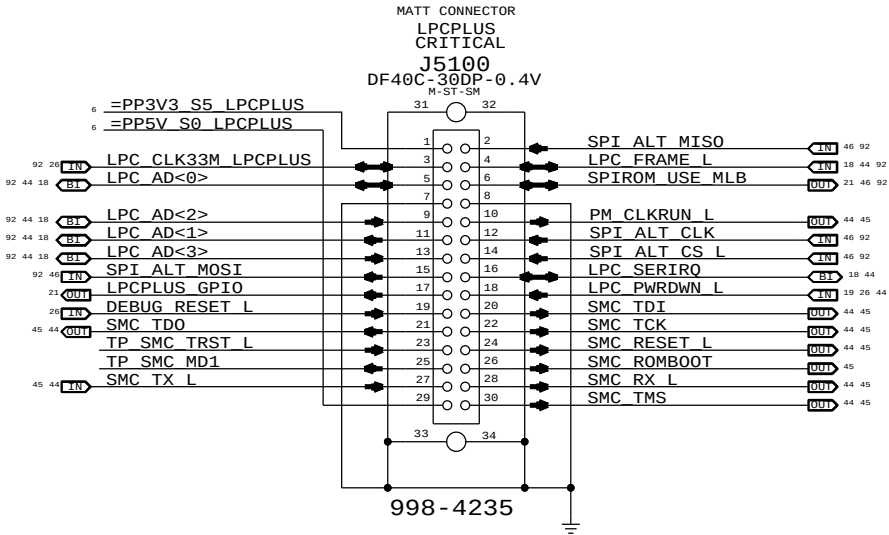
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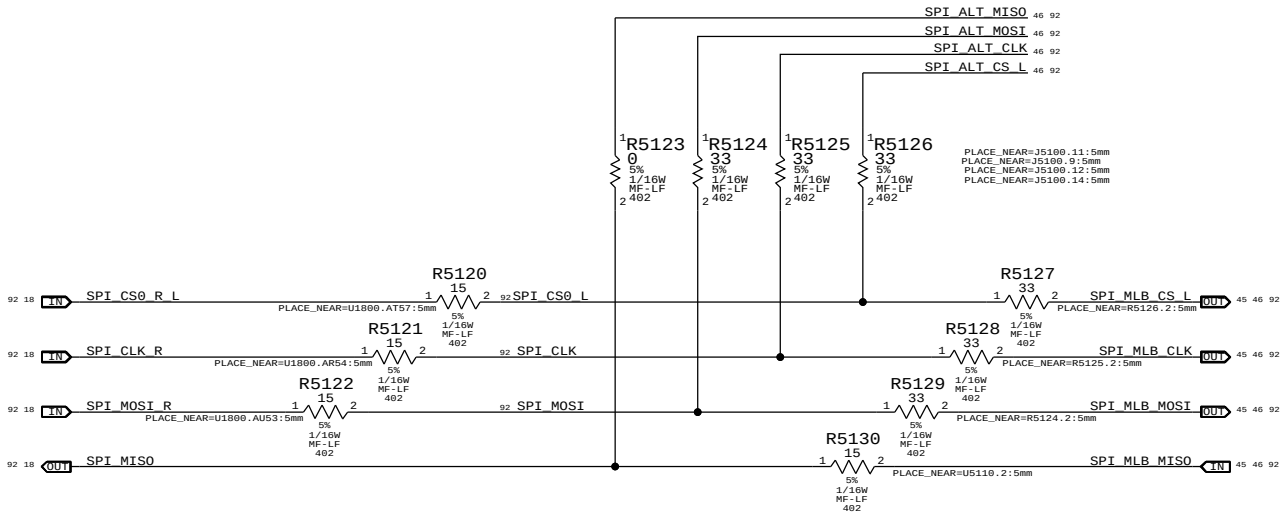
SPI BootROM



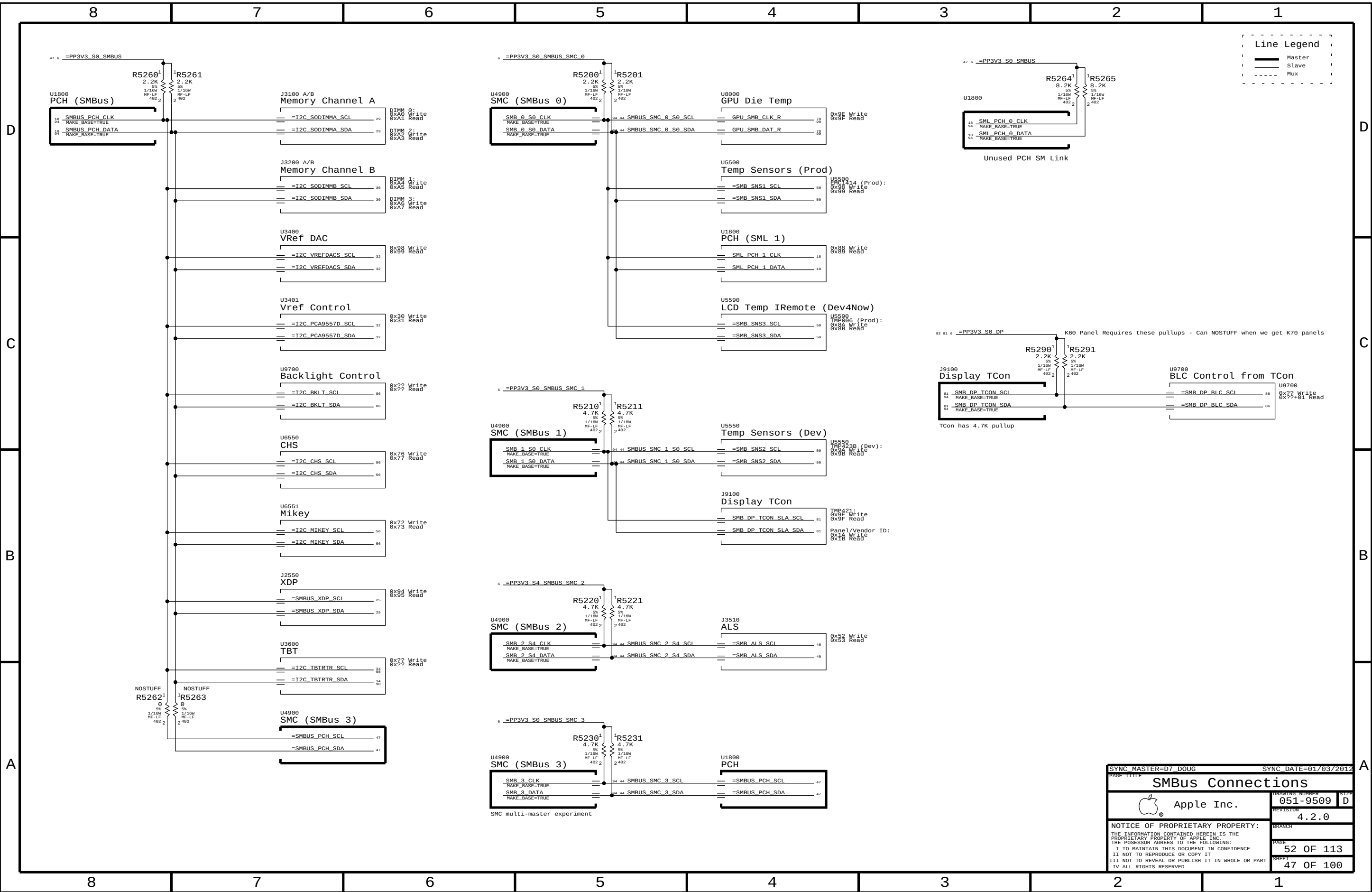
LPC+SPI Connector

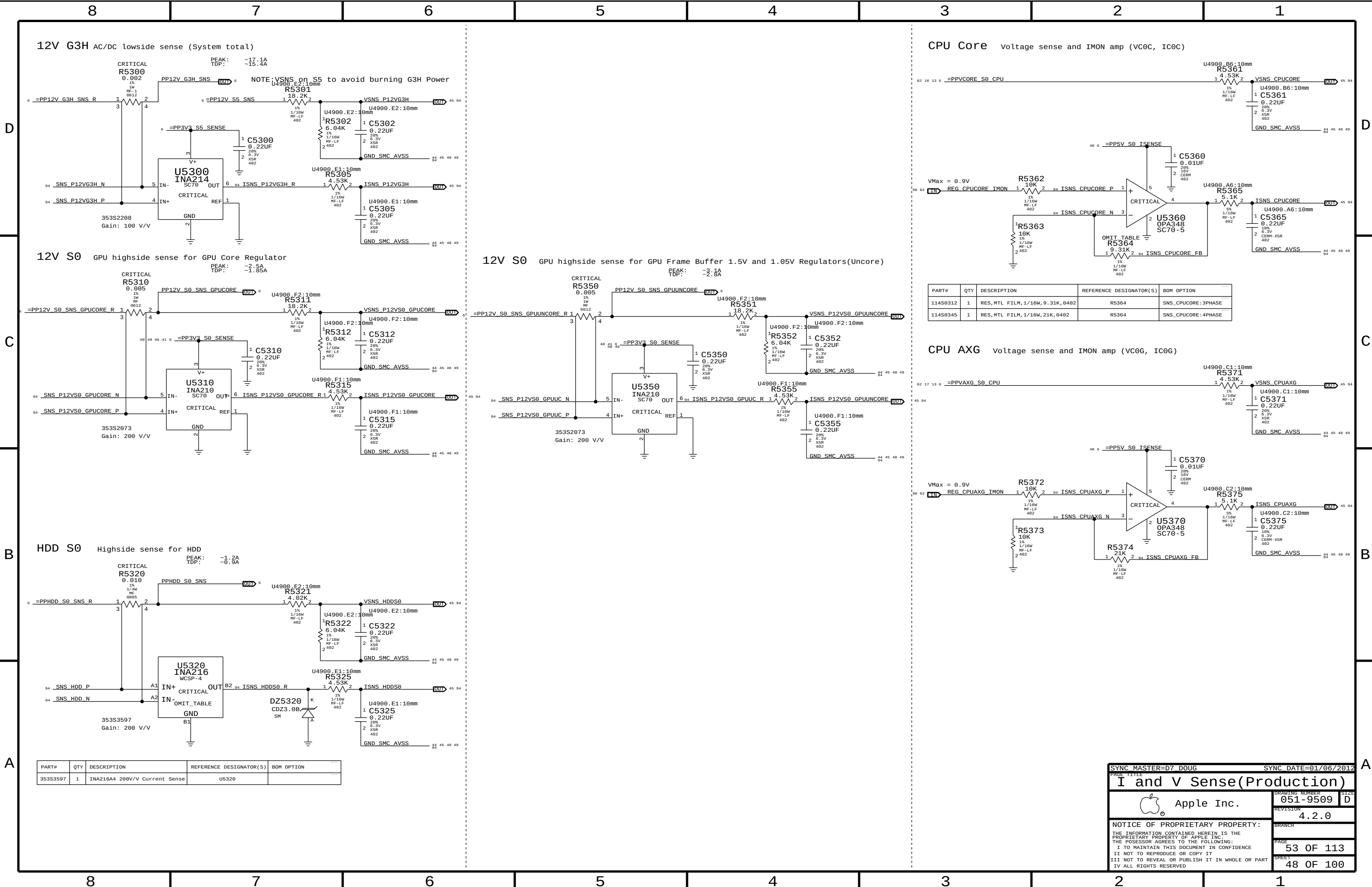


SPI Series Termination

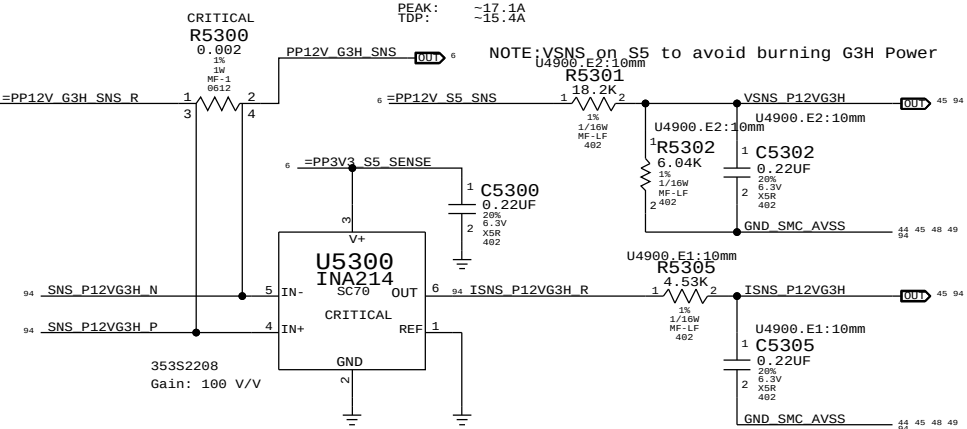


SYNC MASTER=D7 NICK		SYNC DATE=12/13/2011	
PAGE TITLE		SPI and Debug Connector	
Apple Inc.		DRAWING NUMBER	051-9509
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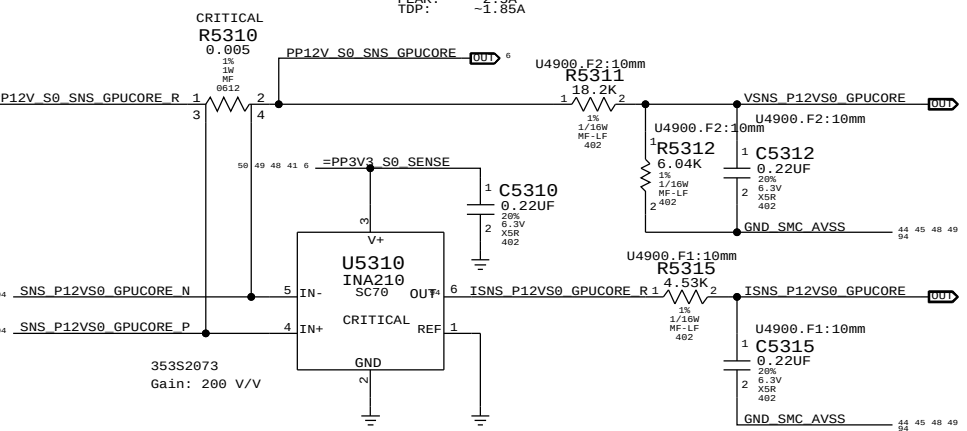




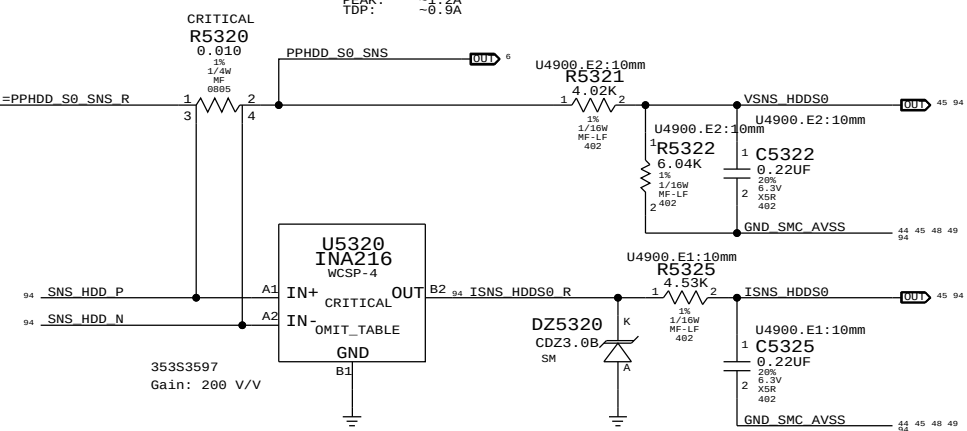
12V G3H AC/DC lowside sense (System total)



12V S0 GPU highside sense for GPU Core Regulator

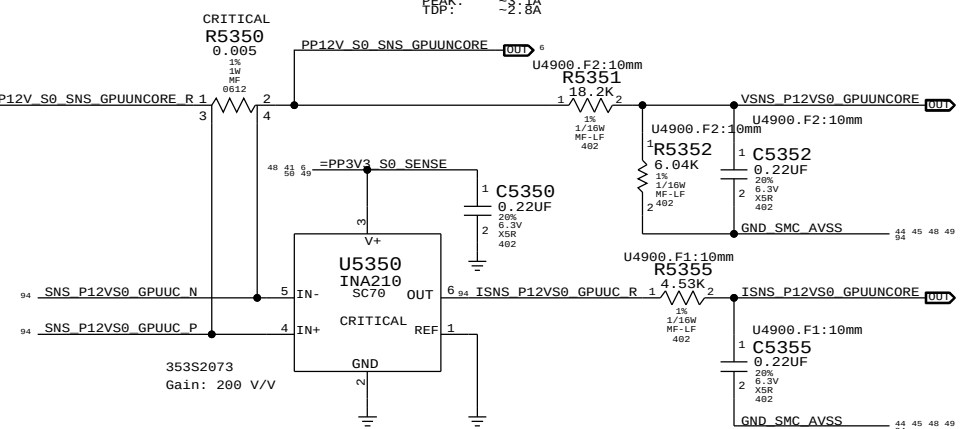


HDD S0 Highside sense for HDD

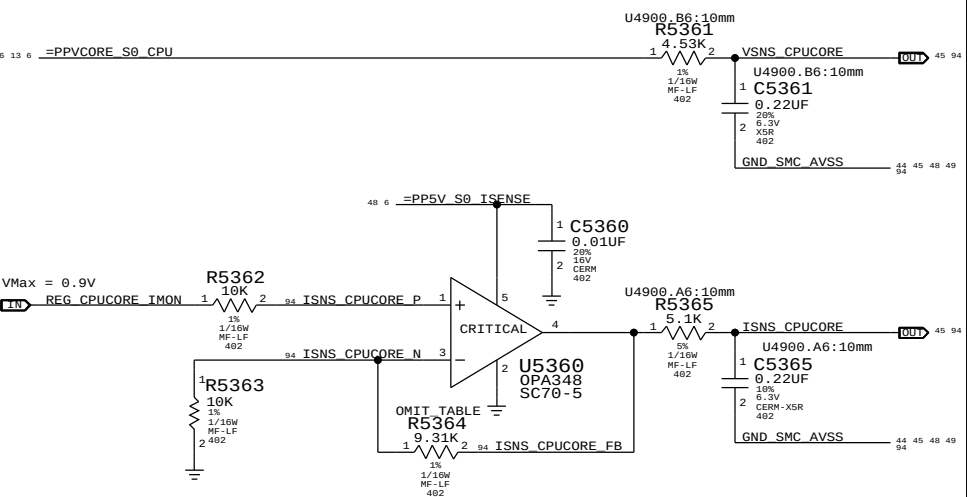


PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
353S3597	1	INA216A4 200V/V Current Sense	U5320	

12V S0 GPU highside sense for GPU Frame Buffer 1.5V and 1.05V Regulators(Uncore)

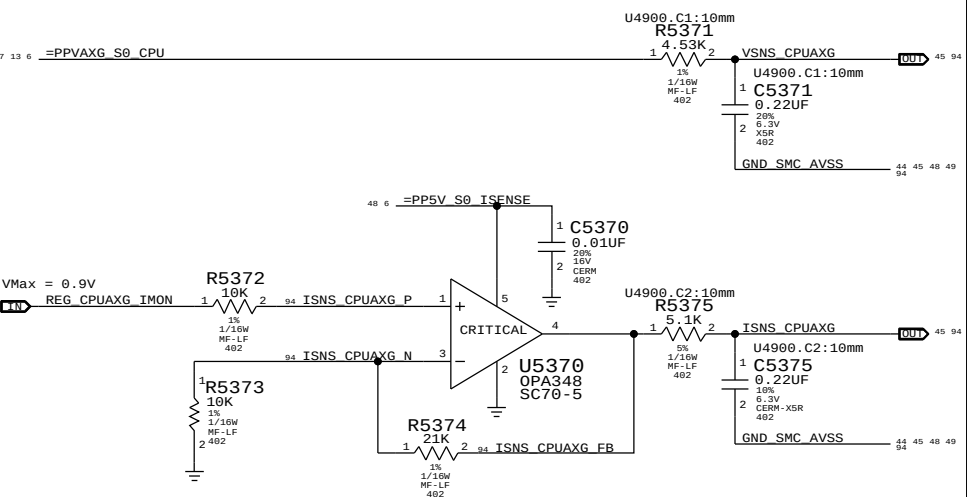


CPU Core voltage sense and IMON amp (VC0C, IC0C)



PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
114S0312	1	RES,MTL FILM,1/16W,9.31K,0402	R5364	SNS_CPUCORE:3PHASE
114S0345	1	RES,MTL FILM,1/16W,21K,0402	R5364	SNS_CPUCORE:4PHASE

CPU AXG voltage sense and IMON amp (VC0G, IC0G)



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I and V Sense(Production)

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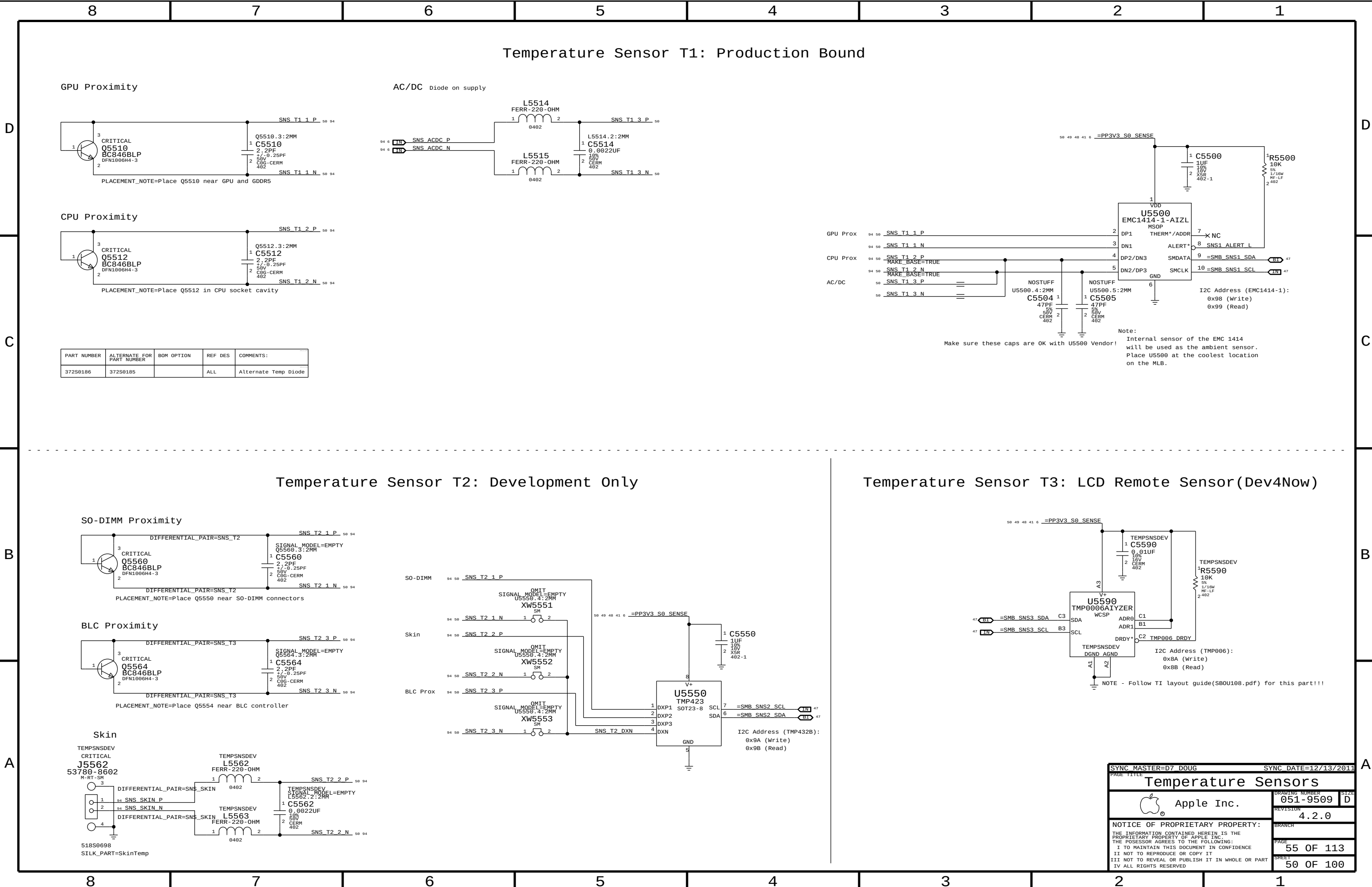
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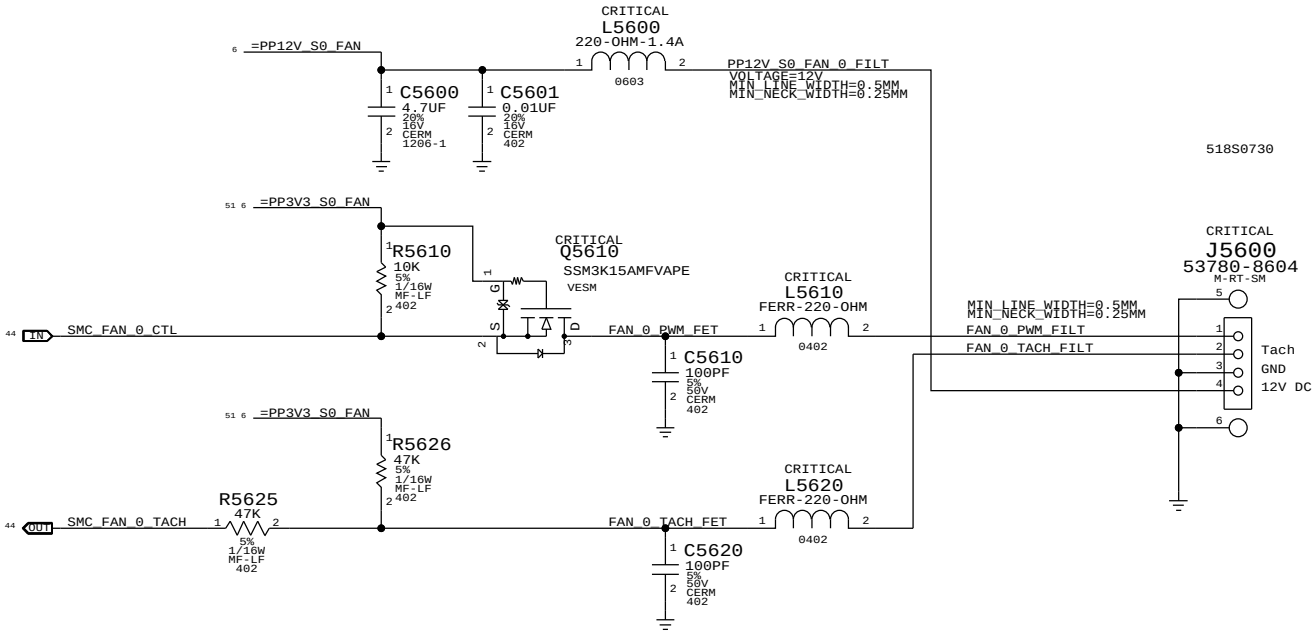
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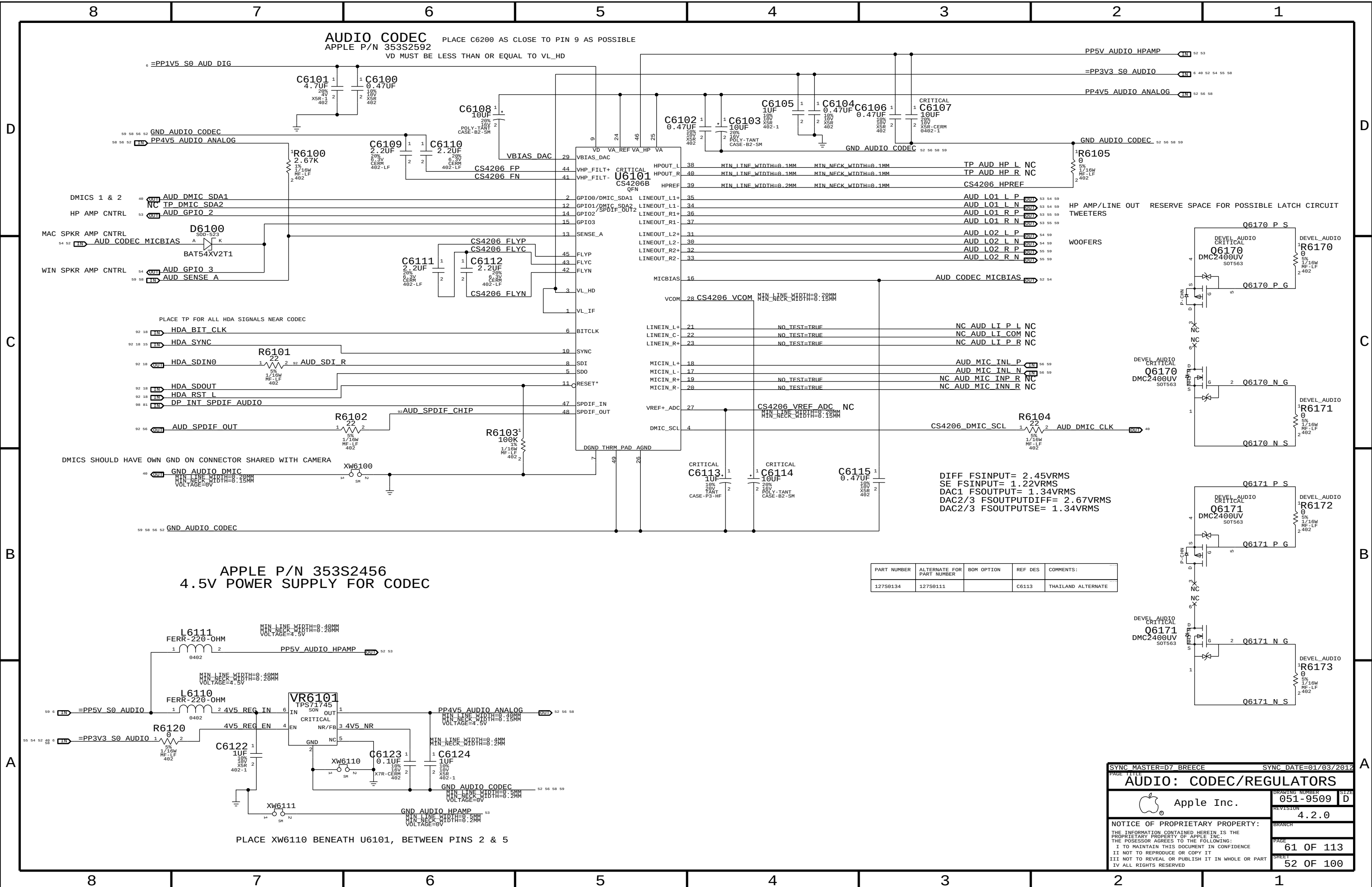
SMC Fan 0 (System)

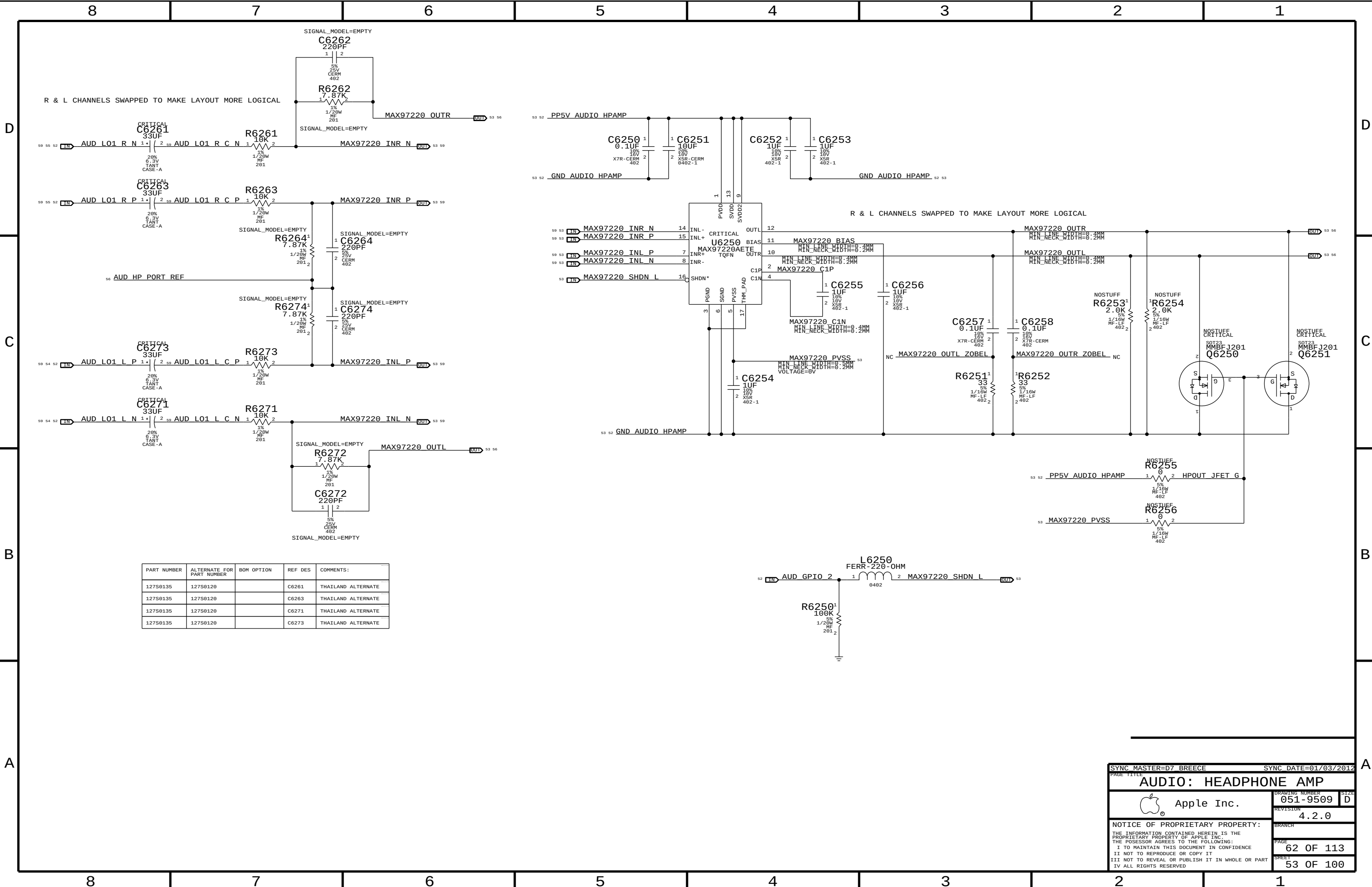
Note:
The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.
This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.
Otherwise, this is simply a pass-FET.



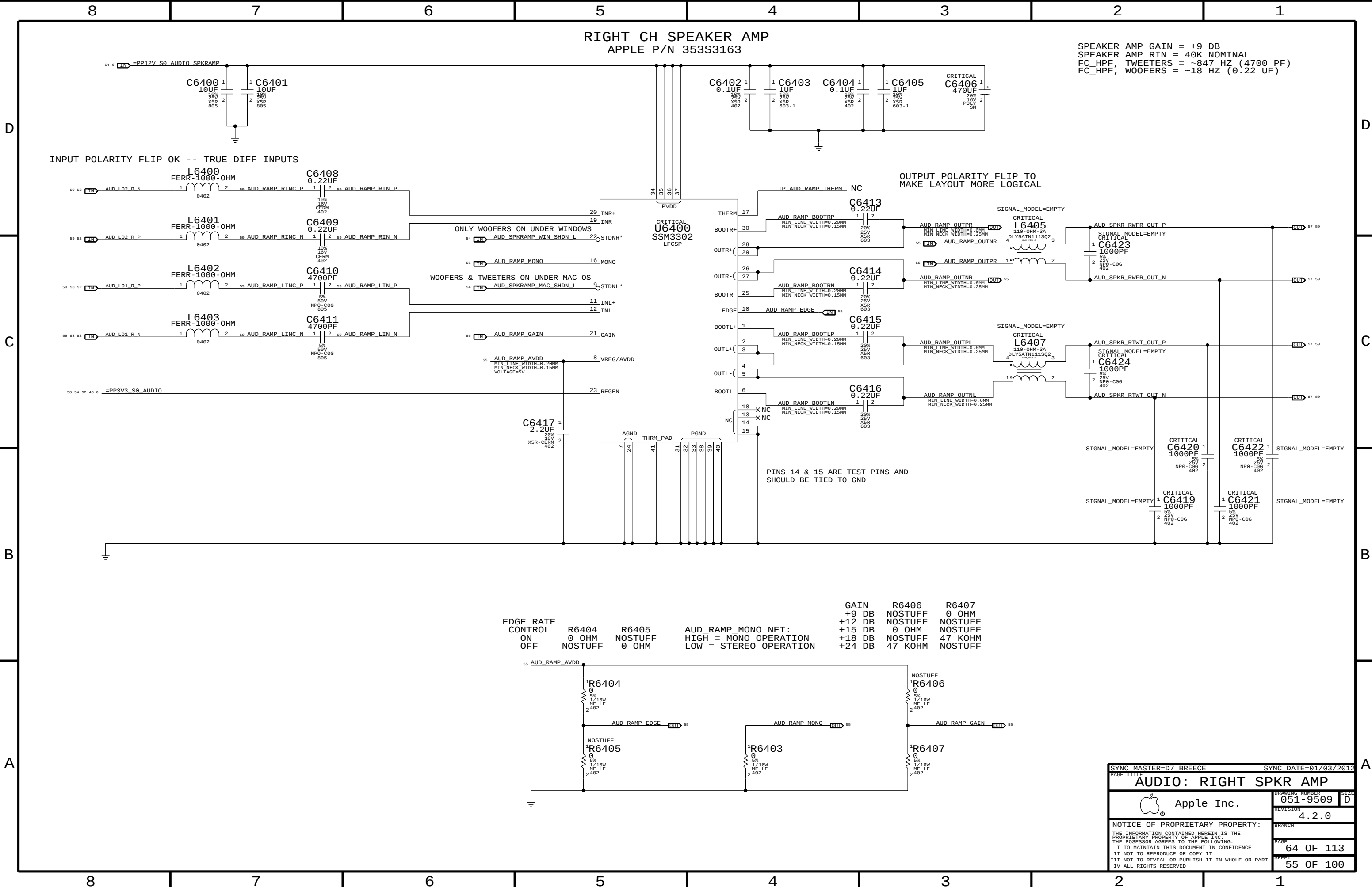
SMC Fan 1 (Unused)







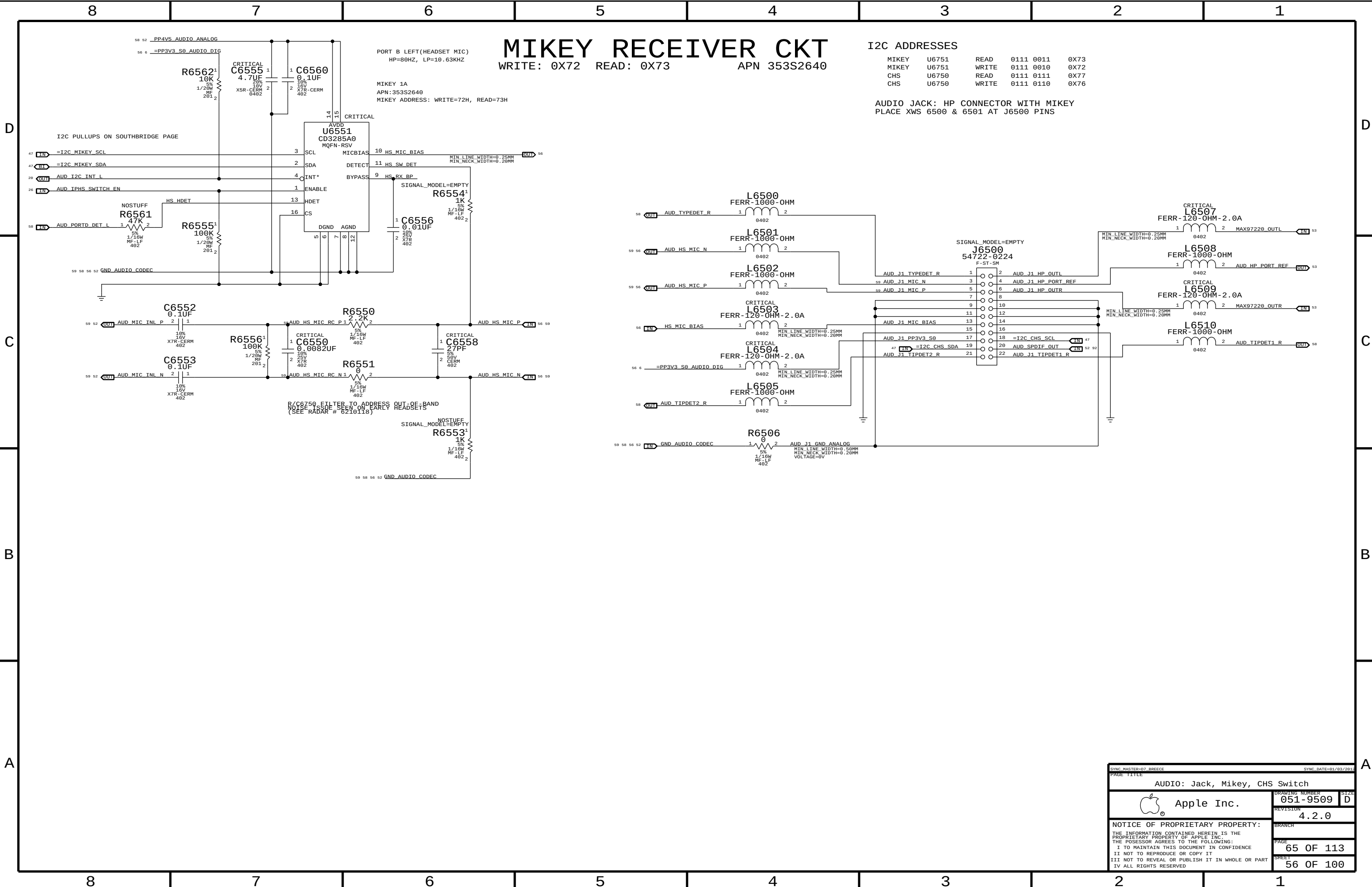
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
127S0135	127S0120		C6261	THAILAND ALTERNATE
127S0135	127S0120		C6263	THAILAND ALTERNATE
127S0135	127S0120		C6271	THAILAND ALTERNATE
127S0135	127S0120		C6273	THAILAND ALTERNATE



SPEAKER AMP GAIN = +9 DB
SPEAKER AMP RIN = 40K NOMINAL
FC_HPF, TWEETERS = ~847 HZ (4700 PF)
FC_HPF, WOOFERS = ~18 HZ (0.22 UF)

EDGE RATE CONTROL ON OFF	R6404	R6405	AUD_RAMP_MONO NET:	GAIN	R6406	R6407
	0 OHM	NOSTUFF	HIGH = MONO OPERATION	+9 DB	NOSTUFF	0 OHM
	NOSTUFF	0 OHM	LOW = STEREO OPERATION	+12 DB	NOSTUFF	NOSTUFF
				+15 DB	0 OHM	NOSTUFF
				+18 DB	NOSTUFF	47 KOHM

SYNC MASTER=D7 BREECE		SYNC DATE=01/03/2012	
AUDIO: RIGHT SPKR AMP			
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2640

I2C ADDRESSES				
MIKEY	U6751	READ	0111 0011	0X73
MIKEY	U6751	WRITE	0111 0010	0X72
CHS	U6750	READ	0111 0111	0X77
CHS	U6750	WRITE	0111 0110	0X76

AUDIO JACK: HP CONNECTOR WITH MIKEY
PLACE XWS 6500 & 6501 AT J6500 PINS

8	7	6	5	4	3	2	1
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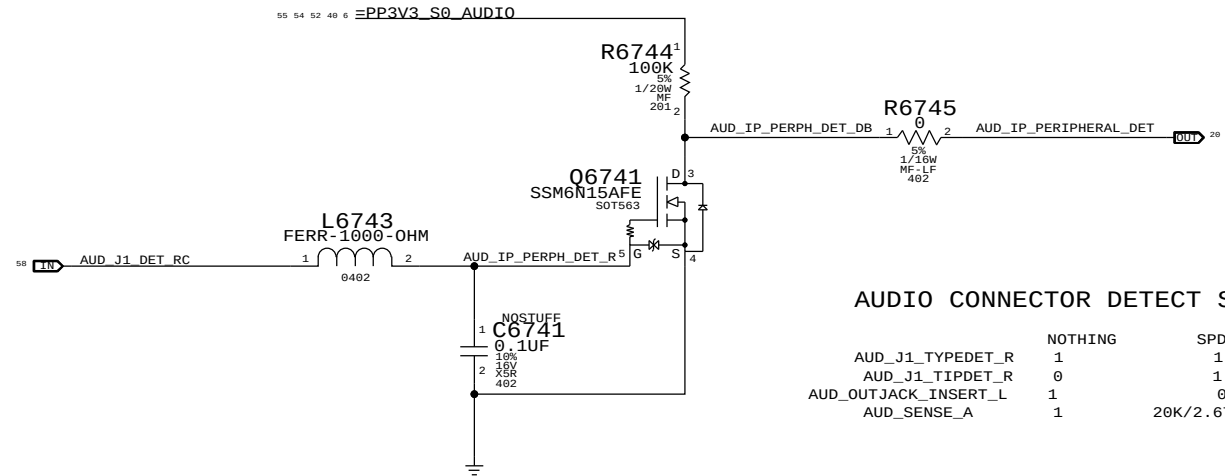
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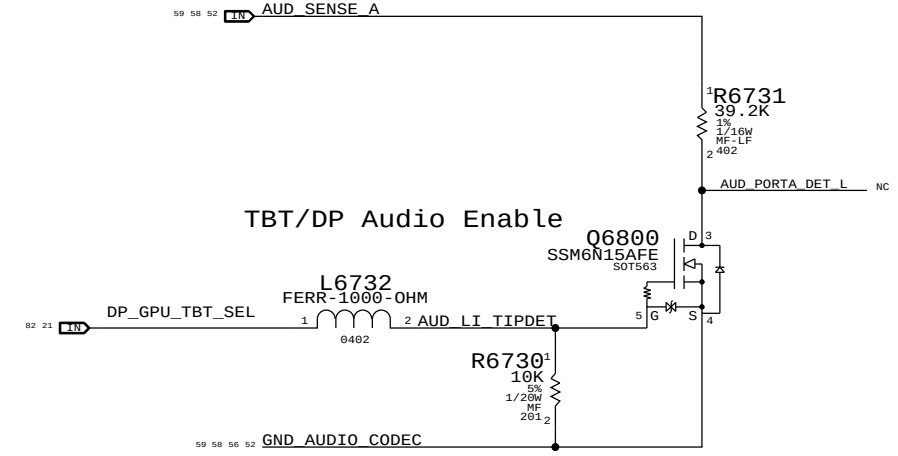
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8	7	6	5	4	3	2	1
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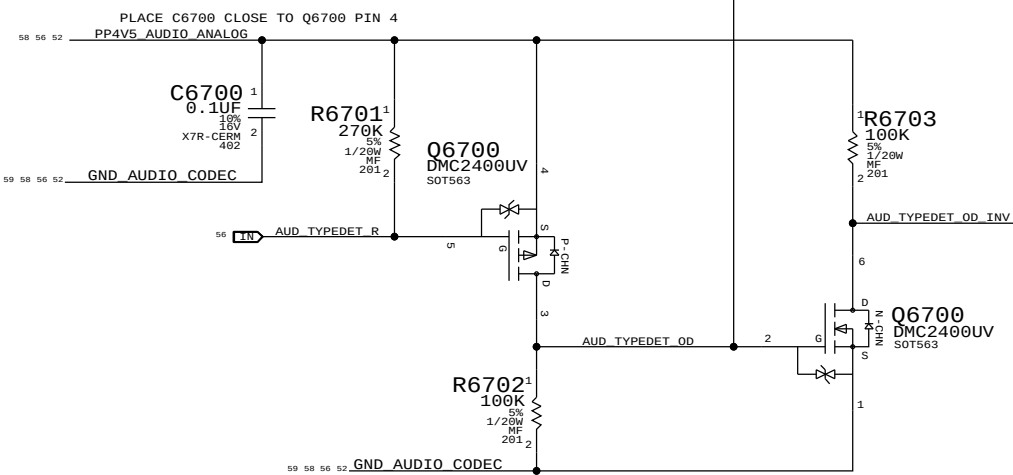
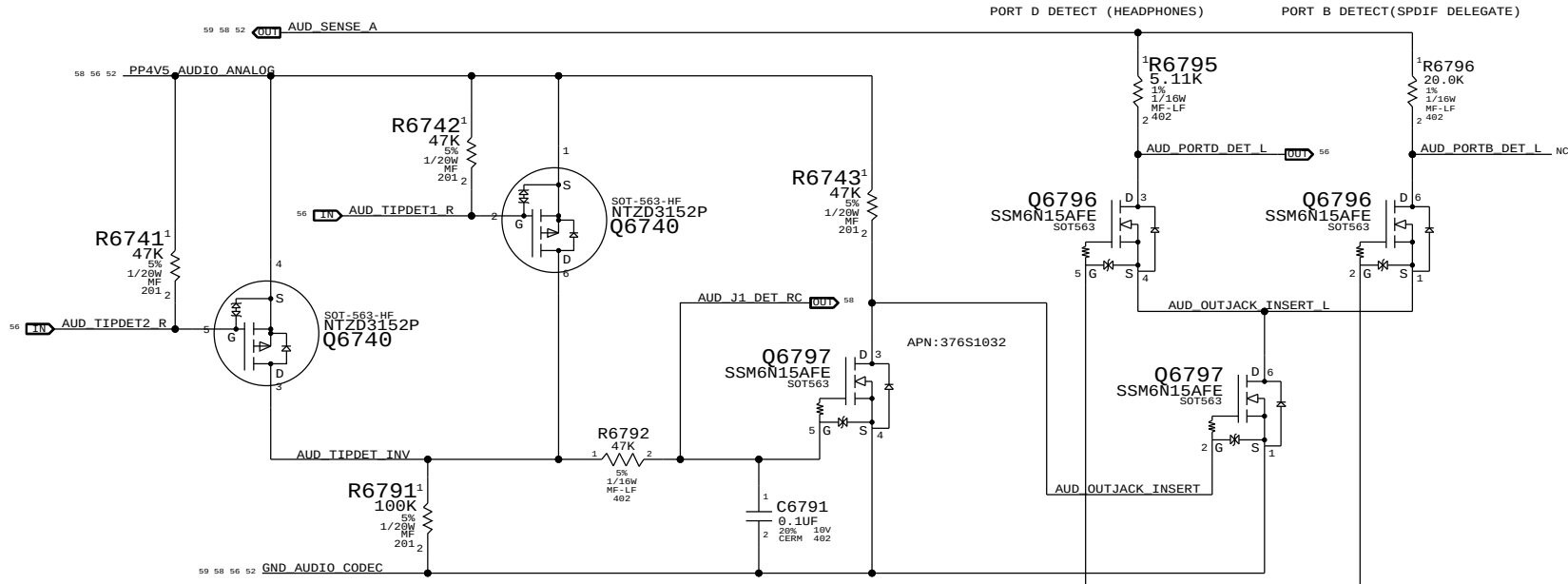
IPHS HS Detect Debounce CKT



LI Insert Detect (DETECT A)



TBT/DP Audio Enable



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SYNC DATE=01/03/2012

AUDIO: Detects/Grounding

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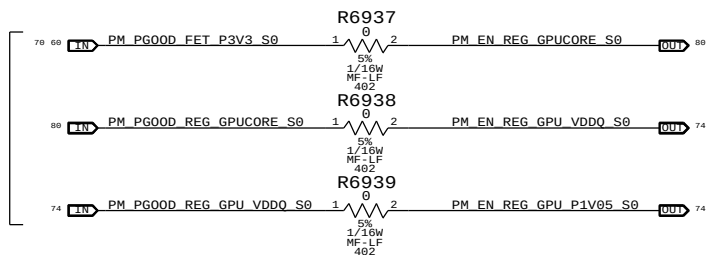
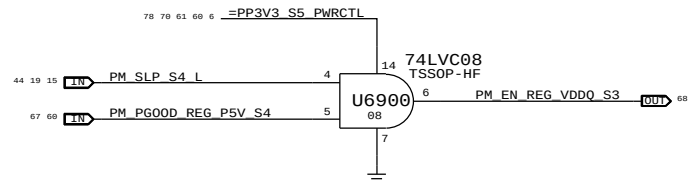
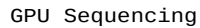
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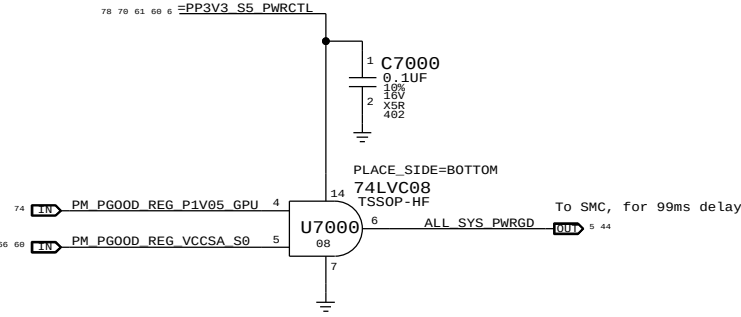
PCH Sequencing



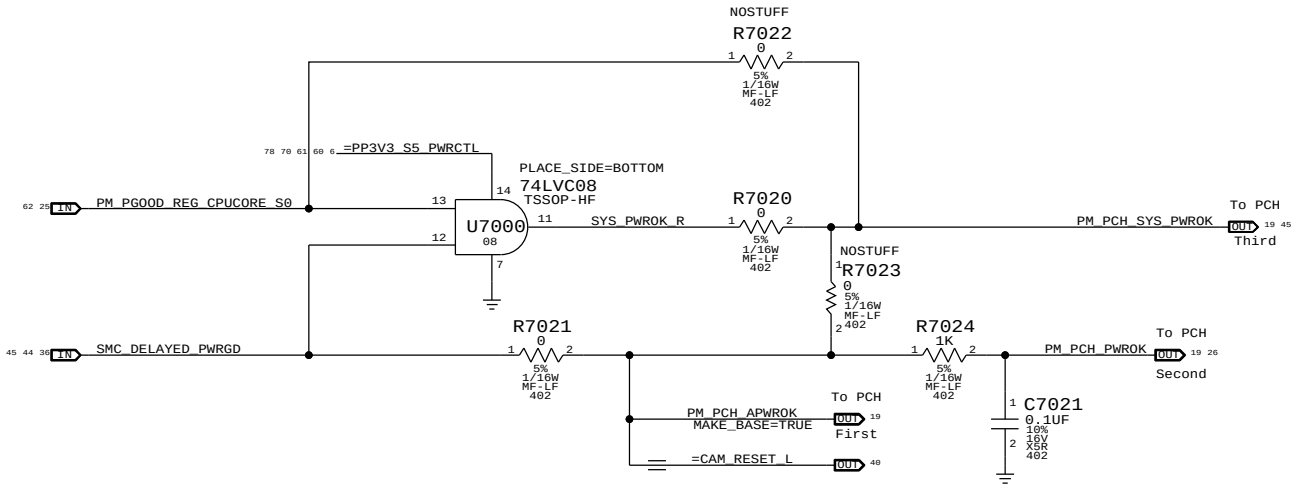
SYNC MASTER=D7 NICK		SYNC DATE=12/13/2013	
PAGE TITLE			
PM Regulator Enables			
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		BRANCH	
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Platform and UnCore Power Good Derive SMC ALL_SYS_PWRGD

Note: GPU power goods are implicitly included because the power goods for VDDQ, DPVDDC, and GPU Core are wired-or together



PCH Power Goods



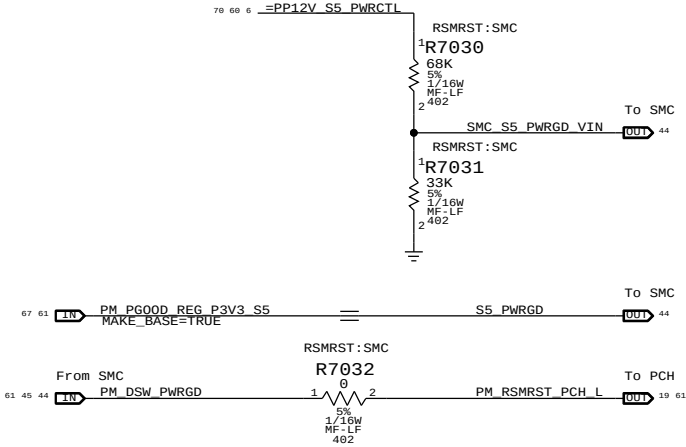
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

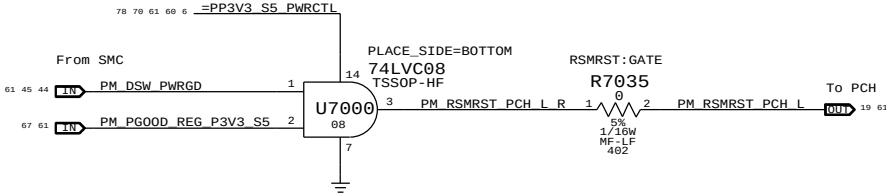
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

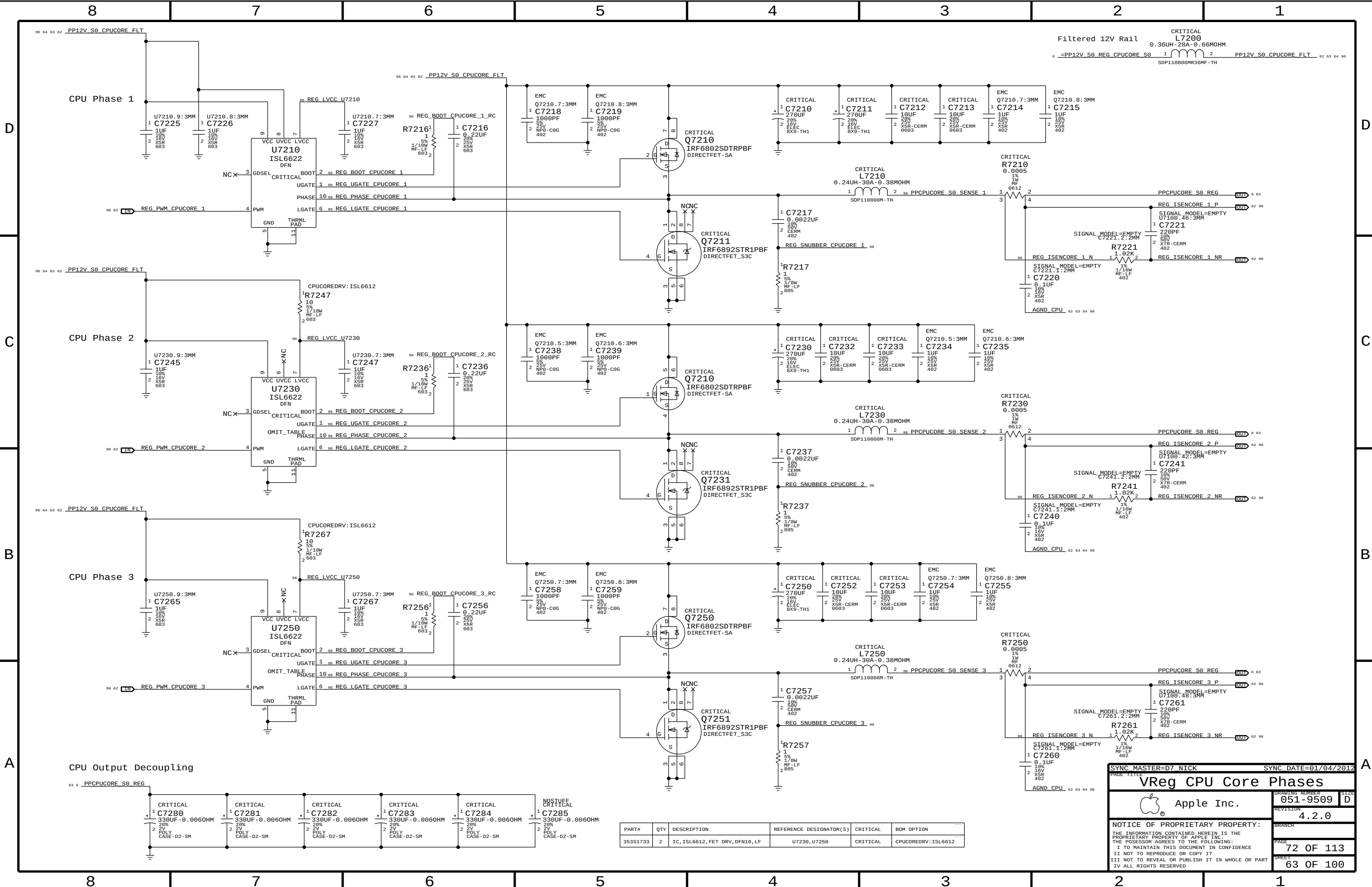
Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC MASTER=D7 NICK		SYNC DATE=12/13/2011	
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S1733	2	IC, ISL6612, FET DRV, DFN10, LF	U7230, U7250	CRITICAL	CPUCOREDRV: ISL6612

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VReg CPU Core Phases

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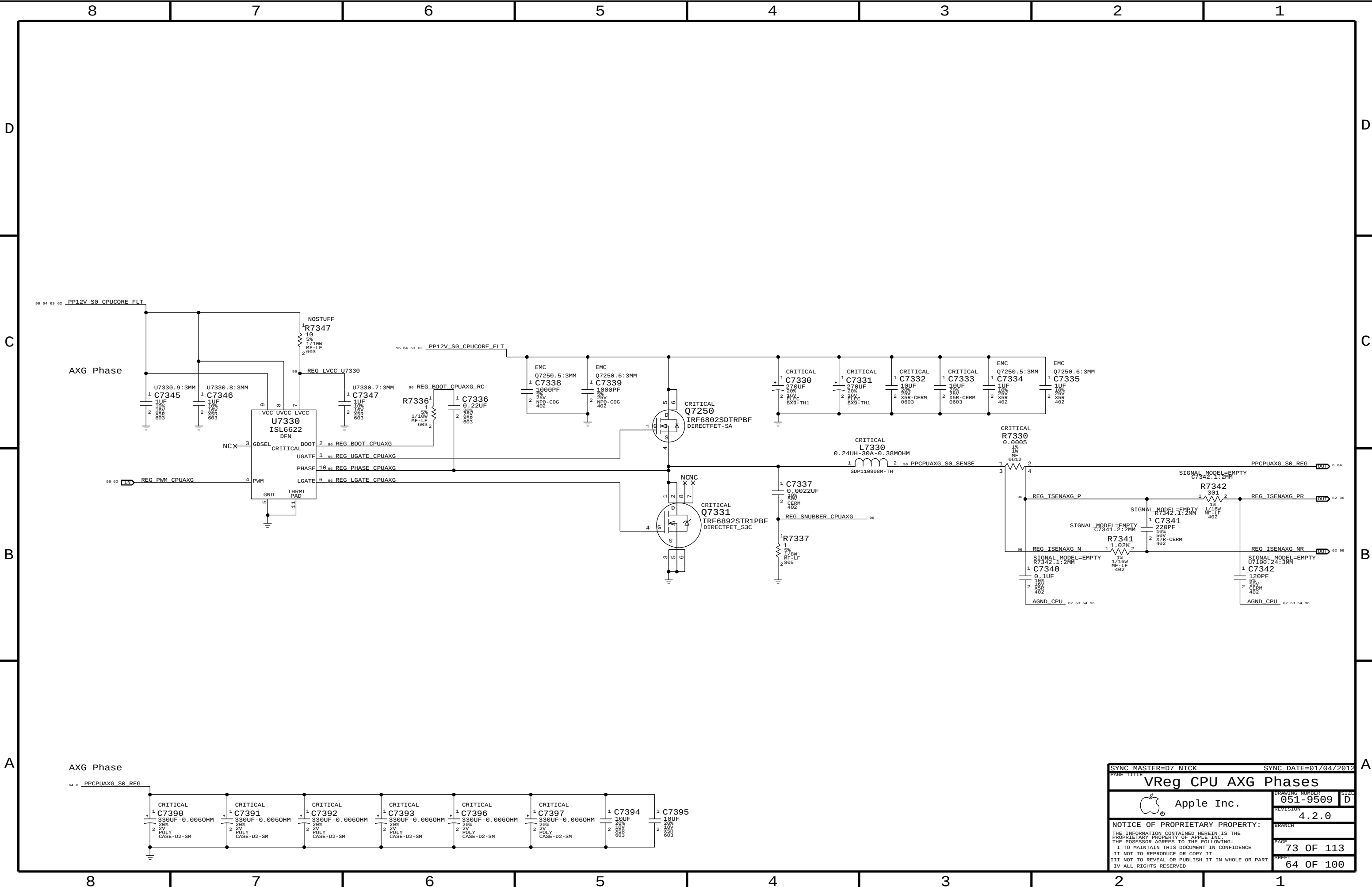
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VReg CPU AXG Phases

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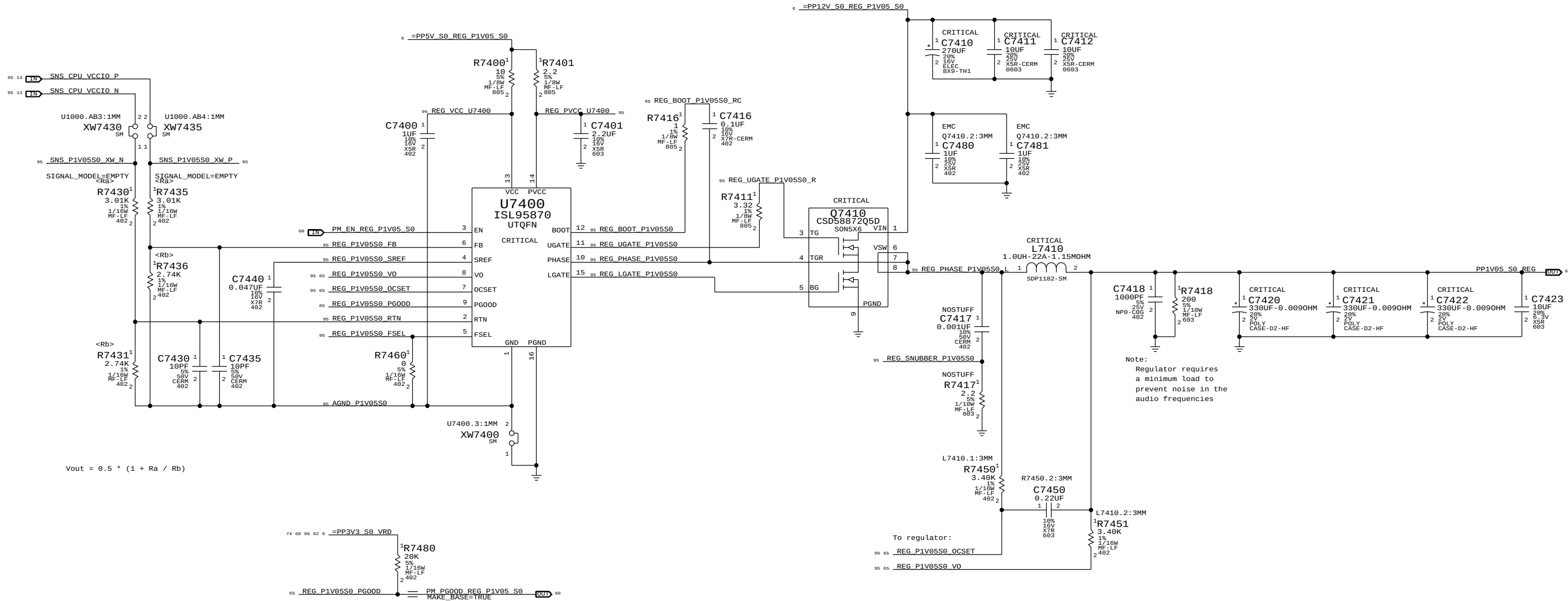
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CPU VccIO/PCH (1.05V) S0 Regulator

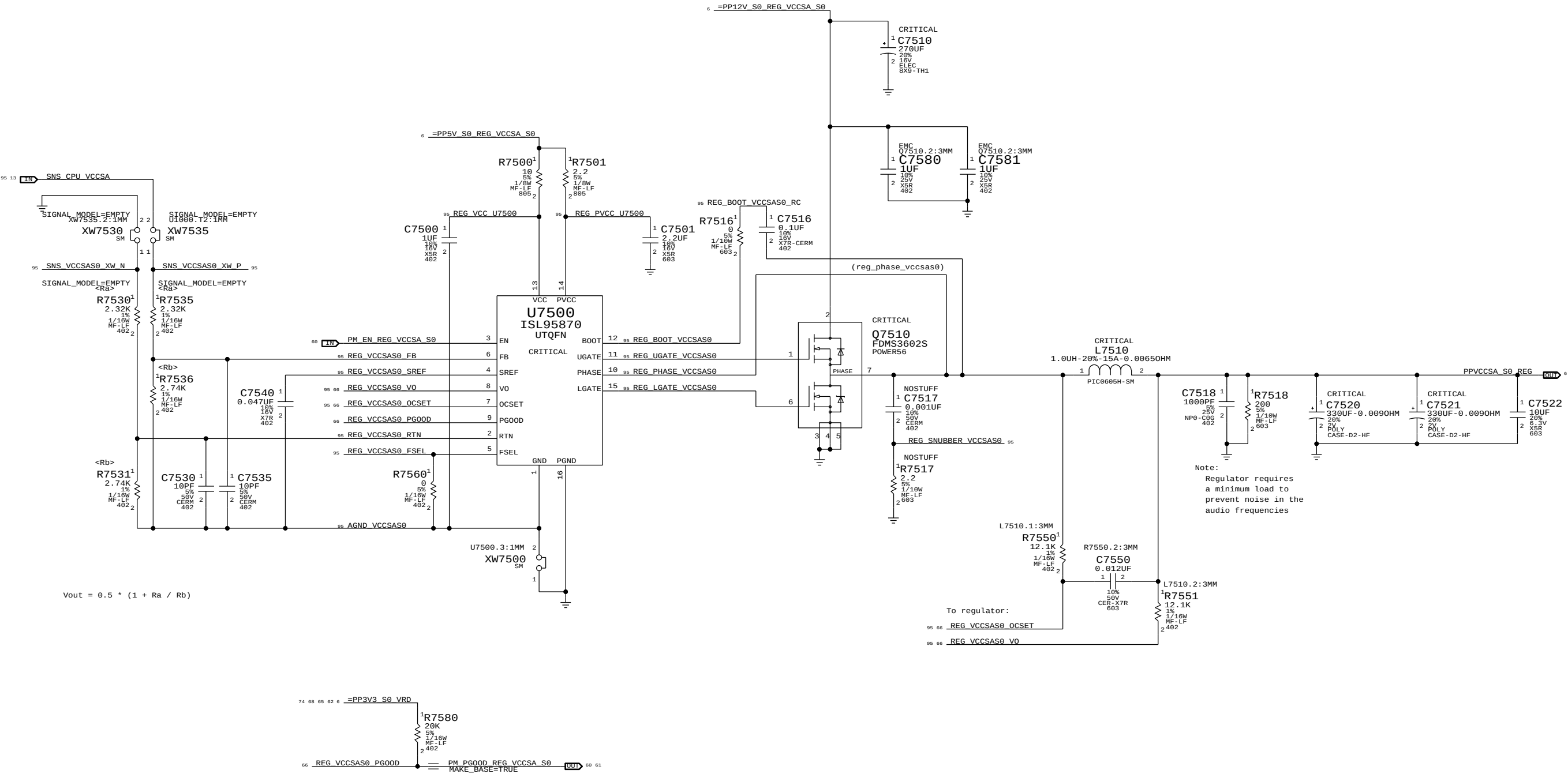
Max avg current: ? A (design)/ 14.38 A (budget)
Max peak current: ? A (design)/ 18.38 A (budget)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



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PAGE TITLE			
VReg CPU/PCH 1.05V S0			
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		PAGE	74 OF 113
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CPU VccSA (0.925V) S0 Regulator

Max avg current: ? A (design)/ 1.51 A (budget)
Max peak current: ? A (design)/ 8.76 A (budget)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



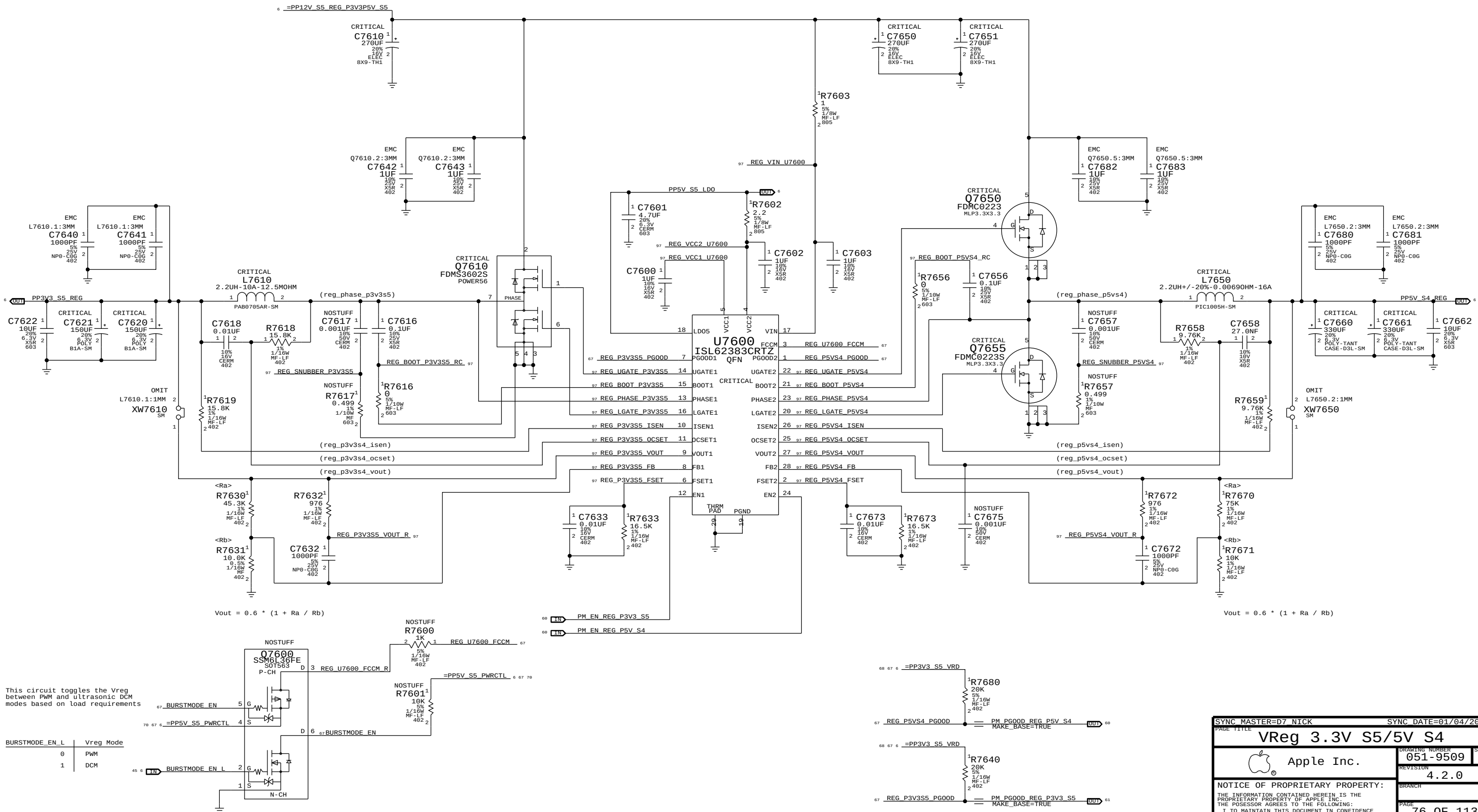
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VReg CPU VccSA S0			
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3.3V S5 Regulator

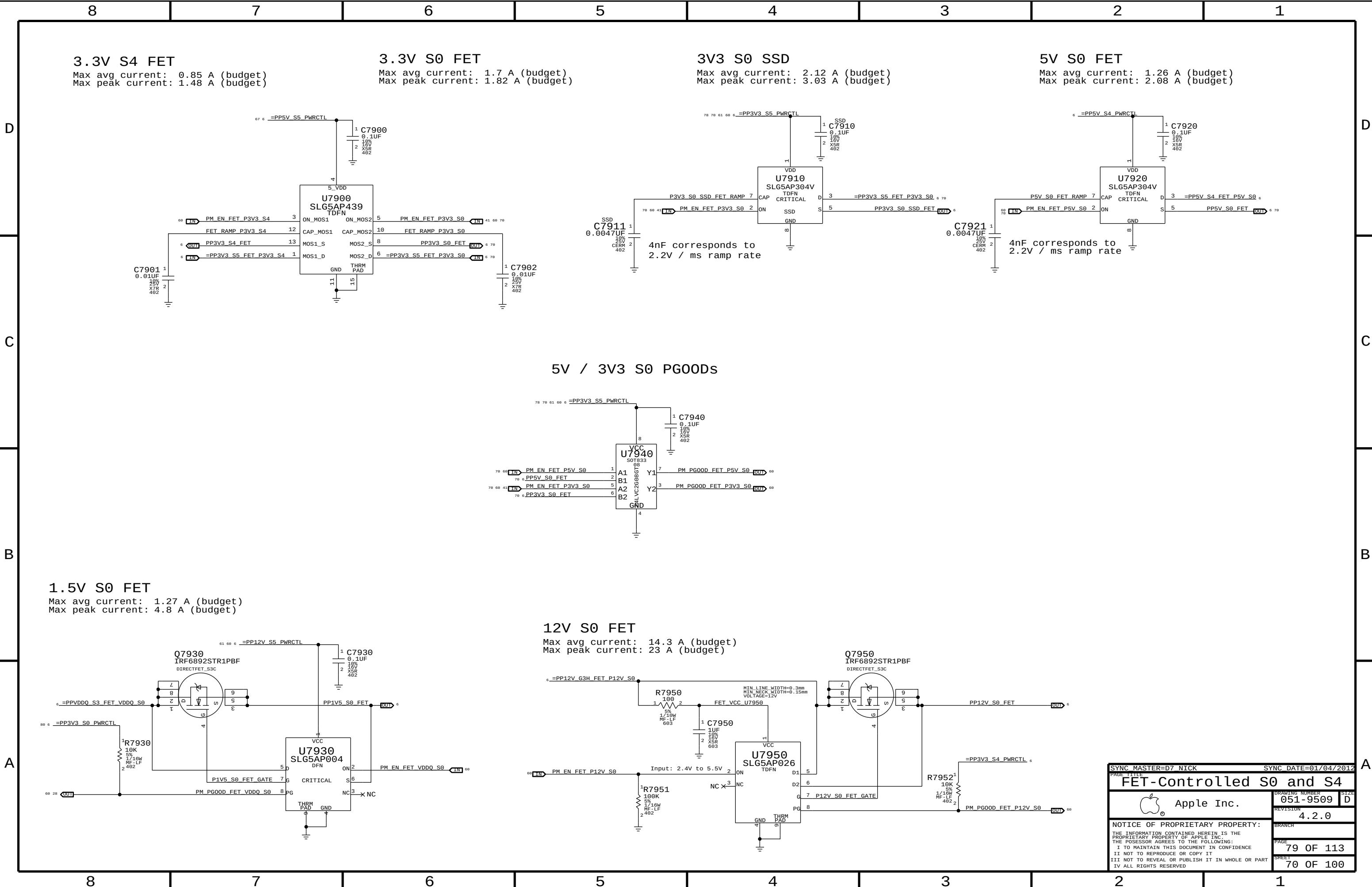
Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



SYNC MASTER=D7 NICK		SYNC DATE=01/04/2012	
PAGE TITLE		VReg 3.3V S5/5V S4	
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3.3V S4 FET

Max avg current: 0.85 A (budget)
Max peak current: 1.48 A (budget)

3.3V S0 FET

Max avg current: 1.7 A (budget)
Max peak current: 1.82 A (budget)

3V3 S0 SSD

Max avg current: 2.12 A (budget)
Max peak current: 3.03 A (budget)

5V S0 FET

Max avg current: 1.26 A (budget)
Max peak current: 2.08 A (budget)

5V / 3V3 S0 PG00Ds

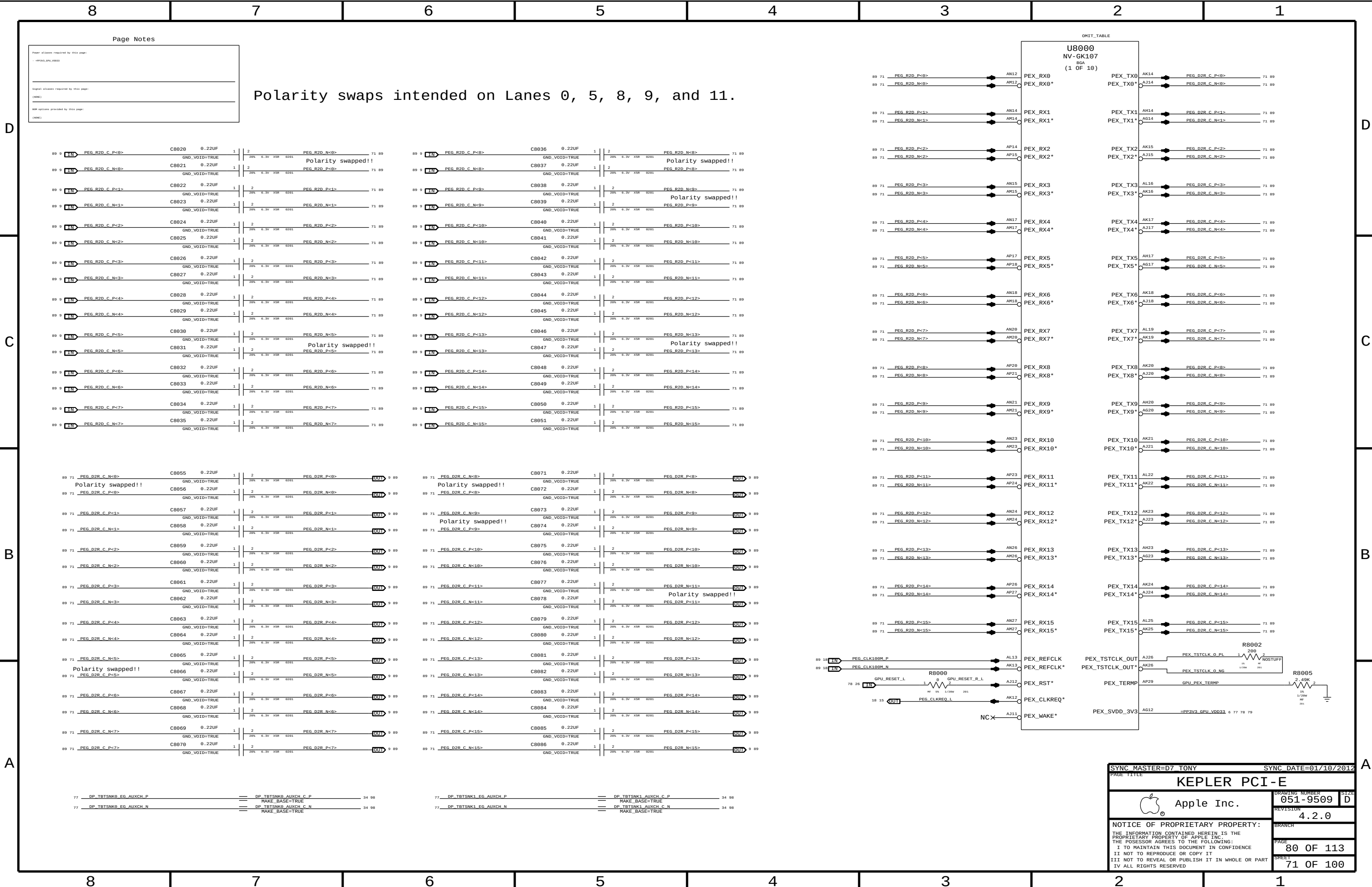
1.5V S0 FET

Max avg current: 1.27 A (budget)
Max peak current: 4.8 A (budget)

12V S0 FET

Max avg current: 14.3 A (budget)
Max peak current: 23 A (budget)

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PAGE TITLE			
FET-Controlled S0 and S4			
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Page Notes

Power aliases required by this page:
- PP3V3_GPU_VDD33

Signal aliases required by this page:
(None)

ROM options provided by this page:
(None)

Polarity swaps intended on Lanes 0, 5, 8, 9, and 11.

SYNC MASTER=D7 TONY

SYNC DATE=01/10/2012

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KEPLER PCI-E

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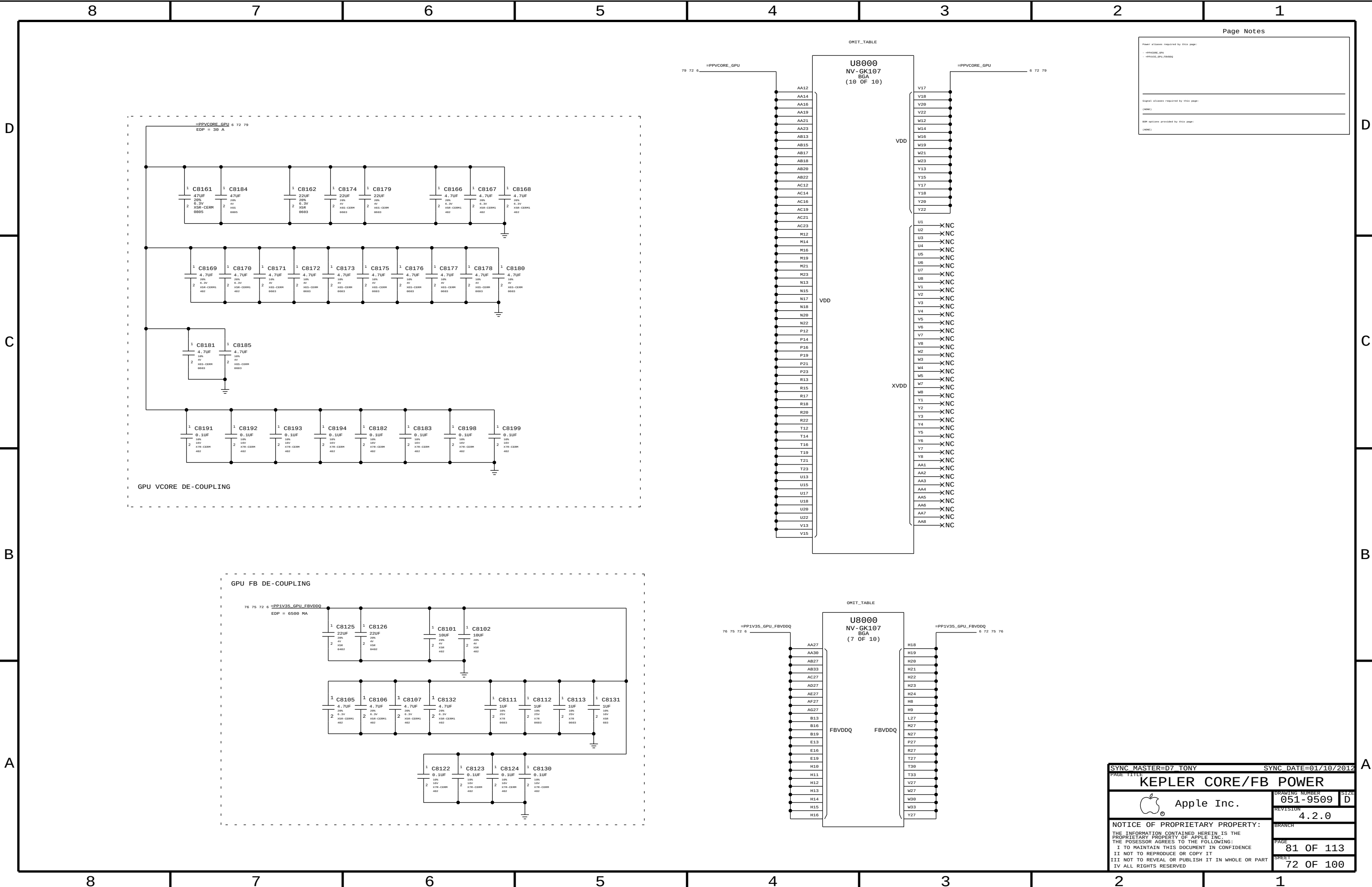
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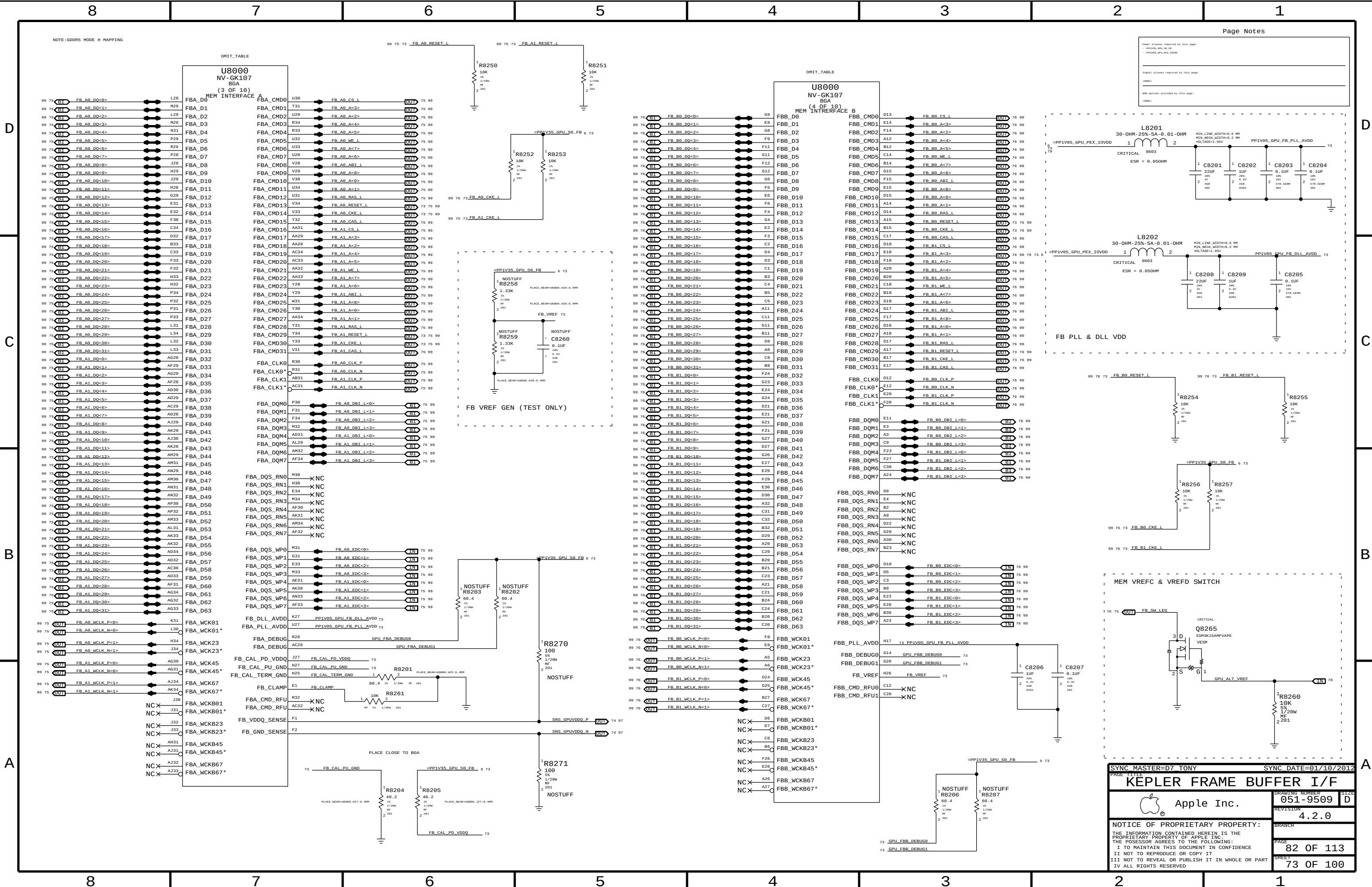
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- PPVCORE_GPU	
- PPVCORE_GPU_FBDQ	
Signal aliases required by this page:	
(NONE)	
BOM options provided by this page:	
(NONE)	

PAGE TITLE		SYNC DATE=01/10/2012	
KEPLER CORE/FB POWER		DRAWING NUMBER	
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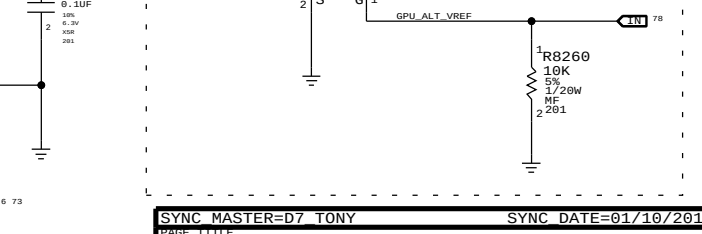
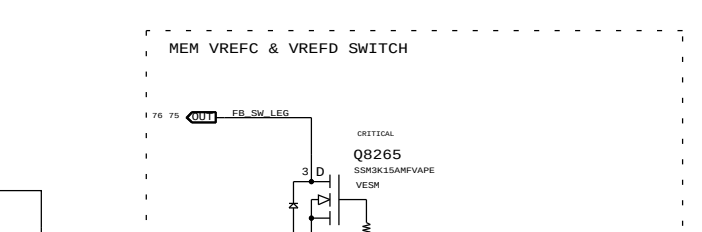
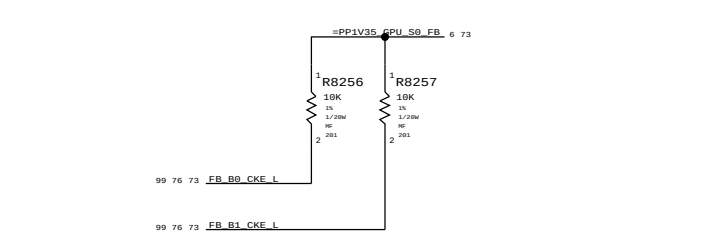
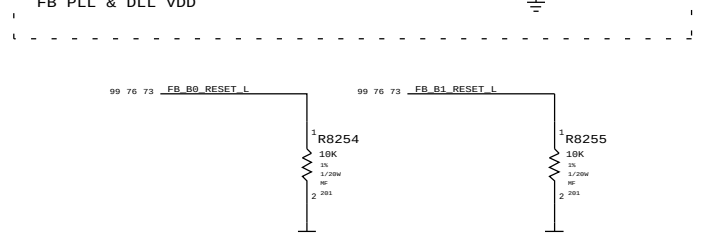
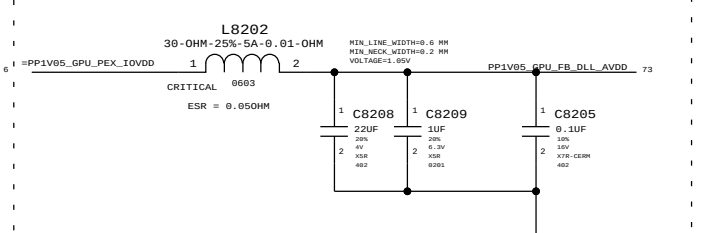
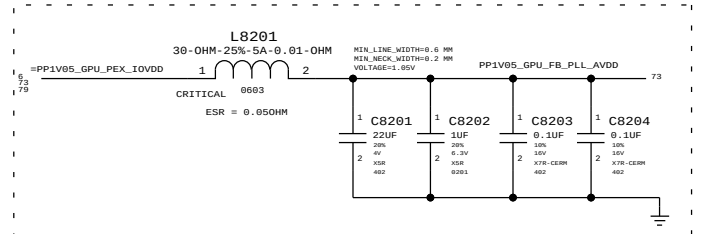


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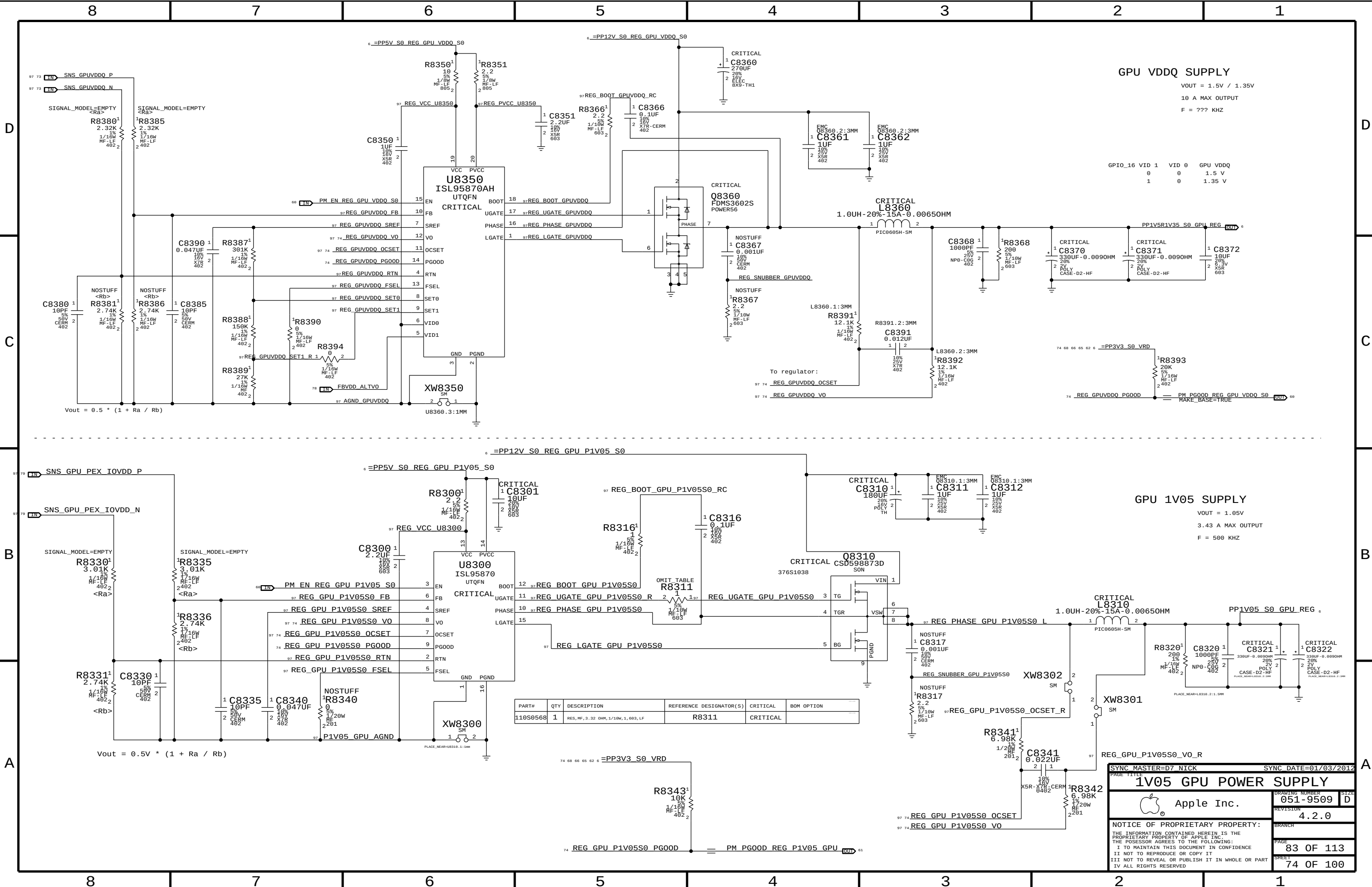
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- PP1V05_GPU_FB
- PP1V05_GPU_PEX

Signal planes required by this page:
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BOM options provided by this page:
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SYNC MASTER=D7 TONY		SYNC DATE=01/10/2012	
KEPLER FRAME BUFFER I/F		DRAWING NUMBER	051-9509
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GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V

10 A MAX OUTPUT

F = ??? KHZ

GPIO_16 VID 1 VID 0 GPU VDDQ

0 0 1.5 V

1 0 1.35 V

GPU 1V05 SUPPLY

VOUT = 1.05V

3.43 A MAX OUTPUT

F = 500 KHZ

SYNC MASTER=D7 NICK SYNC DATE=01/03/2012

1V05 GPU POWER SUPPLY

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DRAWING NUMBER 051-9509 SIZE D

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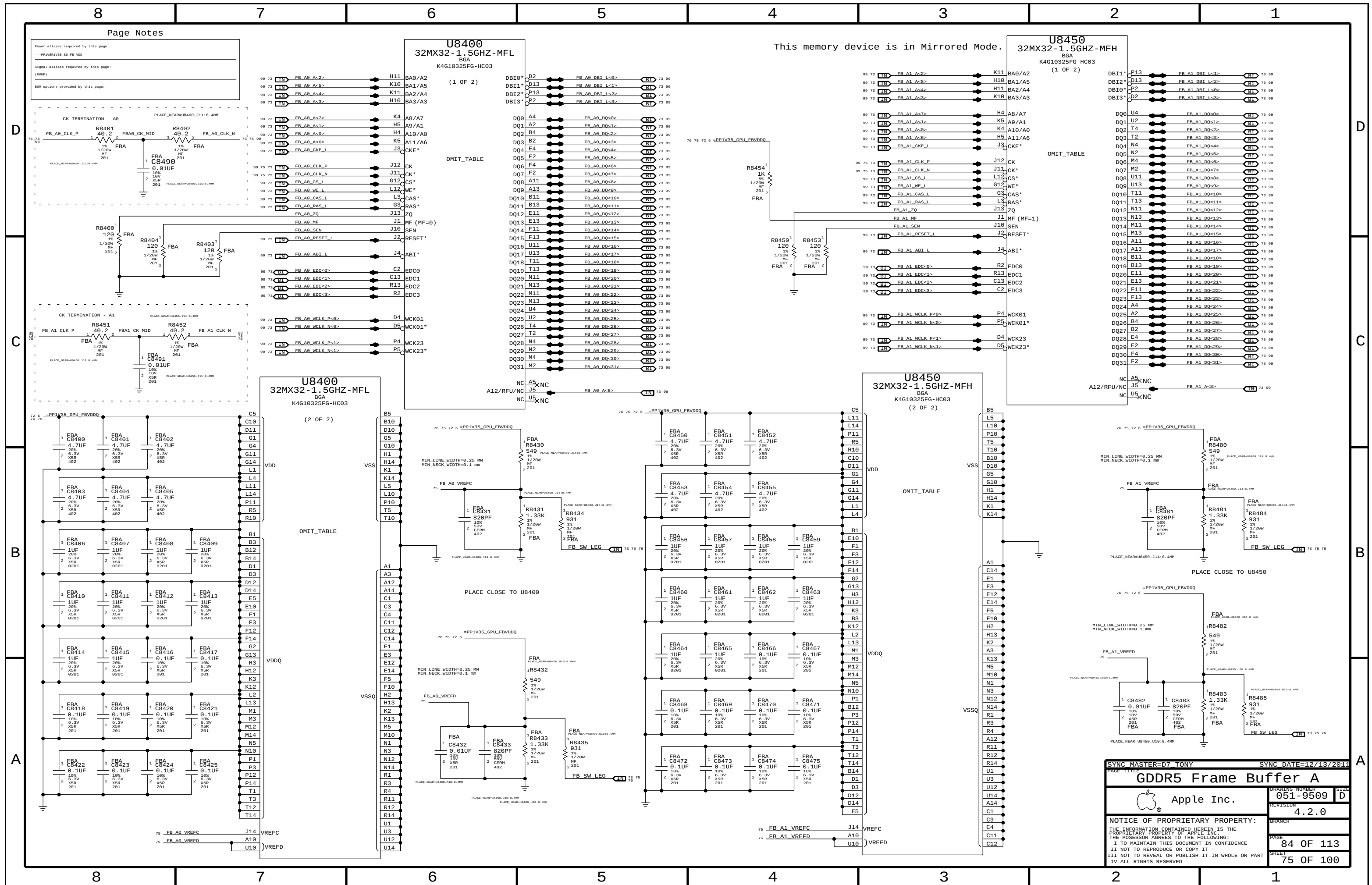
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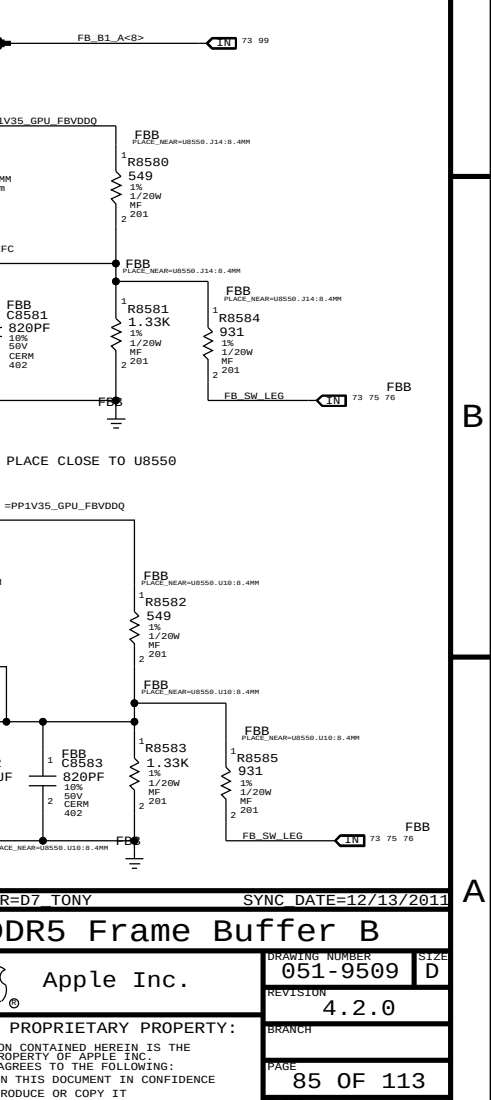
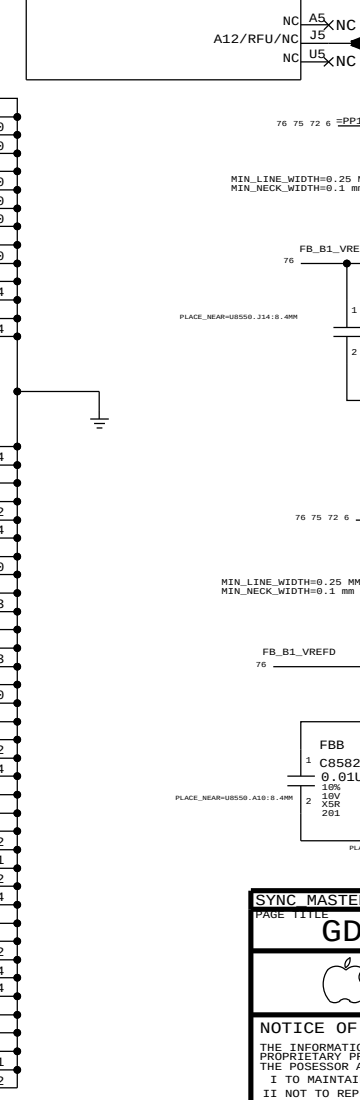
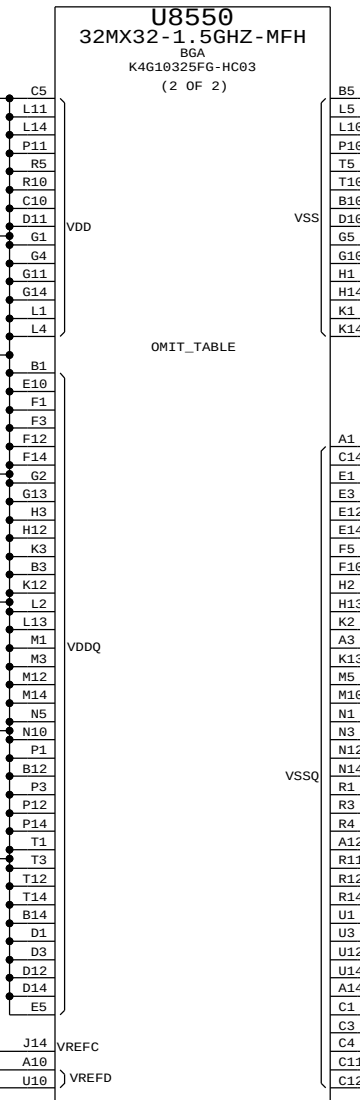
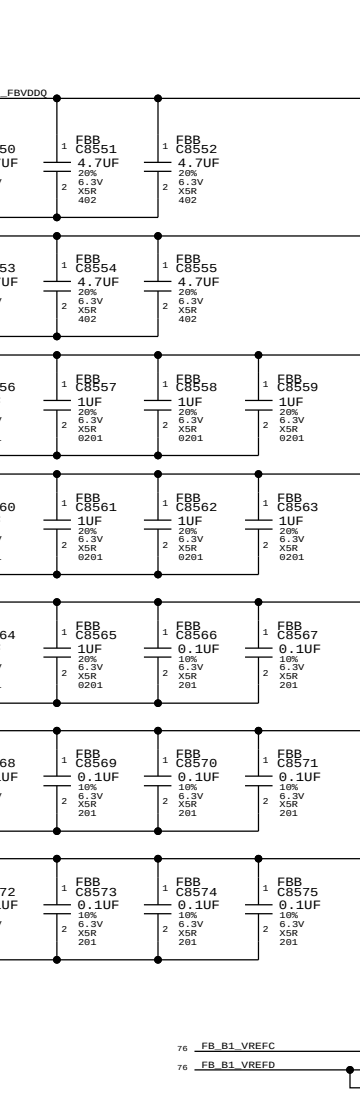
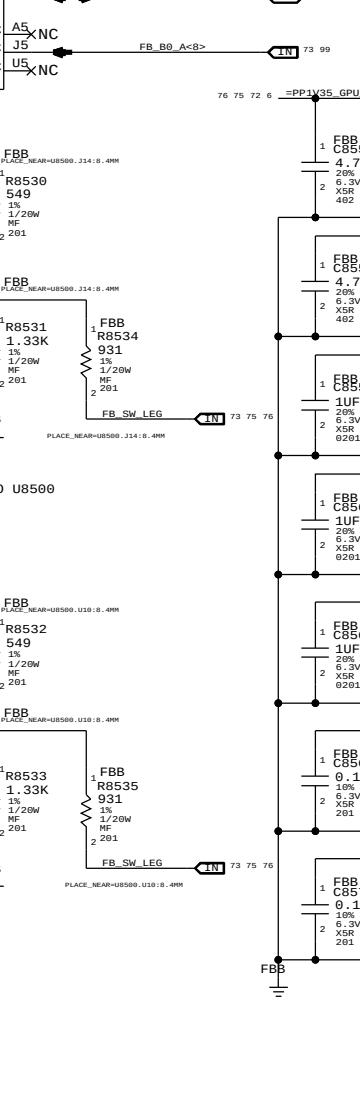
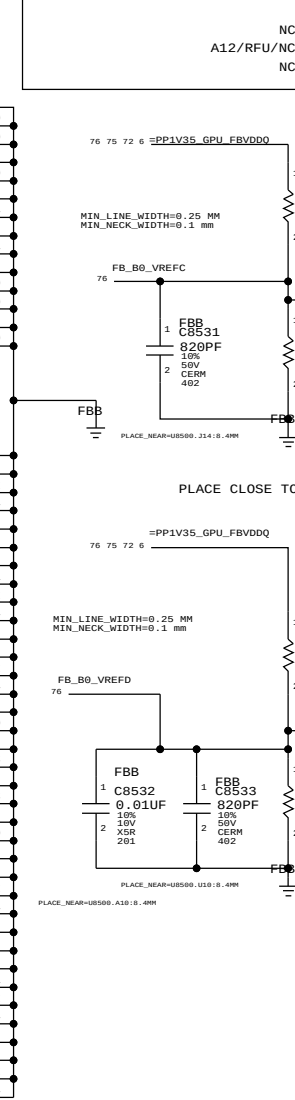
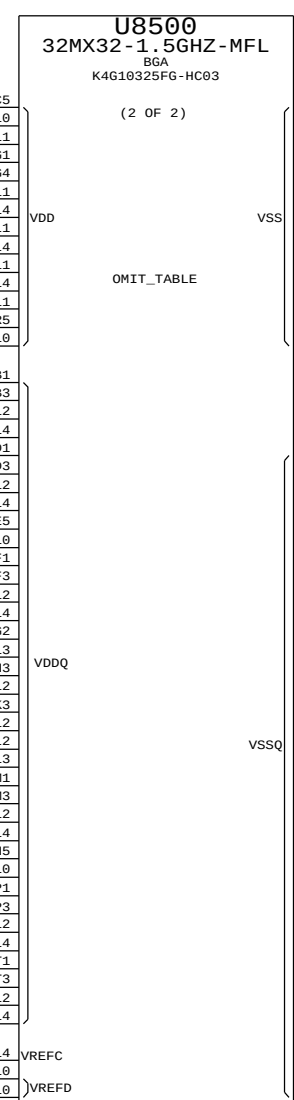
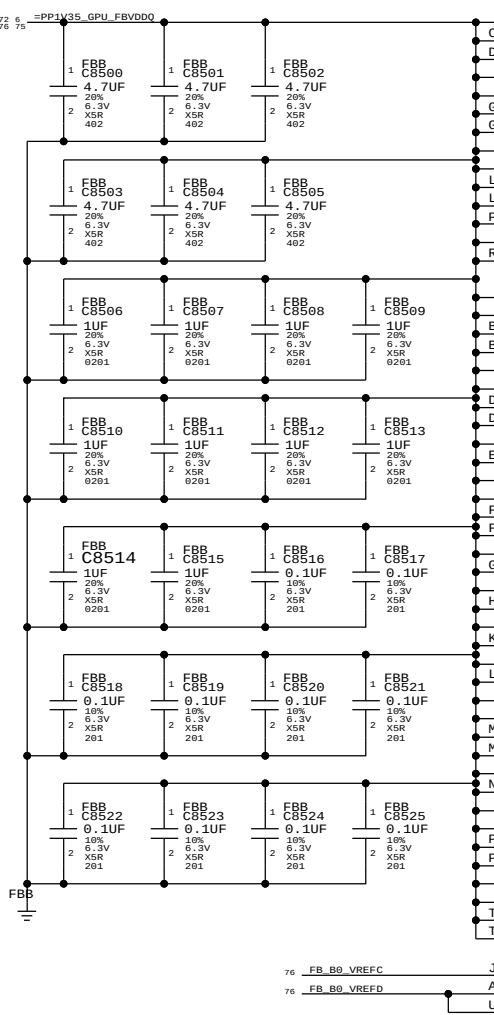
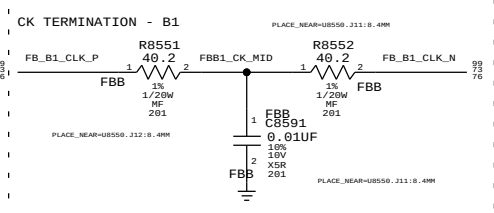
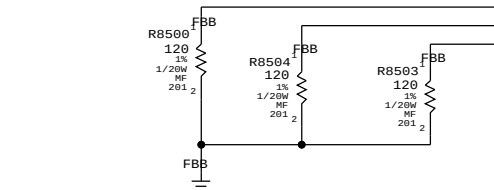
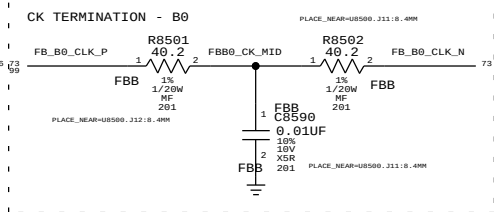
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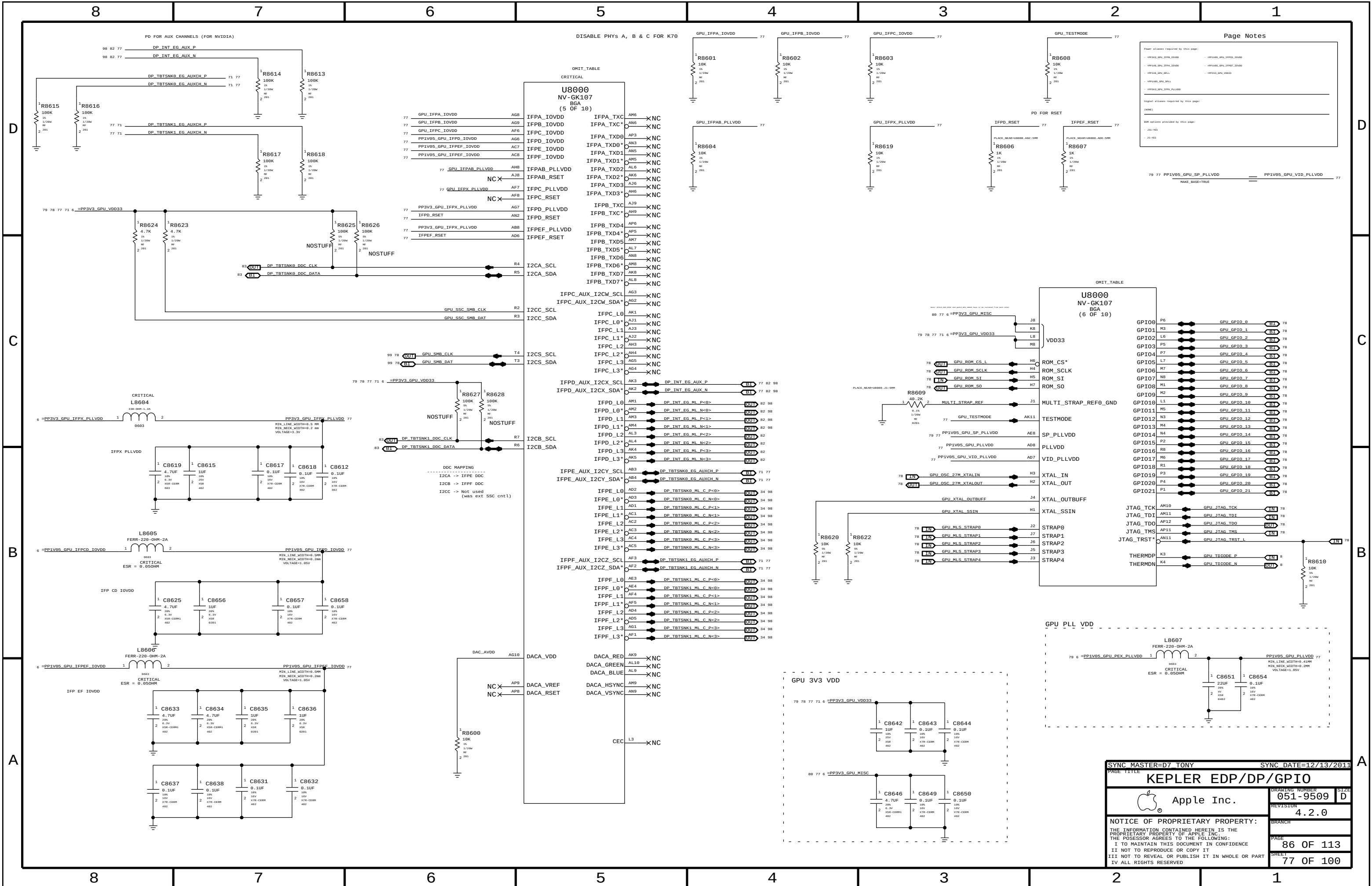
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BOM options provided by this page:



This memory device is in Mirrored Mode.

SYNC MASTER=D7 TONY		SYNC DATE=12/13/2011	
PAGE TITLE			
GDDR5 Frame Buffer B			
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		76 OF 100	



Page Notes

Power allowances required by this page:

- ~ HPPV05_GPU_SPD_VDD033
- ~ HPPV05_GPU_SPD_VDD033
- ~ HPPV05_GPU_SPD_VDD033
- ~ HPPV05_GPU_SPD_VDD033
- ~ HPPV05_GPU_SPD_VDD033
- ~ HPPV05_GPU_SPD_VDD033

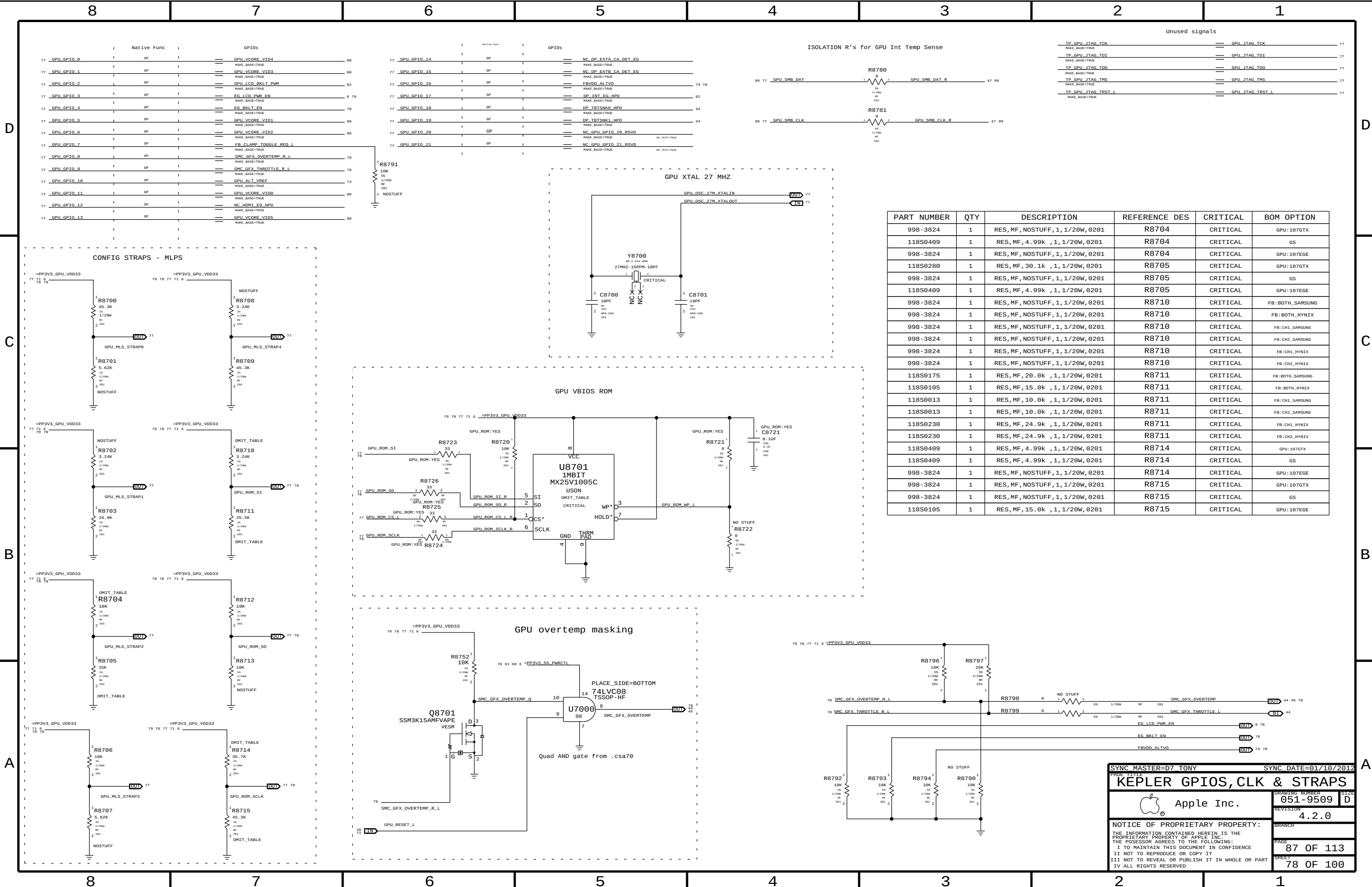
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- ~ HPPV05_GPU_SPD_VDD033

Power options provided by this page:

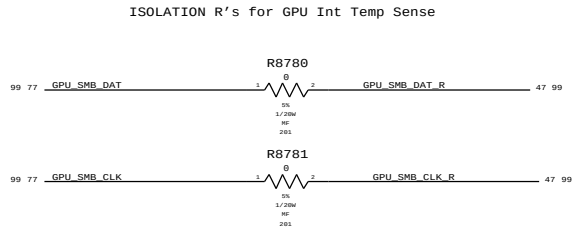
- ~ 20V-VDD
- ~ 20V-VDD

SYNC MASTER=D7 TONY		SYNC DATE=12/13/2011	
PAGE TITLE		KEPLER EDP/DP/GPIO	
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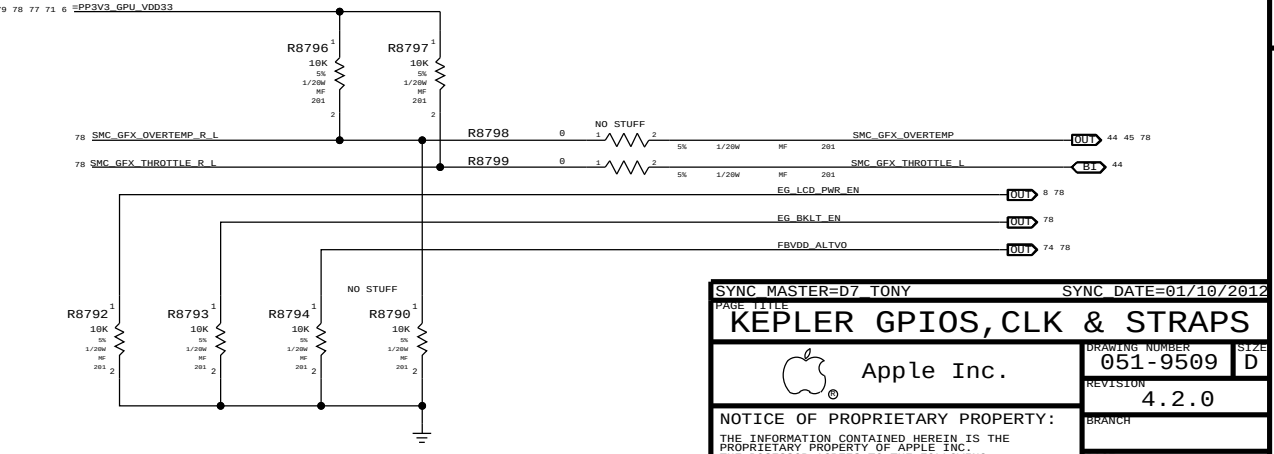
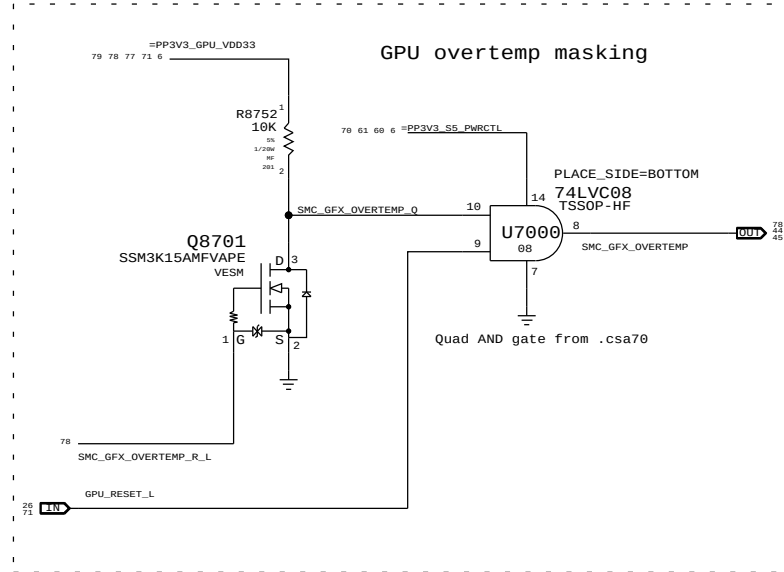
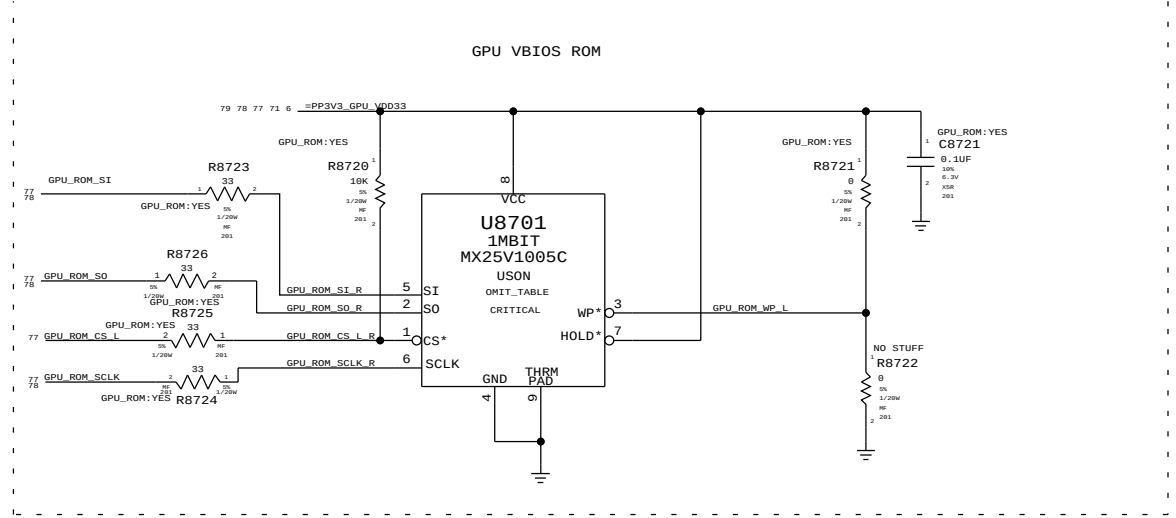
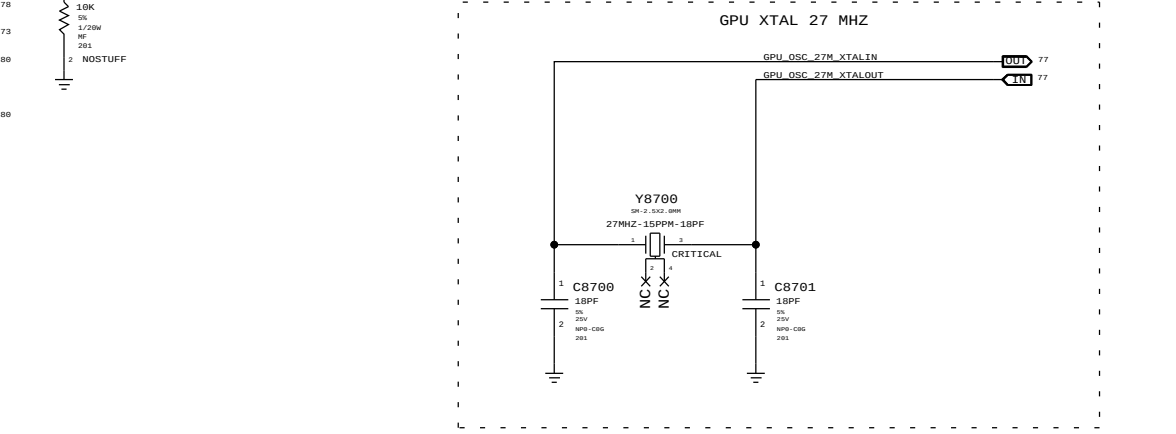
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77 GPU_GPIO_0	GP	==	GPU_VCORE_VID4	80
77 GPU_GPIO_1	GP	==	GPU_VCORE_VID3	80
77 GPU_GPIO_2	GP	==	GPU_LCD_BKLT_PWM	82
77 GPU_GPIO_3	GP	==	EG_LCD_PWR_EN	8 78
77 GPU_GPIO_4	GP	==	EG_BKLT_EN	78
77 GPU_GPIO_5	GP	==	GPU_VCORE_VID1	80
77 GPU_GPIO_6	GP	==	GPU_VCORE_VID2	80
77 GPU_GPIO_7	GP	==	FB_CLAMP_TOGGLE_REQ_L	
77 GPU_GPIO_8	GP	==	SMC_GFX_OVERTEMP_R_L	78
77 GPU_GPIO_9	GP	==	SMC_GFX_THROTTLE_R_L	78
77 GPU_GPIO_10	GP	==	GPU_ALT_VREF	73
77 GPU_GPIO_11	GP	==	GPU_VCORE_VID0	80
77 GPU_GPIO_12	GP	==	NC_HDMI_EG_HPD	
77 GPU_GPIO_13	GP	==	GPU_VCORE_VID5	80

	Native Func		GPIOS	
77 GPU_GPIO_14	GP	==	NC_DP_EXTB_CA_DET_EG	
77 GPU_GPIO_15	GP	==	NC_DP_EXTB_CA_DET_EG	
77 GPU_GPIO_16	GP	==	FBVDD_ALTV0	74 78
77 GPU_GPIO_17	GP	==	DP_INT_EG_HPD	82
77 GPU_GPIO_18	GP	==	DP_TBTSNK0_HPD	34
77 GPU_GPIO_19	GP	==	DP_TBTSNK1_HPD	34
77 GPU_GPIO_20	GP	==	NC_GPU_GPIO_20_RSVD	
77 GPU_GPIO_21	GP	==	NC_GPU_GPIO_21_RSVD	



Unused signals	
TP_GPU_JTAG_TCK	GPU_JTAG_TCK
TP_GPU_JTAG_TDI	GPU_JTAG_TDI
TP_GPU_JTAG_TDO	GPU_JTAG_TDO
TP_GPU_JTAG_TMS	GPU_JTAG_TMS
TP_GPU_JTAG_TRST_L	GPU_JTAG_TRST_L

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8704	CRITICAL	GPU:107GTX
118S0409	1	RES,MF,4.99k,1,1/20W,0201	R8704	CRITICAL	GS
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8704	CRITICAL	GPU:107EGE
118S0280	1	RES,MF,30.1k,1,1/20W,0201	R8705	CRITICAL	GPU:107GTX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8705	CRITICAL	GS
118S0409	1	RES,MF,4.99k,1,1/20W,0201	R8705	CRITICAL	GPU:107EGE
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:BOTH_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:BOTH_HYNIX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH1_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH2_SAMSUNG
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH1_HYNIX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8710	CRITICAL	FB:CH2_HYNIX
118S0175	1	RES,MF,20.0k,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k,1,1/20W,0201	R8711	CRITICAL	FB:BOTH_HYNIX
118S0013	1	RES,MF,10.0k,1,1/20W,0201	R8711	CRITICAL	FB:CH1_SAMSUNG
118S0013	1	RES,MF,10.0k,1,1/20W,0201	R8711	CRITICAL	FB:CH2_SAMSUNG
118S0230	1	RES,MF,24.9k,1,1/20W,0201	R8711	CRITICAL	FB:CH1_HYNIX
118S0230	1	RES,MF,24.9k,1,1/20W,0201	R8711	CRITICAL	FB:CH2_HYNIX
118S0409	1	RES,MF,4.99k,1,1/20W,0201	R8714	CRITICAL	GPU:107GTX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8714	CRITICAL	GPU:107EGE
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8715	CRITICAL	GPU:107GTX
998-3824	1	RES,MF,NOSTUFF,1,1/20W,0201	R8715	CRITICAL	GS
118S0105	1	RES,MF,15.0k,1,1/20W,0201	R8715	CRITICAL	GPU:107EGE



SYNC MASTER=D7 TONY

SYNC DATE=01/10/2012

KEPLER GPIOS, CLK & STRAPS

Apple Inc.

051-9509

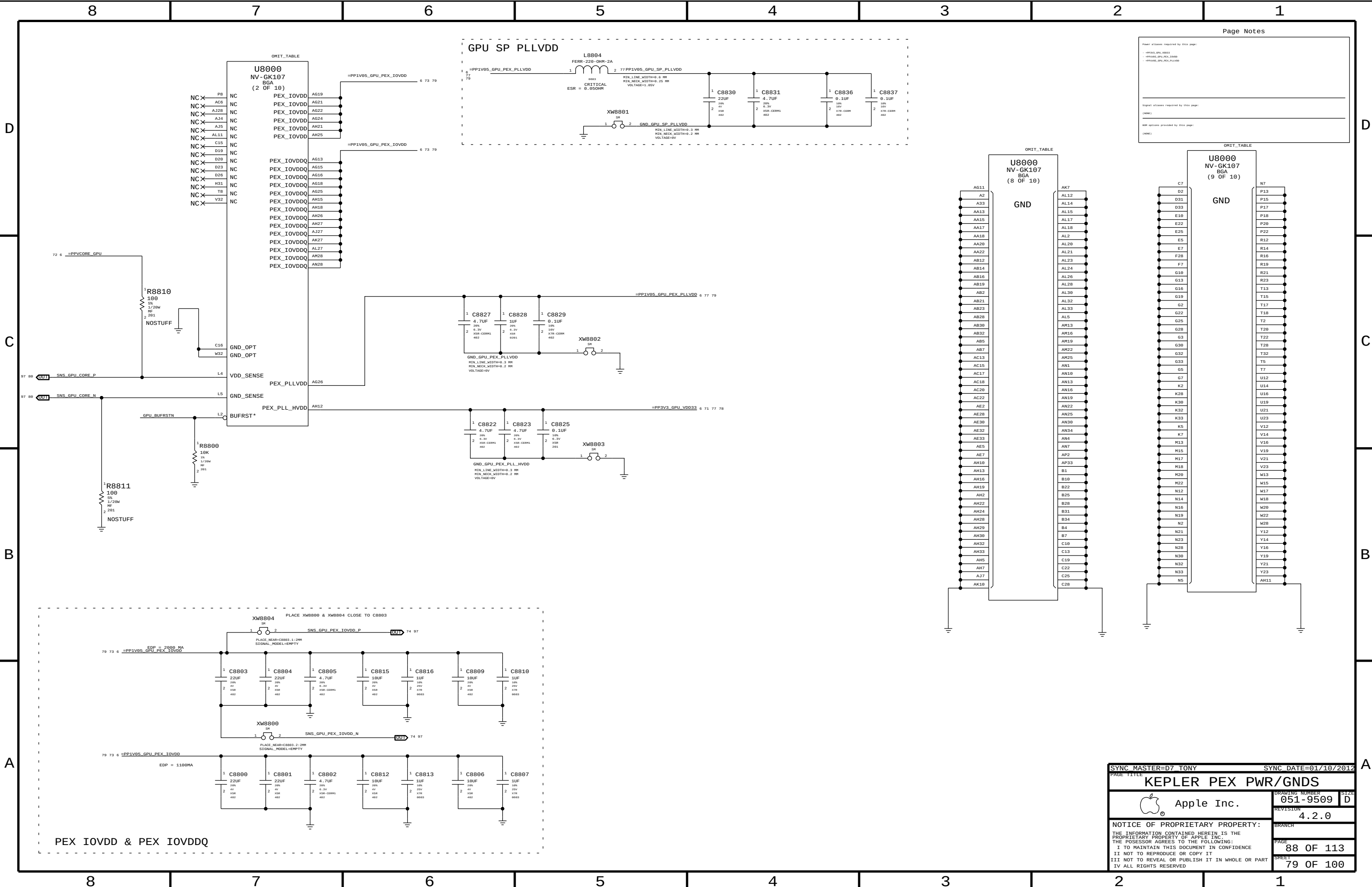
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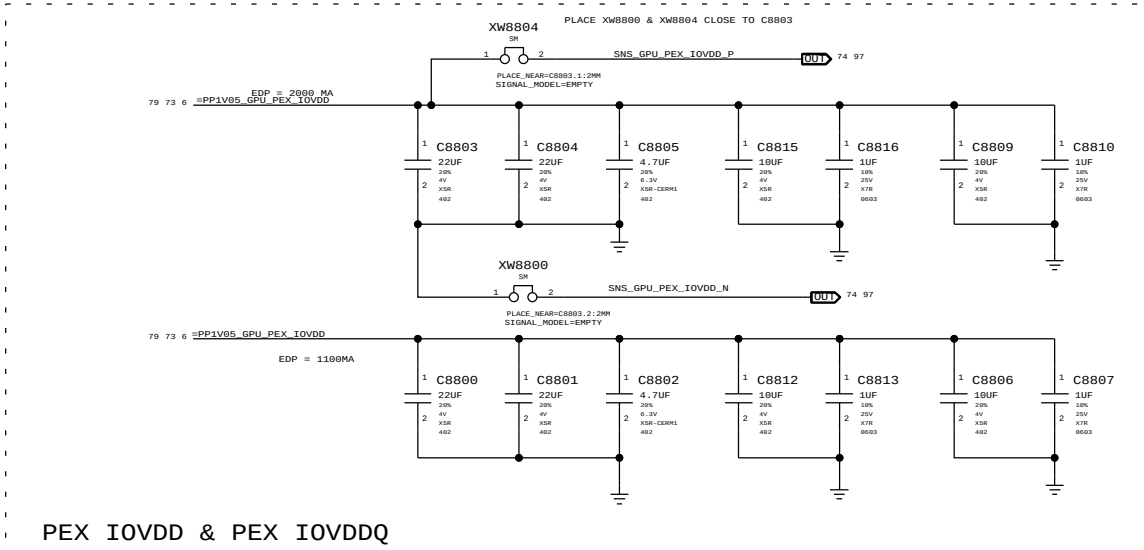
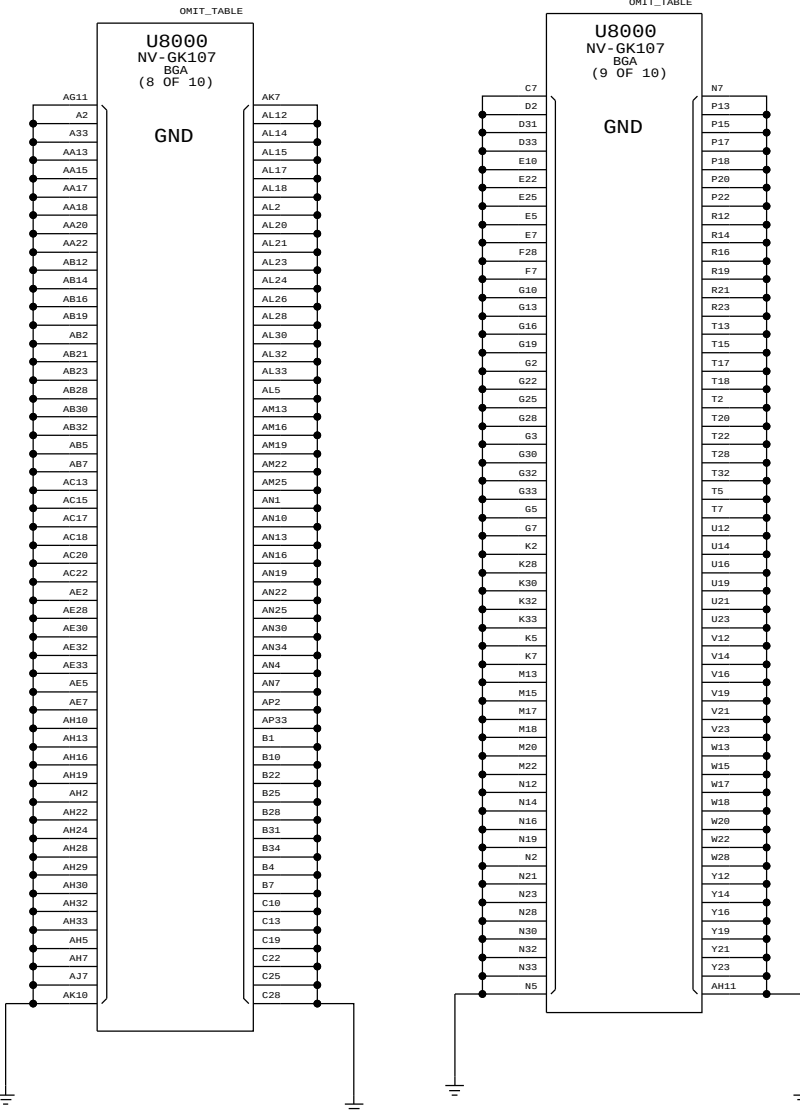
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- PP1V05_GPU_PEX_IOVDD
- PP3V3_GPU_VDD33

Signal aliases required by this page:

(NONE)

SNP options provided by this page:

(NONE)



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SYNC DATE=01/10/2012

KEPLER PEX PWR/GNDS

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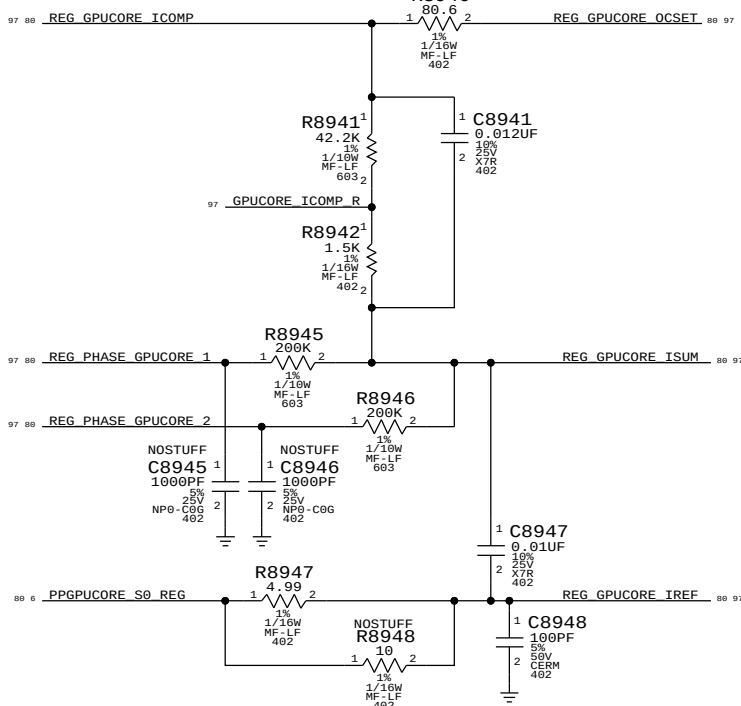
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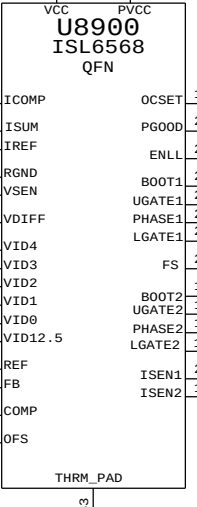
GPU Core S0 Regulator

Max avg current: ? A (design)/? A (budget)
Max peak current: ? A (design)/? A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 290 kHz

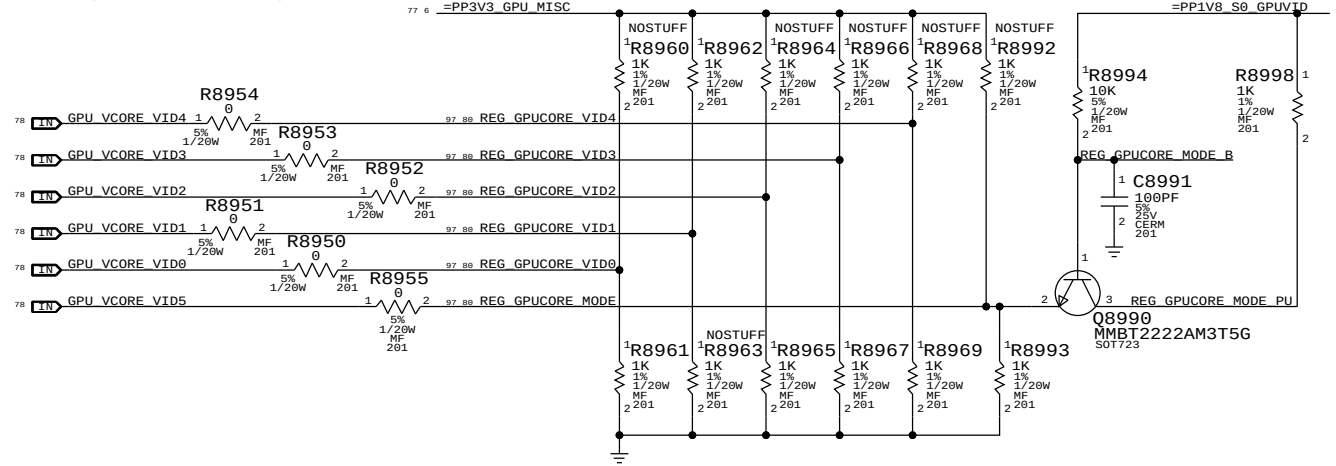
GPU Core current sense



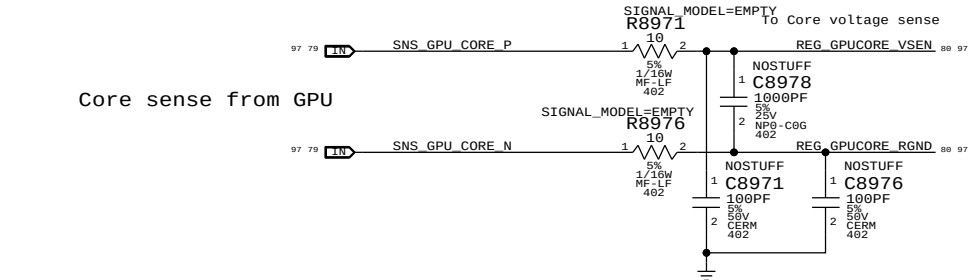
(icomp out) 97 80 REG GPU CORE ICOMP 11 ICOMP
(isum in) 97 80 REG GPU CORE ISUM 12 ISUM
(iref in) 97 80 REG GPU CORE IREF 13 IREF
(vsense in) 97 80 REG GPU CORE RGND 8 RGND
(vsense in) 97 80 REG GPU CORE VSEN 9 VSEN
(vdiff out) 97 80 REG GPU CORE VDIFF 7 VDIFF
(straps) 97 80 REG GPU CORE VID4 22 VID4
(straps) 97 80 REG GPU CORE VID3 21 VID3
(straps) 97 80 REG GPU CORE VID2 30 VID2
(straps) 97 80 REG GPU CORE VID1 31 VID1
(straps) 97 80 REG GPU CORE VID0 32 VID0
(straps) 97 80 REG GPU CORE MODE 1 VID12.5
(fb in) 97 80 REG GPU CORE REF 2 REF
(fb in) 97 80 REG GPU CORE FB 6 FB
(comp out) 97 80 REG GPU CORE COMP 5 COMP
(comp out) 97 80 REG GPU CORE OFS 3 OFS



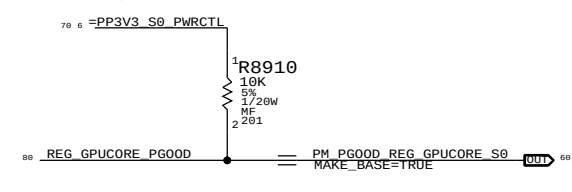
Straps & VID inputs



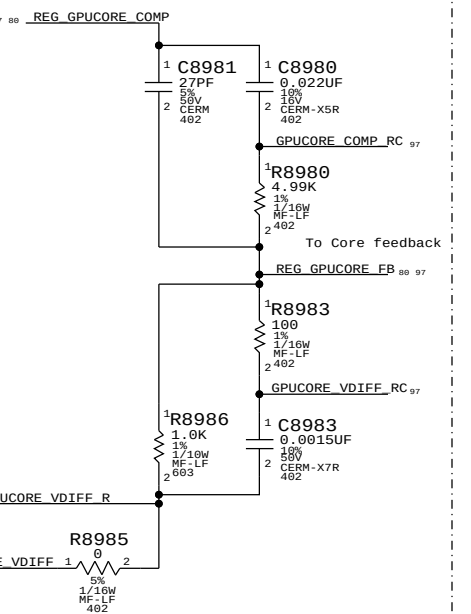
GPU Core voltage sense input



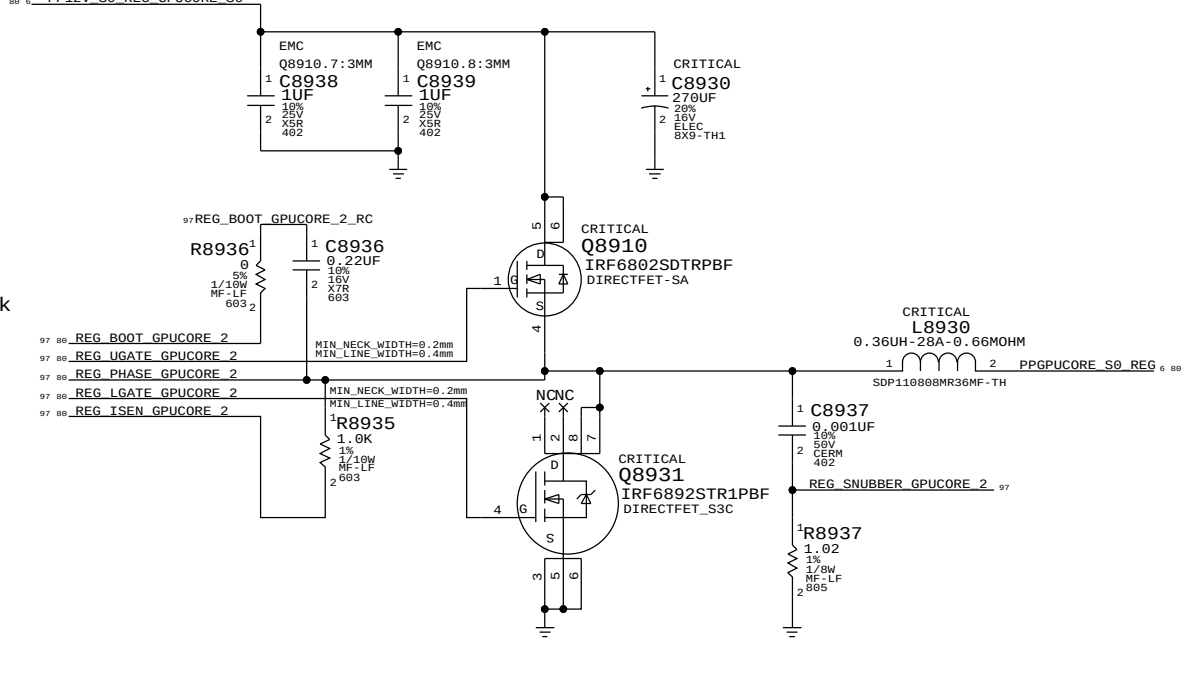
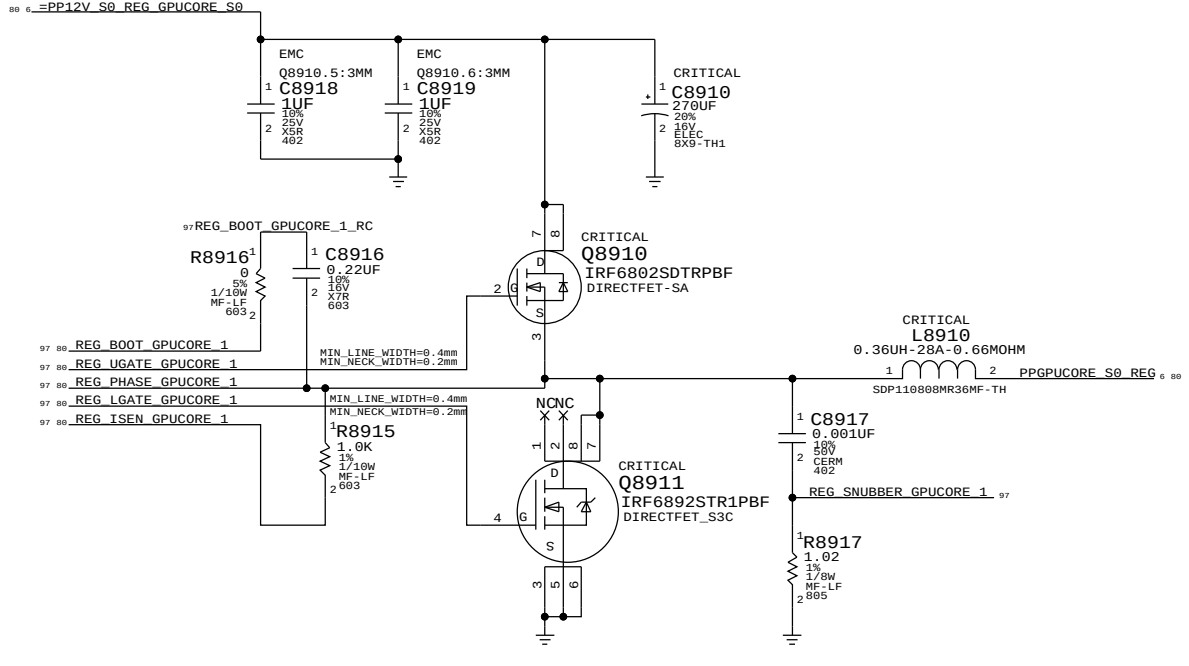
Power goods



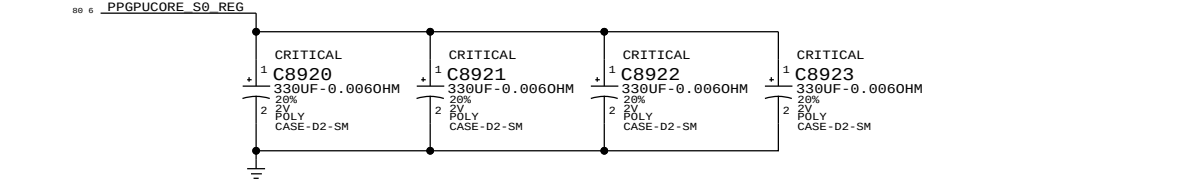
GPU Core compensation and feedback



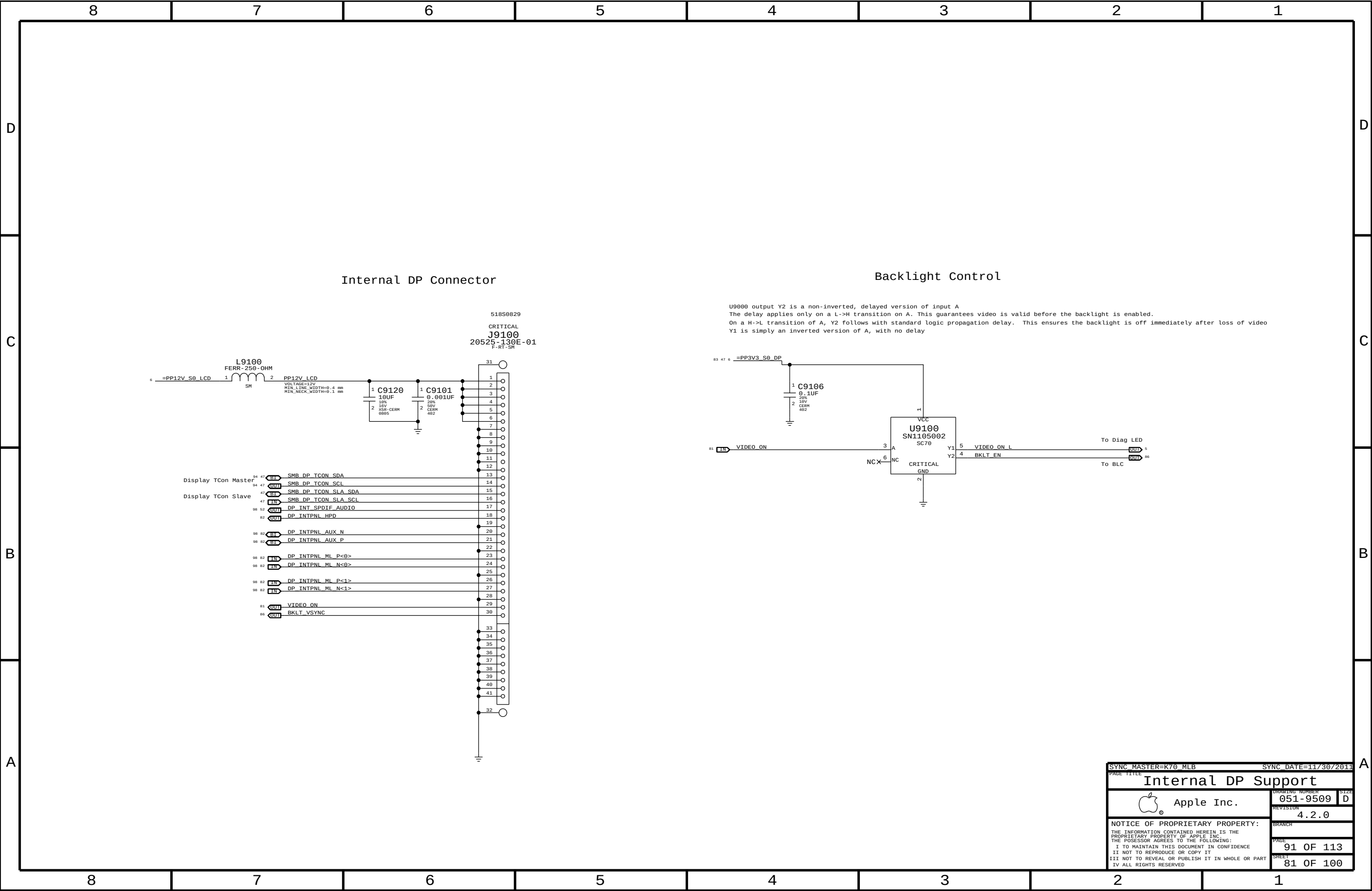
GPU Core Phases



GPU Core Output Decoupling

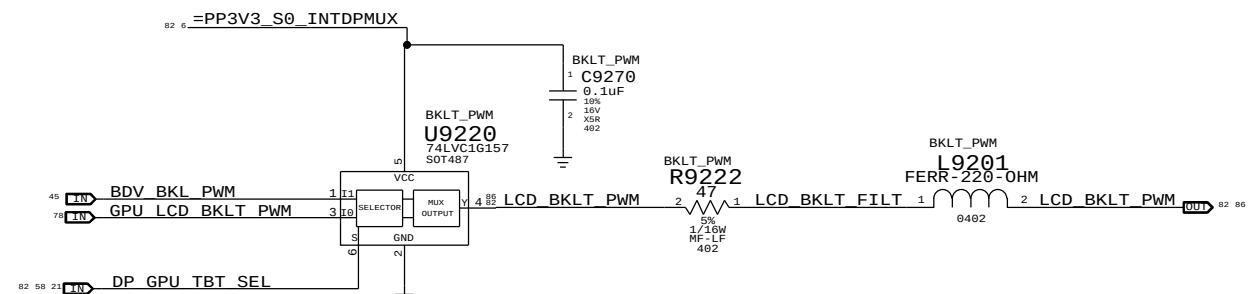
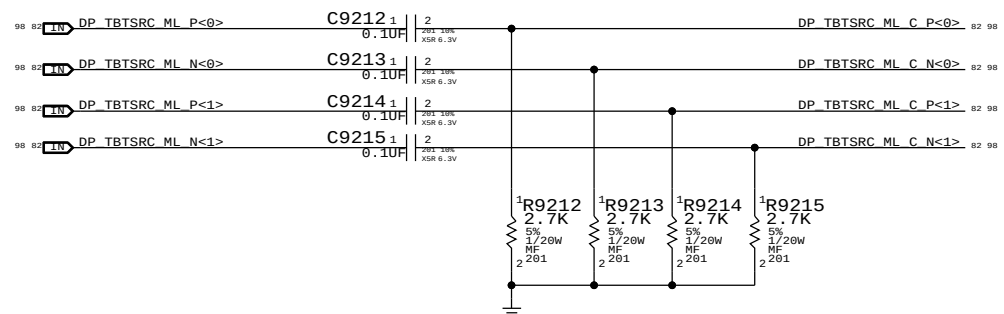
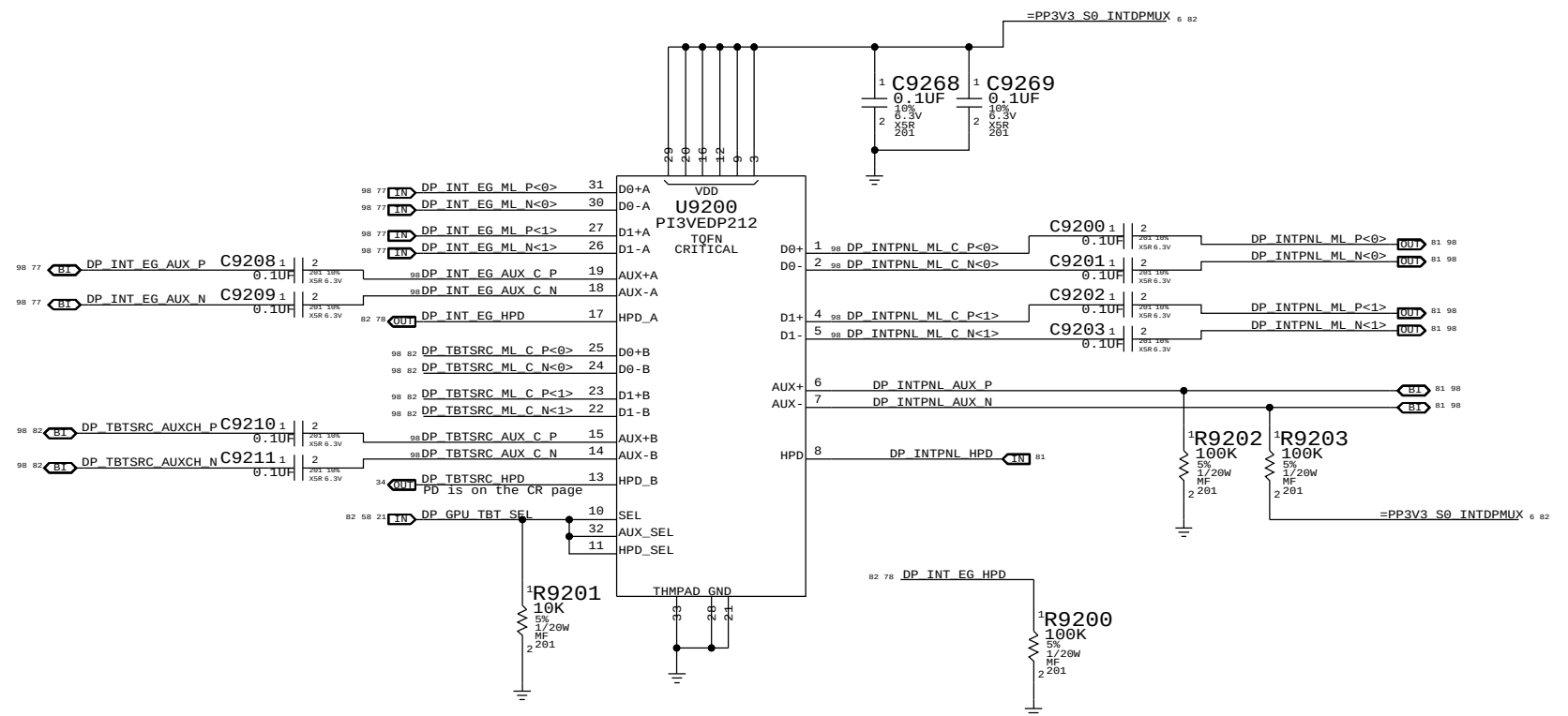


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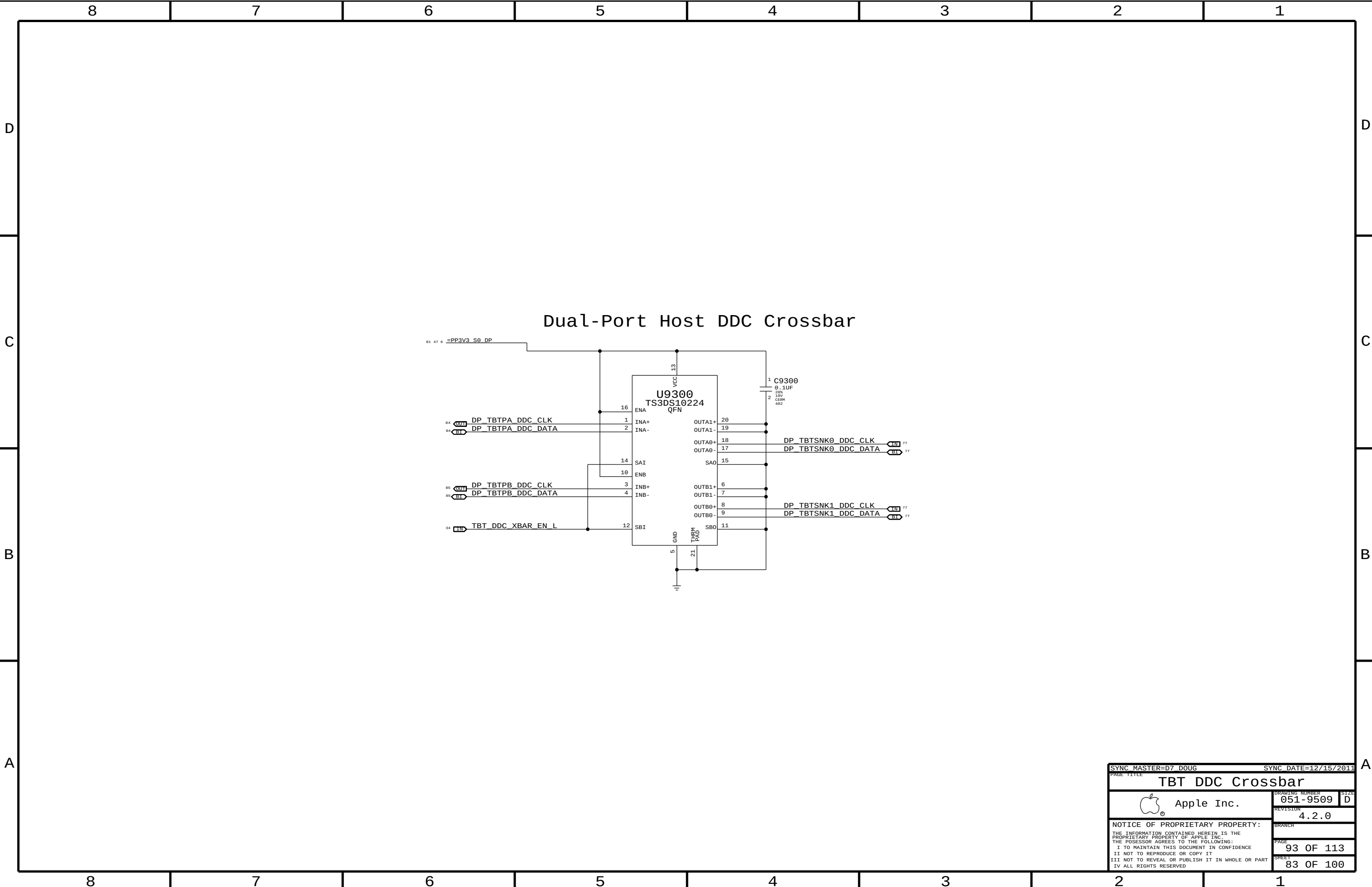


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34	TP	DP	TBTSRC	ML	CN<0>	==		DP	TBTSRC	ML	N<0>	82	98
											NAKE, BASE=TRUE		
34	TP	DP	TBTSRC	ML	CP<1>	==		DP	TBTSRC	ML	P<1>	82	98
34	TP	DP	TBTSRC	ML	CN<1>	==		DP	TBTSRC	ML	N<1>	82	98
											NAKE, BASE=TRUE		
34	TP	DP	TBTSRC	AUXCH	CP	==		DP	TBTSRC	AUXCH	P	82	98
34	TP	DP	TBTSRC	AUXCH	CN	==		DP	TBTSRC	AUXCH	N	82	98
											NAKE, BASE=TRUE		

34		TP DP TBT SRC ML CP<2>	==	NC DP TBT SRC ML P<2>	NO_TEST=TRUE MAKE_BASE=TRUE
34		TP DP TBT SRC ML CN<2>	==	NC DP TBT SRC ML N<2>	NO_TEST=TRUE MAKE_BASE=TRUE
34		TP DP TBT SRC ML CP<3>	==	NC DP TBT SRC ML P<3>	NO_TEST=TRUE MAKE_BASE=TRUE
34		TP DP TBT SRC ML CN<3>	==	NC DP TBT SRC ML N<3>	NO_TEST=TRUE MAKE_BASE=TRUE
77		DP INT EG ML P<2>	==	NC DP INT EG ML P<2>	NO_TEST=TRUE MAKE_BASE=TRUE
77		DP INT EG ML N<2>	==	NC DP INT EG ML N<2>	NO_TEST=TRUE MAKE_BASE=TRUE
77		DP INT EG ML P<3>	==	NC DP INT EG ML P<3>	NO_TEST=TRUE MAKE_BASE=TRUE
77		DP INT EG ML N<3>	==	NC DP INT EG ML N<3>	NO_TEST=TRUE MAKE_BASE=TRUE



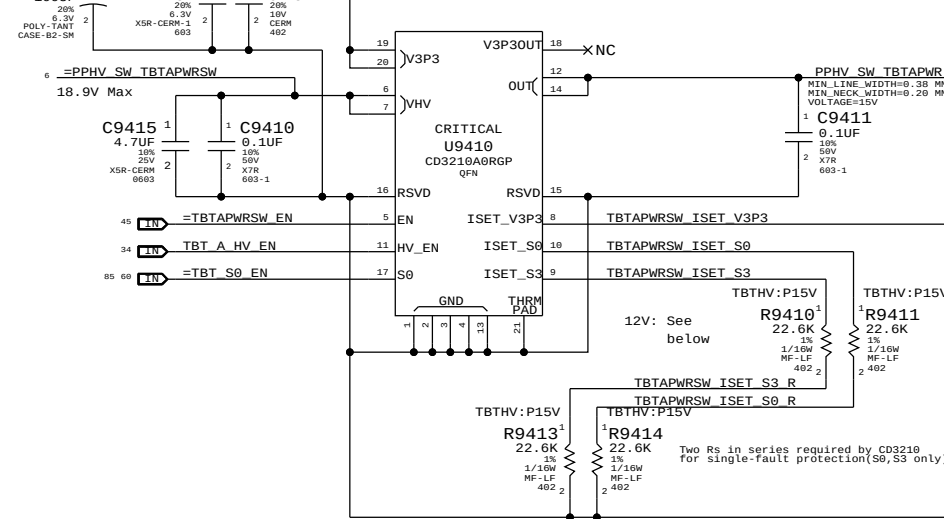
SYNC MASTER=D7 NICK		SYNC DATE=12/14/2011	
PAGE TITLE		Internal DP MUXing	
	Apple Inc.		DRAWING NUMBER 051-9509
			SIZE D
			REVISION 4.2.0
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3.3V/HV Power MUX

V3P3 must be S4 to support wake from Thunderbolt devices.

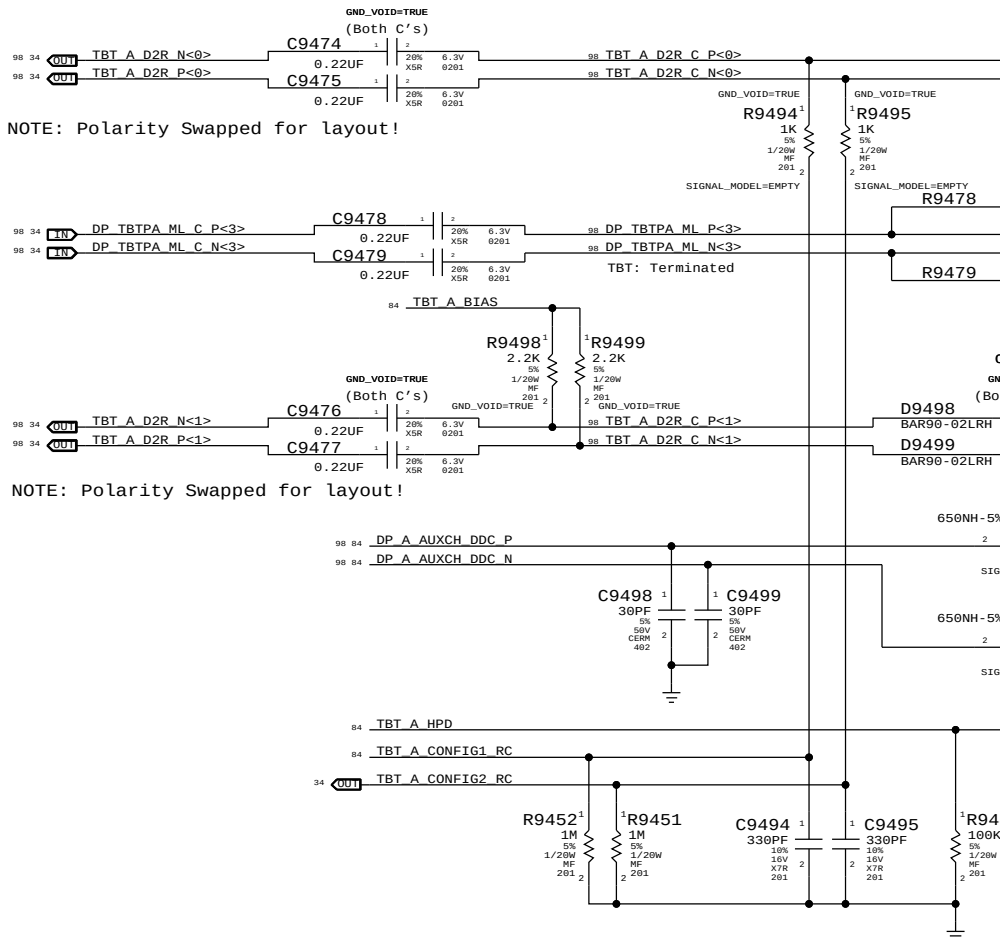
	Nominal	Min	Max
IV3P3	1100mA	1030mA	1200mA
IHVS0	890mA	830mA	930mA (assumes 15V, 12W minimum)
IHVS3	890mA	830mA	930mA (assumes 3S, 9-12.6V, 7.5-11.7W)



For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 9402, SMD, LF	R9410, R9413		TBTHV:P12V
114S0338	2	RES, MTL FILM, 1/16W, 17.8K, 1, 9402, SMD, LF	R9411, R9414		TBTHV:P12V

	Nominal	Min	Max
IHVS0/S3	1120mA	1090mA	1170mA (12W minimum)

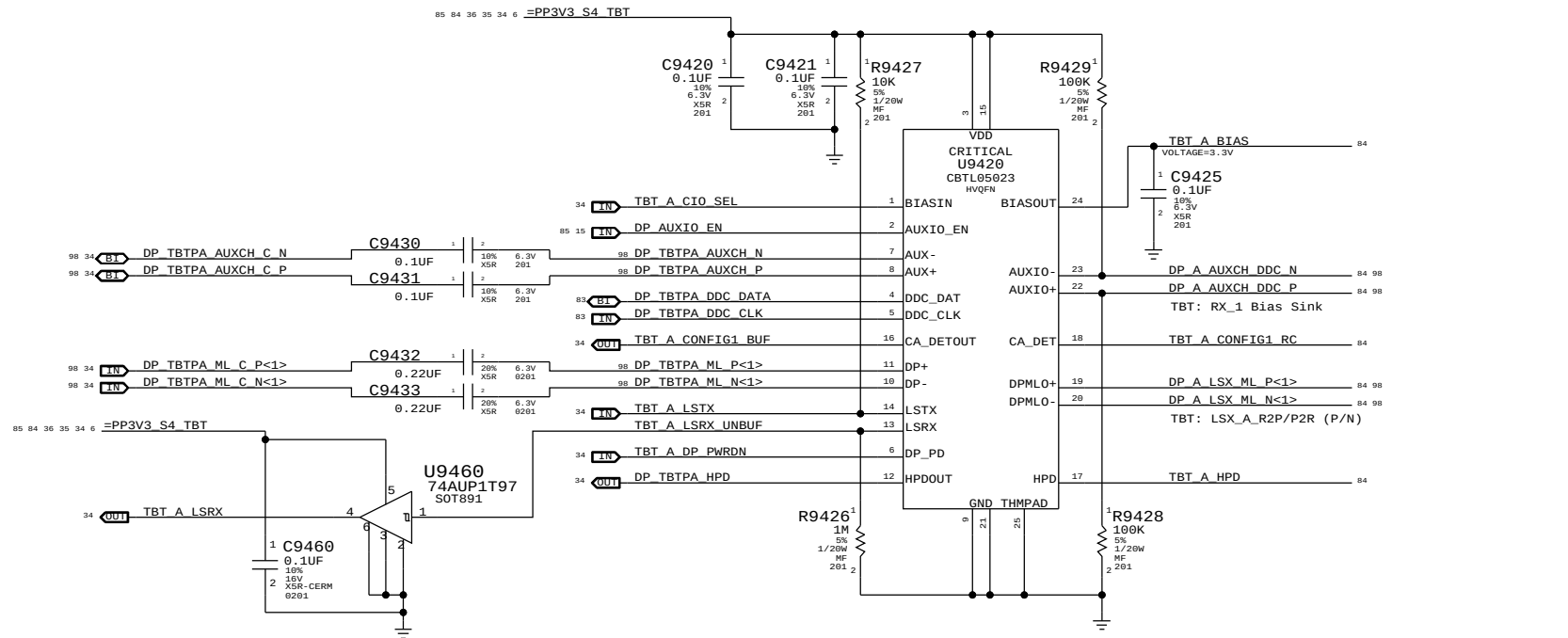


NOTE: Polarity Swapped for layout!

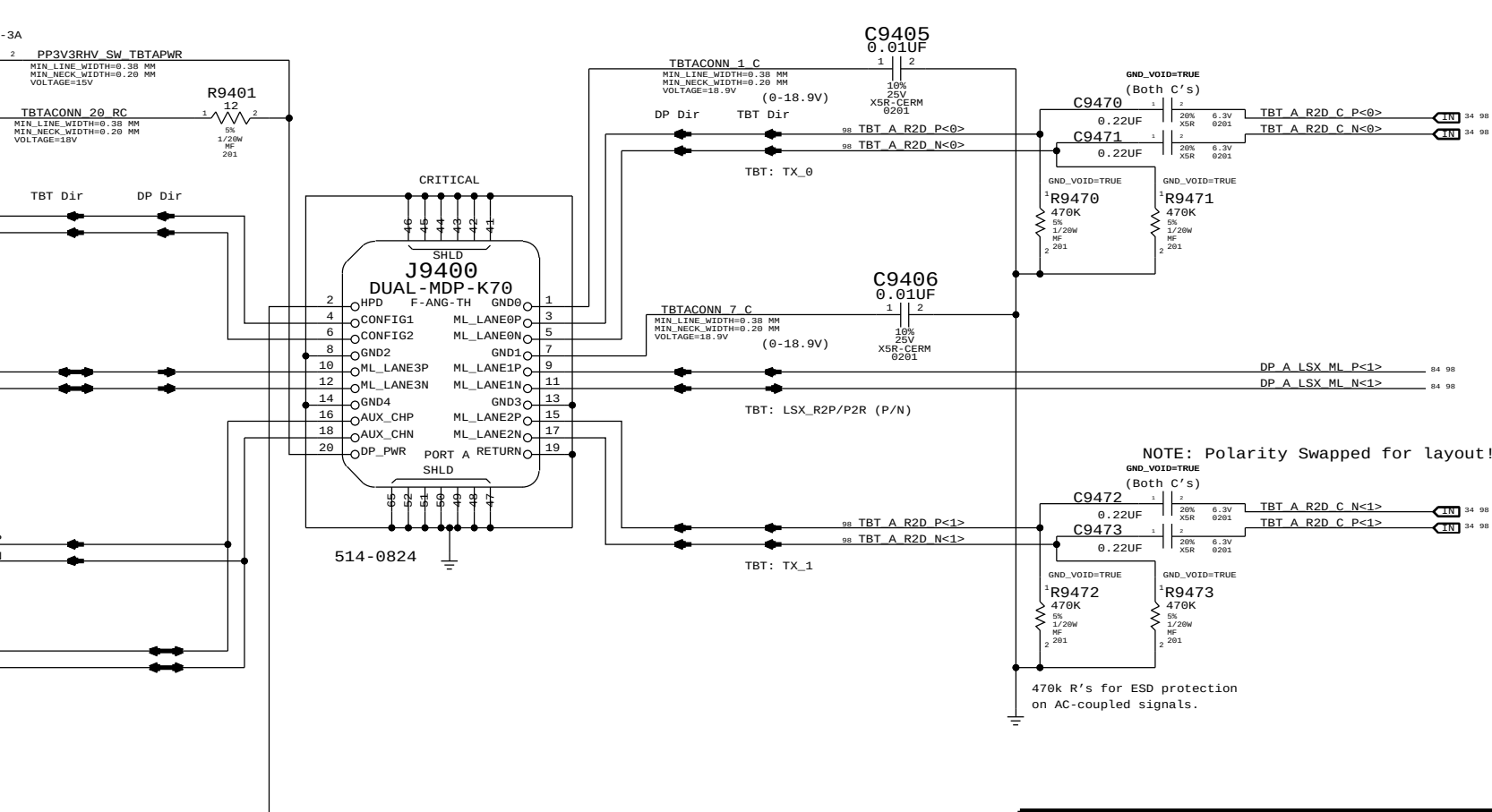
NOTE: Polarity Swapped for layout!

A

Thunderbolt Connector A



Thunderbolt Connector A



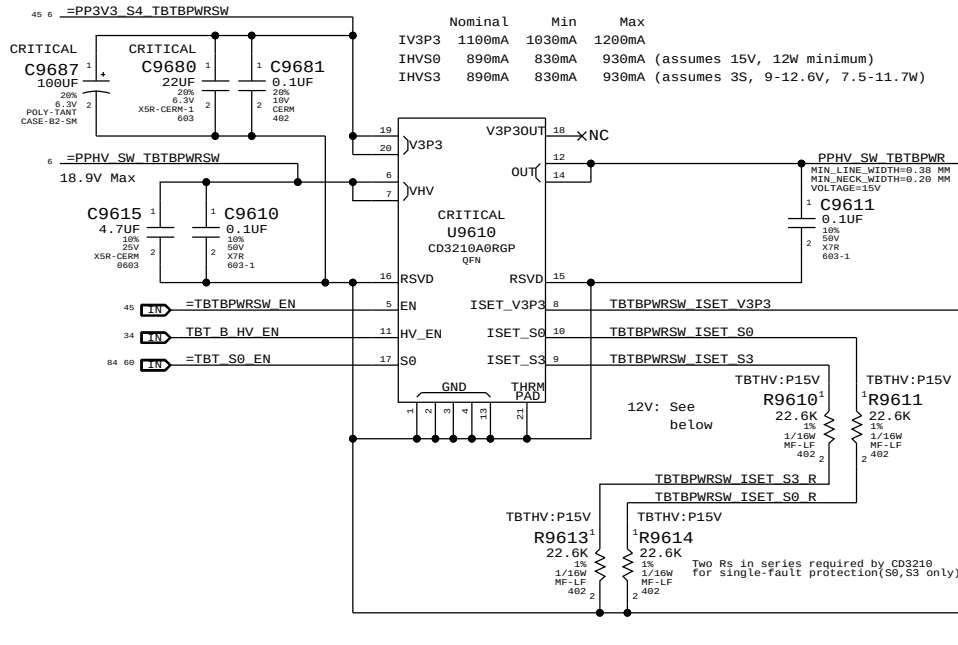
DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC MASTER=D7 DOUG		SYNC DATE=12/15/2011	
PAGE TITLE		Thunderbolt Connector A	
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		REVISION	4.2.0
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3.3V/HV Power MUX

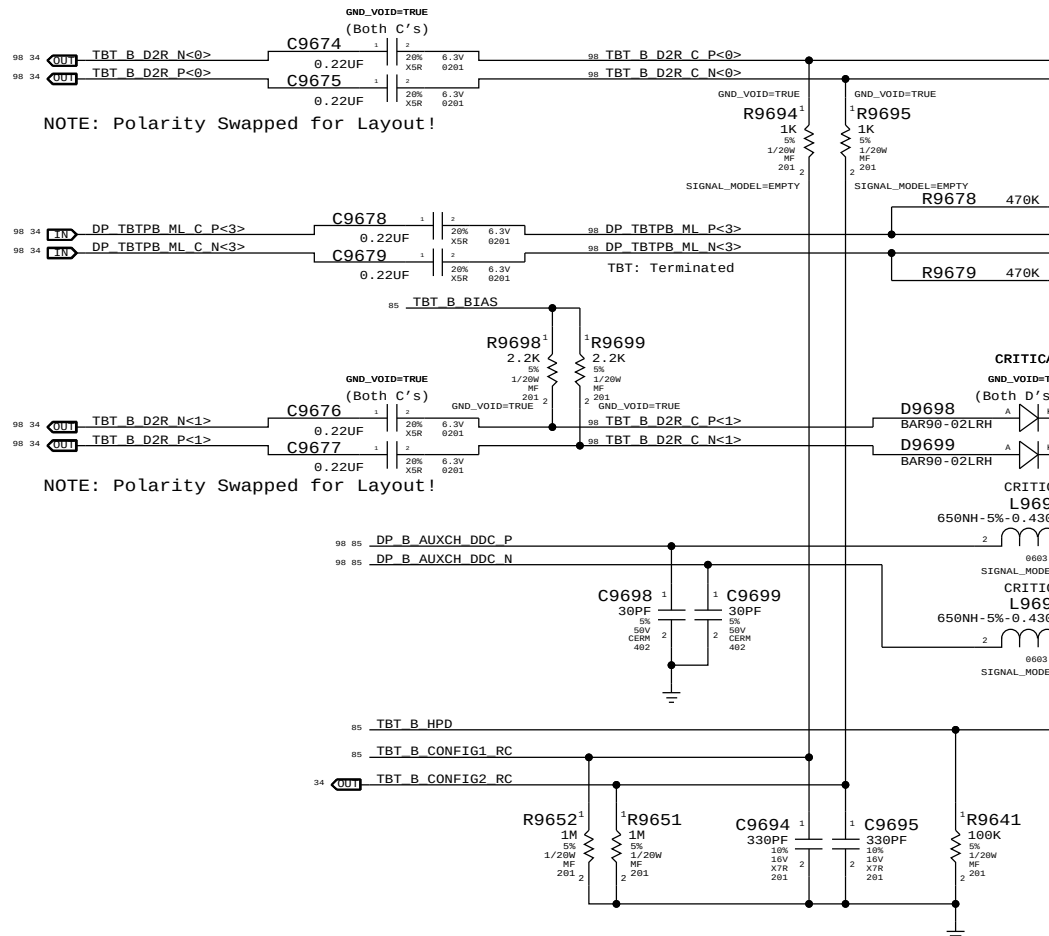
V3P3 must be S4 to support wake from Thunderbolt devices.



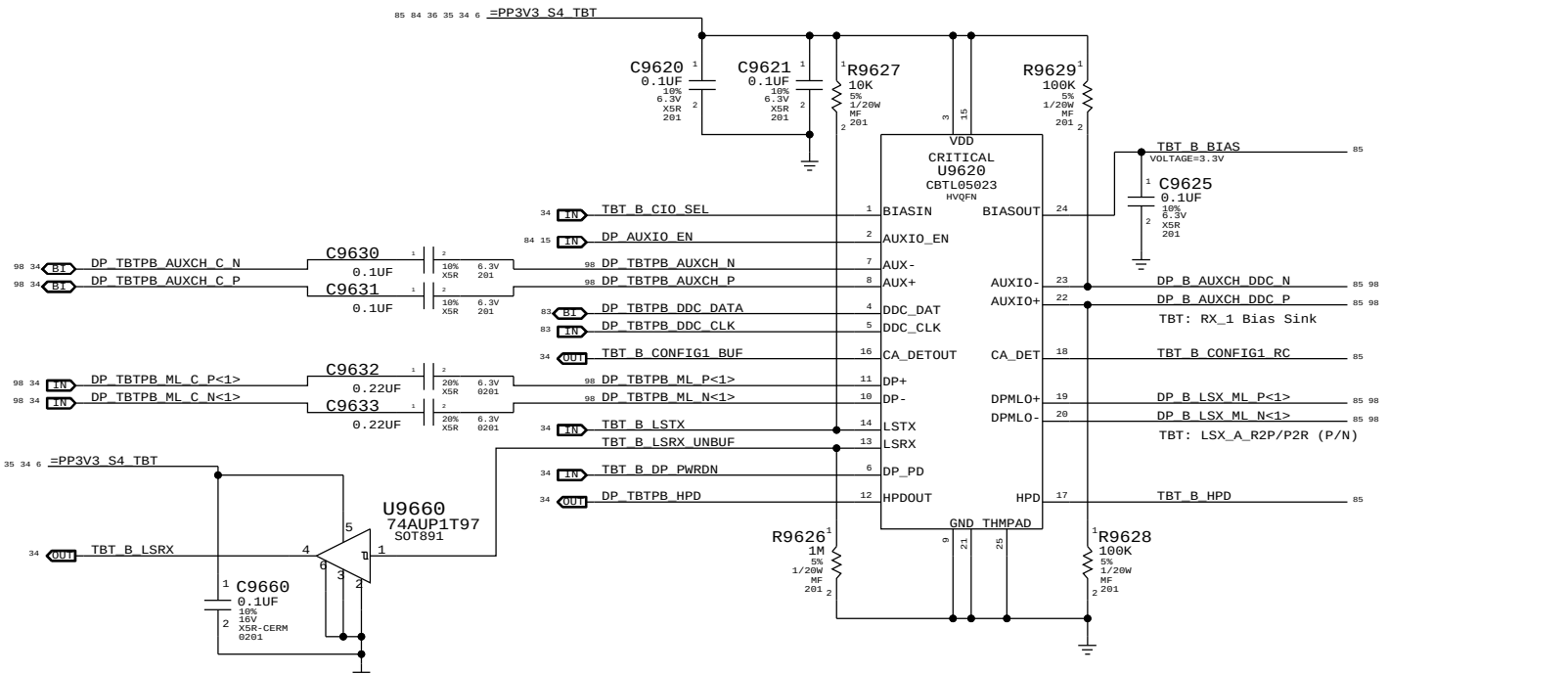
For 12V systems:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9610,R9613		TBTHV:P12V
114S0338	2	RES,MTL FILM,1/16W,17.8K,1,0402,SMD,LF	R9611,R9614		TBTHV:P12V

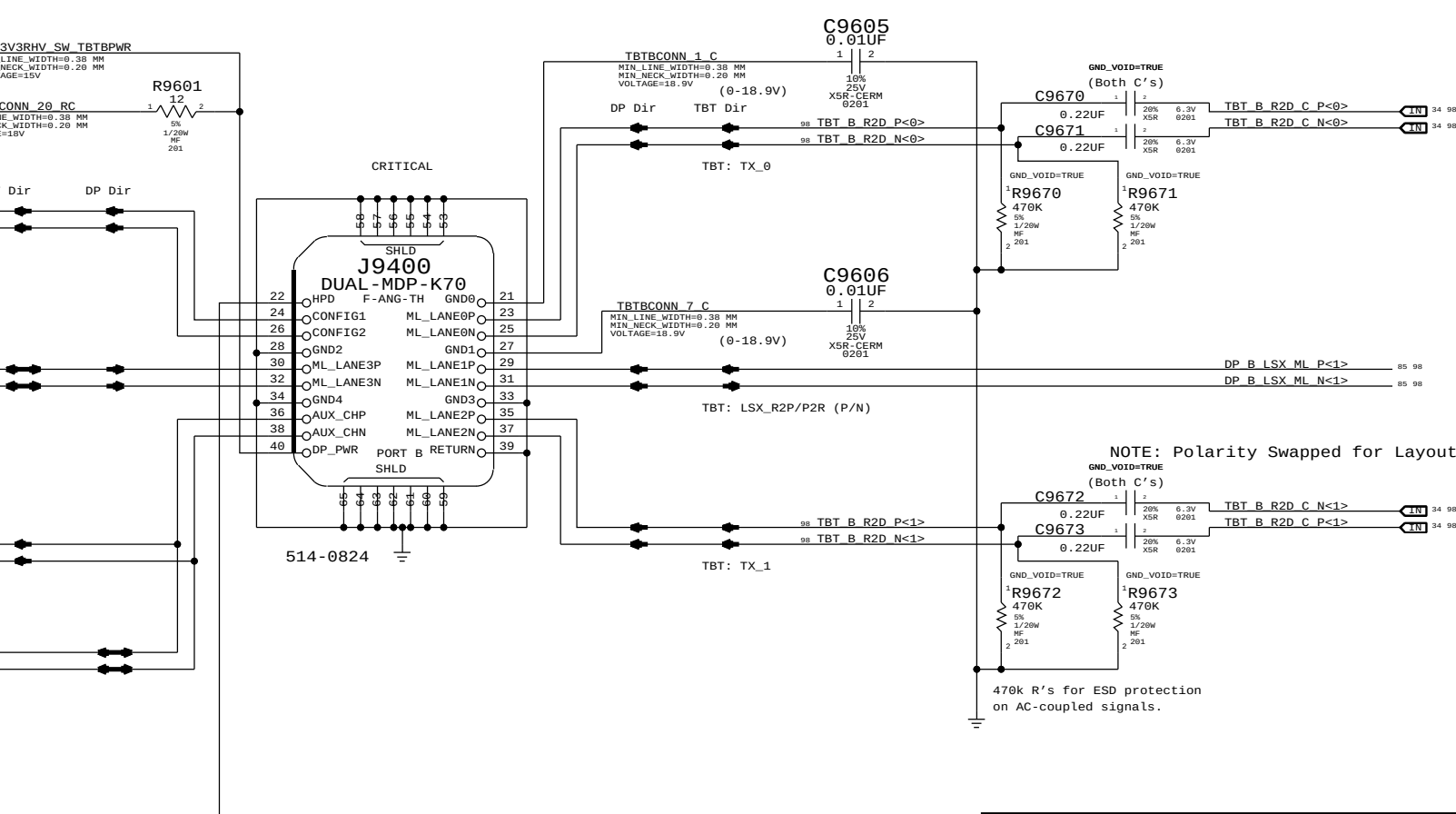
Nominal Min Max
IHVS0/S3 1120mA 1090mA 1170mA (12W minimum)



Thunderbolt Connector B



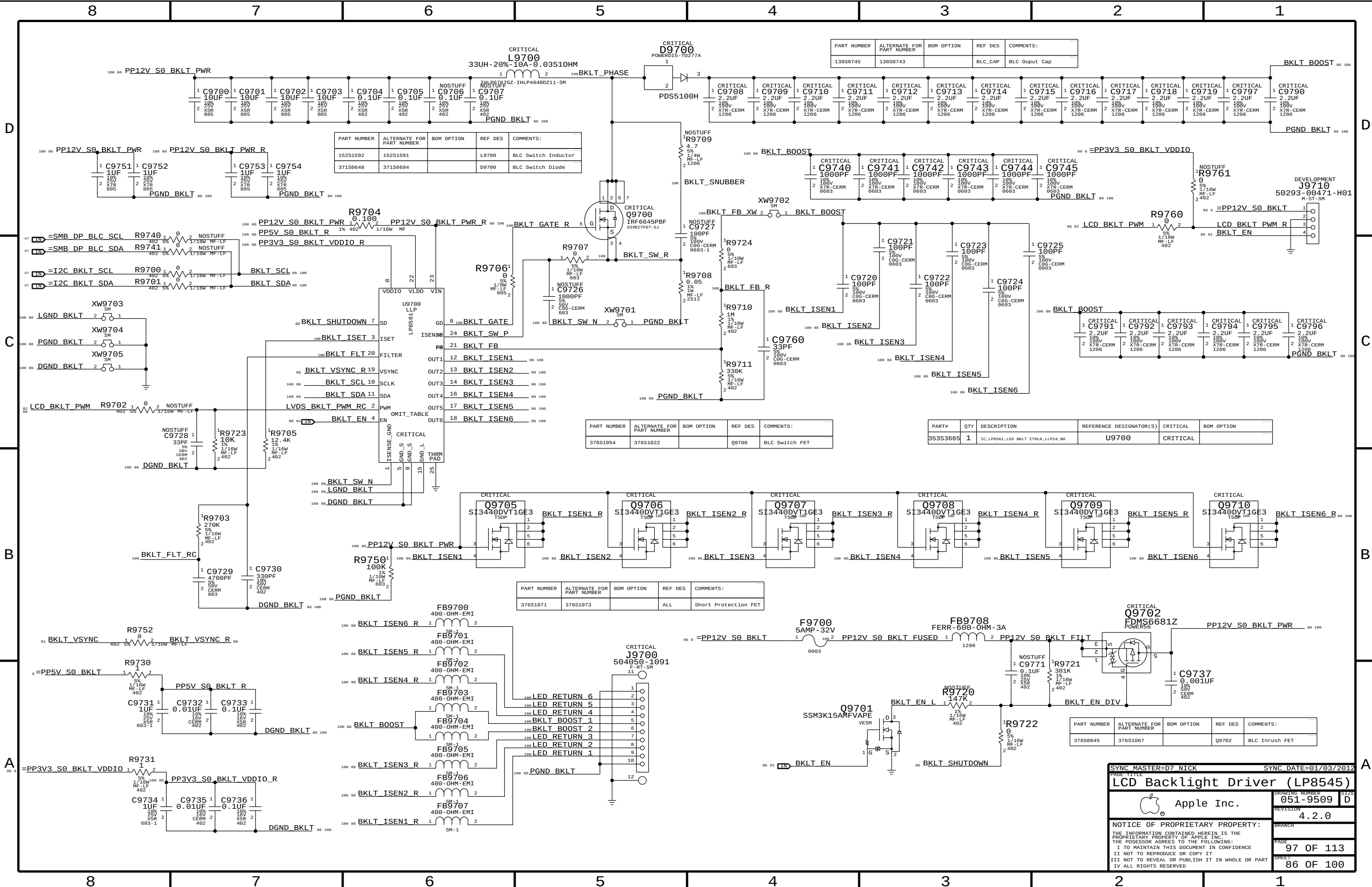
Thunderbolt Connector B



DP Source must pull down HPD input with greater than or equal to 100K (DPv1.1a).

Sink HPD range:
High: 2.0 - 5.0V
Low: 0 - 0.8V

SYNC MASTER=D7 DOUG		SYNC DATE=12/15/2011	
PAGE TITLE		Thunderbolt Connector B	
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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0745	138S0743		BLC_CAP	BLC Output Cap

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S1592	152S1591		L9700	BLC Switch Inductor
371S0648	371S0694		D9700	BLC Switch Diode

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1054	376S1022		Q9700	BLC Switch FET

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
353S3665	1	IC, LP8561, LED BKL CTRLR, LLP24, 80	U9700	CRITICAL	

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1071	376S1073		ALL	Short Protection FET

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S0845	376S1067		Q9702	BLC Inrush FET

SYNC MASTER=D7_NICK

SYNC DATE=01/03/2012

LCD Backlight Driver (LP8545)

Apple Inc.

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DDR3

DDR3-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DDR_34S	*	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=34_OHM_SE	=STANDARD	=STANDARD
DDR_39S	*	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=39_OHM_SE	=STANDARD	=STANDARD
DDR_42S	*	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=STANDARD	=STANDARD
DDR_42S_D		=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	=42_OHM_SE	0.1016 MM	0.1016 MM
DDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
DDR_68D	*	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF	=68_OHM_DIFF

Minimum diff spacing is 4 mil
Table 3-5, Intel Doc# 473718

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
POWER_DDR_P4MM	*	Y	0.400 MM	0.100 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
POWER_DDR	*	POWER_DDR_P4MM
DDR_CLK_PHY	*	DDR_6BD
DDR_CTRL_PHY	*	DDR_39S
DDR_CMD_PHY	*	DDR_34S
DDR_DQ_PHY	*	DDR_42S
DDR_DQS_PHY	*	DDR_42S_D

DDR3 Power-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_DDR	*	=2:1_SPACING	?

DDR3-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DDR_CLK_ISO	*	=5:1_SPACING	?
DDR_CTRL_ISO	*	=3.5:1_SPACING	?
DDR_CTRL2CTRL	*	=2.5:1_SPACING	?
DDR_CMD_ISO	*	=3.5:1_SPACING	?
DDR_CMD2CMD	*	=2:1_SPACING	?
DDR_DATA_ISO	*	=3:1_SPACING	?
DDR_DQ2DQ	*	=2:1_SPACING	900
DDR_DQ2DQS	*	=3:1_SPACING	?
DDR_BL2BL	*	=3:1_SPACING	?
DDR_CH2CH	*	=6.5:1_SPACING	?

Main Segment Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
3-2	4	(diff)	15	19.69	CLK trace spacing controlled by =68_OHM_DIFF
3-3	8	9.84	12	13.78	
3-4	6	7.87	12	13.78	
3-5	8.5	7.87	12	11.81	DQ or DQS to other signals not in the same bytelane (but not ch)
					DQ to DQ in the same bytelane of the same channel
			10	11.81	DQ to DQS in the same bytelane of the same channel
			12	11.81	DQ or DQS in different bytelanes of the same channel
			25	25.59	DQ or DQS in different channels
			-	25.59	DDR3 to any other signal not DDR3

Constraints

Clocks: CK[3:0], CK#[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CLK	*	*	DDR_CLK_ISO

Control: CS#[3:0], CKE[3:0], ODT[3:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CTRL	*	*	DDR_CTRL_ISO
DDR_CTRL	DDR_CTRL	*	DDR_CTRL2CTRL

Command: MA[15:0], RAS#, CAS#, WE# BS[2:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_CMD	*	*	DDR_CMD_ISO
DDR_CMD	DDR_CMD	*	DDR_CMD2CMD

Data: DQS[7:0], DQS#[7:0], DQ[63:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DDR_A_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_A_DQS*	*	*	DDR_DATA_ISO
DDR_B_DQ_BYTE*	*	*	DDR_DATA_ISO
DDR_B_DQS*	*	*	DDR_DATA_ISO
DDR_*_DQ_BYTE*	=SAME	*	DDR_DQ2DQ
DDR_A_DQ_BYTE*	DDR_A_DQS*	*	DDR_DQ2DQS
DDR_A_DQ_BYTE*	DDR_A_DQ_BYTE*	*	DDR_BL2BL
DDR_B_DQ_BYTE*	DDR_B_DQS*	*	DDR_DQ2DQS
DDR_B_DQ_BYTE*	DDR_B_DQ_BYTE*	*	DDR_BL2BL
DDR_A_*	DDR_B_*	*	DDR_CH2CH

See Note (3)

SEE NOTE (1)

See Note (1)

See Note (2)

Note (1):

Deliberately set DQ to DQS spacing to 3:1 to avoid adding complexity to constraints, even though it can be less. Only one rule per channel is needed by trading off a little space.

Note (2):

Intel suggested 25 mil (0.65 mm) spacing for via to channel, and via to pad to two different channels. DDR3 draws about 20 mA per trace with edge rates in the 100s of ps. The main coupling mechanism is capacitive. A 0.65 mm spacing is used for power nets, which draw far more current (inductive coupling however). These rules are far too conservative. To meet these rules, the spacing must be applied to the net.

Note (3):

In order for the constraints `DDR*_DQ_BYTE*` to `=SAME` to win out over `DDR_{A,B}_DQ_BYTE*` to `DDR_{A,B}_DQ_BYTE*` so that the small intra-bytelane spacing is used, the spacing rule `DDR_DQ2DQ` must have a weight greater than `DDR_BL2BL`.

DDR3

Electrical Constraint Set	Physical	Spacing	
Channel A			
FE97 DDR A CLK0	DDR CLK_PHY	DDR CLK	MEM A CLK P<1..0> 12 29
FE98 DDR A CLK0	DDR CLK_PHY	DDR CLK	MEM A CLK N<1..0> 12 29
FE99 DDR A CLK1	DDR CLK_PHY	DDR CLK	MEM A CLK P<3..2> 8 12
FE98 DDR A CLK1	DDR CLK_PHY	DDR CLK	MEM A CLK N<3..2> 8 12
FEA0 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A CKE<1..0> 12 29
FEA0 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A CS L<1..0> 12 29
FEA0 DDR A CTRL0	DDR CTRL_PHY	DDR CTRL	MEM A ODT<1..0> 12 29
FEA1 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A CKE<3..2> 8 12
FEA0 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A CS L<3..2> 8 12
FEA0 DDR A CTRL1	DDR CTRL_PHY	DDR CTRL	MEM A ODT<3..2> 8 12
FEA9 DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A A<15..0> 12 29
FEBA DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A BA<2..0> 12 29
FEBC DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A RAS L 12 29
FEBA DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A CAS L 12 29
FEBA DDR A CMD	DDR CMD_PHY	DDR CMD	MEM A WE L 12 29
FEA9 DDR A DQ BYTE0	DDR DQ_PHY	DDR A DQ BYTE0	MEM A DQ<7..0> 12 31
FEBA DDR A DQ BYTE1	DDR DQ_PHY	DDR A DQ BYTE1	MEM A DQ<15..8> 12 31
FEBC DDR A DQ BYTE2	DDR DQ_PHY	DDR A DQ BYTE2	MEM A DQ<23..16> 12 31
FEA9 DDR A DQ BYTE3	DDR DQ_PHY	DDR A DQ BYTE3	MEM A DQ<31..24> 12 31
FEA9 DDR A DQ BYTE4	DDR DQ_PHY	DDR A DQ BYTE4	MEM A DQ<39..32> 12 31
FEA9 DDR A DQ BYTE5	DDR DQ_PHY	DDR A DQ BYTE5	MEM A DQ<47..40> 12 31
FEA9 DDR A DQ BYTE6	DDR DQ_PHY	DDR A DQ BYTE6	MEM A DQ<55..48> 12 31
FEA9 DDR A DQ BYTE7	DDR DQ_PHY	DDR A DQ BYTE7	MEM A DQ<63..56> 12 31
FEA9 DDR A DQS0	DDR DQS_PHY	DDR A DQS0	MEM A DQS P<0> 12 31
FEA9 DDR A DQS0	DDR DQS_PHY	DDR A DQS0	MEM A DQS N<0> 12 31
FEA9 DDR A DQS1	DDR DQS_PHY	DDR A DQS1	MEM A DQS P<1> 12 31
FEA9 DDR A DQS1	DDR DQS_PHY	DDR A DQS1	MEM A DQS N<1> 12 31
FEA9 DDR A DQS2	DDR DQS_PHY	DDR A DQS2	MEM A DQS P<2> 12 31
FEA9 DDR A DQS2	DDR DQS_PHY	DDR A DQS2	MEM A DQS N<2> 12 31
FEA9 DDR A DQS3	DDR DQS_PHY	DDR A DQS3	MEM A DQS P<3> 12 31
FEA9 DDR A DQS3	DDR DQS_PHY	DDR A DQS3	MEM A DQS N<3> 12 31
FEA9 DDR A DQS4	DDR DQS_PHY	DDR A DQS4	MEM A DQS P<4> 12 31
FEA9 DDR A DQS4	DDR DQS_PHY	DDR A DQS4	MEM A DQS N<4> 12 31
FEA9 DDR A DQS5	DDR DQS_PHY	DDR A DQS5	MEM A DQS P<5> 12 31
FEA9 DDR A DQS5	DDR DQS_PHY	DDR A DQS5	MEM A DQS N<5> 12 31
FEA9 DDR A DQS6	DDR DQS_PHY	DDR A DQS6	MEM A DQS P<6> 12 31
FEA9 DDR A DQS6	DDR DQS_PHY	DDR A DQS6	MEM A DQS N<6> 12 31
FEA9 DDR A DQS7	DDR DQS_PHY	DDR A DQS7	MEM A DQS P<7> 12 31
FEA9 DDR A DQS7	DDR DQS_PHY	DDR A DQS7	MEM A DQS N<7> 12 31
Channel B			
FE97 DDR B CLK0	DDR CLK_PHY	DDR CLK	MEM B CLK P<1..0> 12 30
FE98 DDR B CLK0	DDR CLK_PHY	DDR CLK	MEM B CLK N<1..0> 12 30
FE99 DDR B CLK1	DDR CLK_PHY	DDR CLK	MEM B CLK P<3..2> 8 12
FE98 DDR B CLK1	DDR CLK_PHY	DDR CLK	MEM B CLK N<3..2> 8 12
FEA0 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B CKE<1..0> 12 30
FEA0 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B CS L<1..0> 12 30
FEA0 DDR B CTRL0	DDR CTRL_PHY	DDR CTRL	MEM B ODT<1..0> 12 30
FEA1 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B CKE<3..2> 8 12
FEA0 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B CS L<3..2> 8 12
FEA0 DDR B CTRL1	DDR CTRL_PHY	DDR CTRL	MEM B ODT<3..2> 8 12
FEA9 DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B A<15..0> 12 30
FEBA DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B BA<2..0> 12 30
FEBC DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B RAS L 12 30
FEBA DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B CAS L 12 30
FEBA DDR B CMD	DDR CMD_PHY	DDR CMD	MEM B WE L 12 30
FEA9 DDR B DQ BYTE0	DDR DQ_PHY	DDR B DQ BYTE0	MEM B DQ<7..0> 12 31
FEBA DDR B DQ BYTE1	DDR DQ_PHY	DDR B DQ BYTE1	MEM B DQ<15..8> 12 31
FEBC DDR B DQ BYTE2	DDR DQ_PHY	DDR B DQ BYTE2	MEM B DQ<23..16> 12 31
FEA9 DDR B DQ BYTE3	DDR DQ_PHY	DDR B DQ BYTE3	MEM B DQ<31..24> 12 31
FEA9 DDR B DQ BYTE4	DDR DQ_PHY	DDR B DQ BYTE4	MEM B DQ<39..32> 12 31
FEA9 DDR B DQ BYTE5	DDR DQ_PHY	DDR B DQ BYTE5	MEM B DQ<47..40> 12 31
FEA9 DDR B DQ BYTE6	DDR DQ_PHY	DDR B DQ BYTE6	MEM B DQ<55..48> 12 31
FEA9 DDR B DQ BYTE7	DDR DQ_PHY	DDR B DQ BYTE7	MEM B DQ<63..56> 12 31
FEA9 DDR B DQS0	DDR DQS_PHY	DDR B DQS0	MEM B DQS P<0> 12 31
FEA9 DDR B DQS0	DDR DQS_PHY	DDR B DQS0	MEM B DQS N<0> 12 31
FEA9 DDR B DQS1	DDR DQS_PHY	DDR B DQS1	MEM B DQS P<1> 12 31
FEA9 DDR B DQS1	DDR DQS_PHY	DDR B DQS1	MEM B DQS N<1> 12 31
FEA9 DDR B DQS2	DDR DQS_PHY	DDR B DQS2	MEM B DQS P<2> 12 31
FEA9 DDR B DQS2	DDR DQS_PHY	DDR B DQS2	MEM B DQS N<2> 12 31
FEA9 DDR B DQS3	DDR DQS_PHY	DDR B DQS3	MEM B DQS P<3> 12 31
FEA9 DDR B DQS3	DDR DQS_PHY	DDR B DQS3	MEM B DQS N<3>

SYNCH MASTER=D7 DAVE		SYNCH DATE=12/12/2011	
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DDR3 Constraints		PAGE NO.	
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		SIZE D	
		REVISION 4.2.0	
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PCI Express/DMI

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALL ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
PCIE_COMP	*	Y	0.305 MM	0.105 MM	=STANDARD	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE3_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	PCIE_COMP

PCIe and DMI Compensation Rules (mils)

Table	Imp	Design	Iso	Design	Comments
4-5	50	50	15	15.75	PCIe. Impedance inferred from Table 4-7.
4-7	50	50	8	15.75	DMI. Numbers based on Intel stack-up.

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO
PEG_R2D	PEG_R2D	*	PEG_SAME_DIR
PEG_D2R	PEG_D2R	*	PEG_SAME_DIR
PEG_D2R	PEG_R2D	*	PEG_ALT_DIR
PEG_D2R	*	*	PEG_ISO
PEG_R2D	*	*	PEG_ISO

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PEG_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PEG_SAME_DIR	*	=3.5X_DIELECTRIC	?
PEG_ALT_DIR	*	=7X_DIELECTRIC	?
PEG_ISO	*	=4:1_SPACING	?

PEG Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
4.2.1	80	80	16	15.75	PCIe Gen3. Allow looser spacing for same direction on stripline per Anil

PCIe (CPU)

Electrical Constraint Set	Physical	Spacing	
x16 Graphics			
E4E1 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D P<15> 71
E4E2 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D N<15> 71
E4E3 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D C P<15> 9 71
E4E4 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D C N<15> 9 71
E4E5 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R P<15> 9 71
E4E6 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R N<15> 9 71
E4E7	PC1E3_PHY	PEG_D2R	PEG_D2R C P<15> 71
E4E8	PC1E3_PHY	PEG_D2R	PEG_D2R C N<15> 71
E4E9 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D P<14> 71
E4E0 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D N<14> 71
E4E1 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D C P<14> 9 71
E4E2 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D C N<14> 9 71
E4E3 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R P<14> 9 71
E4E4 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R N<14> 9 71
E4E5	PC1E3_PHY	PEG_D2R	PEG_D2R C P<14> 71
E4E6	PC1E3_PHY	PEG_D2R	PEG_D2R C N<14> 71
E4E7 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D P<13> 71
E4E8 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D N<13> 71
E4E9	PC1E3_PHY	PEG_R2D	PEG_R2D C P<13> 9 71
E4E0	PC1E3_PHY	PEG_R2D	PEG_R2D C N<13> 9 71
E4E1 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R P<13> 9 71
E4E2 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R N<13> 9 71
E4E3	PC1E3_PHY	PEG_D2R	PEG_D2R C P<13> 71
E4E4	PC1E3_PHY	PEG_D2R	PEG_D2R C N<13> 71
E4E5 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D P<12> 71
E4E6 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D N<12> 71
E4E7	PC1E3_PHY	PEG_R2D	PEG_R2D C P<12> 9 71
E4E8	PC1E3_PHY	PEG_R2D	PEG_R2D C N<12> 9 71
E4E9 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R P<12> 9 71
E4E0 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R N<12> 9 71
E4E1	PC1E3_PHY	PEG_D2R	PEG_D2R C P<12> 71
E4E2	PC1E3_PHY	PEG_D2R	PEG_D2R C N<12> 71
E4E3 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D P<11> 71
E4E4 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D N<11> 71
E4E5	PC1E3_PHY	PEG_R2D	PEG_R2D C P<11> 9 71
E4E6	PC1E3_PHY	PEG_R2D	PEG_R2D C N<11> 9 71
E4E7 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R P<11> 9 71
E4E8 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R N<11> 9 71
E4E9	PC1E3_PHY	PEG_D2R	PEG_D2R C P<11> 71
E4E0	PC1E3_PHY	PEG_D2R	PEG_D2R C N<11> 71
E4E1 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D P<10> 71
E4E2 PC1E_GEN3_R2D	PC1E3_PHY	PEG_R2D	PEG_R2D N<10> 71
E4E3	PC1E3_PHY	PEG_R2D	PEG_R2D C P<10> 9 71
E4E4	PC1E3_PHY	PEG_R2D	PEG_R2D C N<10> 9 71
E4E5 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R P<10> 9 71
E4E6 PC1E_GEN3_D2R	PC1E3_PHY	PEG_D2R	PEG_D2R N<10> 9 71
E4E7	PC1E3_PHY	PEG_D2R	PEG_D2R C P<10> 71
E4E8	PC1E3_PHY	PEG_D2R	PEG_D2R C N<10> 71
E4E9 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D P<9> 71
E4E0 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D N<9> 71
E4E1	PC1E3_PHY	PEG_R2D	PEG_R2D C P<9> 9 71
E4E2	PC1E3_PHY	PEG_R2D	PEG_R2D C N<9> 9 71
E4E3 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R P<9> 9 71
E4E4 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R N<9> 9 71
E4E5	PC1E3_PHY	PEG_D2R	PEG_D2R C P<9> 71
E4E6	PC1E3_PHY	PEG_D2R	PEG_D2R C N<9> 71
E4E7 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D P<8> 71
E4E8 PC1E_GEN3_R2D_RVSD	PC1E3_PHY	PEG_R2D	PEG_R2D N<8> 71
E4E9	PC1E3_PHY	PEG_R2D	PEG_R2D C P<8> 9 71
E4E0	PC1E3_PHY	PEG_R2D	PEG_R2D C N<8> 9 71
E4E1 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R P<8> 9 71
E4E2 PC1E_GEN3_D2R_RVSD	PC1E3_PHY	PEG_D2R	PEG_D2R N<8> 9 71
E4E3	PC1E3_PHY	PEG_D2R	PEG_D2R C P<8> 71
E4E4	PC1E3_PHY	PEG_D2R	PEG_D2R C N<8> 71

PCIe (CPU)

Electrical Constraint Set		Physical	Spacing	
x16 Graphics				
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<7> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<7> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<7> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<7> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R P<7> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R N<7> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<7> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<7> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<6> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<6> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<6> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<6> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R P<6> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R N<6> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<6> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<6> 71
1E000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG R2D P<5> 71
1E000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG R2D N<5> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<5> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<5> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R P<5> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R N<5> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<5> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<5> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<4> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<4> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<4> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<4> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R P<4> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R N<4> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<4> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<4> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<3> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<3> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<3> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<3> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R P<3> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R N<3> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<3> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<3> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<2> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<2> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<2> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<2> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R P<2> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R N<2> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<2> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<2> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D P<1> 71
1E000	PCIE GEN3 R2D	PCIE3_PHY	PEG_R2D	PEG R2D N<1> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<1> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<1> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R P<1> 9 71
1E000	PCIE GEN3 D2R	PCIE3_PHY	PEG_D2R	PEG D2R N<1> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<1> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<1> 71
1E000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG R2D P<0> 71
1E000	PCIE GEN3 R2D RVSD	PCIE3_PHY	PEG_R2D	PEG R2D N<0> 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C P<0> 9 71
1E000		PCIE3_PHY	PEG_R2D	PEG R2D C N<0> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R P<0> 9 71
1E000	PCIE GEN3 D2R RVSD	PCIE3_PHY	PEG_D2R	PEG D2R N<0> 9 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C P<0> 71
1E000		PCIE3_PHY	PEG_D2R	PEG D2R C N<0> 71
CPU PCIe Clocks				
1E000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG CLK100M P 18 71
1E000	PEG_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PEG CLK100M N 18 71
CPU PCIe Compensation				
1E000		COMP_PCIE_PHY	COMP_PCIE	CPU_PEG_COMP 18</

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_85D
COMP_DMI_PHY	*	50_OHM_SE

PCie-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_SAME_DIR	*	=3.5X_DIELECTRIC	?
PCIE_ALT_DIR	*	=7X_DIELECTRIC	?
PCIE_ISO	*	=4:1_SPACING	?

TBT x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_TBT_R2D	PCIE_TBT_R2D	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_D2R	*	PCIE_SAME_DIR
PCIE_TBT_D2R	PCIE_TBT_R2D	*	PCIE_ALT_DIR
PCIE_TBT_D2R	*	*	PCIE_ISO
PCIE_TBT_R2D	*	*	PCIE_ISO

SSD x2 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_SSD_R2D	PCIE_SSD_R2D	*	PCIE_SAME_DIR
PCIE_SSD_D2R	PCIE_SSD_D2R	*	PCIE_SAME_DIR
PCIE_SSD_D2R	PCIE_SSD_R2D	*	PCIE_ALT_DIR
PCIE_SSD_D2R	*	*	PCIE_ISO
PCIE_SSD_R2D	*	*	PCIE_ISO

PCH x1 PCIE Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO

DMI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DMI_SAME_DIR	TOP,BOTTOM	=5X_DIELECTRIC	?
DMI_SAME_DIR	*	=4X_DIELECTRIC	?
DMI_ALT_DIR	*	=5X_DIELECTRIC	?
DMI_ISO	*	=4X_DIELECTRIC	?

DMI x4 PCIE Spacing Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DMI_N2S	DMI_N2S	*	DMI_SAME_DIR
DMI_S2N	DMI_S2N	*	DMI_SAME_DIR
DMI_N2S	DMI_S2N	*	DMI_ALT_DIR
DMI_N2S	*	*	DMI_ISO
DMI_S2N	*	*	DMI_ISO

PCie (PCH)

Electrical Constraint Set	Physical	Spacing
x4 Thunderbolt		
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_R2D	PCIE_PHY	PCIE_TBT_R2D
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_GEN2_D2R	PCIE_PHY	PCIE_TBT_D2R
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
x2 SSD		
PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_CONN_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_CONN_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_R2D_MUX_SSD	PCIE_PHY	PCIE_SSD_R2D
PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_GEN2_D2R_MUX_SSD	PCIE_PHY	PCIE_SSD_D2R
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 AirPort		
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
x1 Caesar IV		
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_R2D	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_GEN2_D2R	PCIE_PHY	PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE

DMI

Electrical Constraint Set	Physical	Spacing
DMI		
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_N2S	PCIE_PHY	DMI_N2S
DMI_S2N	PCIE_PHY	DMI_S2N
DMI_S2N	PCIE_PHY	DMI_S2N
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
DMI Compensation		
	COMP_DMT_PHY	COMP_PCIE

SATA

SATA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SATA_PHY	*	SATA_90D
COMP_SATA_PHY	*	SATA_50S

SATA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ISO	*	=6:1_SPACING	?
COMP_SATA_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA	*	*	SATA_ISO
COMP_SATA	*	*	COMP_SATA_ISO

FDI

FDI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
FDI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI	*	Y	0.25 MM	0.25 MM	3 MM	=STANDARD	=STANDARD
FDI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
FDI_DIFF_PHY	*	FDI_85D
FDI_SE_PHY	*	FDI_50S
COMP_FDI_PHY	*	COMP_FDI

FDI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
FDI_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
FDI	*	*	FDI_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_55S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
CLK_JTAG_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	CLK_JTAG_ISO

SATA

Electrical Constraint Set		Physical	Spacing	
PCH SATA Port 0 (HDD)				
1600	SATA_R2D	SATA_PHY	SATA	SATA HDD R2D P 41
1600	SATA_R2D	SATA_PHY	SATA	SATA HDD R2D N 41
1600	SATA_PHY	SATA	SATA	SATA HDD R2D C P 38 41
1600	SATA_PHY	SATA	SATA	SATA HDD R2D C N 38 41
1600	SATA_D2R	SATA_PHY	SATA	SATA HDD D2R P 38 41
1600	SATA_D2R	SATA_PHY	SATA	SATA HDD D2R N 38 41
1600	SATA_PHY	SATA	SATA	SATA HDD D2R C P 41
1600	SATA_PHY	SATA	SATA	SATA HDD D2R C N 41
PCH SATA Port 1 (SSD)				
1600	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D P 41
1600	SATA_R2D_MUX_SSD	SATA_PHY	SATA	SATA SSD R2D N 41
1600	SATA_PHY	SATA	SATA	SATA SSD R2D C P 38 41
1600	SATA_PHY	SATA	SATA	SATA SSD R2D C N 38 41
1600	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R P 38 41
1600	SATA_D2R_MUX_SSD	SATA_PHY	SATA	SATA SSD D2R N 38 41
1600	SATA_PHY	SATA	SATA	SATA SSD D2R C P 41
1600	SATA_PHY	SATA	SATA	SATA SSD D2R C N 41
SSD PCIe/SATA Mux Output				
1600	PCIe SATA R2D_MUX_CONN	SATA_PHY	SATA	PCIe SATA SSD R2D P 41
1600	PCIe SATA R2D_MUX_CONN	SATA_PHY	SATA	PCIe SATA SSD R2D N 41
1600	PCIe SATA D2R_MUX_CONN	SATA_PHY	SATA	PCIe SATA SSD D2R P 41
1600	PCIe SATA D2R_MUX_CONN	SATA_PHY	SATA	PCIe SATA SSD D2R N 41
PCH SATA Compensation				
1600	COMP_SATA_PHY	COMP_SATA		PCH SATA1COMP 38
1600	COMP_SATA_PHY	COMP_SATA		PCH SATA3COMP 38
1600	COMP_SATA_PHY	COMP_SATA		PCH SATA3RBIAS 38

FDI

Electrical Constraint Set		Physical	Spacing	
FDI				
LE80B	FDI_TX	EDT_DIFX_PHY	FDI	CPU_FDI_TX_P<7..0> 8 10
LE80D	FDI_TX	EDT_DIFX_PHY	FDI	CPU_FDI_TX_N<7..0> 8 10
LE80E		FDI_SE_PHY	FDI	CPU_FDI_FSYNC<1..0> 8 10
LE80F		FDI_SE_PHY	FDI	CPU_FDI_LSYNC<1..0> 8 10
LE80G		FDI_SE_PHY	FDI	CPU_FDI_INT 8 10
FDI Compensation				
LE80H		COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO 10

XDP

Electrical Constraint Set	Physical	Spacing		
CPU XDP				
PCB1	XDP_PHY	XDP	XDP BPM L<7..0>	11 25
PCB2	XDP_PHY	XDP	CPU CFG<17..0>	6 10 15 25
PCB3 TTP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU CLK100M P	11 15
PCB4 TTP_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE	ITPCPU CLK100M N	11 15
PCB5	CLK_PCIE_PHY	CLK_PCIE	ITPXPDP CLK100M P	15 18 25
PCB6	CLK_PCIE_PHY	CLK_PCIE	ITPXPDP CLK100M N	15 18 25
PCB7	CLK_PCIE_PHY	CLK_PCIE	XDP CPU CLK100M P	25
PCB8	CLK_PCIE_PHY	CLK_PCIE	XDP CPU CLK100M N	25
PCB9	XDP_PHY	CLK_JTAG	XDP CPU TCK	11 25
PCB10	XDP_PHY	XDP	XDP CPU TMS	11 25
PCB11	XDP_PHY	XDP	XDP CPU TDI	11 25
PCB12	XDP_PHY	XDP	XDP CPU TDO	11 25
PCH XDP				
PCB13	XDP_PHY	CLK_JTAG	XDP PCH TCK	18 25
PCB14	XDP_PHY	XDP	XDP PCH TMS	18 25
PCB15	XDP_PHY	XDP	XDP PCH TDI	18 25
PCB16	XDP_PHY	XDP	XDP PCH TDO	18 25

PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCH	*	=4:1_SPACING	?
COMP_PCH	*	=2:1_SPACING	?

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_PCI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCI	*	=2:1_SPACING	?

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=1.5:1_SPACING	?
CLK_LPC	*	=2:1_SPACING	?

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XTAL	*	=4X_DIELECTRIC	?

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=2:1_SPACING	?

PCI

Electrical Constraint Set	Physical	Spacing
PCI Clock		
100M	CLK_PCT_55S	CLK_PCT
100M	CLK_PCT_55S	CLK_PCT

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
	LPC 55S	LPC	LPC_AD<3..0> 18 44 46
	LPC 55S	LPC	LPC_R_AD<3..0> 18
	LPC 55S	LPC	LPC_FRAME_L 18 44 46
	LPC 55S	LPC	LFRAME_L 18
LPC Clocks			
	CLK LPC 55S	CLK LPC	LPC_CLK33M_LPCPLUS 26 46
	CLK LPC 55S	CLK LPC	LPC_CLK33M_LPCPLUS_R 20 26
	CLK LPC 55S	CLK LPC	LPC_CLK33M_SMC 26 44
	CLK LPC 55S	CLK LPC	LPC_CLK33M_SMC_R 20 26

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
PERIOD	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_SB 26
PERIOD	CLK_PCH_55S	CLK_PCH	PCH_CLK25M_XTALIN 18 26
PCH Ref Clock Comp			
PERIOD	PCH_55S	COMP_PCH	PCH_XCLK_RC0MP 18
PCH RTC 32K			
PERIOD	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 18 26
PERIOD	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 18 26
PERIOD	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 26
SMC 32K			
PERIOD	CLK_PCH_55S	CLK_PCH	PM_CLK32K_SUSCLK_R 19 45
PERIOD	CLK_PCH_55S	CLK_PCH	SMC_CLK32K 44 45

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing		
25M Reference Crystal				
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1	26
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2	26
	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R	26
25M Reference Clocks				
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET	26 37
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_ENET_R	26
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT	26 34
	CLK_PCH_55S	CLK_PCH	SYSCLK_CLK25M_TBT_R	34

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H109	HDA_55S	HDA	HDA_BIT_CLK 18 52
H109	HDA_55S	HDA	HDA_BIT_CLK_R 18
H109	HDA_55S	HDA	HDA_RST_L 18 52
H109	HDA_55S	HDA	HDA_RST_R_L 18
H109	HDA_55S	HDA	HDA_SDOUT 18 52
H109	HDA_55S	HDA	HDA_SDOUT_R 18 18
H109	HDA_55S	HDA	HDA_SYNC 18 18 52
H109	HDA_55S	HDA	HDA_SYNC_R 18
H109	HDA_55S	HDA	HDA_SDI0 18 52
H109	HDA_55S	HDA	AUD_SDI_R 52
SPDIF			
H37A		HDA	AUD_SPDIF_CHIP 52
H37A		HDA	AUD_SPDIF_OUT 52 56

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H303	SPI 50S	SPI	SPI CLK R 18 46
H309	SPI 50S	SPI	SPI CLK 46
H305	SPI 50S	SPI	SPI ALT CLK 46
H308	SPI 50S	SPI	SPI SMC CLK 44 45
H307	SPI 50S	SPI	SPI MLB CLK 45 46
H302	SPI 50S	SPI	SPI CS0 R L 18 46
H306	SPI 50S	SPI	SPI CS0 L 46
H310	SPI 50S	SPI	SPI ALT CS L 46
H301	SPI 50S	SPI	SPI SMC CS L 44 45
H304	SPI 50S	SPI	SPI MLB CS L 45 46
H300	SPI 50S	SPI	SPI MOSI R 18 46
H300	SPI 50S	SPI	SPI MOSI 46
H309	SPI 50S	SPI	SPI ALT MOSI 46
H310	SPI 50S	SPI	SPI SMC MOSI 44 45
H311	SPI 50S	SPI	SPI MLB MOSI 45 46
H305	SPI 50S	SPI	SPI MISO 18 46
H308	SPI 50S	SPI	SPI ALT MISO 46
H312	SPI 50S	SPI	SPI SMC MISO 44 45
H313	SPI 50S	SPI	SPI MLB MISO 45 46
H309	SPI 50S	SPI	SPIROM USE MLB 21 46

USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D

USB-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE+TO+LINE SPACING	WEIGHT
USB2_ISO	*	=3:1_SPACING	?
USB2_ISO	TOP,BOTTOM	=3:1_SPACING	?
USB3_ISO	*	=5.5:1_SPACING	?
USB3_ISO	TOP,BOTTOM	=5.5:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB3	*	*	USB3_ISO

Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S
CIV_SPI	*	SPI_55S

CIV-specific Spacing Definitions

Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

Camera Processor-to-Camera Sensor I/F (SMIA/MIPI)

Camera Processor's SMIA Interface Physical Rules

[illegible]

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D

Camera Processor's SMIA Interface Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Design	Comments
12.2.1	90	90	12	11.81	USB 2.0
13.3.1	85	85	20	21.65	USB 3.0

Constraints Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kV isolation

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_N
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_F_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_A_RX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_F_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_C_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_A_TX_C_N
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB_PCH_0_P
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB_PCH_0_N
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB2_EXT_A_MIXED_P
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB2_EXT_A_MIXED_N
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB2_EXT_A_P
FE00	USB2_MIXED_MUX0_CONN	USB2_PHY	USB2	USB2_EXT_A_N
External Port B (J4610)				
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_N
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_F_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_B_RX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_F_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_C_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_B_TX_C_N
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB_PCH_1_P
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB_PCH_1_N
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB_PCH_9_P
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB_PCH_9_N
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB2_EXT_B_MIXED_P
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB2_EXT_B_MIXED_N
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB2_EXT_B_P
FE00	USB2_MIXED_MUX1_CONN	USB2_PHY	USB2	USB2_EXT_B_N
External Port C (J4700)				
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_N
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_F_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_C_RX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_F_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_C_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_C_TX_C_N
FE00	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P
FE00	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N
FE00	USB2_CONN	USB2_PHY	USB2	USB2_EXT_C_P
FE00	USB2_CONN	USB2_PHY	USB2	USB2_EXT_C_N
External Port D (J4710)				
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_N
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_F_P
FE00	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXT_D_RX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_F_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_F_N
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_C_P
FE00	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXT_D_TX_C_N
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB_PCH_3_P
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB_PCH_3_N
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB_PCH_10_P
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB_PCH_10_N
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB2_EXT_D_MIXED_P
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB2_EXT_D_MIXED_N
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB2_EXT_D_P
FE00	USB2_MIXED_MUX2_CONN	USB2_PHY	USB2	USB2_EXT_D_N
Camera (J3510)				
FE00	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_P
FE00	USB2_CONN_TNT	USB2_PHY	USB2	USB_CAMERA_N
PCH USB Compensation				
FE00		PCH_SS	COMP_PCH	PCH_USB_RBIA

RMH Love

Electrical Constraint Set	Physical	Spacing	
USB 2.0 Hub			
IE9P USB2_HUB	USB2_PHY	USB2	USB PCH 7 P 26 27
IE9Q USB2_HUB	USB2_PHY	USB2	USB PCH 7 N 26 27
IE9P USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_P 27 33
IE9Q USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_N 27 33
IE9S USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_MUX_P 33
IE9D USB2_HUB_CONN	USB2_PHY	USB2	USB_BT_MUX_N 33
IE9T USB2_HUB_RES	USB2_PHY	USB2	USB_SMC_P 27 44
IE9V USB2_HUB_RES	USB2_PHY	USB2	USB_SMC_N 27 44
IE9U	USB2_PHY	USB2	USB_HUB_2_P 27
IE9V	USB2_PHY	USB2	USB_HUB_2_N 27
USB 2.0 Hub Compensation			
IE9A	PCH_55S	COMP_PCH	USB_HUB_RBIAS 27
USB 2.0 Hub Crystal			
IE9P	CLK_XTAL	XTAL	USB_HUB_XTAL1 27
IE9Q	CLK_XTAL	XTAL	USB_HUB_XTAL2 27

Et tu Brute?

Electrical Constraint Set	Physical	Spacing	
Ethernet			
U372 ENET_MDT	ENET_DTEF_PHY	ENET_DTEF	ENETCONN_MDI P<3..0> 37 38
U373 ENET_MDT	ENET_DTEF_PHY	ENET_DTEF	ENETCONN_MDI N<3..0> 37 38
U374	ENET_DTEF_PHY	ENET_TRANS	ENETCONN_MDI T P<3..0> 38
U375	ENET_DTEF_PHY	ENET_TRANS	ENETCONN_MDI T N<3..0> 38
U510		ENET_TRANS	ENETCONN_MCT0 38
U520		ENET_TRANS	ENETCONN_MCT1 38
U524		ENET_TRANS	ENETCONN_MCT2 38
U522		ENET_TRANS	ENETCONN_MCT3 38
U526		ENET_TRANS	ENETCONN_MCT_BS 38
U477	ENET_COMP_PHY	COMP_ENET	ENET_RDAC 37
SD			
U479 SD_DATA	SD_PHY	SD	ENET_CR_DATA<7..0> 37
U480	SD_PHY	SD	SDCONN_DATA<7..0> 37 39
U515	SD_PHY	SD	SDCONN_DATA_R <7..0> 39
U483 SD_CMD	SD_PHY	SD	ENET_SD_CMD 37
U487	SD_PHY	SD	SDCONN_CMD 37 39
U516	SD_PHY	SD	SDCONN_CMD_R 39
U488 SD_CLK	SD_PHY	SD	ENET_SD_CLK 37
U489	SD_PHY	SD	SDCONN_CLK 37 39
U490	SD_PHY	SD	SDCONN_CLK_R 39
U491	SD_PHY	SD	ENET_MEDIA_SENSE 15 18
U492	SD_PHY	SD	ENET_SD_DETECT_L 37 39
U493	SD_PHY	SD	SDCONN_WP 37 39
CIV SPI			
U495	CIV_SPI	SPT	ENET_SCLK 37
U496	CIV_SPI	SPT	ENET_MISO 37
U497	CIV_SPI	SPT	ENET_MOSI 37
U498	CIV_SPI	SPT	ENET_CS_L 37

Camera Processor-Camera Sensor I/F

Electrical Constraint Set		Physical	Spacing	
H50P	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_P 40
H50P	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_N 40
H50P	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_F_P 40
H57P	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_DATA_F_N 40
H50S	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_P 40
H50S	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_N 40
H59S	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_F_P 40
H59S	SMIA_DP	SMIA_DIEF_PHY	SMIA_DIEF	SMIA_CLK_F_N 40
H50P		SPT_50S	SPT	CAM_SF_CLK 40
H50P		SPT_50S	SPT	CAM_SF_CLK_R 40
H50P		SPT_50S	SPT	CAM_SF_DIN 40
H31D		SPT_50S	SPT	CAM_SF_DIN_R 40
H31D		SPT_50S	SPT	CAM_SF_CS_L 40
H31S		SPT_50S	SPT	CAM_SF_WP_L 40
H57P		SPT_50S	SPT	CAM_SF_DOUT 40
H31D		SPT_50S	SPT	CAM_SF_DOUT_R 40
H31P		SMB_PHY	SMB	I2C CAMSENSOR_SDA 40
H31P		SMB_PHY	SMB	I2C CAMSENSOR_SCL 40

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USB/Ethernet/SD Constraints			
 Apple Inc.		DRAWING NUMBER 051-9509	SIZE D
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SMBus

SMBus-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMB_PHY	*	SMB_55S

SMBus-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB_ISO	*	=2x_DIELECTRIC	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMB	*	*	SMB_ISO

Sensor

Sensor-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.085 MM

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SNS_DIFF_PHY	*	1:1_DIFFPAIR

Sensor-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SENSE	*	*	SENSE_ISO
SENSE	POWER	*	PWR_P2MM
SENSE	GND	*	GND_P2MM

SMBus

Electrical Constraint Set	Physical	Spacing	
SMC			
H10	SMB_PHY	SMR	SMBUS SMC 0 S0_SCL 44 47
H9	SMB_PHY	SMR	SMBUS SMC 0 S0_SDA 44 47
H11	SMB_PHY	SMB	SMBUS SMC 1 S0_SCL 44 47
H12	SMB_PHY	SMR	SMBUS SMC 1 S0_SDA 44 47
H13	SMB_PHY	SMB	SMBUS SMC 2 S4_SCL 44 47
H14	SMB_PHY	SMB	SMBUS SMC 2 S4_SDA 44 47
H15	SMB_PHY	SMR	SMBUS SMC 3_SCL 44 47
H16	SMB_PHY	SMB	SMBUS SMC 3_SDA 44 47
H17	SMB_PHY	SMR	SMBUS SMC 5 G3H_SCL 44 45
H18	SMB_PHY	SMR	SMBUS SMC 5 G3H_SDA 44 45
PCH			
H21	TBT_T2C_55S	TBT_T2C	SMBUS PCH_CLK 18 47
H22	TBT_T2C_55S	TBT_T2C	SMBUS PCH_DATA 18 47
H23	SMB_PHY	SMB	SML PCH 0_CLK 28 47
H24	SMB_PHY	SMR	SML PCH 0_DATA 18 47
Display TCon			
H50	SMB_PHY	SMR	SMB DP TCON_SCL 47 B1
H51	SMB_PHY	SMR	SMB DP TCON_SDA 47 B1

Temperature Sense

Electrical Contract Set	Physical	Spacing	
EMC1414-1 (Production)			
H59 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 P 50
H58 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 1 N 50
H59 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 P 50
H60 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T1 2 N 50
H65 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC P 6 50
H66 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS ACDC N 6 50
TMP423 (Development)			
T67 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 P 50
T68 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 1 N 50
T69 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 P 50
T70 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 2 N 50
T61 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN P 50
T62 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS SKIN N 50
T73 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 P 50
T72 SNS_TEMP	SNS_DIFF_PHY	SENSE	SNS T2 3 N 50
HDD Out-of-Band			
H75		SENSE	SMC HDD_O0B_TEMP
H75		SENSE	HDD_O0B_TEMP_CONN
H75		SENSE	HDD_O0B_TEMP_FILT
H76		SENSE	HDD_O0B_TEMP_R
SSD Out-of-Band			
H77		SENSE	SMC SSD_O0B_TEMP
H77		SENSE	SMC SSD_TEMP_CTL
H79		SENSE	SSD_O0B_TEMP

SMC

Electrical Constraint Set	Physical	Spacing		
SMC				
E1	CLK_XTAL	XTAL	SMC_XTAL	44 45
E2	CLK_XTAL	XTAL	SMC_EXTAL	44 45

Current/Voltage Sense

Electrical Constraint Set	Physical	Spacing	
Common			
E13		SENSE	GND_SMC_AVSS 44
12V S5 (System Total)			
E14 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_P 48
E15 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VG3H_N 48
E16		SENSE	ISNS_P12VG3H_R 48
E17		SENSE	ISNS_P12VG3H 45
E18		SENSE	VSNS_P12VG3H 45
12V S0 (GPU Core)			
E22 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_P 48
E23 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUCORE_N 48
E24		SENSE	ISNS_P12VS0_GPUCORE_R 48
E25		SENSE	ISNS_P12VS0_GPUCORE 45
E26		SENSE	VSNS_P12VS0_GPUCORE 45
HDD			
E100 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_HDD_P 48
E101 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_HDD_N 48
E102		SENSE	ISNS_HDDS0_R 48
E103		SENSE	ISNS_HDDS0 45
E104		SENSE	VSNS_HDDS0 45
SSD			
E105 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_SSD_P 49
E106 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_SSD_N 49
E107		SENSE	ISNS_SSDS0_R 49
E108		SENSE	ISNS_SSDS0 45
E109		SENSE	VSNS_SSDS0 45
VDDQ S3 (DDR)			
E88 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_P 49
E89 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_VDDQS3_DDR_N 49
E90		SENSE	ISNS_VDDQS3_DDR_R 49
E91		SENSE	ISNS_VDDQS3_DDR 45
E92		SENSE	VSNS_VDDQS3_DDR 45
VDDQ S0 (GPU Uncore)			
E37 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUUC_P 48
E38 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	SNS_P12VS0_GPUUC_N 48
E39		SENSE	ISNS_P12VS0_GPUUC_R 48
E40		SENSE	ISNS_P12VS0_GPUUNCORE 45
E41		SENSE	VSNS_P12VS0_GPUUNCORE 45
CPU Core			
E42 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	ISNS_CPUCORE_P 48
E43 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	ISNS_CPUCORE_N 48
E44		SENSE	ISNS_CPUCORE_FB 48
E45		SENSE	ISNS_CPUCORE 45
E46		SENSE	VSNS_CPUCORE 45
CPU AXG			
E47 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	ISNS_CPUAXG_P 48
E48 SNS_CURRENT SNS_DIFF_PHY	SNS_DIFF_PHY	SENSE	ISNS_CPUAXG_N 48
E49		SENSE	ISNS_CPUAXG_FB 48
E50		SENSE	ISNS_CPUAXG 45
E51		SENSE	VSNS_CPUAXG 45

DC - DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GND_P3MM	*	Y	0.300 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.150 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GND	*	GND_P5MM
GND	BGA	GND_P3MM
POWER	*	POWER_P6MM
POWER	BGA	POWER_P3MM
VR_CTL_PHY	*	POWER_P3MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_DIDT_PHY	*	POWER_P6MM
VR_DIDT_PHY	BGA	STANDARD

Power-specific Spacing Definitions

Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?
GND_ISO	*	=STANDARD	?

Constraints

Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?

DC-DC Baddies

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_SWITCH	*	*	SWNODE_ISO
VR_SWITCH	*	BGA	BGA_P1MM
VR_SWITCH	VR_SWITCH	*	SWNODE_SW2SW
VR_SWITCH	POWER	*	SWNODE_SW2PWR
VR_SWITCH	GND	*	SWNODE_SW2GND

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

DC-DC Control

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
VR_CTL	*	*	VR_CTL_ISO
VR_VID	*	*	VR_VID_ISO

VDDQ S3 (1.5V)/VTT S0

Physical	Spacing	Voltage	IDT	No_TEST	
Input Bus					
E43 POWER	POWER	5V			REG_V5IN_U7700
Local Ground					
E44 GND	GND	0V			AGND_VDDQS3
VDDQ S3					
E45 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3
E47 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VDDQS3_L
E46 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_VDDQS3
E48 VR_D1DT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_VDDQS3_RC
E49 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3
E50 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VDDQS3_R
E51 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VDDQS3
E53 VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VDDQS3
E54 POWER	POWER	1.5V			PPVDDQ_S3_SENSE
E53 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VDDQSNS
E55 VR_CTL_PHY	VR_CTL				REG_VDDQS3_VREF
E56 VR_CTL_PHY	VR_CTL				REG_VDDQS3_REFIN
E57 VR_CTL_PHY	VR_CTL				REG_VDDQS3_MODE
E58 VR_CTL_PHY	VR_CTL				REG_VDDQS3_TRIP
E59 VR_CTL_PHY	VR_CTL				LDO_DDRVTT0_SNS
Output Bus					
E60 POWER	POWER	1.5V			PPVDDQ_S3
E61 POWER_DDR	POWER_DDR	0.75V			PPDDRVT1_S3
E62 POWER_DDR	POWER_DDR	0.75V			PPDDRVT1_S0
FET Switched					
E63 POWER	POWER	1.5V			PP1V5_S0
Sensed					
E64 POWER	POWER	1.5V			PPVDDQ_S3_DDR

CPU VccIO/ PCH 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	IDT	NO_TEST	
Input Bus						
E1	POWER	POWER	5V			REG_VCC_U7400 65
E2	POWER	POWER	5V			REG_PVCC_U7400 65
Local Ground						
E3	GND	GND	6V			AGND_P1V05S0 65
1.05V S0						
E5	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0 65
E6	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_P1V05S0_L 65
E7	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_BOOT_P1V05S0 65
E8	VR_DTDT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_BOOT_P1V05S0_RC 65
E9	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0 65
E10	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_P1V05S0_R 65
E11	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_P1V05S0 65
E12	VR_DTDT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_P1V05S0 65
E13	VR_CTL_PHY	VR_CTL				REG_P1V05S0_OCSET 65
E14	VR_CTL_PHY	VR_CTL				REG_P1V05S0_V0 65
E15 VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_P 13 65
E16 VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE				SNS_CPU_VCCIO_N 13 65
E17	SNS_DIFF_PHY	SENSE				SNS_P1V05S0_XW_P 65
E18	SNS_DIFF_PHY	SENSE				SNS_P1V05S0_XW_N 65
E19		SENSE				REG_P1V05S0_FB 65
E20		SENSE				REG_P1V05S0_RTN 65
E21	VR_CTL_PHY	VR_CTL				REG_P1V05S0_SREF 65
E22	VR_CTL_PHY	VR_CTL				REG_P1V05S0_FSEL 65
Output Bus						
E23	POWER	POWER	1.05V			PP1V05_S0 6
FET Switched						
E24	POWER	POWER	1.05V			PP1V05_TBTLIC 6
E25	POWER	POWER	1.05V			PP1V05_TBTCIO 6

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	IDT	NO_TEST	
Input Bus						
E24	POWER	POWER	5V			REG_VCC_U7500
E25	POWER	POWER	5V			REG_PVCC_U7500
Local Ground						
E26	GND	GND	0V			AGND_VCCSAS0
VCCIO						
E27	VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_PHASE_VCCSAS0
E28	VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_B00T_VCCSAS0
E29	VR_D1DT_PHY	VR_SWITCH	12V	TRUE	TRUE	REG_B00T_VCCSAS0_RC
E31	VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_UGATE_VCCSAS0
E30	VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_LGATE_VCCSAS0
E33	VR_D1DT_PHY	VR_SWITCH	12V	TRUE		REG_SNUBBER_VCCSAS0
E32	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_OCSSET
E35	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_VO
E34		SENSE				SNS_CPU_VCCSA
E37	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE			SNS_VCCSAS0_XW_P
E38	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE			SNS_VCCSAS0_XW_N
E39		SENSE				REG_VCCSAS0_FB
E40		SENSE				REG_VCCSAS0_RTN
E41	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_SREF
E43	VR_CTL_PHY	VR_CTL				REG_VCCSAS0_FSEL
Output Bus						
E42	POWER	POWER	0_925V			PPVCCSA_S0

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Thunderbolt

Thunderbolt-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBT_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD
TBTD_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Thunderbolt-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE+TO+LINE SPACING	WEIGHT
TBT_I2C	*	=2x_DIELECTRIC	?
TBT_SPI	*	=2x_DIELECTRIC	?
TBTDP	*	=5x_DIELECTRIC	?
TBTDP	TOP, BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius's T29 Routing Notes

DisplayPort

DP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	0.08MM	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

DP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DISPLAYPORT	*	=3:1_SPACING	?

Pairs should be within 100 mils of clock length.

Max length of DisplayPort traces: 12 inches

DisplayPort intra-pair matching should be 5 ps. Inter-pair matching should be within 150 ps.

DisplayPort AUX channel intra-pair matching should be 5 ps. No relationship to other signals.

TBT IC Net Properties

Electrical Constraint Set	Physical	Spacing		
EE990 DP TBTSNK0 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK0 ML C P<3...0>	34 77
EE990 DP TBTSNK0 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK0 ML C N<3...0>	34 77
EE990 DP TBTSNK0 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK0 ML P<3...0>	34
EE990 DP TBTSNK0 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK0 ML N<3...0>	34
EE990 DP TBTSNK0 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK0 AUXCH C P	34 71
EE990 DP TBTSNK0 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK0 AUXCH C N	34 71
EE990 DP TBTSNK0 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK0 AUXCH P	34
EE990 DP TBTSNK0 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK0 AUXCH N	34
EE990 DP TBTSNK1 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK1 ML C P<3...0>	34 77
EE990 DP TBTSNK1 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK1 ML C N<3...0>	34 77
EE990 DP TBTSNK1 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK1 ML P<3...0>	34
EE990 DP TBTSNK1 ML	DP 85D	DTSPPLAYPORT	DP TBTSNK1 ML N<3...0>	34
EE990 DP TBTSNK1 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK1 AUXCH C P	34 71
EE990 DP TBTSNK1 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK1 AUXCH C N	34 71
EE990 DP TBTSNK1 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK1 AUXCH P	34
EE990 DP TBTSNK1 AUX	DP 85D	DTSPPLAYPORT	DP TBTSNK1 AUXCH N	34
EE990 DP INTNPN TRT ML MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC ML P<3...0>	82
EE990 DP INTNPN TRT ML MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC ML N<3...0>	82
EE990 DP INTNPN TRT ML MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC ML C P<3...0>	82
EE990 DP INTNPN TRT ML MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC ML C N<3...0>	82
EE990 DP INTNPN TRT AUX MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC AUXCH P	82
EE990 DP INTNPN TRT AUX MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC AUXCH N	82
EE990 DP INTNPN TRT AUX MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC AUX C P	82
EE990 DP INTNPN TRT AUX MIX	DP 85D	DTSPPLAYPORT	DP TBTSRC AUX C N	82
EE990 TBT I2C 55S	TBT I2C		=I2C TBTIRTR_SCL	34 47
EE990 TBT I2C 55S	TBT I2C		=I2C TBTIRTR_SDA	34 47
EE990 TBT_SPT_CLK	TBT_SPT 55S	TBT_SPT	TBT SPI_CLK	34
EE990 TBT_SPT_MOSI	TBT_SPT 55S	TBT_SPT	TBT SPI_MOSI	34
EE990 TBT_SPT_MISO	TBT_SPT 55S	TBT_SPT	TBT SPI_MISO	34
EE990 TBT_SPT_CS_1	TBT_SPT 55S	TBT_SPT	TBT SPI_CS_1	34

*: Only used on hosts supporting T29 video-in

DisplayPort

Electrical Constraint Set		Physical	Spacing	
Graphics Source				
FE05P	DP INTPNL EG MI_MUX	DP SSD	DISPLAYPORT	DP INT EG ML P<1..0> 77 82
FE05S	DP INTPNL EG MI_MUX	DP SSD	DISPLAYPORT	DP INT EG ML N<1..0> 77 82
FE06P	DP INTPNL EG AUX_MUX	DP SSD	DISPLAYPORT	DP INT EG AUX P 77 82
FE06S	DP INTPNL EG AUX_MUX	DP SSD	DISPLAYPORT	DP INT EG AUX N 77 82
FE07P		DP SSD	DISPLAYPORT	DP INT EG AUX C P 82
FE07S		DP SSD	DISPLAYPORT	DP INT EG AUX C N 82
Internal Panel				
FE08P		DP SSD	DISPLAYPORT	DP INTPNL ML C P<3..0> 82
FE08S		DP SSD	DISPLAYPORT	DP INTPNL ML C N<3..0> 82
FE09P	DP INTPNL MI_CONN	DP SSD	DISPLAYPORT	DP INTPNL ML P<3..0> 81 82
FE09S	DP INTPNL MI_CONN	DP SSD	DISPLAYPORT	DP INTPNL ML N<3..0> 81 82
FE10P	DP INTPNL AUX_CONN	DP SSD	DISPLAYPORT	DP INTPNL AUX P 81 82
FE10S	DP INTPNL AUX_CONN	DP SSD	DISPLAYPORT	DP INTPNL AUX N 81 82
Internal DP SPDIF				
FE10P			HDA	DP INT SPDIF AUDIO 52 81

TBT/DP Net Properties

Electrical Constraint Set	Physical	Spacing	
Port A			
E521 TBT_A_R2D	TRTDP_960	TRTDP	TBT_A_R2D_C_P<1..0> 34 84
E522 TBT_A_R2D	TRTDP_960	TRTDP	TBT_A_R2D_C_N<1..0> 34 84
E523	TRTDP_960	TRTDP	TBT_A_R2D_P<1..0> 84
E524	TRTDP_960	TRTDP	TBT_A_R2D_N<1..0> 84
E525 DP_TBTPA_M1_1	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_P<1> 34 84
E526 DP_TBTPA_M1_1	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_N<1> 34 84
E527 DP_TBTPA_M1_3	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_P<3> 34 84
E528 DP_TBTPA_M1_3	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_N<3> 34 84
E529	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_P<1> 84
E530	DP_850	DISPLAYPORT	DP_TBTPA_M1_C_N<1> 84
E531	DP_850	DISPLAYPORT	DP_TBTPA_M1_N<1> 84
E532	DP_850	DISPLAYPORT	DP_TBTPA_M1_P<3> 84
E533	DP_850	DISPLAYPORT	DP_TBTPA_M1_N<3> 84
E534 DP_A_LSX	DP_850	DISPLAYPORT	DP_A_LSX_M1_P<1> 84
E535 DP_A_LSX	DP_850	DISPLAYPORT	DP_A_LSX_M1_N<1> 84
E536	TRTDP_960	TRTDP	TBT_A_D2R_C_P<1..0> 84
E537	TRTDP_960	TRTDP	TBT_A_D2R_C_N<1..0> 84
E538 TBT_A_D2R1	TRTDP_960	TRTDP	TBT_A_D2R_P<1> 34 84
E539 TBT_A_D2R1	TRTDP_960	TRTDP	TBT_A_D2R_N<1> 34 84
E540 TBT_A_D2R6	TRTDP_960	TRTDP	TBT_A_D2R_P<6> 34 84
E541 TBT_A_D2R6	TRTDP_960	TRTDP	TBT_A_D2R_N<6> 34 84
E542 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_P 34 84
E543 TBT_A_AUXCH	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_C_N 34 84
E544	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_P 84
E545	DP_850	DISPLAYPORT	DP_TBTPA_AUXCH_N 84
E546 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_P 84
E547 DP_A_AUXCH_DDC	DP_850	DISPLAYPORT	DP_A_AUXCH_DDC_N 84
E548	TRTDP_960	TRTDP	TBT_A_D2R1_AUXDDC_P 84
E549	TRTDP_960	TRTDP	TBT_A_D2R1_AUXDDC_N 84
Port B			
E550 TBT_B_R2D	TRTDP_960	TRTDP	TBT_B_R2D_C_P<1..0> 34 85
E551 TBT_B_R2D	TRTDP_960	TRTDP	TBT_B_R2D_C_N<1..0> 34 85
E552	TRTDP_960	TRTDP	TBT_B_R2D_P<1..0> 85
E553	TRTDP_960	TRTDP	TBT_B_R2D_N<1..0> 85
E554 DP_TBTPB_M1_1	DP_850	DISPLAYPORT	DP_TBTPB_M1_C_P<1> 34 85
E555 DP_TBTPB_M1_1	DP_850	DISPLAYPORT	DP_TBTPB_M1_C_N<1> 34 85
E556 DP_TBTPB_M1_3	DP_850	DISPLAYPORT	DP_TBTPB_M1_C_P<3> 34 85
E557 DP_TBTPB_M1_3	DP_850	DISPLAYPORT	DP_TBTPB_M1_C_N<3> 34 85
E558	DP_850	DISPLAYPORT	DP_TBTPB_M1_P<1> 85
E559	DP_850	DISPLAYPORT	DP_TBTPB_M1_N<1> 85
E560	DP_850	DISPLAYPORT	DP_TBTPB_M1_P<3> 85
E561	DP_850	DISPLAYPORT	DP_TBTPB_M1_N<3> 85
E562 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_M1_C_P<1> 85
E563 DP_B_LSX	DP_850	DISPLAYPORT	DP_B_LSX_M1_N<1> 85
E564	TRTDP_960	TRTDP	TBT_B_D2R_C_P<1..0> 85
E565	TRTDP_960	TRTDP	TBT_B_D2R_C_N<1..0> 85
E566 TBT_B_D2R1	TRTDP_960	TRTDP	TBT_B_D2R_P<1> 34 85
E567 TBT_B_D2R1	TRTDP_960	TRTDP	TBT_B_D2R_N<1> 34 85
E568 TBT_B_D2R6	TRTDP_960	TRTDP	TBT_B_D2R_P<6> 34 85
E569 TBT_B_D2R6	TRTDP_960	TRTDP	TBT_B_D2R_N<6> 34 85
E570 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_P 34 85
E571 TBT_B_AUXCH	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_C_N 34 85
E572	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_P 85
E573	DP_850	DISPLAYPORT	DP_TBTPB_AUXCH_N 85
E574 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_P 85
E575 DP_B_AUXCH_DDC	DP_850	DISPLAYPORT	DP_B_AUXCH_DDC_N 85
E576	TRTDP_960	TRTDP	TBT_B_D2R1_AUXDDC_P 85
E577	TRTDP_960	TRTDP	TBT_B_D2R1_AUXDDC_N 85

SYNC MASTER=D7 NICK

SYNC DATE=12/13/2011

PAGE TITLE

TBT/DP Constraints



Apple Inc.

DRAWING NUMBER

051-9509

SIZE

D

REVISION

4.2.0

BRANCH

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GDDR5

GDDR5-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
GDDR_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
GDDR_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	12 MM	=STANDARD	=STANDARD
GDDR_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
GDDR_MA_PHY	*	GDDR_45S
GDDR_ADBI_PHY	*	GDDR_45S
GDDR_CTRL_PHY	*	GDDR_45S
GDDR_CLK_PHY	*	GDDR_80D
GDDR_DQ_PHY	*	GDDR_45S
GDDR_EDC_PHY	*	GDDR_45S
GDDR_DBI_PHY	*	GDDR_45S
GDDR_WCK_PHY	*	GDDR_80D

Main Segment Min Spacing Rules for 4.5 Gbps or Less (AMD Doc# 49919)

[illegible]

GDDR5-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_ISO	*	=5x_DIELECTRIC	?
GDDR_ISO	TOP, BOTTOM	=5x_DIELECTRIC	?
GDDR_MA2MA	*	=2x_DIELECTRIC	?
GDDR_MA2MA	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	*	=2x_DIELECTRIC	?
GDDR_ADBI2ADBI	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	*	=2x_DIELECTRIC	?
GDDR_CTRL2CTRL	TOP, BOTTOM	=2x_DIELECTRIC	?
GDDR_CLK2CLK	*	=5x_DIELECTRIC	?
GDDR_CLK2CLK	TOP, BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GDDR_DQ2DQ	*	=3x_DIELECTRIC	?
GDDR_DQ2DQ	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_EDC_ISO	*	=7x_DIELECTRIC	?
GDDR_EDC_ISO	TOP, BOTTOM	=7x_DIELECTRIC	?
GDDR_EDC2EDC	*	=7x_DIELECTRIC	?
GDDR_EDC2EDC	TOP, BOTTOM	=7x_DIELECTRIC	?
GDDR_DBI2DBI	*	=3x_DIELECTRIC	?
GDDR_DBI2DBI	TOP, BOTTOM	=3x_DIELECTRIC	?
GDDR_WCK2WCK	*	=5x_DIELECTRIC	?
GDDR_WCK2WCK	TOP, BOTTOM	=5x_DIELECTRIC	?

Constraints ($x \in \{A, B\}, y \in \{0, 1\}$)

Memory Address: MAXy[8:0]

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_MA	*	*	GDDR_ISO
GDDR_*_*_MA	=SAME	*	GDDR_MA2MA

Address Dynamic Bus Inversion: ADBIx

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_ADBI	*	*	GDDR_ISO
GDDR_*_*_ADBI	=SAME	*	GDDR_ADBI2ADBI

Control: Reset, CKExy, CSxy, WExy, RASxy, CASxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	*	*	GDDR_ISO
GDDR_*_*_CTRL	=SAME	*	GDDR_CTRL2CTRL

Clock: CKxy

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GDDR_*_*_CLK	*	*	GDDR_ISO
GDDR_*_*_CLK	=SAME	*	GDDR_CLK2CLK

GDDR5 Frame Buffer A

Electrical Constraint Set	Physical	Spacing		
Memory Address				
FE09 GDDR A0 MA	GDDR_MA_PHY	GDDR_A_0_MA	FB A0 A<8..0>	73 76
FE0A GDDR A1 MA	GDDR_MA_PHY	GDDR_A_1_MA	FB A1 A<8..0>	73 76
Address Dynamic Bus Inv				
FE0E GDDR A0 ADBT	GDDR_ADBT_PHY	GDDR_A_0_ADBT	FB A0 ABT I	73 76
FE0F GDDR A1 ADBT	GDDR_ADBT_PHY	GDDR_A_1_ADBT	FB A1 ABT I	73 76
Control				
FE09 GDDR A0_CKE	GDDR_CTRI_PHY	GDDR_A_0_CTRI	FB A0 CKE L	73 76
FE0A GDDR A0_CTRI	GDDR_CTRI_PHY	GDDR_A_0_CTRI	FB A0 CS L	73 76
FE0B GDDR A0_CTRI	GDDR_CTRI_PHY	GDDR_A_0_CTRI	FB A0 WE L	73 76
FE0C GDDR A0_CTRI	GDDR_CTRI_PHY	GDDR_A_0_CTRI	FB A0 CAS L	73 76
FE0D GDDR A0_CTRI	GDDR_CTRI_PHY	GDDR_A_0_CTRI	FB A0 RAS L	73 76
FE0E GDDR A1_CKE	GDDR_CTRI_PHY	GDDR_A_1_CTRI	FB A1 CKE L	73 76
FE0F GDDR A1_CTRI	GDDR_CTRI_PHY	GDDR_A_1_CTRI	FB A1 CS L	73 76
FE10 GDDR A1_CTRI	GDDR_CTRI_PHY	GDDR_A_1_CTRI	FB A1 WE L	73 76
FE11 GDDR A1_CTRI	GDDR_CTRI_PHY	GDDR_A_1_CTRI	FB A1 CAS L	73 76
FE12 GDDR A1_CTRI	GDDR_CTRI_PHY	GDDR_A_1_CTRI	FB A1 RAS L	73 76
Clock				
FE09 GDDR A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK P	73 76
FE0A GDDR A0_CLK	GDDR_CLK_PHY	GDDR_A_0_CLK	FB A0 CLK N	73 76
FE0B GDDR A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK P	73 76
FE0C GDDR A1_CLK	GDDR_CLK_PHY	GDDR_A_1_CLK	FB A1 CLK N	73 76
Data				
FE09 GDDR A0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<7..0>	73 76
FE0A GDDR A0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<15..8>	73 76
FE0B GDDR A0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<23..16>	73 76
FE0C GDDR A0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_0_DQ	FB A0 DQ<31..24>	73 76
FE0D GDDR A1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<7..0>	73 76
FE0E GDDR A1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<15..8>	73 76
FE0F GDDR A1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<23..16>	73 76
FE10 GDDR A1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_A_1_DQ	FB A1 DQ<31..24>	73 76
Error Detection				
FE0A GDDR A0_EDC0	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<0>	73 76
FE0B GDDR A0_EDC1	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<1>	73 76
FE0C GDDR A0_EDC2	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<2>	73 76
FE0D GDDR A0_EDC3	GDDR_EDC_PHY	GDDR_A_0_EDC	FB A0 EDC<3>	73 76
FE0E GDDR A1_EDC0	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<0>	73 76
FE0F GDDR A1_EDC1	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<1>	73 76
FE10 GDDR A1_EDC2	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<2>	73 76
FE11 GDDR A1_EDC3	GDDR_EDC_PHY	GDDR_A_1_EDC	FB A1 EDC<3>	73 76
Data Dynamic Bus Inv				
FE09 GDDR A0_DBI0	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI I<0>	73 76
FE0A GDDR A0_DBI1	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI I<1>	73 76
FE0B GDDR A0_DBI2	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI I<2>	73 76
FE0C GDDR A0_DBI3	GDDR_DBI_PHY	GDDR_A_0_DBI	FB A0 DBI I<3>	73 76
FE0D GDDR A1_DBI0	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI I<0>	73 76
FE0E GDDR A1_DBI1	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI I<1>	73 76
FE0F GDDR A1_DBI2	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI I<2>	73 76
FE10 GDDR A1_DBI3	GDDR_DBI_PHY	GDDR_A_1_DBI	FB A1 DBI I<3>	73 76
Forwarded Clock				
FE0B GDDR A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK P<0>	73 76
FE0C GDDR A0_WCK0	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK N<0>	73 76
FE0D GDDR A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK P<1>	73 76
FE0E GDDR A0_WCK1	GDDR_WCK_PHY	GDDR_A_0_WCK	FB A0 WCLK N<1>	73 76
FE0F GDDR A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK P<0>	73 76
FE10 GDDR A1_WCK0	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK N<0>	73 76
FE11 GDDR A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK P<1>	73 76
FE12 GDDR A1_WCK1	GDDR_WCK_PHY	GDDR_A_1_WCK	FB A1 WCLK N<1>	73 76

GPU Misc.

Electrical Constraint Set	Physical	Spacing		
SMB				
	SMB_PHY	SMB	GPU_SMB_CLK	77 78
	SMB_PHY	SMB	GPU_SMB_DAT	77 78
	SMB_PHY	SMB	GPU_SMB_CLK_R	47 78
	SMB_PHY	SMB	GPU_SMB_DAT_R	47 78

GDDR5 Frame Buffer B

Electrical Constraint Set	Physical	Spacing	
Memory Address			
FE00 GDDR_B0_MA	GDDR_MA_PHY	GDDR_R_0_MA	FB_B0 A<8...0>
FE01 GDDR_B1_MA	GDDR_MA_PHY	GDDR_R_1_MA	FB_B1 A<8...0>
Address Dynamic Bus Inv			
FE00 GDDR_B0_ADBT	GDDR_ADBT_PHY	GDDR_R_0_ADBT	FB_B0 ABT I
FE01 GDDR_B1_ADBT	GDDR_ADBT_PHY	GDDR_R_1_ADBT	FB_B1 ABT I
Control			
FE00 GDDR_B0_CKE	GDDR_CTRI_PHY	GDDR_R_0_CTRI	FB_B0 CKE L
FE01 GDDR_B0_CTRI	GDDR_CTRI_PHY	GDDR_R_0_CTRI	FB_B0 CS L
FE02 GDDR_B0_CTRI	GDDR_CTRI_PHY	GDDR_R_0_CTRI	FB_B0 WE L
FE03 GDDR_B0_CTRI	GDDR_CTRI_PHY	GDDR_R_0_CTRI	FB_B0 CAS L
FE04 GDDR_B0_CTRI	GDDR_CTRI_PHY	GDDR_R_0_CTRI	FB_B0 RAS L
FE05 GDDR_B1_CKE	GDDR_CTRI_PHY	GDDR_R_1_CTRI	FB_B1 CKE L
FE06 GDDR_B1_CTRI	GDDR_CTRI_PHY	GDDR_R_1_CTRI	FB_B1 CS L
FE07 GDDR_B1_CTRI	GDDR_CTRI_PHY	GDDR_R_1_CTRI	FB_B1 WE L
FE08 GDDR_B1_CTRI	GDDR_CTRI_PHY	GDDR_R_1_CTRI	FB_B1 CAS L
FE09 GDDR_B1_CTRI	GDDR_CTRI_PHY	GDDR_R_1_CTRI	FB_B1 RAS L
Clock			
FE00 GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_R_0_CLK	FB_B0 CLK P
FE01 GDDR_B0_CLK	GDDR_CLK_PHY	GDDR_R_0_CLK	FB_B0 CLK N
FE02 GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_R_1_CLK	FB_B1 CLK P
FE03 GDDR_B1_CLK	GDDR_CLK_PHY	GDDR_R_1_CLK	FB_B1 CLK N
Data			
FE00 GDDR_B0_DQ_BYTE0	GDDR_DQ_PHY	GDDR_R_0_DQ	FB_B0 DQ<7...0>
FE01 GDDR_B0_DQ_BYTE1	GDDR_DQ_PHY	GDDR_R_0_DQ	FB_B0 DQ<15...8>
FE02 GDDR_B0_DQ_BYTE2	GDDR_DQ_PHY	GDDR_R_0_DQ	FB_B0 DQ<23...16>
FE03 GDDR_B0_DQ_BYTE3	GDDR_DQ_PHY	GDDR_R_0_DQ	FB_B0 DQ<31...24>
FE04 GDDR_B1_DQ_BYTE0	GDDR_DQ_PHY	GDDR_R_1_DQ	FB_B1 DQ<7...0>
FE05 GDDR_B1_DQ_BYTE1	GDDR_DQ_PHY	GDDR_R_1_DQ	FB_B1 DQ<15...8>
FE06 GDDR_B1_DQ_BYTE2	GDDR_DQ_PHY	GDDR_R_1_DQ	FB_B1 DQ<23...16>
FE07 GDDR_B1_DQ_BYTE3	GDDR_DQ_PHY	GDDR_R_1_DQ	FB_B1 DQ<31...24>
Error Detection			
FE00 GDDR_B0_EDC0	GDDR_EDC_PHY	GDDR_R_0_EDC	FB_B0 EDC<0>
FE01 GDDR_B0_EDC1	GDDR_EDC_PHY	GDDR_R_0_EDC	FB_B0 EDC<1>
FE02 GDDR_B0_EDC2	GDDR_EDC_PHY	GDDR_R_0_EDC	FB_B0 EDC<2>
FE03 GDDR_B0_EDC3	GDDR_EDC_PHY	GDDR_R_0_EDC	FB_B0 EDC<3>
FE04 GDDR_B1_EDC0	GDDR_EDC_PHY	GDDR_R_1_EDC	FB_B1 EDC<0>
FE05 GDDR_B1_EDC1	GDDR_EDC_PHY	GDDR_R_1_EDC	FB_B1 EDC<1>
FE06 GDDR_B1_EDC2	GDDR_EDC_PHY	GDDR_R_1_EDC	FB_B1 EDC<2>
FE07 GDDR_B1_EDC3	GDDR_EDC_PHY	GDDR_R_1_EDC	FB_B1 EDC<3>
Data Dynamic Bus Inv			
FE00 GDDR_B0_DBT0	GDDR_DBT_PHY	GDDR_R_0_DBT	FB_B0 DBI L<0>
FE01 GDDR_B0_DBT1	GDDR_DBT_PHY	GDDR_R_0_DBT	FB_B0 DBI L<1>
FE02 GDDR_B0_DBT2	GDDR_DBT_PHY	GDDR_R_0_DBT	FB_B0 DBI L<2>
FE03 GDDR_B0_DBT3	GDDR_DBT_PHY	GDDR_R_0_DBT	FB_B0 DBI L<3>
FE04 GDDR_B1_DBT0	GDDR_DBT_PHY	GDDR_R_1_DBT	FB_B1 DBI L<0>
FE05 GDDR_B1_DBT1	GDDR_DBT_PHY	GDDR_R_1_DBT	FB_B1 DBI L<1>
FE06 GDDR_B1_DBT2	GDDR_DBT_PHY	GDDR_R_1_DBT	FB_B1 DBI L<2>
FE07 GDDR_B1_DBT3	GDDR_DBT_PHY	GDDR_R_1_DBT	FB_B1 DBI L<3>
Forwarded Clock			
FE00 GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_R_0_WCK	FB_B0 WCLK P<0>
FE01 GDDR_B0_WCK0	GDDR_WCK_PHY	GDDR_R_0_WCK	FB_B0 WCLK N<0>
FE02 GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_R_0_WCK	FB_B0 WCLK P<1>
FE03 GDDR_B0_WCK1	GDDR_WCK_PHY	GDDR_R_0_WCK	FB_B0 WCLK N<1>
FE04 GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_R_1_WCK	FB_B1 WCLK P<0>
FE05 GDDR_B1_WCK0	GDDR_WCK_PHY	GDDR_R_1_WCK	FB_B1 WCLK N<0>
FE06 GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_R_1_WCK	FB_B1 WCLK P<1>
FE07 GDDR_B1_WCK1	GDDR_WCK_PHY	GDDR_R_1_WCK	FB_B1 WCLK N<1>

Frame Buffer Reset

Electrical Constraint Set		Physical	Spacing	
Reset				
R269	GDDR_A0_RESET	GDDR_S0S	GDDR_CTRL1	FB_A0 RESET L
R270	GDDR_A1_RESET	GDDR_S0S	GDDR_CTRL1	FB_A1 RESET L
R271	GDDR_B0_RESET	GDDR_S0S	GDDR_CTRL1	FB_B0 RESET L
R272	GDDR_B1_RESET	GDDR_S0S	GDDR_CTRL1	FB_B1 RESET L

SYNCH MASTER=D7 DAVE		SYNCH DATE=12/12/2011	
PAGE TITLE			
GDDR5/GPU Constraints			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-9509	D
		REVISION	
		4.2.0	
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